



FINANCIAL ADVISORY SERVICES FOR DORJILUNG HYDROPOWER PROJECT

DRAFT PRELIMINARY INVESTMENT REPORT

Draft for Discussion
May 2025

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY	1
2. PROGRESS SO FAR	3
3. OUR APPROACH TO THE STUDY	6
4. PROFILE OF DORJILUNG HYDROPOWER PLANT	8
5. INDIAN POWER MARKET SCENARIO	12
6. INDIAN POWER MARKET – OFFTAKE ASSESSMENT	21
7. BANGLADESH POWER SECTOR LANDSCAPE	101
8. REGULATORY LANDSCAPE IN INDIA AND BANGLADESH	105
9. ELECTRICITY EXPORT RISK ANALYSIS.....	116
10. TRANSMISSION INFRASTRUCTURE BETWEEN INDIA, BHUTAN AND BANGLADESH ..	119
11. EVOLVING TARIFF TRENDS IN THE INDIAN ELECTRICITY MARKET	121
12. IDENTIFYING THE TRENDS OF HYDROPOWER FINANCING AND INVESTMENT IN BHUTAN	123
13. PROJECT RISKS - BENCHMARKING AND MITIGATIONS.....	127
14. EVALUATION OF FINANCING STRUCTURE MODELS FOR DHPP	133
15. NEXT STEPS.....	143

LIST OF FIGURES

Figure 1: Generational profile of DHPP at P50 and P90 probabilities.....	8
Figure 2: Projected Demand and Supply in Bhutan Power Market after commissioning of DHPP	10
Figure 3: Map of BBINS (not to scale).....	10
Figure 4: India Electricity Demand and Supply	12
Figure 5: India Electricity Peak Demand and Peak Supply	12
Figure 6: Future Electricity Demand Drivers in India	13
Figure 7: Source wise Installed Capacity (in MW) in India	14
Figure 8: India Electricity Sector Forecast Observations	14
Figure 9: India's RPO trajectory till 2030.....	15
Figure 10: State wise RPO targets (in GWh) by FY 2031-32.....	16
Figure 11: Northern Region monthly energy profile of FY 2024-25	17
Figure 12: Southern Region monthly energy profile of FY 2024-25	18
Figure 13: Western Region monthly energy profile of FY 2024-25	18
Figure 14: Eastern Region monthly energy profile of FY 2024-25	19
Figure 15: North-Eastern Region monthly energy profile of FY 2024-25	20
Figure 16: Fuel wise capacity mix of Maharashtra	24
Figure 17: Snapshot of PRAAPTI – Maharashtra	24
Figure 18: Capacity Addition Plan for next 10 years of Maharashtra.....	25
Figure 19: Merit Order Analysis of Maharashtra	26
Figure 20: Fuel wise capacity mix of Gujarat	27
Figure 21: Snapshot of PRAAPTI - Gujarat.....	27
Figure 22: Capacity Addition Plan for next 10 years of Gujarat.....	28
Figure 23: Merit Order Analysis of Gujarat	29
Figure 24: Fuel-wise capacity mix of Madhya Pradesh.....	30
Figure 25: Snapshot of PRAAPTI- Madhya Pradesh	30
Figure 26: Capacity Addition plan for next 10 years of Madhya Pradesh	31
Figure 27: Merit Order Analysis of Madhya Pradesh	32
Figure 28: Fuel-wise capacity mix of Chhattisgarh.....	33
Figure 29: Snapshot of PRAAPTI -Chhattisgarh	34
Figure 30: Capacity Addition Plan for next 10 years of Chhattisgarh	35
Figure 31: Merit Order Analysis of Chhattisgarh	35
Figure 32: Fuel-wise capacity mix of West Bengal	37
Figure 33: Snapshot of PRAAPTI - West Bengal.....	38
Figure 34: Capacity addition plans for next 10 years of West Bengal	39
Figure 35: Merit Order Analysis of West Bengal.....	39
Figure 36: Fuel-wise capacity mix of Jharkhand	40
Figure 37: Snapshot of PRAAPTI - Jharkhand.....	41
Figure 38: Capacity addition plan for next 10 years of Jharkhand	42
Figure 39: Merit Order Analysis of Jharkhand	42
Figure 40: Fuel-wise capacity mix of Bihar	43
Figure 41: Snapshot of PRAAPTI -Bihar.....	44
Figure 42: Capacity addition plan for next 10 years of Bihar	45
Figure 43: Merit Order Analysis of Bihar.....	45
Figure 44: Fuel-wise capacity mix of Odisha	46
Figure 45: Snapshot of PRAAPTI - Odisha	47
Figure 46: Capacity addition plan for next 10 years of Odisha.....	48
Figure 47: Merit Order Analysis of Odisha	48
Figure 48: Fuel-wise capacity mix of Assam*	50
Figure 49: Snapshot of PRAAPTI - Assam	50
Figure 50: Capacity Addition Plan for next 10 years of Assam	51
Figure 51: Merit Order Analysis of Assam	52

Figure 52: Fuel-wise capacity mix of Uttar Pradesh*	55
Figure 53: Snapshot of PRAAPTI - Uttar Pradesh	55
Figure 54: Capacity Addition Plan for next 10 years of Uttar Pradesh	56
Figure 55: Merit Order Analysis of Uttar Pradesh	57
Figure 56: Capacity Mix of Rajasthan*	58
Figure 57: Snapshot of PRAAPTI - Rajasthan	59
Figure 58: Capacity addition plan for next 10 years of Rajasthan	60
Figure 59: Merit Order Analysis of Rajasthan	60
Figure 60: Fuel-wise capacity mix of Punjab*	61
Figure 61: Snapshot of PRAAPTI - Punjab	62
Figure 62: Capacity Addition Plan for next 10 years of Punjab	63
Figure 63: Merit Order Analysis of Punjab	63
Figure 64: Fuel-wise capacity mix of Haryana*	64
Figure 65: Snapshot of PRAAPTI - Haryana	65
Figure 66: Capacity Addition Plan for next 10 years of Haryana	66
Figure 67: Merit Order Analysis of Haryana	66
Figure 68: Capacity Mix of BRPL – Delhi*	67
Figure 69: Capacity Mix of BYPL- Delhi*	68
Figure 70: Capacity Mix of TPDDL - Delhi*	68
Figure 71: Snapshot of PRAAPTI - Delhi	69
Figure 72: Capacity Addition Plan for next 10 years of BRPL	70
Figure 73: Capacity Addition Plan for next 10 years of BYPL	70
Figure 74: Capacity Addition Plan for next 10 years of TPDDL	71
Figure 75: Merit Order Analysis – BRPL	71
Figure 76: Merit Order Analysis – BYPL	72
Figure 77: Merit Order Analysis - TPDDL	72
Figure 78: Fuel-wise capacity mix of Tamil Nadu*	75
Figure 79: Snapshot of PRAAPTI Portal - Tamil Nadu	75
Figure 80: Capacity Addition for the next 10 years of Tamil Nadu	76
Figure 81: Merit Order Analysis of Tamil Nadu	77
Figure 82: Fuel-wise capacity mix of Telangana*	78
Figure 83: Snapshot of PRAAPTI Portal - Telangana	78
Figure 84: Capacity Addition for next 10 years of Telangana	79
Figure 85: Merit Order Analysis of Telangana	80
Figure 86: Fuel-wise capacity mix of Andhra Pradesh*	81
Figure 87: Snapshot of PRAAPTI portal – Andhra Pradesh	81
Figure 88: Capacity Addition Plan for next 10 years of Andhra Pradesh	82
Figure 89: Merit Order Analysis of Andhra Pradesh	83
Figure 90: Fuel-wise capacity mix of Karnataka*	84
Figure 91: Snapshot of PRAAPTI portal – Karnataka	85
Figure 92: Capacity Addition Plan for the next 8 years of Karnataka	86
Figure 93: Merit Order Analysis of Karnataka	86
Figure 94: Key Drivers impacting Short/Medium Term Power Markets	89
Figure 95: Market Segments in the Short/Medium -Term Power Market	89
Figure 96: Share of different market segments in Short-Term Power Transactions in 2023-24	89
Figure 97: Trend of volume of transaction through Traders and Power Exchanges	90
Figure 98: Trend of weighted average price of transaction through Traders and Power Exchanges	90
Figure 99: Power sold in IEX (DAM and RTM) by Nepal and Bhutan	91
Figure 100: Monthly FSV and MCP in DAM of IEX during last 3 financial years	92
Figure 101: Monthly FSV and MCP in RTM of IEX during last 3 financial years	92
Figure 102: Hourly FSV and MCP in DAM of IEX during last 3 financial years from May to November	93
Figure 103: Hourly FSV and MCP in RTM of IEX during last 3 financial years from May to November	93
Figure 104: Category wise Electricity Consumption in TWh in India	96
Figure 105: ABR (in INR/kWh) of top 10 States for Commercial Consumers	98

Figure 106: ABR (in INR/kWh) of top 10 States for Industrial Consumers.....	98
Figure 107: List of Offtake Arrangement of Hydro Projects	99
Figure 108 : Power Sector Snapshot of Bangladesh	101
Figure 109 : Electricity Peak Demand and Peak Supply (GW).....	101
Figure 110 : Installed Capacity trend of Bangladesh by fuel type (MW).....	102
Figure 111: Per Unit cost of electricity generation and purchase by plant type in last 3 years (INR/kWh)	103
Figure 112 : Per Unit cost of fuel for BPDB's electricity generation in last 3 years (INR/kWh).....	103
Figure 113 : Electricity Demand and Supply forecast (in TWh)	104
Figure 114: Key Instruments guiding Bangladesh Cross Border Power Trade	107
Figure 115: Source wise Electricity tariff (INR/kWh) in last 8 years	121
Figure 116: Funding structure of Ruzizi III Hydroelectric PowerStation - Africa	138
Figure 117: Funding structure of Nam Ngiep 1 Hydropower Project – Laos	139
Figure 118: Funding structure of Xayaburi Dam Project - Laos.....	141

LIST OF TABLES

Table 1: Study Focus Areas and Objectives	6
Table 2: Detailed Approach & Methodology	6
Table 3: Month-wise Generational Profile of DHPP at P-90	9
Table 4: Month-wise Generational Profile of DHPP at P-50	9
Table 5: India: Electricity demand forecast*	13
Table 6: Region wise projection for Demand and Supply	16
Table 7: Summary of Region wise Energy and Peak Deficits in india	20
Table 8: Parameters considered for Analyzing DISCOMs	21
Table 9 : State-wise Gross Input Energy	22
Table 10: State-level DISCOM's Gross Input Energy	22
Table 11: Energy Demand of Maharashtra	23
Table 12: Performance Report Snapshot	25
Table 13: Energy Demand of Gujarat	26
Table 14: Performance Report Snapshot	28
Table 15: Energy Demand of Madhya Pradesh	29
Table 16: Performance Snapshot Report	31
Table 17: Energy Demand of Chhattisgarh	33
Table 18: Performance Report Snapshot	34
Table 19: State-wise Gross Input Energy	36
Table 20: State-level DISCOM's Gross Input Energy	36
Table 21: Energy Demand of West Bengal	37
Table 22 : Performance Report Snapshot	38
Table 23: Energy Demand of Jharkhand	40
Table 24: Performance Report Snapshot	41
Table 25: Energy Demand of Bihar	43
Table 26: Performance Report Snapshot	44
Table 27: Energy Demand of Odisha	46
Table 28: Performance Snapshot Report	47
Table 29: State-wise Gross Input Energy	49
Table 30: Energy Demand of Assam	49
Table 31: Performance Report Snapshot	51
Table 32: State-wise Gross Input Energy	53
Table 33: State-level DISCOM's Gross Input Energy	53
Table 34: Energy Demand of Uttar Pradesh	54
Table 35: Performance Snapshot Report	56
Table 36: Energy Demand of Rajasthan	58
Table 37: Performance Snapshot Report	59
Table 38: Energy Demand of Punjab	61
Table 39: Performance Snapshot Report	62
Table 40: Energy Demand of Haryana	64
Table 41: Performance Snapshot Report	65
Table 42: Energy Demand of Delhi	67
Table 43: Performance Snapshot Report	69
Table 44: State-wise Gross Input Energy	73
Table 45: State-level DISCOM's Gross Input Energy	73
Table 46: Energy Demand of Tamil Nadu	74
Table 47: Performance Snapshot Report	75
Table 48 : Energy Demand of Telengana	77
Table 49: Performance Snapshot Report	78
Table 50: Energy Demand of Andhra Pradesh	80
Table 51: Performance Snapshot Report	82
Table 52: Energy Demand of Karnataka	84
Table 53: Performance Snapshot Report	85

Table 54: Summary of Discom Offtake Analysis	87
Table 55: Power Exchange Transactions of Bhutan recorded in IEX (DAM and RTM)	91
Table 56: Bilateral transactions in short and medium term	94
Table 57: Power procurement pattern in the Short term bilateral transactions	95
Table 58: Share of Renewable Energy of major C&I consumers in India	96
Table 59: The detailed list of key Hydro project is provided in the table below	99
Table 60: Import of power by Bangladesh in FY2024	102
Table 61: BIMSTEC MoU Agreements	105
Table 62: Normative Parameters as per CERC Tariff Regulation, 2024-29.....	109
Table 63: Key Parameters of similar Hydro Projects in India	110
Table 64: Electricity Export Risk Analysis	116
Table 65: Power Trade Agreements between South Asia	119
Table 66: Details of Transmission Infrastructure between India and Bangladesh	120
Table 67: Evolving Trends affecting Pricing In India.....	121
Table 68: Envisaged Funding Structure of the Ruzizi III Hydroelectric PowerStation Project.....	137
Table 69: Funding Commitments by Development Agencies/ Multilateral Organizations	138
Table 70: Funding structure of Nam Ngiep 1 Hydropower Project– Laos	139
Table 71: Funding structure of Xayaburi Dam Project – Laos	140
Table 72: Suitability of Different Financing Structures for DHPP's Development	142

ABBREVIATIONS

Acronym	Definition
ABR	Aggregate Billing Rate
ACS	Average Cost of Supply
ADB	Asian Development Bank
ADF	Asian Development Fund
ADFB	African Development Fund Board
AFC	Annual Fixed Charges
AIIB	Asian Infrastructure Investment Bank
AKFED	Aga Khan Fund for Economic Development
APCPDCL	Andhra Pradesh Central Power Distribution Company Limited
APDCL	Assam Power Distribution Company Limited
APEPDCL	Andhra Pradesh Eastern Power Distribution Company Limited
APR	Annual Performance Review
APSPDCL	Andhra Pradesh Southern Power Distribution Company Limited
ARR	Aggregate Revenue Requirement
ASA	Average System Availability
AVVNL	Ajmer Vidyut Vitran Nigam Limited
B2B	Business to Business
BAU	Business As Usual
BBIN	Bangladesh, Bhutan, India and Nepal
BBINS	Bangladesh, Bhutan, India, Nepal and Sri Lanka
BDT	Bangladeshi Taka
BEA	Bhutan Electricity Authority
BEC	Bulk Electric Consumer
BERC	Bihar Electricity Regulatory Commission
BESCOM	Bangalore Electricity Supply Company
BESS	Battery Energy Storage System
BGICC	Bharat Gas Infrastructure Corporation
BHP	Bangladesh Energy Regulatory Commission
BIMSTEC	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
BLTPS	Bharat Aluminium Company Limited Thermal Power Station
BMP	Best Management Practices
BOOT	Build Own Operate Transfer
BOT	Build Operate Transfer
BPC	Bhutan Power Corporation
BPDB	Bangladesh Power Development Board
BRPL	BSES Rajdhani Power Limited
BRSR	Business Responsibility and Sustainability Report

Acronym	Definition
BSES	Bombay Suburban Electric Supply
BSPHCL	Bihar State Power Holding Company Limited
BYPL	BSES Yamuna Power Limited
CAGR	Compound Annual Growth Rate
CBET	Cross Border Electricity Trade
CCPP	Combined Cycle Power Plant
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CESCO	Central Electricity Supply Company
CHESCOM	Chamundeshwari Electricity Supply Corporation Limited
CHMP	Capital Health Management Plan
CHP	Combined Heat and Power
CO2	Carbon Dioxide
COD	Commercial Operation Date
COP26	26th Conference of the Parties
CSPDCL	Chhattisgarh State Power Distribution Company Limited
CSS	Central Sector Scheme
DAM	Day Ahead market
DBFOT	Design Build Finance Operate Transfer
DECC	Department of Energy and Climate Change
DERC	Delhi Electricity Regulatory Commission
DGPC	Druk Green Power Corporation
DGVCL	Dakshin Gujarat Vij Company Limited
DHBVN	Dakshin Haryana Bijli Vitran Nigam
DHEP	Doyang Hydro Electric Plant
DHI	Druk Holdings & Investments
DHP	Dagachhu Hydropower Plant
DHPP	Dorjilung Hydropower Project
DoE	Department of Energy
DISCOM	Distribution Company
DPR	Detailed Project Report
DSCR	Debt Service Coverage Ratio
DSRA	Debt Service Reserve Account
DTTILLP	Deloitte Touche Tohmatsu India LLP
DTTL	Deloitte Touche Tohmatsu Limited
DVC	Damodar Valley Corporation
DVVNL	Dakshinanchal Vidyut Vitran Nigam Limited
EFMP	Environmental Flow Management Plan
EGAT	Electricity Generating Authority of Thailand
EGCO	Electricity Generating Public Company Limited
EIB	European Investment Bank
EPC	Engineering, Procurement, and Construction

Acronym	Definition
EPS	Electric Power Survey
ERA	Energy Regulatory Authority
ESIA	Environmental and Social Impact Assessment
EUR	EURO
EURIBOR	Euro Interbank Offered Rate
FDI	Foreign Direct Investment
FI	Financial Institution
FSV	Final Scheduled volume
FY	Financial Year
G2G	Government-to-Government
GDAM	Green Day Ahead Market
GDP	Gross Domestic Product
GoA	Government of Austria
GESCOM	Gulbarga Electricity Supply Company
GHG	Greenhouse Gas
GMR	Geometrical Mean Radius
GOI	Government of India
GRIDCO	Grid Corporation of Odisha Limited
GSECL	Gujarat State Electricity Corporation Limited
GT	Gas Turbine
GTPS	Gas Turbine Power Station
GUVNL	Gujarat Urja Vikas Nigam Limited
GW	Gigawatt
HDSC	Hydro Debt Service Coverage
HEP	Hydro Electric Power
HESCOM	Hubli Electricity Supply Company
HFO	Heavy Fuel Oil
HPGCL	Haryana Power Generation Corporation Limited
HPO	Hydropower Purchase Obligations
HPP	Hydropower projects
HPPCL	Himachal Pradesh Power Corporation Limited
HPX	Hindustan Power Exchange Limited
HSPL	Himachal Sorang Power Limited
HV	High Voltage
HVAC	High Voltage Alternating Current
HVDC	High Voltage Direct Current
IBRD	International Bank for Reconstruction and Development
ICR	Interest Coverage Ratio
ICRA	Investment Information and Credit Rating Agency
IDA	International Development Association
IDC	Interest During Construction
IEA	International Energy Agency

Acronym	Definition
IEDC	Interest Expenses During Construction
IEPMP	Integrated Energy and Power Master Plan
IEX	Indian Energy Exchange
IFC	International Finance Corporation
IFIs	International Financial Institutions
INR	Indian Rupee
InVIT	Infrastructure Investment Trust
IPS	Industrial Promotion Services
IRR	Internal Rate of Return
ISGS	Inter-State Generating Station
JBIC	Japan Bank for International Cooperation
JBVNL	Jharkhand Bijli Vitran Nigam Limited
JICA	Japan International Cooperation Agency
JV	Joint Venture
KESCO	Kanpur Electricity Supply Company
KFAED	Kuwait Fund for Arab Economic Development
KHP	Kurichhu Hydropower plant
KHPL	Khorlochhu Hydropower
KPIC	Kshema Power and Infrastructure Company Private Limited
KW	Kilowatt
LALRP	Land Acquisition and Livelihood Restoration Plan
LC	Letter of Credit
LGBR	Load Generation Balance Report
LHPP	Large Hydro Power Plant
LLCR	Loan Life Coverage Ratio
LLP	Limited Liability Partnership
LS	Lower Stage
MBED	Market-Based Economic Dispatch
MCLR	Marginal Cost of Funds-based Lending Rate
MCP	Market Clearing Price
MDB	Multilateral Development Bank
MGVCL	Madhya Gujarat Vij Company Limited
MIPA	Moveable and Immovable Property Act
MLA	Multilateral Agencies
Mn	Million
MoENR	Ministry of Energy & Natural Resources
MoF	Ministry of Finance
MoU	Memorandum of Understanding
MNRE	Ministry of New and Renewable Energy
MPMKVVCL	Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Limited

Acronym	Definition
MPPKVVCL	Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited
MPPMCL	Madhya Pradesh Power Management Company Limited
MSEDCL	Maharashtra State Electricity Distribution Company Limited
MSTC	Metal Scrap Trade Corporation
MT	Million Tonnes
MU	Million Units
MV	Medium Voltage
MVVNL	Madhyanchal Vidyut Vitaran Nigam Limited
MW	Megawatt
MYT	Multi Year Tariff
NBPDC	North Bihar Power Distribution Company Ltd
NEA	Nepal Electricity Authority
NEEPCO	North Eastern Electric Power Corporation Limited
NESCO	North Eastern Electricity Supply Company of Odisha Limited
NHP	Nikachhu Hydropower Plant
NHPC	National Hydroelectric Power Corporation
NLC	National Land Commission
NOC	No Objection Clearances
NPL	Nabha Power Limited
NPPF	National Pension and Provident Fund of Bhutan
NTGMP	National Transmission Grid Master Plan
Nu	Ngultrum
NTPC	National Thermal Power Corporation
NVVN	NTPC Vidyut Vyapar Nigam Ltd
OCF	Operational Cash Flow
OCR	Ordinary Capital Resources
OPEC	Organization of the Petroleum Exporting Countries
PAF	Plant Availability Factor
PAT	Profit After Tax
PFC	Power Finance Corporation
PGVCL	Paschim Gujarat Vij Company Limited
PLCR	Project Life Coverage Ratio
PLF	Plant Load Factor
PLN	Perusahaan Listrik Negara
PMF	Probable Maximum Flood
PPA	Power Purchase Agreement
PSA	Power Sales Agreement
PSP	Pumped Storage Plant
PTC	Power Trading Corporation
PVVNL	Paschimanchal Vidyut Vitran Nigam Limited

Acronym	Definition
RE	Renewable Energy
REL	Ruzizi III Energy Limited
RGOB	Royal Government of Bhutan
RMA	Royal Monetary Authority
RPO	Renewable Purchase Obligation
RSPCL	Rajasthan State Power Corporation Limited
RTM	Real-Time Market
RTS	Roof-Top Solar
RZB	Raiffeisen Bank International
SAARC	South Asian Association for Regional Cooperation
SAIL	Steel Authority of India Limited
SASEC	South Asia Subregional Economic Cooperation
SBI	State bank of India
SBPDCL	South Bihar Power Distribution Company Ltd
SECI	Solar Energy Corporation of India
SERC	State Electricity Regulatory Commission
SHDP	Sustainable Hydropower Development Project
SHPP	Small hydropower projects
SJVNL	Satluj Jal Vidyut Nigam Limited
SLDC	State Load Dispatch Centre
SNP	SN Power AS
SO	System Operator
SOFR	Secured Overnight Financing Rate
SPV	Special Purpose Vehicle
TANGEDCO	Tamil Nadu Generation and Distribution Corporation
THDC	Tehri Hydro Development Corporation Limited
THPP	Tala Hydropower Plant
TNEB	Tamil Nadu Electricity Board
TNW	Tangible Net Worth
TOL	Total Outside Liabilities
TPCL	Tata Power Company Limited
TPDDL	Tata Power Delhi Distribution Limited
TPS	Thermal Power Station
TPTCL	Tata Power Trading Company Limited
TSERC	Telangana State Electricity Regulatory Commission
UDAY	Ujwal DISCOM Assurance Yojana
UHBVN	Uttar Haryana Bijli Vitran Nigam
UPPCL	Uttar Pradesh Power Corporation Limited
USD	United States Dollar
US	Upper Stage
VAT	Value Added Tax
VRE	Variable Renewable Energy

Acronym	Definition
WACC	Weighted Average Cost of Capital
WBSEDCL	West Bengal State Electricity Distribution Company Limited
WESCO	Western Electricity Supply Company of Orissa
WR	Western Region

1. Executive Summary

About DHPP	This sections sets the context for this study by assessing the generation profile curve of DHPP under both P50 and P90 scenario and understanding the monthly quantum available for exports in South Asian region which is 85% of the total generation occurring in the month from May to November. The key countries for potential offtake of power from DHPP is India due to its scale and also Bangladesh due to already established cross border trade between India and Bangladesh.
Indian Power Market Scenario	The offtake assessment begins with an analysis of the Indian power market, where demand–supply is projected to grow steadily at approximately 4% annually until FY 2042. This is followed by a deeper regional-level demand–supply profiling to understand energy and peak deficits across different regions. Such insights are critical before evaluating various offtake options, including through State DISCOMs, sale through merchant markets or C&I consumers.
Indian Power Market- Offtake Assessment	Offtake options have been thoroughly examined through a detailed analysis of Indian DISCOMs, evaluating the performance of individual states and their respective DISCOMs. This assessment centers on key parameters to develop a comprehensive understanding of each DISCOM’s viability and reliability as an offtake partner. In addition to long-term DISCOM arrangements, participation in short-term power markets and bilateral trades has also been considered as alternative offtake channels. Furthermore, the growth potential of the Commercial & Industrial (C&I) segment has been analyzed by examining electricity consumption patterns of major corporate houses alongside the average industrial billing rates across states. This analysis provides insight into the market opportunity within this segment. Additionally, we have benchmarked offtake structures of several major hydro projects, ranging from government-to-government (G2G) to fully private models. It was observed that government-backed projects in hydro rich state such as Himachal Pradesh may allocate up to 30% of their offtake through the short-term market, whereas private hydro projects may rely on the short-term market for as much as 100% of their power sales.
Bangladesh Power Sector Landscape	The power market assessment of Bangladesh shows that the the country has historically been reliant on gas which contributes over ~50% of its total generation capacity. Considering the depleting natural gas reserves and to ensure energy security for the country the Government of Bangladesh has also been increasing power imports from neighbouring countries over the years.
Regulatory Landscape in South Asia	In this chapter, we examine the regulatory landscape across the South Asian region, with a particular focus on cross-border electricity trade (CBET) regulations and the tariff determination processes. The Indian regulatory framework has been analyzed in detail, including case studies highlighting instances where regulators have disallowed certain cost components—providing important insights into the potential regulatory risks. Additionally, the tariff determination process in Bangladesh has also been studied to identify regulatory considerations. These insights help outline the key regulatory cautions that DHPP should be aware of when planning power exports to these markets.

Export Risk Analysis	Cross border electricity trade entails several key risks that need careful management. Curtailment risk arises from transmission constraints and demand fluctuations, potentially limiting the ability to fully evacuate power. Currency risk may also be significant due to revenue and cost mismatches when dealing with foreign exchange fluctuations in cross-border transactions. Additionally, participation in the merchant power market exposes projects to price volatility and market liquidity challenges.
Transmission Infrastructure in South Asia	The transmission infrastructure section provides a comprehensive overview of the current status and future plans for transmission lines connecting India, Bhutan, and Bangladesh. This study includes key existing interconnections, their capacity, and planned upgrades or new projects aimed at enhancing cross-border power flow. Understanding this infrastructure is critical for developing an effective evacuation strategy for DHPP, ensuring that generated power can be reliably transmitted to offtake markets while minimizing bottlenecks and curtailment risks.
Evolving Tariff Trends in the Indian Electricity Market	In this section we have assessed the historical tariff trends in India as well as the major developments in the sector governing future pricing outlooks.
Evaluation of hydropower financing trends, risks and potential structure models for DHPP	To determine an optimal financial structure for DHPP, a study of Bhutan's hydropower financing evolution was conducted. Historically, Bhutan relied on bilateral funding from India for major projects like Chukha, Kurichhu, Tala, and Mangdechhu. Post-2010, DGPC began diversifying funding sources to reduce dependence on sovereign support, engaging multilateral institutions, private investors, and export credit agencies. The Nikachhu project marked Bhutan's first fully public-funded hydropower initiative. International case studies such as Ruzizi III (Africa) and Nam Ngiep 1 (Laos) were also analyzed. Lenders typically assess technical, environmental, geological, and commercial aspects before financing. Legally, DHPP can borrow from foreign lenders with prior government approval under Bhutan's MIPA Act, which also governs the enforceability of security interests and step-in rights. The financing process is expected to take 5–7 months post-PPA finalization. Key risks—such as land acquisition, geological surprises, hydrology shocks, and cost overruns—have been identified, with mitigation strategies in place based on lessons from similar projects.

2. Progress So Far

2.1. Key insights from Inception Report

The Inception Report offered a comprehensive overview of Bhutan's macroeconomic environment and the power sector, detailing its institutional framework and key policies. It highlighted the importance and unique features of the Dorjilung Hydropower Project (DHPP), while also analyzing the key dynamics of South Asian regional power market. The report also covered Bhutan's evolving financial strategy for large hydropower projects, noting a shift from traditional bilateral support to a more diversified approach involving joint ventures and partnerships with international financial institution.

Bhutan's economy is significantly driven by its hydropower sector, which contributes substantially to GDP, export revenues, and overall fiscal stability. The country possesses a vast hydropower potential, estimated at 37,000 MW¹, with approximately 2,453 MW currently installed. The development of this hydropower potential has demonstrably influenced key macroeconomic indicators. Historical data over the past two decades indicate a consistent positive correlation between the commissioning of major hydropower projects and subsequent increases in GDP, typically ranging from 2% to 4%². A notable instance is the Tala Hydropower Plant, the commissioning of which precipitated a 10% rise in GDP. Such projects not only augment domestic electricity supply and enhance export revenues but also catalyze broader economic activity, thereby contributing to an improved trade balance. This pronounced reliance on hydropower development, while economically beneficial, also presents distinct financial considerations. The nation's debt-to-GDP ratio is elevated, standing at approximately 109.8%, with investments in hydropower projects constituting a significant component of the external debt portfolio (64%). However, the operational profitability of these hydropower assets has proven sufficient to service the associated debt obligations, as evidenced by a Hydro Debt Service Coverage (HDSC) ratio of 1.9 in Fiscal Year 2023, which is projected to improve further. Furthermore, hydropower generation contributes directly to governmental revenues through the provision of royalty energy, representing 21.1% of total tax revenue in 2023, thereby supporting public fiscal capacity.

To effectively govern and strategically expand this pivotal sector, Bhutan has established a comprehensive institutional and regulatory architecture. A significant restructuring of the power sector occurred in 2002, which unbundled the previously vertically integrated system, clearly demarcating responsibilities for policy formulation, regulatory oversight, and operational management. The Ministry of Energy & Natural Resources, through its Department of Energy, assumes the central role in policy development and strategic planning. Regulatory functions are vested in the Energy Regulatory Authority (formerly the Bhutan Electricity Authority), while state-owned enterprises, notably Druk Green Power Corporation Limited (DGPC) for generation and Bhutan Power Corporation Limited (BPC) for transmission and distribution, manage operational aspects. This framework is further reinforced by a suite of policies aimed at promoting sustainable hydropower development, encouraging the diversification of renewable energy sources, and ensuring the establishment of equitable and transparent tariff mechanisms.

The efficacy of these institutional arrangements is continually tested by evolving sectoral dynamics, particularly the rapid escalation in domestic electricity demand. Over the last five years, domestic consumption has increased at an exceptional rate of approximately 38%³, with threefold increase in peak demand to 1,026 MW³ in FY 2024. This surge is primarily attributable to the expansion of energy-intensive industries and the development of new industrial parks. While Bhutan's overall hydropower generation has remained relatively stable, the absence of significant new capacity additions in recent years, juxtaposed with rising domestic requirements, has necessitated an increase in electricity imports from India, particularly during the lean winter months when hydrological conditions reduce domestic generation capacity.

¹ [Bhutan Energy Directory 2022](#)

² [World Bank, GDP Growth, 2005-2011 and 2012-2024](#)

³ DGPC Additional Data Workbook

In response to these evolving demand-supply dynamics and to harness its untapped potential, Bhutan has formulated ambitious strategic initiatives for capacity augmentation. The national energy plan envisages the addition of approximately 23,500 MW of hydropower capacity and 5,000 MW of solar capacity by the Fiscal Year 2040⁴. Central to this expansion strategy are projects such as the Dorjilung Hydropower Project (DHPP). With an expected installed capacity of 1,125 MW, DHPP is designed not only to address domestic energy deficits during periods of low supply but also to generate a substantial surplus for export. The project's strategic location on the Kurichhu river, coupled with a meticulously planned transmission evacuation system, is intended to ensure the reliable conveyance of generated power to both domestic consumers and regional export markets.

The ultimate viability and economic contribution of such large-scale projects are intrinsically linked to the ability to secure stable and remunerative offtake arrangements within the broader South Asian regional power market. Bhutan's hydropower exports are strategically positioned to cater to the escalating energy demands of neighboring countries. India, for instance, is pursuing an aggressive expansion of its renewable energy capacity and has instituted Renewable Purchase Obligations, including specific mandates for hydropower, thereby creating a receptive market for Bhutanese electricity. Similarly, Bangladesh, which currently relies heavily on expensive imported fossil fuels, presents a significant potential market for cleaner and more cost-effective hydropower imports. The emergence of Nepal as another regional hydropower exporter introduces a competitive element that underscores the importance of timely project execution and market engagement for Bhutan.

The substantial capital investment required for the development of these large-scale hydropower facilities necessitates financing strategies. The capacity of Bhutan's domestic banking sector is insufficient to independently finance projects of this magnitude. Consequently, Bhutan has progressively diversified its financing modalities, moving beyond traditional bilateral arrangements to embrace joint ventures, public-private partnerships (PPPs), and concessional funding from international financial institutions such as the World Bank and the Asian Development Bank, as well as commercial lending from Indian banks. This multifaceted financing approach, which is also envisaged for the Dorjilung Hydropower Project, aims to achieve a balance between ensuring financial sustainability and facilitating the accelerated development of the nation's hydropower resources.

DGPC, the sole hydropower operator in Bhutan, is a wholly owned subsidiary of Druk Holdings & Investments – a Royal Government of Bhutan (RGoB) company incorporated for holding all government owned enterprises in Bhutan across multiple sectors. India based rating agency ICRA, subsidiary of Moody's Ratings, has assigned DGPC with an issuer rating of BBB+ with a stable outlook on March 13, 2024.

DGPC currently operates at an installed capacity of 2,453 MW with its projects housed under holding company i.e. DGPC (2,209 MW) and standalone subsidiaries (224 MW- Dahgachhu Hydropower Corporation Limited and Tangsbji Hydro Energy Ltd.). Druk Hydro Energy Limited (100% subsidiary) was incorporated in December 2021 to house small and medium scale hydropower projects. To support the hydropower projects, the company has incorporated Bhutan Automation Engineering Limited and Bhutan Hydro Power Services Ltd. to offer a portfolio of critical and specialized services.

DGPC has developed comprehensive business plan for multiple hydropower and solar projects to expand its footprint in the renewable energy sector and increase its capacity over the next 5 years. The financial structuring of these projects is based on an optimal debt-to-equity mix, ensuring financial sustainability while facilitating efficient capital mobilization.

The company currently has a total operating capacity of 2,453 MW hydropower projects housed either directly under DGPC or DGPC controlled subsidiaries/joint ventures. The existing operational capacity is projected to increase to 3,547 MW, that is almost ~1.5x times of the existing capacity over the period

⁴ DGPC Energy Data Excel Workbook

of 5 years i.e., till 2030. This increase in operational capacity constitutes capacity addition of 494 MW and 600 MW in SHPP and LHPP (Khorluchhu HPP) respectively.

While hydropower remains the backbone of Bhutan's energy sector, DGPC is actively diversifying into solar energy to create a more resilient and sustainable energy mix. Recognizing the potential of solar power as a complimentary source of hydropower, DGPC has initiated the development of solar projects across locations. This diversification enhances energy security by ensuring stable supply during dry seasons when hydropower generation may fluctuate. DGPC has projected to develop a total of 965 MW of the operational solar capacity by 2030. The estimated project cost of these 16 solar projects is c. Nu 59,138 Nu with debt equity ratio of 70:30.

Under its five-year plan, DGPC envisages to undertake a capex of ~ Nu. 498,538 million encompassing development of 8.9 GW of under-construction/to be operational SHPPs, LHPPs and Solar power plants. On a cumulative basis this capex shall be funded on a Debt:Equity Ratio of 70:30, translating into a debt and equity infusion requirement of Nu. 348,883 million and Nu. 140,656 million respectively. Further, part of the equity requirement i.e. Nu. 50,658 million is expected to be funded by JV partners in certain projects, balance Nu. 89,998 million shall have to be sourced by DGPC from its resources.

DGPC has established strong financial relationships with multiple banking institutions, including domestic and foreign banks, Indian banks, and multilateral agencies. These strategic partnerships play a crucial role in securing debt and equity financing for the overall expansion of DGPC's projects. Predominantly DGPC's hydropower projects have been funded with support from Sovereign Partners. Over the years, DGPC has been actively diversifying its financing structure to reduce reliance on bilateral sovereign funding. Subsequently, DGPC has utilized a mix of funding from Commercial Banks, Multilateral Organizations and Strategic Partners (PPP mode) to finance its operational/under-construction projects.

Various routes have also been suggested to DGPC for arranging funds for pipeline projects like green Bonds, InVITs, concession-based model, PPP model and refinancing of existing projects. These routes have been suggested keeping in view the financial suitability and objectives of DGPC that are, lowered cost of capital, broadened investor base, reduced reliance on RGoB for fund raise, maintaining health GDP-Debt ratio and long term financing aligned to project's life cycle.

Dorjilung Hydropower plant is expected to be developed on a Public-Private Partnership model with World bank as the anchor. Tata Power Corporation Limited along with its trading arm is envisaged to be the off-taker for power exports and strategic partner with non-controlling stake in the project SPV. Envisaged financing structure is similar to Dagachhu Hydropower plant and financing structures being adopted in international market.

DHPP is expected to be developed at an estimated total cost of \$1.7 billion. Funding mix for the project is yet to be determined. It has been observed that in the past and internationally, financing mix of 70:30 debt equity ratio is considered reasonable for projects of such nature, depending on tariff and financing viability. World Bank group, through its entities would be participating in part-financing of the project. Part of the funding for the project would be contributed to by Strategic partner and commercial debt financiers.

3. Our Approach to the Study

3.1.Our Approach towards the study:

This Preliminary Investment Report focuses on identifying potential off-take options and financial analysis for Bhutan’s Dorjilung Hydropower Plant. Our approach segments the study under six key steps outlined below –

Table 1: Study Focus Areas and Objectives

KEY FOCUS AREA	OBJECTIVES
DHPP Generation Profile and Strategic Importance	Provide detailed technical overview of DHPP's generation capabilities and articulate the strategic importance of the project for Bhutan's energy security, export earnings, and regional energy cooperation
Indian Power Market Analysis - Demand, Supply, Hydropower's Role	Evaluate India as the primary export market for DHPP, understanding its electricity landscape, demand-supply dynamics, and the evolving role of hydropower
Offtake Options and Discom Level Assessment in India	Identify and evaluate potential offtakers in the Indian power market
Bangladesh Power Market Assessment	Evaluate Bangladesh as a potential secondary export market for DHPP, considering its energy needs, import strategies, and the evolving framework for regional power trade
Regulatory, Risk and Tariff Framework in Export Markets	Provide clear understanding of the tariff determination regulations and prevailing regulatory challenges in CBET including India and Bangladesh, which will directly impact the pricing and bankability of power exported from DHPP. Additionally, study of export risk consideration for DHPP.
Transmission Infrastructure and Regional Energy Cooperation	Assess the adequacy of existing and planned transmission infrastructure for facilitating power exports from DHPP and to highlight the broader context of regional energy cooperation in South Asia

In accordance to these key steps’ objectives, our approach has been structured into two key sections:

Offtake Analysis	Financial Analysis
------------------	--------------------

3.2.Detailed Approach & Methodology

Table 2: Detailed Approach & Methodology

OFFTAKE ANALYSIS	Our Understanding on DHPP	• Analysis of generation profile under P-50 and P-90 scenarios. • Significance of the study
	Indian Power Market Analysis	• Comprehensive analysis of the Indian power market has been undertaken to evaluate both national and regional demand-supply dynamics • Critical insights into the evolving energy landscape • Regional assessment to identify potential offtake opportunities for surplus generation with examination of monthly demand and supply patterns across key Indian states.

		Profiling of Indian Offtake Options On basis of Gross Input Energy, the largest States from each region have been selected for detailed DISCOM-level analysis. From each selected state, DISCOMs are evaluated on Installed Capacity, Average Cost of Supply (ACS), Average Revenue realized (ARR), Gap between ACS-ARR, Payable days, profitability, Uncontracted HPO Obligations and Merit Order Analysis of Power Purchase Cost
		- Power trading through both exchange and bilateral routes - Analyze major corporate houses, identifying their net-zero targets, and evaluating their future sustainability plans in alignment with national Renewable Purchase Obligation (RPO) and Hydro Purchase Obligation (HPO) mandates
	Bangladesh Power Sector Overview	- Reviewed Bangladesh's power sector overview: peak demand growth installed capacity, and fuel mix - Analyzed the power import trends of Bangladesh - Assessed role of the Bangladesh Power Development Board as single buyer and its generation/purchase cost structures. - Reviewed Bangladesh's Integrated Energy and Power Master Plan (IEPMP) 2023 for future demand and supply and need for hydropower import
	Regulation, Risk and Transmission	- Examine regulatory framework for tariff determination of hydropower generation in South Asia, with focus on CBET regulations and regulatory landscape in India and Bangladesh - Case studies and tariff comparisons of various hydropower plants - Identify challenges encountered during tariff determination - Recommend areas of optimization - Study of export risk in context of DHPP.

FINANCIAL ANALYSIS	Optimal financing structure	- Study of financing structure prevailing in Bhutan for HPPs and its evolution - two phases namely Bilateral mode of funding (1988-2010) and diversification in financing structures to reduce reliance on bilateral sovereign funding (2010 and onwards). - Significant transformation in Bhutan's hydro journey - Funding structures of other similar international hydropower projects in Asia and Africa continents
	Considerations for India/international lenders	- Broadly divided into three sub-sections namely, key appraisal requirements by lenders for financing a hydropower project, security enforcement plan and financing timeline - First sub-section - discusses as to how banks / financiers typically require a thorough and comprehensive appraisal of under-construction hydropower projects to assess their financial viability and risk profile before extending loans or investments. DHPP's preparedness towards various appraisal aspects and requirements. - Next sub-section - security enforcement by Indian and international lenders. Details on DGPC obtaining a legal opinion for Tangsibji Hydro Energy Limited (THyE) Project from Bhutan Law Services regarding enforceability and validity of the proposed security package in Bhutan.
	Project risks-benchmarking and mitigation	- Range of material risks, including Land acquisition issues, geological surprises, construction delays leading to time and cost overruns, and regulatory challenges. - Devising mitigation strategies

4. Profile of Dorjilung Hydropower Plant

Bhutan's electricity sector is a cornerstone of its socio-economic development, with hydropower being the dominant source. The country has an installed hydropower capacity of approximately 2,453 MW⁵, the sector not only ensures reliable and affordable electricity for domestic use but also generates significant revenue through exports, primarily to India. Bhutan's hydropower potential is estimated at 37,000 MW, with 33,000 MW⁶ considered technologically and economically feasible.

Bhutan's electricity demand has surged at a compound annual growth rate (CAGR) of approximately 31% from FY2021 to FY2024, driven by rapid electrification, economic growth, and the government's push to transform the country into an industrial and service powerhouse. The sales mix includes both domestic demand and electricity exports to India. Domestic demand has grown at an exceptional rate of around 38% over the past five years, with peak demand tripling from 374 MW in FY2020 to 1,026 MW in FY2024.⁷ This increase is primarily due to the rising high voltage (HV) and medium voltage (MV) demand from energy-intensive industries and the expansion of the Jigmeling and Motanga industrial parks along the India-Bhutan border. Bhutan's energy demand is projected to reach 20,968 GWh, requiring approximately 7,141 MW of capacity addition by FY2030.⁸

4.1. Generation profile of Dorjilung Hydropower Plant

Bhutan is set to harness its substantial yet untapped hydropower resources with the Dorjilung Hydropower Plant, which will have an anticipated capacity of 1,125 MW, making it one of the nation's largest hydropower projects. According to the Detailed Project Report (DPR), the expected commissioning of the project will be in FY2031, with an annual design energy output of 4,504 GWh. The project is expected to enhance Bhutan's domestic power supply and boost its export potential.

At P90 probability, Total anticipated generation capacity as per DPR is 4,504 GWh out of which 85% is available for export whereas 15% would be held for domestic consumption.

Generational Profile of DHPP⁹

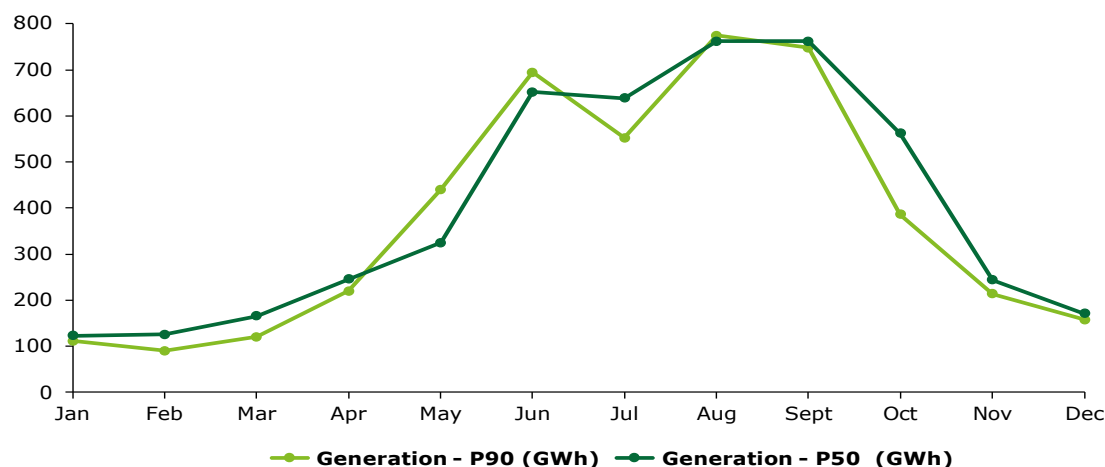


Figure 1: Generational profile of DHPP at P50 and P90 probabilities

⁵ [DGPC Website](#)

⁶ Bhutan Energy Directory, Department of Energy Ministry of Energy and Natural Resources, 2022

⁷ DGPC Additional Databook Excel

⁸ DGPC Energy Databook Excel

⁹ DHPP Generation Data P50 and P90 Excel

Table 3: Month-wise Generational Profile of DHPP at P-90¹⁰

Generation - P90					
Month	Generation (GWh)	Aux Loss (1.12%)	Energy net (GWh)	Domestic Allocation (GWh)	Energy Available for Export (GWh)
Jan	111	1.24	110	110	-
Feb	90	1.00	89	89	-
Mar	120	1.35	119	119	-
Apr	219	2.46	217	217	-
May	440	4.93	435	-	435
Jun	695	7.78	687	-	687
Jul	552	6.18	545	-	545
Aug	774	8.67	765	-	765
Sept	747	88.37	739	-	739
Oct	386	4.32	381	-	381
Nov	214	2.39	211	-	211
Dec	158	1.76	156	156	-
Total	4504	50.45	4,453	690	3,764

At P50 probability, Total anticipated generation capacity as per DPR is 4,772 GWh out of which 83% is available for export whereas 17% would be held for domestic consumption.

Table 4: Month-wise Generational Profile of DHPP at P-50¹¹

Generation - P50					
Month	Generation (GWh)	Aux Loss (1.12%)	Energy net (GWh)	Domestic Allocation (GWh)	Energy Available for Export (GWh)
Jan	123	1.38	122	122	-
Feb	125	1.40	124	124	-
Mar	165	1.85	163	163	-
Apr	245	2.75	243	243	-
May	325	3.64	321	-	321
Jun	651	7.29	644	-	644
Jul	638	7.15	631	-	631
Aug	761	8.53	753	-	753
Sept	762	8.53	753	-	753
Oct	562	6.29	556	-	556
Nov	244	2.73	241	-	241
Dec	171	1.91	169	169	-
Total	4,772	53.45	4,719	820	3,899

During the non-monsoon period (January to April), power generation capacity is expected to experience a notable decline, necessitating its allocation primarily for domestic consumption. Following April, projections indicate increase in generation capacity, which will be directed towards meeting export demands. Detailed Project Report has majorly focused on the quantum part which is available for export.

4.2. Need to Identify Offtake Markets for DHPP

Apart from DHPP, Bhutan is also expected to add significant installed capacity for electricity generation. The total generation of electricity is projected to rise from 27,371 GWh to 76,042 GWh from FY 2031

¹⁰ Source: Detailed Project Report for DHPP

¹¹ Source: Information shared by Client (DGPC)

to FY 2040, growing at CAGR of 13.6%. Concurrently, the anticipated load is expected to increase from 22,061 GWh to 34,757 GWh from FY 2031 to FY 2040, growing at CAGR of 5.8%.

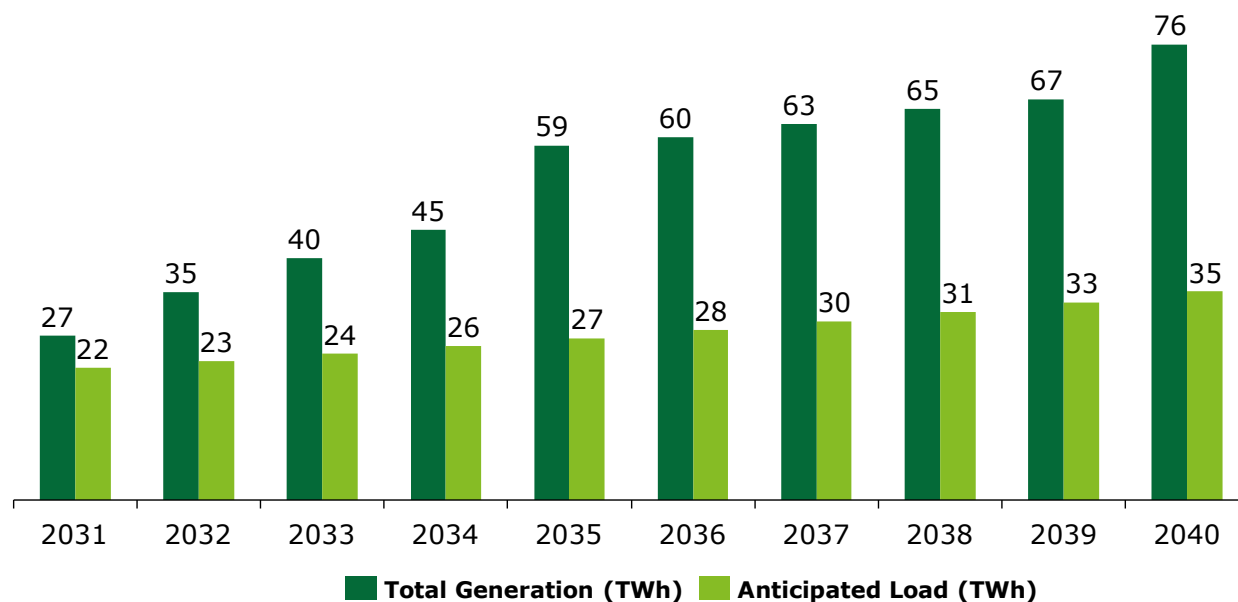


Figure 2: Projected Demand and Supply in Bhutan Power Market after commissioning of DHPP¹²

The projected surplus in generation capacity offers a valuable opportunity for regional energy trade. This excess electricity can be strategically utilized through offtake arrangements with neighboring countries, fostering regional cooperation, and contributing to Bhutan's economic development.

South Asian power sector is undergoing significant transformation due to rising energy demand. In the past two decades, energy demand in South Asia has surged by more than 50%¹³, primarily driven by population growth and industrial expansion. This region heavily relies on energy imports, with approximately two-thirds of its primary energy being imported, making it vulnerable to global energy price fluctuations. To address these challenges, there is a strong push towards integrating renewable energy sources, such as hydropower, solar, and wind, to reduce dependence on fossil fuels and mitigate greenhouse gas emissions.

The BBIN region within South Asia comprising of Bangladesh, Bhutan, India, Nepal are already partnering within on cross border electricity trade. Most of the agreements and trade have been bilateral in nature with a few trilateral agreements emerging in recent years. Sri Lanka too may be connected to the Indian grid in a few years taking the trade potential to BBINS (with Sri Lanka). Both India and Bangladesh present opportunities for offtake in the region due to issues such as reliance on thermal fuel and shift towards variable renewable energy in India, liquid and imported fuels in Bangladesh. These are discussed in detail in subsequent sections.

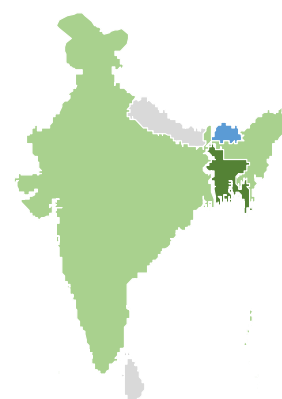


Figure 3: Map of BBINS (not to scale)

India and Bhutan have a strong partnership in the hydropower sector, significantly benefiting Bhutan's economy and India's energy security. Bhutan exports 70% of its generated electricity to India, with India providing financial and technical assistance for various projects. In 2006, both countries signed a framework agreement to develop and trade hydropower, committing India to import at least 5,000 MW

¹² DGPC Energy Data

¹³ An integrated electricity market in South Asia is key to energy security, World Bank Blogs

by 2020, later revised to 10,000 MW in 2009. India, as the world's third-largest power consumer, benefits significantly from Bhutan's surplus electricity. A joint initiative between Government of India and State's "Power for All"¹⁴ has accelerated capacity expansion, with Bhutan playing a crucial role as both a buyer and seller of power. Evaluating the DISCOMs in the Indian power market is essential for assessing offtake options for DHPP in Indian power market.

Similar to India, Bangladesh is also emerging as a key market for cross border power trading in the region. It's installed power capacity stands at 27,424 MW¹⁵, with only 2.78% from renewables, mainly off-grid solar. The country's heavy reliance on depleting natural gas reserves and rising energy demand underscore the need for diversification. Bangladesh is already engaged in cross-border electricity trade, importing ~2.5 GW¹⁶ (primarily from India and marginal volume from Nepal). Given its rising electricity demand and commitment to clean energy, Bangladesh presents a viable offtake market for Bhutanese hydropower.

The consistent growth in both electricity generation and domestic load in Bhutan highlights the country's strong capacity to efficiently meet its projected energy demands. This dependable supply, combined with a significant surplus in generation capacity, presents a strategic opportunity to expand electricity exports. Leveraging this surplus through well-structured offtake arrangements with neighboring countries can maximize Bhutan's export potential while contributing meaningfully to regional energy security.

¹⁴ Power for All, Ministry of Power

¹⁵ Bangladesh Power Development Board

¹⁶ [Annual Report, Bangladesh Power Development Board](#)

5. Indian Power Market Scenario

5.1. Electricity Landscape in India

India is the largest electricity market in South Asia, and a potential market for power export from the Dorjilung hydropower project. The country has a deep history of cross border electricity trade with Bhutan, strong existing cross border transmission interconnections, and favourable policy and regulatory frameworks enabling such power trade.

In this context, this study has been conducted to understand the existing and future electricity demand, including seasonal trends in India under a range of scenarios, and identification of the likely mix of generation (capacity and energy) required to meet the forecasted electricity demand in the country.

One of the major drivers of growth in electricity demand is economic growth; and in this regard, India's GDP has grown by 8.2% in FY 2023-24¹⁷ while same was 7.2% in FY 2022-23. The Government of India (GoI) has also set a target of becoming a \$5 trillion economy by 2024-25.¹⁸

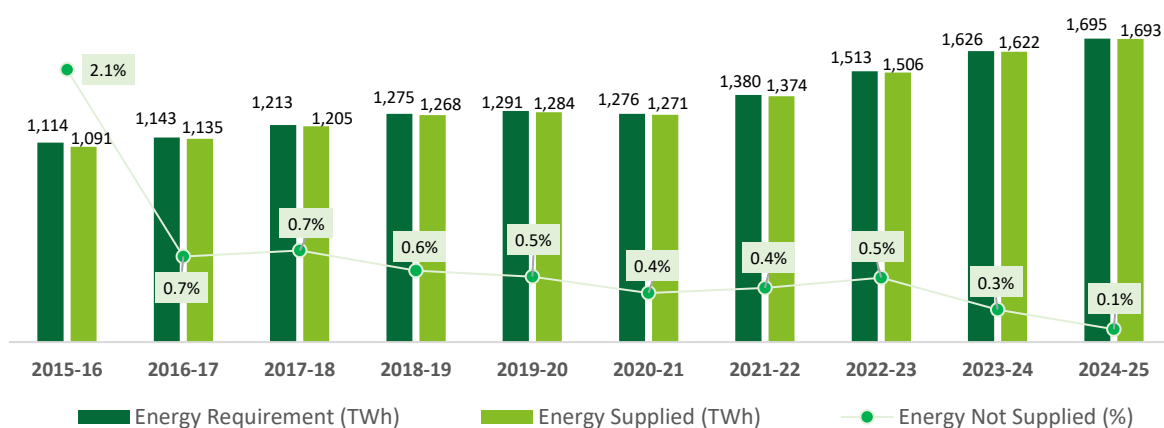


Figure 4: India Electricity Demand and Supply

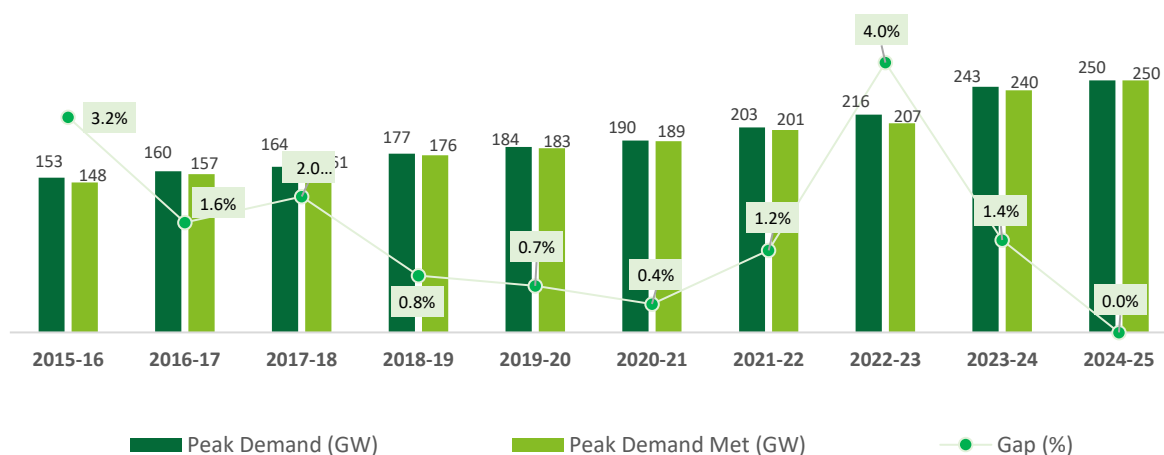


Figure 5: India Electricity Peak Demand and Peak Supply

¹⁷ Source: [Press Information Bureau, GoI](#)

¹⁸ Source: [Press Information Bureau, GoI](#)

In the past nine years, from FY2016 to FY 2025, electricity demand in India has grown at a compound average growth rate of 4.77%, i.e., from 1,114 TWh to 1,695 TWh, while at the same time, peak electricity demand in India grew at a compound average growth rate of 5.7%, i.e., from ~148 GW in FY 2015-16 to ~249 GW in FY 2024-25¹⁹.

5.1.1. Electricity Demand Forecast

The Central Electricity Authority (CEA) publishes periodic forecasts for demand in the country. The energy and peak demand are expected to increase at a compound rate of 4%-5% per annum through to 2042 from the current level of around 1,695 TWh and 249 GW as on 31st March 2025.

Table 5: India: Electricity demand forecast*

As per CEA 20th EPS Vol 1	Scenarios	2021-22 (Base)	2023-24 (P)	2026-27 (P)	2031-32 (P)	2036-37 (P)	2041-42 (P)
Energy Requirement (TWh)	Moderate Scenario	1,382	1,552	1,849	2,297	2,796	3,343
	Optimistic Scenario		1,572	1,908	2,474	3,095	3,776
Peak Demand (GW)	Moderate Scenario	203	226	264	329	405	488
	Optimistic Scenario		230	277	366	466	575
Base Corrected Scenario	Scenarios	2021-22 (A)	2024-25(A)**	2026-27 (P)	2031-32 (P)	2036-37 (P)	2041-42 (P)
Energy Requirement (TWh)	Moderate Scenario	1,382	1,695	1,905	2,362	2,874	3,430
	Optimistic Scenario			1,930	2,498	3,128	3,824
Peak Demand (GW)	Moderate Scenario	203	249	277	343	423	510
	Optimistic Scenario			282	372	472	583

*Demand is Ex-Bus; Base corrected (P): CAGR is as per considerations in CEA 20th EPS report; **As per LGBR 2025-26 Report

The key demand drivers which are envisaged to influence the overall electricity demand in India are mentioned below:



Figure 6: Future Electricity Demand Drivers in India

¹⁹ Source: Monthly Power Supply Reports, CEA

5.1.2. Electricity Supply Forecast

The overall capacity addition in India stands at 475 GW, out of which thermal energy continues to play a significant role in India's electricity generation, having accounted for 51%²⁰ of the total electricity generation in FY 2024-25 (refer figure below). The Government of India (GoI) has set an ambitious plan of 500 GW renewable energy installed capacity by 2030 and in this regard has already achieved ~220 GW as on May 2025²¹ (~44% of 2030 target).

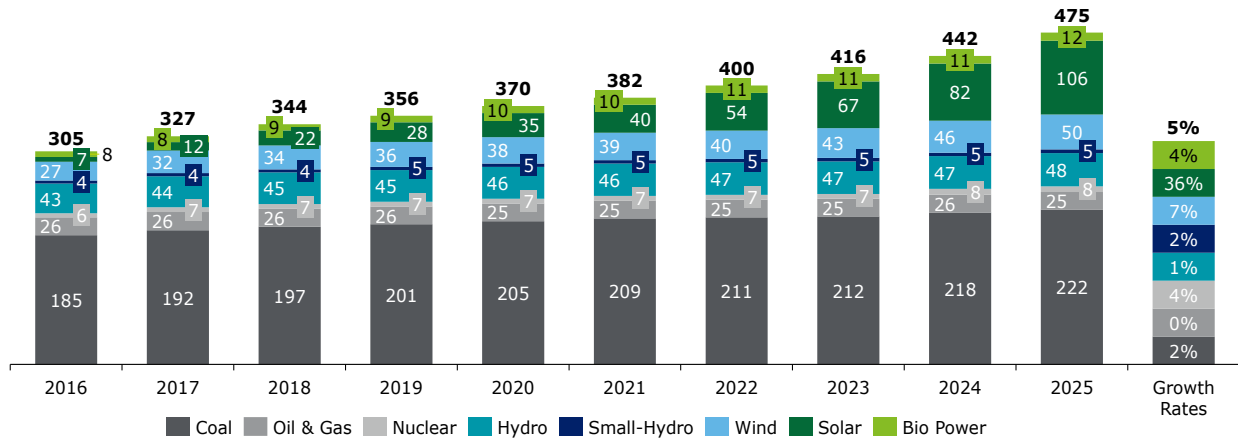


Figure 7: Source wise Installed Capacity (in MW) in India²²

Electricity supply forecast by CEA and some other industry reports were reviewed. Key observations are as follows^{23,24}:

India will experience steady growth in the electricity demand in the years up to 2050.

The share of coal based thermal power generation capacity in the year 2030 would be around 32-34% in the energy mix as against the present share of ~51%.

The share of coal-based power generation capacity is further expected to reduce to ~13% of total installed capacity by 2050 under Economic Transition Scenario (ETS) wherein cost-based technology changes taken in consideration.

RE penetration (particularly solar and wind) will accelerate because of aggressive policies and targets, coupled with lower capital costs and easier access to finance. The capacity share of renewables will grow from the current level of 44% to ~51-53% by 2030 while wind and solar alone would account for ~57% in 2050 under ETS scenario.

Hydropower development has slowed down in the last decade due to combination of technical and safeguard challenges. This is also reflected in CEA projections, as only ~7-15 GW of Hydro capacity addition is projected by 2029-30 from current Hydro installed capacity of ~47 GW as of FY 2024-25.

Cross-border power trade will become increasingly important to meet hydropower capacity targets, to underpin decarbonisation efforts and to provide balancing power requirements for integrating RE into the grid.

Figure 8: India Electricity Sector Forecast Observations

The evolving energy landscape in India highlights the significant shift towards renewable energy sources, driven by ambitious targets and supportive policies. This transition not only promises a more sustainable

²⁰ Source: [National Power Portal](#), May 2025

²¹ Source: [National Power Portal](#), May 2025

²² Source: [India Climate & Energy Dashboard](#)

²³ Source: [National Power Portal](#), May 2025

²⁴ Source: [New Energy Outlook, Bloomberg, 2023](#)

future but also underscores the importance of strategic planning and international cooperation in achieving energy security and environmental goals.

5.1.3. Role of Hydropower

India's hydropower sector has seen significant decline in recent decades: the share of hydropower in the total capacity has reduced from 50% in the 1960s to around 10% as of May 2025. This decline can be attributed largely to the relatively high development risk associated with hydropower, coupled with the proliferation of domestic coal resources and more recent shift towards cheaper solar power.

The continued push for adding renewable sources in the energy mix is creating infrastructural challenges for maintaining the national grid as the variable nature of renewable energy sources are inherently accompanied by intermittency in generation. This necessitates the development of storage solutions, which can rapidly respond to changes in variable renewable energy (VRE) output, thereby stabilizing the grid. Storage hydro plants are expected to play a pivotal role in grid balancing requirements, and also in providing ancillary services such as black start, voltage support, over-frequency reserve, instantaneous reserve and frequency keeping.

Lastly, hydropower is expected to resume its role as base load plant due to the obligated reduction in coal plants to meet GoI's net zero targets. Hydroelectric power projects and Pumped Storage Projects with an aggregate capacity of ~9.9 GW are under construction in the country as on April 2025²⁵. The hydro capacity is projected to increase from 47 GW in FY 2024-25 to ~54-62 GW by 2029-30. Currently, PSPs with aggregate capacity of ~4 GW are under construction in the country while CEA in its revised report on optimal generation mix in 2030, has projected installed capacity of around ~19-25 GW of PSP by 2029-30 under different scenarios.

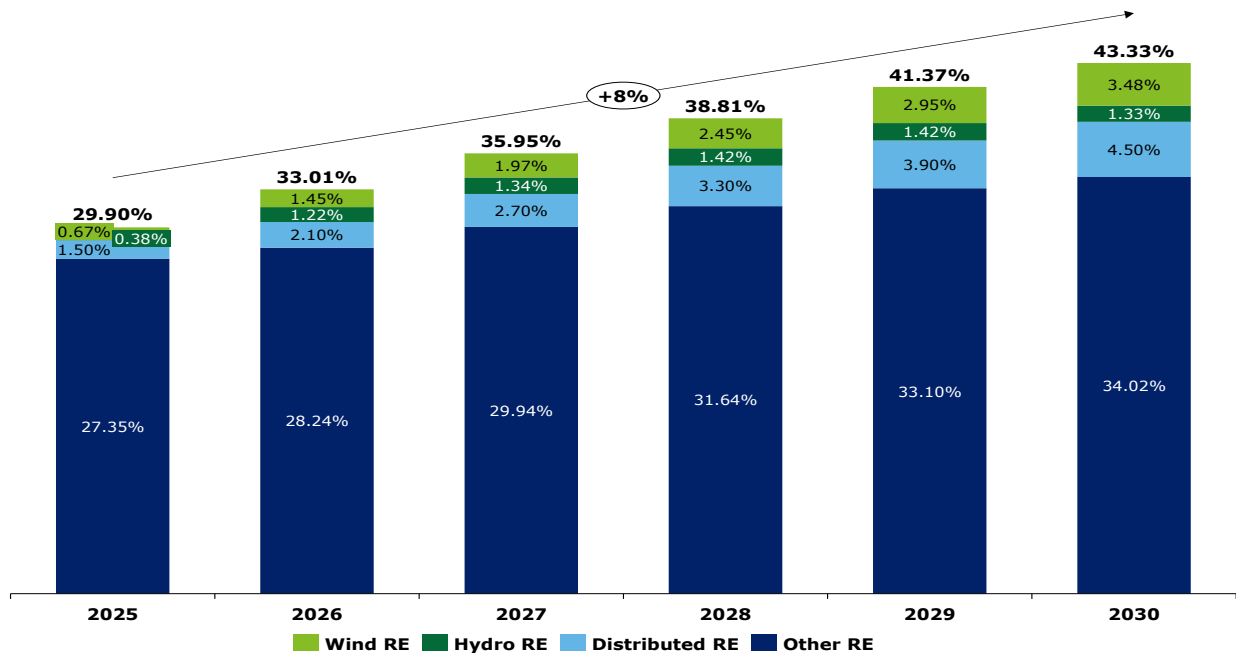


Figure 9: India's RPO trajectory till 2030

The Government of India, through recent policy updates and the revised Renewable Purchase Obligation (RPO) trajectory notified by the Ministry of Power, has broadened the definition of renewable energy to include large hydropower projects (LHPs) commissioned after March 8, 2019. As part of the new structure, the Hydro Purchase Obligation (HPO) has been created as a dedicated sub-category within the overall RPO targets, enabling States to meet their renewable energy obligations through procurement of hydropower. As a result of this policy guidance, distribution utilities are mandated to

²⁵ Source: [Hydroelectric Projects under Construction, CEA, April 2025](#)

purchase a certain proportion of their electricity requirements from hydropower plants. The RPO trajectory²⁶ as notified by GoI is provided in the figure below:-

Several States have been assigned significant RPO targets for the year 2031-32, reflecting their contribution towards India's broader clean energy goals:

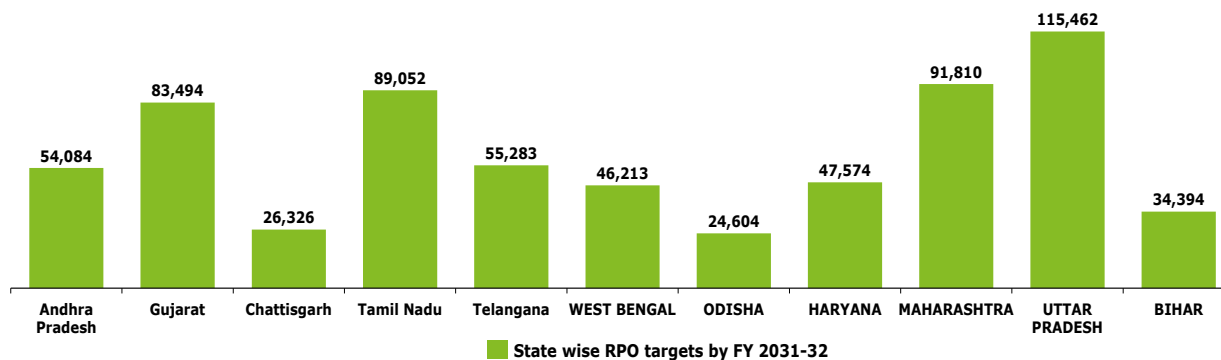


Figure 10: State wise RPO targets (in GWh) by FY 2031-32²⁷

Cross-border power trade is therefore expected to play a key role in reducing the dependency on fossil fuels, providing flexibility to the grid and meeting RPO/HPO obligation. India will likely increasingly leverage the benefit of hydropower (and particularly storage hydro) from neighbouring nations such as Bhutan and Nepal for peaking and balancing services.

In this context, it is important to further deep dive and study the offtake options to further align the generation profile of DHPP with electricity landscape of different regions in India.

5.2. Region wise Electricity Demand and Supply

As India is predominantly a sub-tropical country, cooling loads during the period May-October dominate the heating loads during the period November – February. Demand profiles also vary regionally. The region-wise electricity demand projections for the period 2016-17 to 2031-32 is presented in the table below:

Table 6: Region wise projection for Demand and Supply²⁸

Region	Energy Requirement (in TWh)				Peak Demand (in GW)			
	2016-17 (A)	2021-22 (A)	2026-27 (P)	2031-32 (P)	2016-17 (A)	2021-22 (A)	2026-27 (P)	2031-32 (P)
Northern Region	356	418	592	773	55	73	97	127
Western Region	352	428	596	763	50	65	89	114
Southern Region	307	351	460	596	44	61	80	107
Eastern Region	128	164	232	308	20	26	37	50

²⁶ Source: [Ministry of Power, GoI, 2023](#)

²⁷ Resource RPO trajectory based on forecasted demand as per Resource Adequacy Study of different States. Some states are yet to adopt the RPO set by Ministry of Power, India but are expected to align with the national trajectory in the long run

²⁸ CAGR is as per considerations in CEA 20th EPS report

North-Eastern Region	15	18	24	32	2	3	4	6
All India	1160	1382	1907	2473	161	203	277	366

5.2.1.1. Northern Region Electricity Demand and Supply

In the Northern Region, Chandigarh, Delhi, Haryana and Punjab had closely met the Energy Requirement in full. Himachal Pradesh, UT of J&K and Ladakh, Rajasthan, Uttar Pradesh and Uttarakhand experienced marginal gap between Energy Requirement and Energy Supplied in the range of 0.2% to 0.4%. However, the gap between the Peak Demand and Peak Met is 0.9%. The month wise regional profile is provided below:-

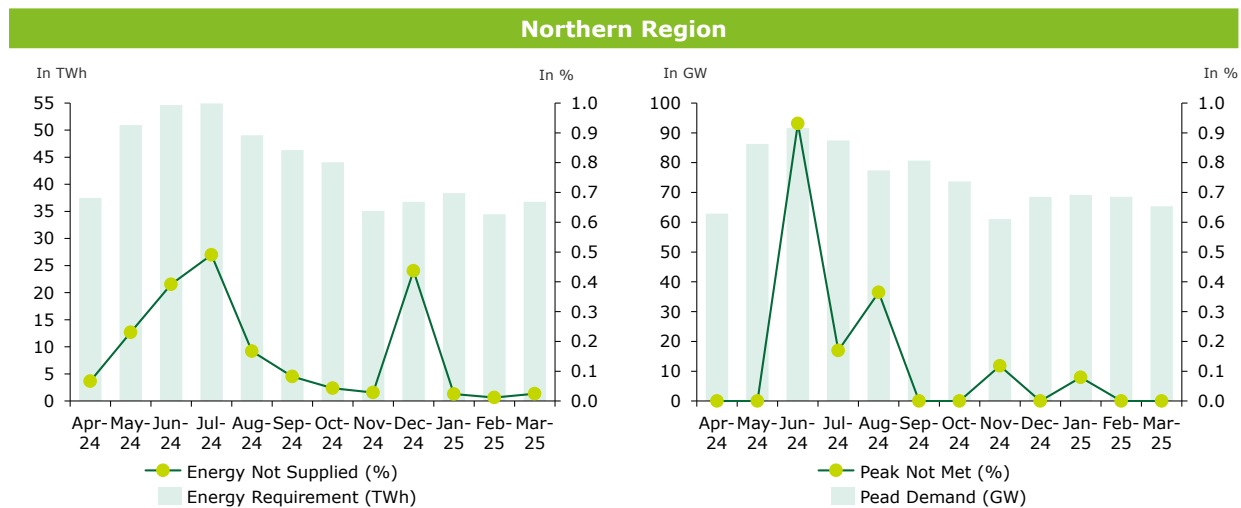


Figure 11: Northern Region monthly energy profile of FY 2024-25

- In the Northern Region, of the total 952 GWh energy deficit, approximately 75% occurs between May and September majorly due to hot summer months.
- The peak deficit in FY 2024 occurred between June and August, with the highest deficit of 845 MW recorded in June. This period aligns with the maximum generation phase for hydro projects, making it an ideal time to leverage hydroelectric power.

5.2.1.2. Southern Region Electricity Demand and Supply

In Southern Region, all the States and UTs i.e. Andhra Pradesh, Karnataka Kerala, Tamil Nadu, Puducherry, and Telangana closely met the Energy Requirement. However, Southern region has not faced a gap between the Peak Demand and Peak Met. The month wise regional profile is provided below:-

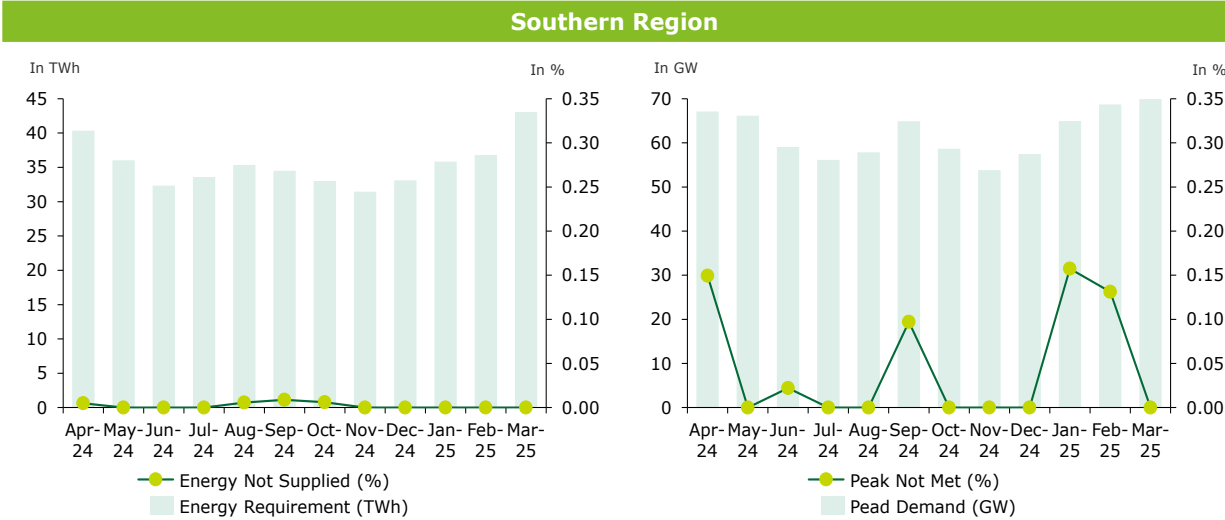


Figure 12: Southern Region monthly energy profile of FY 2024-25

➤ In FY 2024, the Southern Region experienced negligible energy deficits across most months. In FY 2024, the Southern Region experienced negligible energy deficits primarily due to a strong focus on renewable energy and a well-managed power supply system across most months. However, peak deficits of up to 100 MW were observed in January, February, and April.

5.2.1.3. Western Region Electricity Demand and Supply

In the Western Region, Gujarat, Maharashtra, DDD

NH and Goa had met the Energy Requirement in full while Madhya Pradesh and Chhattisgarh had experienced a marginal gap of 0.1% between Energy Requirement and Energy Supplied. However, Western had faced a gap of 0.4% between the Peak Demand and Peak Met.

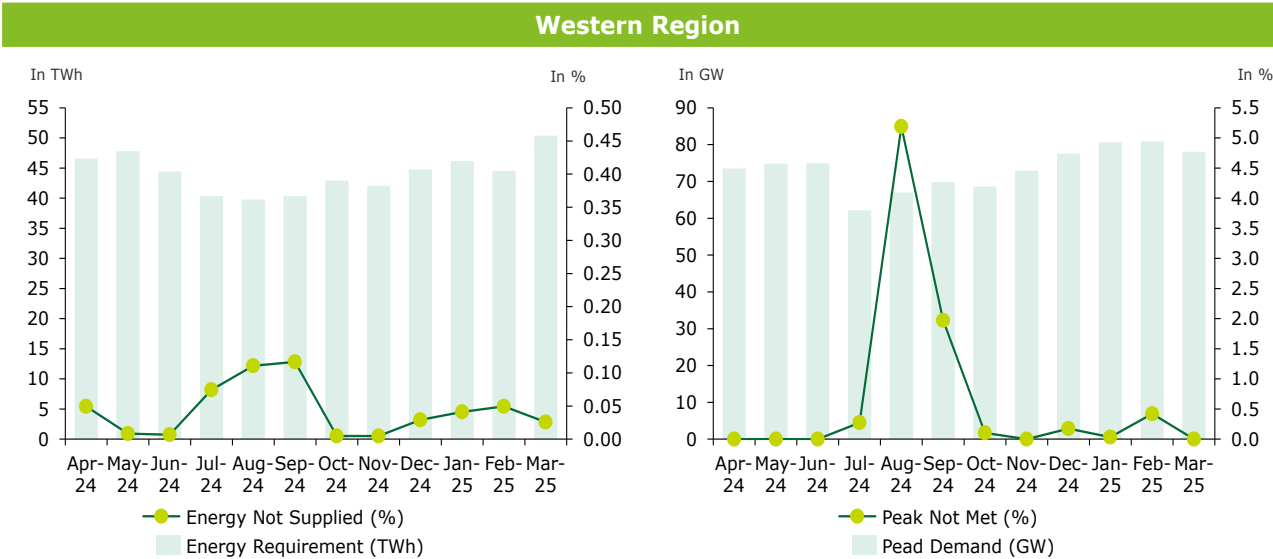


Figure 13: Western Region monthly energy profile of FY 2024-25

- In the Western Region, of the total 223 GWh energy deficit, approximately 55% occurs between July and September.
- The peak deficit is the highest among all other regions, standing at 3,306 MW in August 2024 and 1,350 MW in September. This period aligns with the maximum generation phase for hydro projects, making it an ideal time to leverage hydroelectric power.

5.2.1.4. Eastern Region wise Electricity Peak Demand and Peak Supply

In the Eastern Region, Sikkim and DVC met the Energy Requirement almost in full. West Bengal, Bihar, Jharkhand and Odisha experienced marginal Energy deficits in the range of 0.1% to 0.5%. However, Eastern had faced a gap of 0.3% between the Peak Demand and Peak Met.

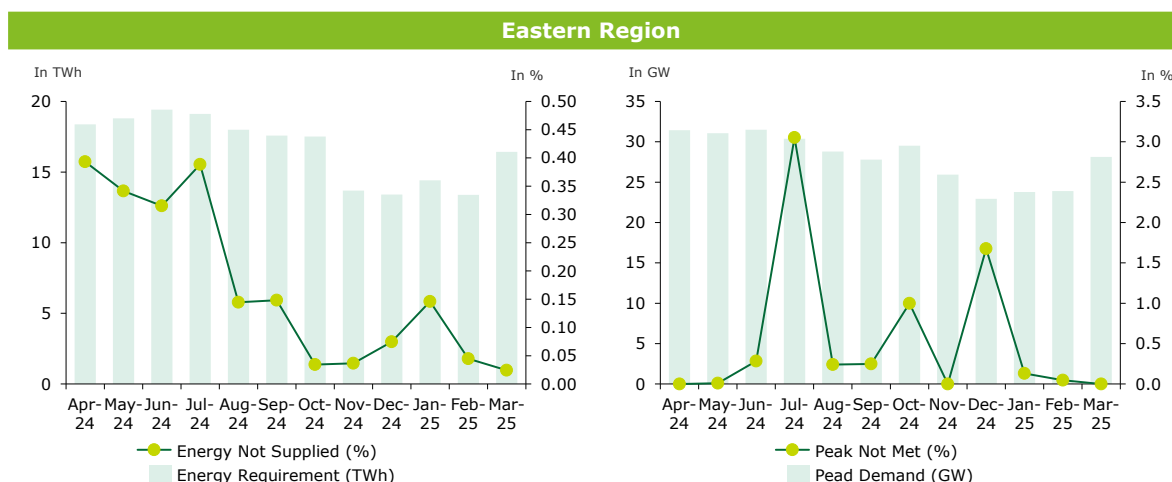


Figure 14: Eastern Region monthly energy profile of FY 2024-25

- In FY 2024, the Eastern Region experienced minimal energy deficits throughout most months. However, peak demand typically occurs between April and July due to hot, humid summer months, with the deficit reaching a peak of 900 MW in July. Consequently, this aligns with the generation profile of hydroelectric plants during these months.

5.2.1.5. North-Eastern Region wise Electricity Peak Demand and Peak Supply

In the North-Eastern Region, Arunachal Pradesh, Assam, Meghalaya, Mizoram, Nagaland, and Tripura completely met the Energy Requirement. However, Manipur experienced the energy deficit of 1.0%. North-Eastern Regions had faced a gap of 1.1% respectively between the Peak Demand and Peak Met.

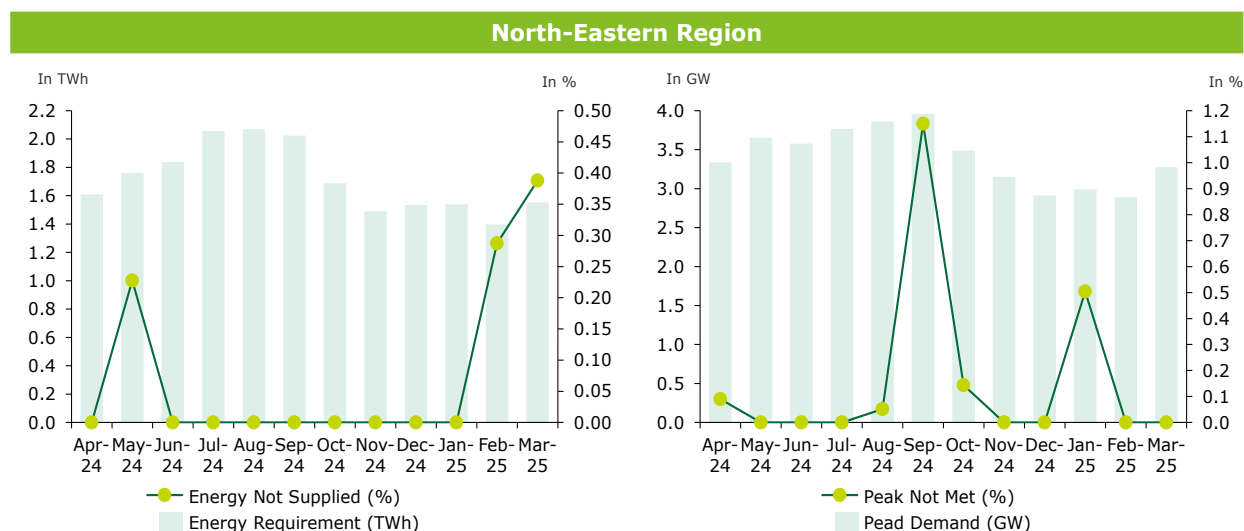


Figure 15: North-Eastern Region monthly energy profile of FY 2024-25

- In FY 2024, the North-Eastern Region experienced minimal energy deficits throughout most months. In FY 2024, the North-Eastern Region experienced minimal energy deficits throughout most months. The Peak Demand was ~3,914 MW with deficit of 45 MW in the month of September.

In summary, during the export period of DHPP, from May to November, the Northern region experiences the highest energy deficit, while the Western region faces the greatest peak demand deficit. The details are summarized in the table below:

Table 7: Summary of Region wise Energy and Peak Deficits in India

Region	Energy Deficits between May to November 2025	Peak Demand Deficits between May to November 2025
Northern	745 GWh	1,346 MW
Southern	17 GWh	76 MW
Western	133 GWh	4,898 MW
Eastern	261 GWh	1,422 MW
North-Eastern	6 GWh	52 MW

In response to the evolving dynamics of the regional power market in India, we plan to conduct a thorough assessment of the potential offtake opportunities for DHPP. This analysis will focus on major DISCOMs, evaluating their Gross Input Energy, Demand and Generation profiles, merit order, and retirement plans for existing assets. Additionally, we will assess the opportunity for DHPP to participate in power exchanges, open access arrangements, and potential integration with C&I consumers. The objective is to identify the most viable and strategic offtake pathways that align with the project's operational characteristics and market conditions, ensuring the optimal integration of DHPP.

6. Indian Power Market – Offtake Assessment

This report explores three primary methods for selling Bhutanese hydropower in India:

1. Long-term Power Purchase Agreements (PPAs) with Distribution Companies (DISCOMs)
2. Short Term Power Market Analysis through
 - Indian Power Market Exchange
 - Bilateral Trades
3. Targeting large corporate houses like Commercial and Industrial (C&I) segments

6.1. Offtake Analysis at Indian DISCOM level

India's power sector is highly diversified, offering numerous opportunities to bridge the gap between electricity demand and supply. The country hosts around 70 Distribution Companies (DISCOMs) with significant variation in energy input, operational scale, and market reach.

For the purpose of offtake analysis, a filtering approach has been adopted to focus on State's with high Gross Input Energy. This method ensures that the analysis targets larger markets where the potential for offtake arrangements is more substantial. Prioritizing high-demand regions and major State's as well as DISCOMs allows for a more strategic assessment of export opportunities and enhances the relevance of the findings in the context of regional energy planning and cross-border trade.

Further for analyzing each DISCOMs, certain parameters has been selected. The rationale for selecting parameters has been stated below:

Table 8: Parameters considered for Analyzing DISCOMs

Parameters	Rationale for Parameter Selection
Average Cost of Supply	Average Cost of Supply has been considered as one of the parameters as it reflects the cost incurred by DISCOMs to procure and supply electricity.
Average Revenue Realized	Average Revenue Realized (ARR) reflects the average amount of revenue collected per unit of electricity supplied, on a subsidy-received basis. It is one of the crucial metrics for evaluating the financial performance of a DISCOM. A higher ARR, compared to the ACS, suggests better financial health and profitability.
Gap between Average Cost of Supply vs Average Revenue Realized	Average Cost of Supply (ACS) and Average Revenue Realized (ARR) is a key indicator of DISCOM's of financial health and profitability. A smaller gap suggests that the DISCOM is recovering its costs effectively, which is vital for maintaining operations without incurring losses.
Resource Adequacy Report	Resource adequacy covers future as well projected demand of different states and utilities in India. This metric gives overview of requirement of quantum of power required from different sources like wind, hydro and solar which needs to be furnished.
Profit After Tax (PAT)	Positive PAT indicates financial health and efficient operations, making DISCOMs attractive to investors and ensuring they can meet contractual obligations. This stability reduces risk and enhances consumer confidence in the reliability of electricity supply.
Payable Days	High Payable days indicates that DISCOMs take longer to pay their bills, which can signal cash flow issues and potential payment delays.

Region-wise detailed analysis has been performed, refer below sections:

6.1.1. Western Region

This region is one of the most industrialized and power-intensive zones in the country and contributes a large share of India's thermal power, especially coal-based generation. DISCOMs in Gujarat and Maharashtra have a history of engaging in long-term power purchase agreements, including with Bhutanese projects like Tala and Chhukha. List of States of Western Region have been stated below²⁹:

Table 9 : State-wise Gross Input Energy³⁰

State	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Maharashtra	172,082	39%
Gujarat	125,896	28%
Madhya Pradesh	96,499	22%
Chhattisgarh	45,453	10%
Goa	5,517	1%
Dadra & Nagar Haveli, Daman & Diu	-	-

Among the above-mentioned States, Maharashtra, Gujarat, Madhya Pradesh, and Chhattisgarh has been selected further for detailed level Analysis as it covers approximately 99% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 10: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Maharashtra	Maharashtra State Electricity Distribution Company Limited (MSEDCL)	1,66,970	91%
	Adani Electricity Mumbai Limited	10,779	6%
	Brihan Mumbai Electric Supply & Transport Undertaking	5,112	3%
	Tata Power Company Limited	-	0%
	Total of Maharashtra	1,82,861	
Madhya Pradesh	Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Ltd	34,017	35%
	Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Ltd	33,522	35%
	Madhya Pradesh Poorv Kshetra Vidyut Vitaran Company Ltd	28,961	30%
	Total of Madhya Pradesh	96,500	
Gujarat	Paschim Gujarat Vij Company Limited	46,544	37%
	Uttar Gujarat Vij Company Limited	33,057	26%
	Dakshin Gujarat Vij Company Limited	31,515	25%
	Madhya Gujarat Vij Company Limited	14,781	12%
	Torrent Power Limited	-	0%
	Total of Gujarat	1,25,897	

²⁹ Central Electricity Authority

³⁰ Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Chhattisgarh	Chhattisgarh State Power Distribution Company Limited	45,453	100%
	Total of Chhattisgarh	45,453	

From the above-mentioned list, DISCOM has been selected from each State based on its contribution of atleast 90% of the State's total Gross Input Energy is most to that State, refer below mentioned table for the analysis of DISCOMs:

1. Region: West

State: Maharashtra

Maharashtra has historically made substantial investments in rural electrification, renewable energy development, and capacity expansion. In FY 2025, the state recorded a peak electricity demand of approximately 30.7 GW³¹. The state has an installed generation capacity of 53.3 GW³², comprising 55% from fossil fuel-based projects, 43% from renewable energy sources (RES), including hydro, and remaining 2% from nuclear projects. Maharashtra is among the leading states in renewable energy adoption, hosting more than 10.4% of India's renewable energy capacity. The state's energy requirement and peak demand from FY 2021-22 to FY 2031-32 has been detailed below-

Table 11: Energy Demand of Maharashtra

Particulars	Units	2021-22 (Actual) ³³	2024-25 (Actual) ³⁴	2026-27 (Projected) ³⁵	2031-32 (Projected)
Energy Requirement	TWh	170	203	231	265
Peak Demand	GW	26	31	35	45

Fuel Mix of Maharashtra:³⁶

Maharashtra's contracted power capacity reached 35 GW by the end of March 2023, of this capacity, 62% is derived from fossil fuel-based sources, primarily coal and natural gas and the rest is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, small hydro, wind and other RE. This composition reflects the state's continued reliance on conventional energy sources while also indicating a growing commitment to renewable energy integration.

The projected contracted capacity for the year FY 2033-34 is 100 GW, out of which 72% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at CAGR of 11% over the period of next 10 years Capacity wise generation mix from FY 2023-34 has been illustrated below:

³¹ Load Generation Balance Report, 2025

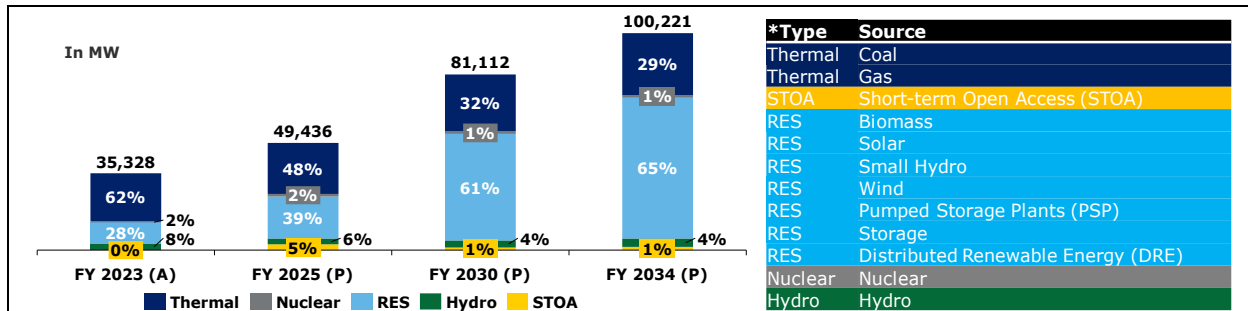
³² Government of India, Central Electricity Authority, 2024

³³ Load Generation Balance Report, 2022

³⁴ Load Generation Balance Report, 2025

³⁵ National Electricity Plan 2022-32

³⁶ Resource Adequacy Report, Central Electricity Report, GOI, 2024

Figure 16: Fuel wise capacity mix of Maharashtra³⁷

Financial Health of Maharashtra's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 26th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 40% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 60%, are those that are still within their respective due dates but have not yet been fully or partially paid :³⁸

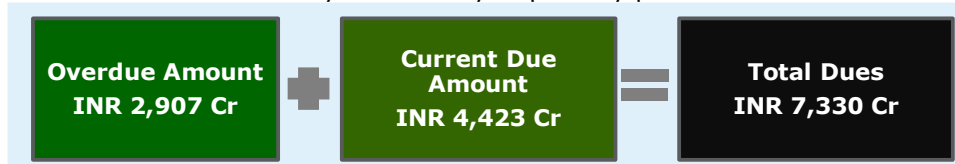


Figure 17: Snapshot of PRAAPTI – Maharashtra

Utility Name : Maharashtra State Electricity Distribution Co. Ltd. (MSEDCL):

Maharashtra State Electricity Distribution Company Limited (MSEDCL) is the one of the largest power distribution company both in the country and in Asia, serving over 27.43 million³⁹ consumers. Established in 2005, MSEDCL operates an extensive network to ensure reliable power supply, contributing significantly to the state's energy infrastructure. MSEDCL serves a diverse consumer base, with the industrial sector accounting for approximately 42% of total electricity consumption. This is followed by the agricultural sector at 22%, domestic consumers at 20%, and the commercial segment at 9%, with the remaining share attributed to other sectors.⁴⁰

Performance History of MSEDCL:

In FY 2023–24, MSEDCL reported consistent increase in gross input energy over the past four years at CAGR of 8% similarly, gross energy sold has also demonstrated steady growth of 10%. The Average Cost of Supply (ACS) declined marginally to INR 8.96 per kWh while Average Revenue realized (ARR) has increased to INR 8.67 per kWh in FY 2023–24, narrowing the gap between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past four years, a decline was observed over in FY 2023–24. Payable Days, while still high, they have steadily decreased over the past three years and improved compared to the national average (excluding private DISCOMs) of 130 days in FY 2023–24.⁴¹

³⁷ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

³⁸ Dashboard for Maharashtra, PRAAPTI, as accessed on 26th May 2025

³⁹ India Climate & Energy Dashboard

⁴⁰ India Climate & Energy Dashboard- Maharashtra

⁴¹ Report on Performance of Power Utilities, Power Finance Corporation, 2023–24

Table 12: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ⁴²	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	105,484	116,663	126,741	139,470
Gross Input Energy	GWh	132,485	144,235	155,096	166,970
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.91	7.83	9.32	8.96
Average Revenue Realized (ARR)	INR /kWh	7.37	7.60	7.60	8.67
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.54)	(0.23)	(1.72)	(0.29)
PAT (Profit/(Loss))	INR in Crore	(1,322)	280	(5,199)	(4,517)
Payable Days	Days	178	177	113	109

Capacity Addition Plans of Maharashtra (MSEDCL):

MSEDCL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2023 to 29% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. As per Resource Adequacy report, MSEDCL has plans to increase its capacity mix by 1,063 MW through hydropower till 2034. The state's planned capacity addition for next 10 years has been stated below:

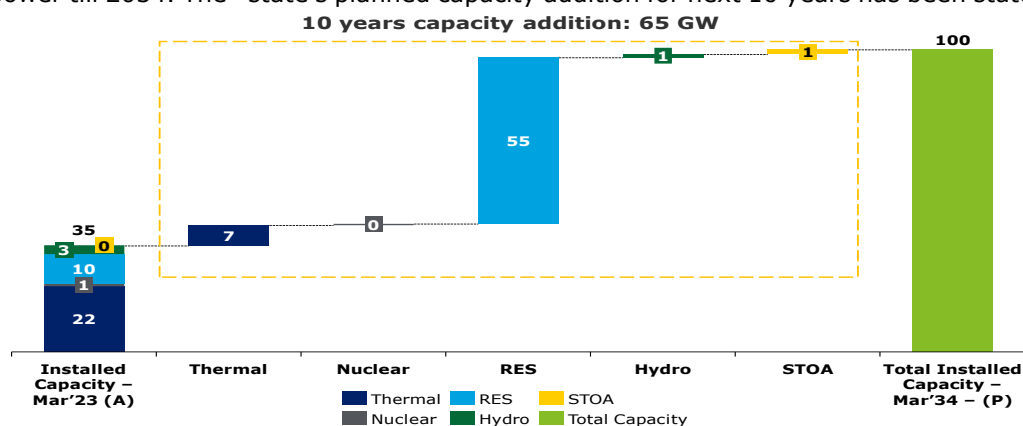


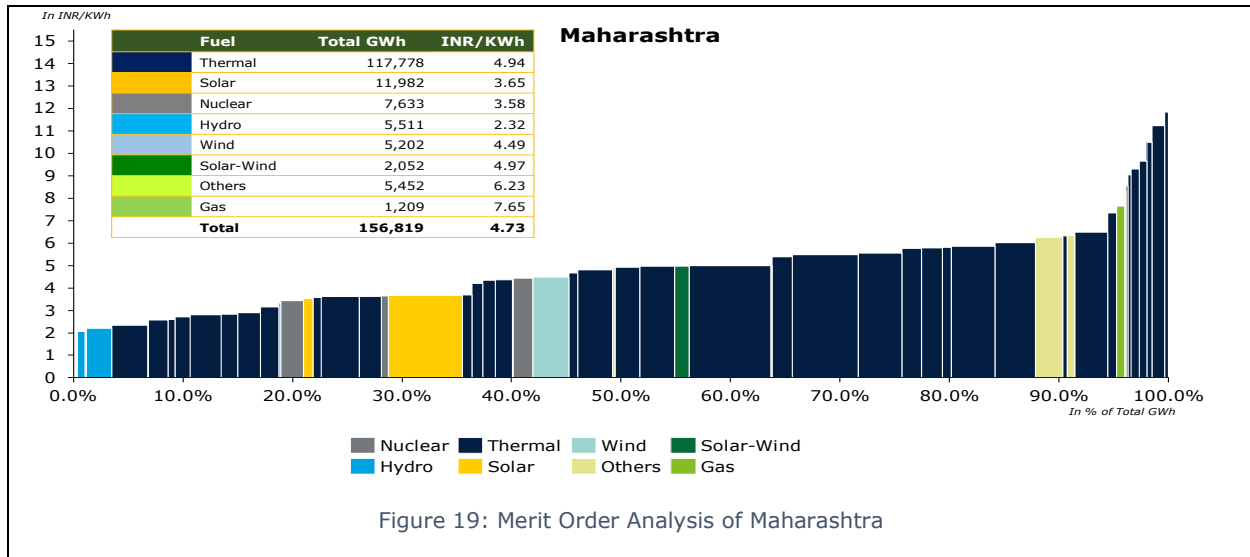
Figure 18: Capacity Addition Plan for next 10 years of Maharashtra

Merit Order Analysis ⁴³:-

Maharashtra's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 156,819 GWh. Of this, thermal power accounts for approximately 75% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 7.97 per kWh. Notably, some of the older thermal assets, such as the Parli- Unit6&7 thermal Power Station operated by MAHAGENCO, report a generation cost of INR 9.64 per kWh and Parli Replacement U8 reports generation cost of INR 11.85 per kWh. Similarly, the gas-based GTPS Uran plant had tariff of at INR 7.65 per kWh, while some cogeneration and biomass projects average around INR 6.98 per kWh.

⁴² Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

⁴³ MYT Order, Maharashtra State Electricity Commission, 2024



1. Region: West

State: Gujarat

Gujarat is one of the India's most industrialized state, has made significant investments in renewable energy, resulting in a substantial share of solar and wind power in its overall energy mix, with peak electricity demand of approximately 25.6 GW⁴⁴ in FY 2025. The state has an installed generation capacity of approximately 57 GW⁴⁵, with about 42% from fossil fuel-based sources, 56% from renewable energy sources (RES), including hydro, and the remaining 2% from nuclear power.

Table 13: Energy Demand of Gujarat

Particulars	Units	2021-22 (Actual) ⁴⁶	2024-25 (Actual) ⁴⁷	2026-27 (Projected) ⁴⁸	2031-32 (Projected) ⁴⁹
Energy Requirement	TWh	124	152	183	243
Peak Demand	GW	19	26	38	36

Fuel Mix of Gujarat⁵⁰

Gujarat's contracted power capacity reached 33 GW by the end of March 2024, of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas and the rest is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, small hydro, wind and other RE.

The projected contracted capacity for the year 2034-35 is 118 GW, out of which 83% through non-fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at CAGR of 10% over the next 10 years. Capacity wise generation mix from 2023-35 has been illustrated below:

⁴⁴ Load Generation Balance Report, 2025

⁴⁵ Government of India, Central Electricity Authority, 2024

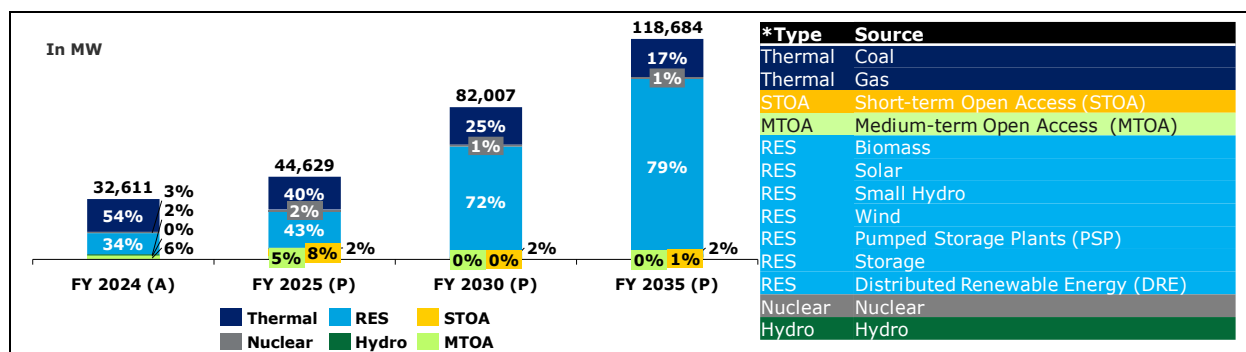
⁴⁶ Load Generation Balance Report, 2022

⁴⁷ Load Generation Balance Report, 2025

⁴⁸ National Electricity Plan 2022-32

⁴⁹ National Electricity Plan 2022-32

⁵⁰ Resource Adequacy Report of Gujarat

Figure 20: Fuel wise capacity mix of Gujarat⁵¹

Financial Health of Gujarat's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 50% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 50%, are those that are still within their respective due dates but have not yet been fully or partially paid : ⁵²

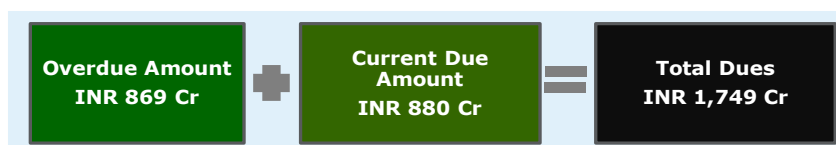


Figure 21: Snapshot of PRAAPTI - Gujarat

Utility Name 1: Gujarat Urja Vikas Nigam Limited (GUVNL)

Gujarat's power distribution is managed primarily by four State-owned distribution companies (DISCOMs) under the umbrella of Gujarat Urja Vikas Nigam Limited (GUVNL). Gujarat Urja Vikas Nigam Limited (GUVNL), established in 2004, is a State-owned holding company responsible for managing Gujarat's power sector following the unbundling of the Gujarat Electricity Board.:

- Dakshin Gujarat Vij Company Limited (DGVCL)
- Madhya Gujarat Vij Company Limited (MGVCL)
- Paschim Gujarat Vij Company Limited (PGVCL)
- Uttar Gujarat Vij Company Limited (UGVCL)

GUVNL serving over 17.47 million⁵³ consumers, across a diverse customer base. The industrial sector accounts for approximately 65% of total electricity consumption, followed by the agricultural sector at 15%, domestic consumers at 12%, the commercial segment at 5%, and the remaining share attributed to other sectors. ⁵⁴

Parameter History of GUVNL:

In FY 2023-24, GUVNL reported consistent increase in gross input energy over the past four years at CAGR of 10% similarly, gross energy sold has also demonstrated steady growth of 11%. The Average Revenue Realized (ARR) consistently remained slightly above the Average Cost of Supply (ACS), resulting in a positive gap that significantly widened in FY 2023-24. This improvement contributed to

⁵¹ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

⁵² Dashboard for Gujarat, PRAAPTI, as accessed on 27th May 2025

⁵³ India Climate & Energy Dashboard

⁵⁴ India Climate & Energy Dashboard- Gujarat

a sharp rise in profitability, which reached ₹4,339 crore in FY 2023-24. Additionally, the reduction in payable days to just 3 suggests enhanced financial discipline and timely payments.⁵⁵

Table 14: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ⁵⁶	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	79,680	95,327	104,078	109,205
Gross Input Energy	GWh	94,585	110,622	121,872	125,896
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	6.12	6.46	7.23	7.12
Average Revenue Realized (ARR)	INR /kWh	6.20	6.52	7.25	7.78
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	0.08	0.06	0.02	0.67
PAT (Profit /(Loss))	INR in Crore	429	371	147	4,339
Payable Days	Days	-	-	-	3

Capacity Addition Plans of GUVNL:⁵⁷

Gujarat's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 54% in FY 2024 to 17% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. As per Resource Adequacy report, GUVNL has plans to increase its capacity mix by 1,098 MW through hydropower till 2035. The state's planned capacity addition for next 10 years has been stated below:

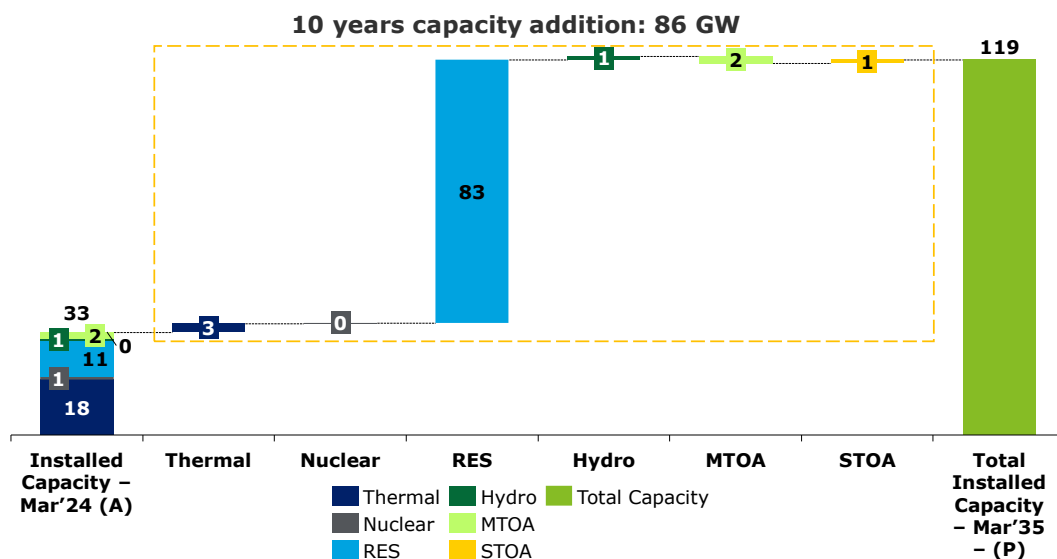


Figure 22: Capacity Addition Plan for next 10 years of Gujarat

Merit Order Analysis:-⁵⁸

Gujarat's energy generation landscape is characterized by a diverse mix of sources, with a total generation of 149,290 GWh. Thermal power remains the leading contributor, generating 59,993 GWh.

⁵⁵ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

⁵⁶ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

⁵⁷ Resource Adequacy Report, Central Electricity Report, GOI, 2024

⁵⁸ Truing up for FY 2022-23 and Determination of Tariff for FY 2024-25, Gujarat Electricity Regulatory Commission, 2024

According to the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal, lignite, and competitively bid sources, with an average cost of INR 11.02 per kWh. Older thermal assets such as NTPC Khargone and NTPC Solapur report significantly high generation costs of INR 30.25 and INR 25.83 per kWh, respectively, mainly due to elevated fixed charges. Similarly, the gas-based plant operated by Gujarat State Energy Generation reports a high generation cost of INR 20.60 per kWh. Lignite-based plants, including GSECL Kutch Lignite and GSECL BLTPS, average around INR 7.57 per kWh. Additionally, competitively bid power from Adani Power Limited is priced at INR 13.42 per kWh, further contributing to the high cost profile of the upper quartile.

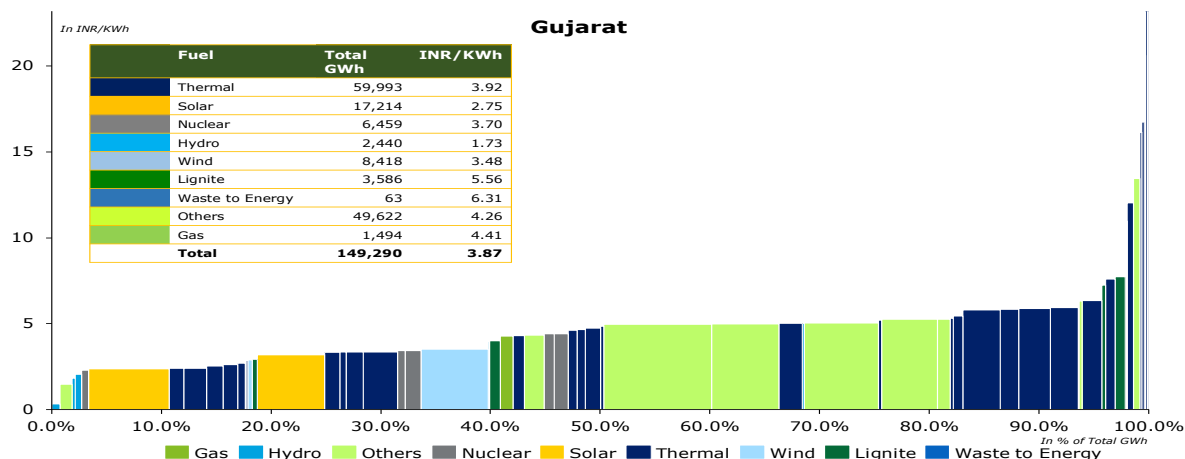


Figure 23: Merit Order Analysis of Gujarat

1. Region: West

State: Madhya Pradesh

In FY 2025, Madhya Pradesh recorded a peak electricity demand of approximately 19.37 GW⁵⁹, of which 19.18 GW was met, resulting in a peak deficit of around 1%. The state has an installed generation capacity of about 28.6 GW⁶⁰, comprising approximately 57% from fossil fuel-based sources, 41.3% from renewable energy sources (RES), including hydro, and the remaining 1.7%⁶¹ from nuclear power. Madhya Pradesh is steadily advancing in renewable energy adoption and significantly contributing to India's overall renewable energy capacity.

Table 15: Energy Demand of Madhya Pradesh

Particulars	Units	2021-22 (Actual) ⁶²	2024-25 (Actual)	2026-27 (Projected) ⁶³	2031-32 (Projected)
Energy Requirement	TWh	85	104	129	169
Peak Demand	GW	16	19	22	27

Fuel Mix of Madhya Pradesh⁶⁴

Madhya Pradesh's contracted power capacity stood at 25 GW with 52% derived from fossil fuel-based generation, primarily coal and natural gas and the rest is derived from non-fossil fuel-based sources, including nuclear, bioenergy, hydro, solar, wind and other RE.

⁵⁹ Load Generation Balance Report, 2025

⁶⁰ All India Installed Capacity, Central Electricity Authority, 2024

⁶¹ Resource Adequacy Report of Madhya Pradesh

⁶² Load Generation Balance Report, 2022

⁶³ National Electricity Plan 2022-32

⁶⁴ Resource Adequacy Report of Madhya Pradesh

The projected contracted capacity for the year 2034-35 is 82 GW, out of which 73% is anticipated to come from non-fossil fuel sources. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:

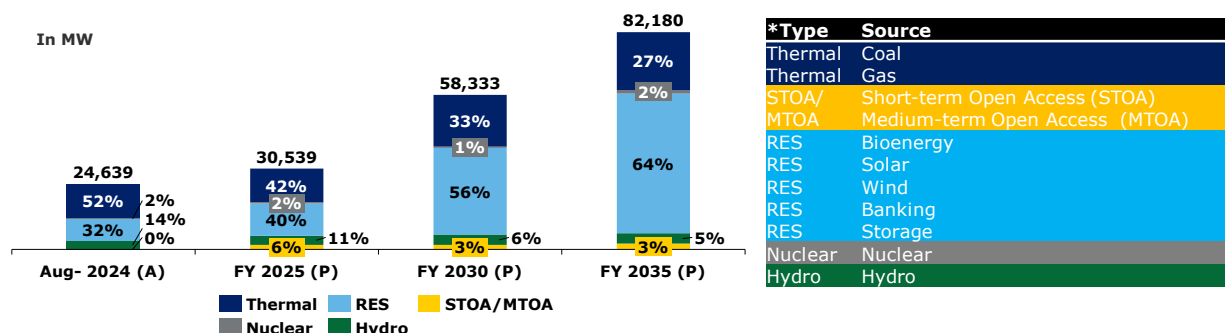


Figure 24: Fuel-wise capacity mix of Madhya Pradesh⁶⁵

Financial Health of Madhya Pradesh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 37% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 63%, are those that are still within their respective due dates but have not yet been fully or partially paid :⁶⁶

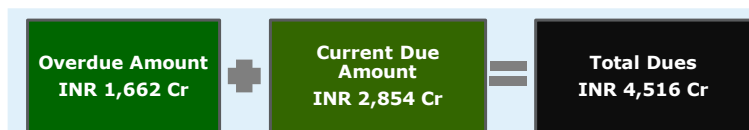


Figure 25: Snapshot of PRAAPTI- Madhya Pradesh

Utility Name 1: M.P. Power Management Company Limited (MPPMCL)

MP Power Management Company has been designated as the holding company for all electricity distribution companies (DISCOMs) in Madhya Pradesh. The state's electricity distribution is managed by three regional DISCOMs:

- Madhya Pradesh Poorv Kshetra Vidyut Vitaran Company Limited (MPPoKVVCL) – East
- Madhya Pradesh Madhya Kshetra Vidyut Vitaran Company Limited (MPMKVVCL) – Central
- Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited (MPPaKVVCL) – West

These DISCOMs serve over 18.06 million⁶⁷ consumers across the eastern, central, and western regions of the state. In FY 2022–23, the agriculture sector accounted for the largest share of electricity consumption at 37%, followed by the industrial sector at 25%, and domestic consumption at 24% and remaining attributed to commercial and other sectors.⁶⁸

Performance History of MPPMCL:

Over the past four years, gross energy sold has shown consistent growth at CAGR of 7% similarly, gross input energy has increased steadily at 8%. The average cost of supply (ACS) has fluctuated slightly but increased to INR 8.22 per kWh in FY 2023–24, while the average revenue realized (ARR)

⁶⁵ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

⁶⁶ Dashboard for Madhya Pradesh, PRAAPTI- as accessed on 27th May 2025.

⁶⁷ India Climate & Energy Dashboard

⁶⁸ India Climate & Energy Dashboard- Madhya Pradesh

has plateaued at INR 7.54 per kWh, persistent gap between ACS and ARR has been noted. PAT has remained negative over past four years. Additionally, payable days have remained high, hovering above 200 days, which is higher than the national average (excluding private DISCOMs) payable days of 130⁶⁹.

Table 16: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ⁷⁰	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	58,828	63,058	67,794	71,365
Gross Input Energy	GWh	80,523	83,100	89,958	96,499
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	8.01	7.94	7.92	8.22
Average Revenue Realized (ARR)	Rs/kWh	7.30	7.41	7.54	7.54
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(0.71)	(0.53)	(0.38)	(0.68)
PAT (Profit/ (Loss))	Rs in Crore	(4,152)	(3,315)	(2,585)	(4,886)
Payable Days	Days	230	217	210	211

Capacity Addition Plans of MPPMCL:

Madhya Pradesh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 42% in FY 2025 to 27% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. Additionally, Madhya Pradesh Pashchim Kshetra Vidyut Vitaran Company Limited (MPPKVCL) has planned to increase its capacity by 339 MW through hydropower till 2035.

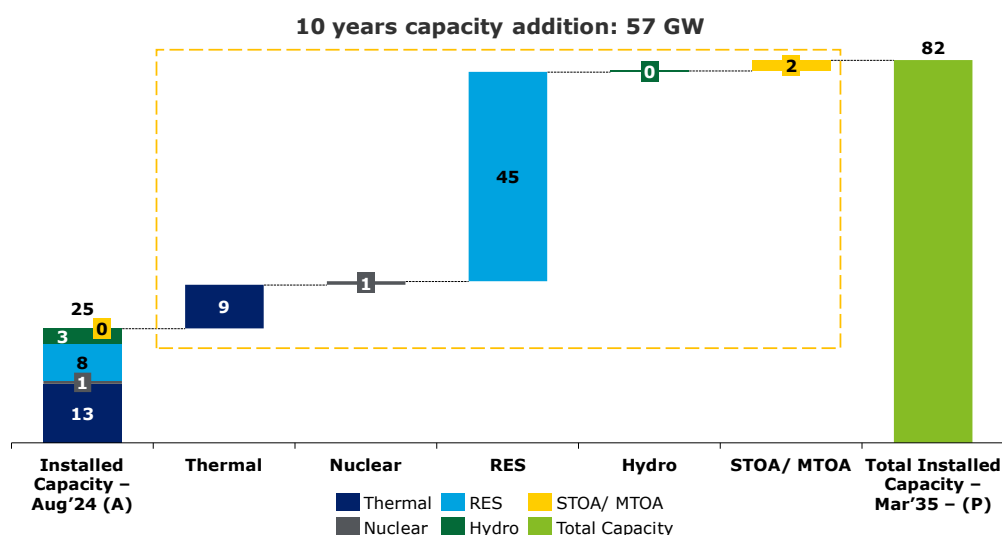


Figure 26: Capacity Addition plan for next 10 years of Madhya Pradesh

Merit Order Analysis ⁷¹:-

Madhya Pradesh's energy generation landscape is characterized by a diverse mix of sources, with thermal power continuing to dominate due to its relatively low average cost of INR 3.21 per kWh. Solar energy stands out as a key renewable contributor, offering both substantial output and cost

⁶⁹ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

⁷⁰ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

⁷¹ MYT Order, Madhya Pradesh Electricity Regulatory Commission

efficiency at INR 3.33 per kWh. According to the approved ARR tariff for FY 2025–26, the fourth quartile (75%–100%) of power purchase cost is primarily composed of thermal power plants, with an average generation cost of INR 4.49 per kWh. However, some older thermal assets, such as NTPC Khargone and NTPC Lara, report significantly higher generation costs of INR 8.18 and INR 7.53 per kWh, respectively. The nuclear plant at KAPP Kakrapar is the most expensive, with a generation cost of INR 11.90 per kWh. Others include non-solar renewable sources contribute at an average tariff of INR 4.36 per kWh, reflecting a moderate cost profile within the state's energy mix.

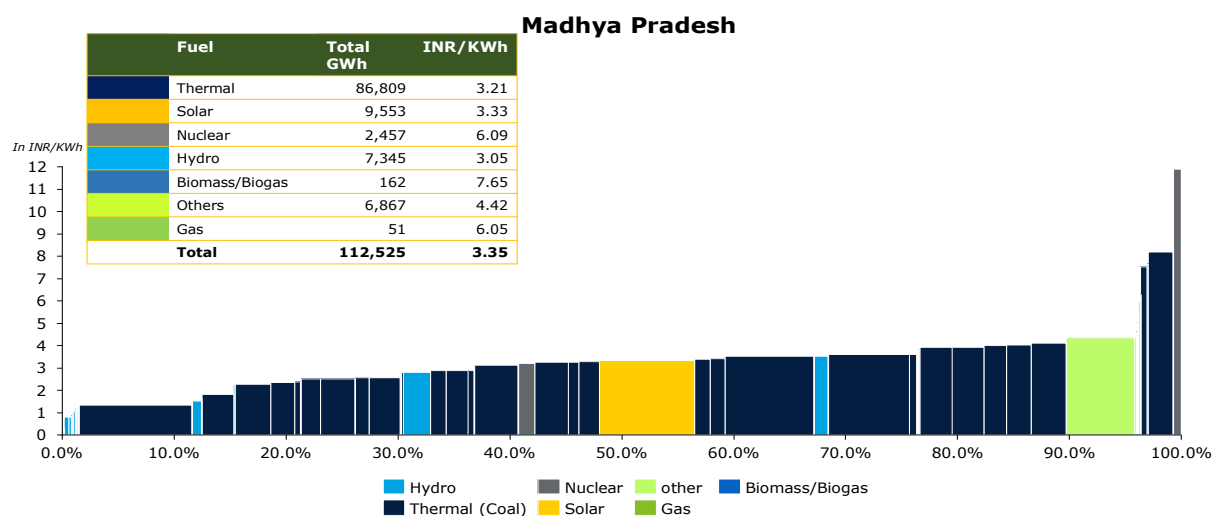


Figure 27: Merit Order Analysis of Madhya Pradesh

1. Region: West

State: Chhattisgarh

In FY 2025, Chhattisgarh recorded a peak electricity demand of approximately 6.5 GW⁷², which has met without any surplus or deficit. The state has an installed generation capacity of around 14.33 GW, with about 85.3% derived from fossil fuel-based sources, 13.8% from renewable energy sources (RES), including hydro, and the remaining 0.9% from nuclear power⁷³. The state has made notable progress in renewable energy adoption, with over 3,000 MW⁷⁴ of installed renewable capacity, contributing around 2.5% to India's total renewable energy portfolio.

Table 17: Energy Demand of Chhattisgarh

Particulars	Units	2021-22 (Actual) ⁷⁵	2024-25 (Actual)	2026-27 (Projected) ⁷⁶	2031-32 (Projected) ⁷⁷
Energy Requirement	TWh	32	43	47	64
Peak Demand	GW	5	7	7	10

Fuel Mix of Chhattisgarh⁷⁸

As of March 2024, Chhattisgarh's contracted power capacity stood at 9 GW with 67% derived from fossil fuel-based sources primarily coal and the rest from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, hybrid, wind power and other RE technologies.

The projected contracted capacity for the year 2034-35 is 35 GW, with 72% expected from non-fossil fuel and 28% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 11% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:

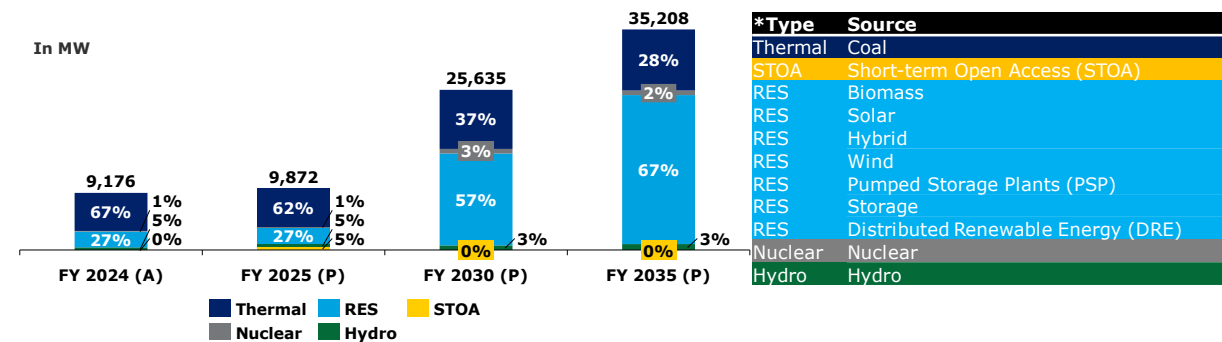


Figure 28: Fuel-wise capacity mix of Chhattisgarh⁷⁹

Financial Health of Chhattisgarh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 12% of the total, refer to payments that have surpassed their due dates and remain either partially or fully

⁷² Load Generation Balance Report, 2025

⁷³ Government of India, Central Electricity Authority, 2024

⁷⁴ The Role of Renewable Energy in Electricity Production in Chhattisgarh, SSR Journal of Engineering and Technology (SSRJET), 2024

⁷⁵ Load Generation Balance Report, 2022

⁷⁶ National Electricity Plan 2022-32

⁷⁷ National Electricity Plan 2022-32

⁷⁸ Resource Adequacy report of Chhattisgarh

⁷⁹ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

unpaid. In contrast, Current dues, accounting for the remaining 88%, are those that are still within their respective due dates but have not yet been fully or partially paid :⁸⁰



Figure 29: Snapshot of PRAAPTI -Chhattisgarh

Utility Name 1: Chhattisgarh State Power Distribution Company Limited (CSPDCL)

Chhattisgarh's electricity distribution is primarily managed by the Chhattisgarh State Power Distribution Company Limited (CSPDCL), a State-owned utility responsible for supplying power across urban and rural areas, serving over 6.31 million⁸¹ consumers. CSPDCL serves a diverse consumer base, with the industrial sector accounting for approximately 63% of total electricity consumption. This is followed by the agricultural sector at 15%, domestic consumers at 14%, and the commercial segment at 4%, with the remaining share attributed to other sectors.⁸²

Performance History of CSPDCL:

Chhattisgarh has demonstrated consistent growth over the past four years, whereas gross energy sold increasing at CAGR of 10%, while 10% similarly gross input energy has grown also risen steadily at 9%. The average cost of supply (ACS) peaked at INR 6.61 per /kWh in FY 2022-23 but declined to INR 6.13 per /kWh in FY 2023-24. Notably, the gap between ACS and Meanwhile, the average revenue realized (ARR) turned positive for improved to INR 6.14/kWh, resulting in a marginal surplus of INR 0.01/kWh—the first time in four years, reaching INR 0.01 per kWh. positive gap in four years. Profit After Tax (PAT), which had remained negative in previous years, showed a turnaround, turned positive in FY 2023-24 with a profit of INR 41 crore in FY 2023-24., indicating a potential financial recovery. Additionally, payable days reduced significantly from 202 to 108, which is notably better than the national average (excluding private DISCOMs) of 130 days, indicating improved financial discipline and liquidity.⁸³

Table 18: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ⁸⁴	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	29,244	31,124	32,483	38,515
Gross Input Energy	GWh	35,302	37,951	39,325	45,453
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	5.70	6.11	6.61	6.13
Average Revenue Realized (ARR)	INR /kWh	5.56	5.72	6.26	6.14
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.14)	(0.39)	(0.35)	0.01
PAT (Profit/(Loss))	INR in Crore	(420)	(1,214)	(1,133)	41
Payable Days	Days	202	181	114	108

⁸⁰ Dashboard for Chhattisgarh, PRAAPTI- as accessed on 27th May 2025

⁸¹ India Climate & Energy Dashboard

⁸² India Climate & Energy Dashboard- Chhattisgarh

⁸³ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

⁸⁴ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of CSPDCL:

Chhattisgarh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2025 to 28% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. Additionally, CSPDCL had planned to purchase 568 MW through hydropower till 2035.

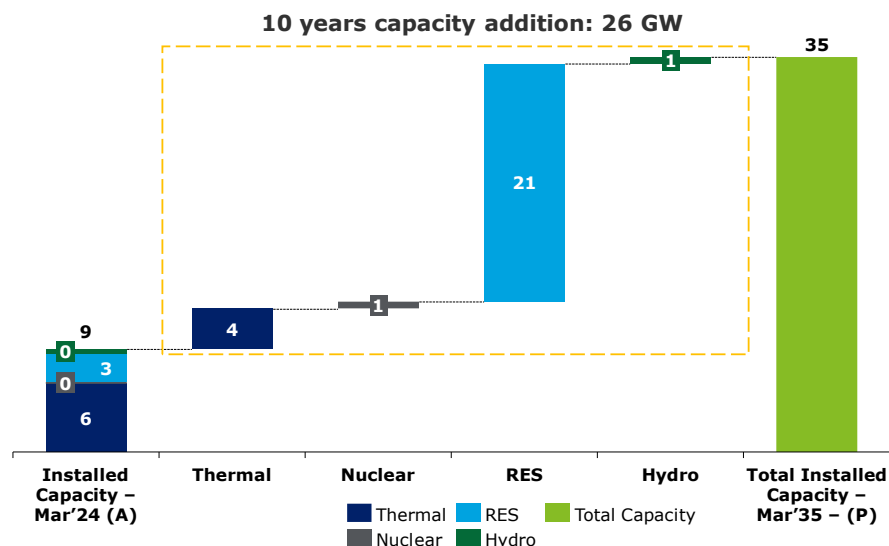


Figure 30: Capacity Addition Plan for next 10 years of Chhattisgarh

Merit Order Analysis⁸⁵:

Chhattisgarh's power generation is dominated by thermal energy, contributing over 80% of the total at an average cost of INR 3.69 per kWh. Solar and other renewables contribute smaller shares but are generally more cost-effective, while non-solar renewables are the most expensive at INR 5.08 per kWh. As per the FY 2025–26 tariff order, the costliest quartile (75%–100%) is mainly thermal, averaging at INR 4.18 per kWh, with older assets like NTPC SAIL reaching INR 4.34 per kWh. Additionally, Hydropower and biomass are the priciest, averaging at INR 7.53 per kWh.

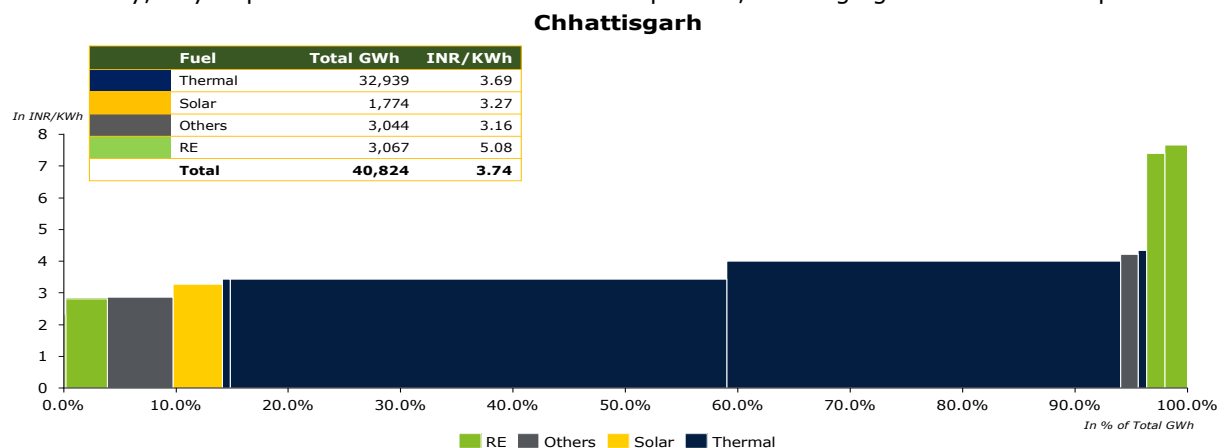


Figure 31: Merit Order Analysis of Chhattisgarh

⁸⁵ Tariff Order for FY 2024-25, Chhattisgarh State Electricity Regulatory Commission

6.1.1. Eastern Region

The Eastern region is a key part of India's energy landscape, with abundant coal reserves and major thermal power plants, especially in Jharkhand and Odisha. West Bengal and Odisha have relatively efficient DISCOMs with a track record of long-term power purchase agreements, including with Bhutanese hydropower projects, and are actively integrating renewables. Bihar and Jharkhand face higher losses and financial stress but are undergoing reforms under RDSS and UDAY. The region's proximity to Bhutan and established transmission links makes it well-positioned for hydropower imports to support grid balancing and renewable integration. List of All States of Eastern have been Stated below⁸⁶:

Table 19: State-wise Gross Input Energy⁸⁷

State	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
West Bengal	53,974	35%
Bihar	47,296	31%
Odisha	35,444	23%
Jharkhand	15,785	10%
Sikkim	1,076	1%
Andaman & Nicobar Islands	378	0%

Among the above-mentioned States, West Bengal, Bihar, Odisha and Jharkhand has been selected further for detailed level Analysis as it covers approximately 99% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 20: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
West Bengal	West Bengal State Electricity Distribution Company Limited	53,974	98%
	India Power Corporation Limited	943	2%
	Total of West Bengal	54,917	
Odisha	Tata Power Western Odisha Distribution Limited (erstwhile WESCO)	12,752	36%
	Tata Power Central Odisha Distribution Limited (erstwhile CESCO)	11,299	32%
	Tata Power Northern Odisha Distribution Limited (erstwhile NESCO)	7,047	20%
	Tata Power Southern Odisha Distribution Limited (erstwhile SouthCO)	4,345	12%
	Total of Odisha	35,443	
Jharkhand	Jharkhand Bijli Vitran Nigam Limited	15,785	100%
	Total of Jharkhand	15,785	
Bihar	South Bihar Power Distribution Company Limited	26,259	56%
	North Bihar Power Distribution Company Limited	21,037	44%
	Total of Bihar	47,296	

⁸⁶ Central Electricity Authority

⁸⁷ Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

2. Region: East

State: West Bengal

West Bengal recorded peak electricity demand of 12.65 GW in FY 2025. The state has an installed generation capacity of 11 GW⁸⁸, with 80% from fossil fuel-based sources and the rest from renewable energy sources (RES), including hydro. West Bengal is a key player in India's electricity sector, with WBSEDCL providing power to over 96% of the state. Though the state is 10th biggest electricity user in India, it used only about of 2112 MW renewable energy potential, showing big chances for growths⁸⁹.

Table 21: Energy Demand of West Bengal

Particulars	Units	2021-22 (Actual) ⁹⁰	2024-25 (Actual) ⁹¹	2026-27 (Projected) ⁹²	2031-32 (Projected) ⁹³
Energy Requirement	TWh	54	71	76	99
Peak Demand	GW	9	13	13	17

Fuel Mix of West Bengal⁹⁴

West Bengal's contracted power capacity reached 11 GW as of March 2024 with 74% is derived from fossil fuel-based sources, primarily coal and the remaining from non-fossil fuel-based energy, including hydro, solar, battery, wind, PSP and Other RE.

The projected contracted capacity for the year 2034-35 is 56 GW, with 75% derived from non-fossil fuel sources and 25% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 14% over the next 10 years. Capacity wise generation mix from FY 2024-35 has been illustrated below:

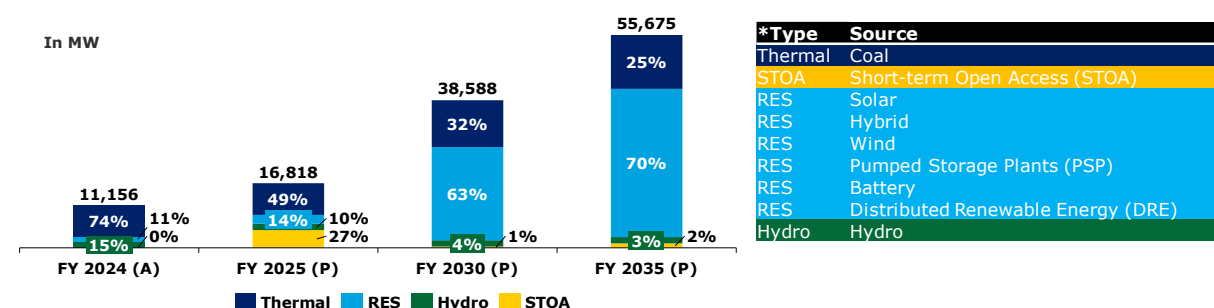


Figure 32: Fuel-wise capacity mix of West Bengal⁹⁵

Financial Health of West Bengal's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 42% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 58%, are those that are still within their respective due dates but have not yet been fully or partially paid :⁹⁶

⁸⁸ All India Installed Capacity, Central Electricity Authority of India, 2025

⁸⁹ Indian Climate & Energy Dashboard

⁹⁰ Load Generation Balance Report, 2022

⁹¹ Load Generation Balance Report, 2025

⁹² National Electricity Plan 2022-32

⁹³ National Electricity Plan 2022-32

⁹⁴ Resource Adequacy Report, Central Electricity Report, GOI, 2024

⁹⁵ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

⁹⁶ Dashboard for West Bengal, PRAAPTI- as accessed on 27th May 2025

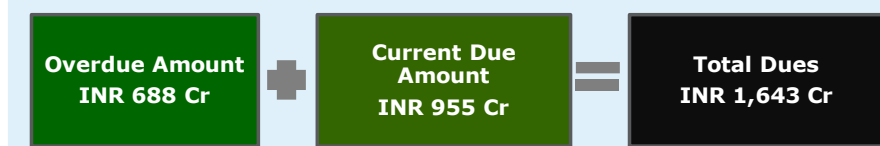


Figure 33: Snapshot of PRAAPTI - West Bengal

Utility Name : West Bengal State Electricity Distribution Company Limited (WBSEDCL)

WBSEDCL is responsible for supplying electricity to 23.10 million consumers across West Bengal. The company is focused on ensuring reliable power distribution and continuous infrastructure development to deliver consistent and efficient services. WBSEDCL caters to a diverse range of consumer categories, with the industrial sector accounting for around 41% of total electricity consumption. This is followed by the domestic consumers at 32%, commercial segment at 12%, agricultural sector at 3%, and the remaining share attributed to other sectors.⁹⁷

Performance History of WBSEDCL:

In FY 2023–24, WBSEDCL reported consistent increase in gross input energy over the past four years at CAGR of 7% similarly, gross energy sold has also demonstrated steady growth of 8%. The average cost of supply (ACS) has shown fluctuations during the period, peaking in FY 2023–24. Concurrently the average revenue realized (ARR) has also increased, helping to narrow the gap between cost and revenue. Although the gap between ARR vs ACS remained negative in most years, the deficit has reduced significantly. Profitability has shown a positive turnaround, moving from losses in FY 2020–21 and FY 2021–22 to profits in FY 2022–23 and FY 2023–24. Payable days, though declining over the past three years, rose again in FY 2023–24 and remain above the national average (excluding private DISCOMs) of 130 days.⁹⁸

Table 22 : Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ⁹⁹	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	34,522	43,054	41,888	43,636
Gross Input Energy	GWh	44,221	51,253	51,958	53,974
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	7.84	6.21	7.53	7.86
Average Revenue Realized (ARR)	Rs/kWh	6.61	6.45	7.14	7.65
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(1.23)	0.24	(0.40)	(0.21)
PAT (Profit/(Loss))	Rs in Crore	(190)	(205)	21	109
Payable Days	Days	175	147	143	162

Capacity Addition Plans of WBSEDCL:

WBSEDCL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 74% in FY 2024 to 25% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise.

⁹⁷ India Climate & Energy Dashboard⁹⁸ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24⁹⁹ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

As per Resource Adequacy report, WBSEDCL has plans to increase its capacity mix by 90 MW through hydropower till 2035. The state's planned capacity addition for next 10 years has been stated below:¹⁰⁰

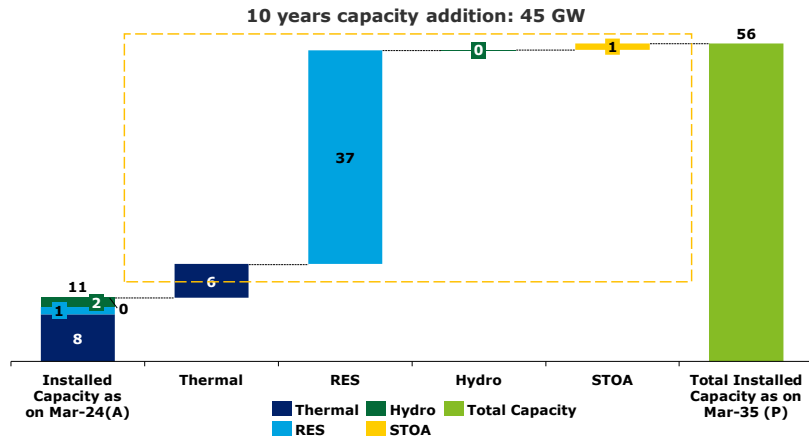


Figure 34: Capacity addition plans for next 10 years of West Bengal

Merit Order Analysis¹⁰¹:

West Bengal's energy generation landscape is characterized by a diverse mix of sources, with a total generation of 56,589 GWh. Thermal power remains the leading contributor, generating 47,609 GWh. According to the approved tariff order for FY 2023-24, the fourth quartile (75%-100%) of the state's power procurement is primarily composed of thermal and hydropower plant, with an average cost of INR 5.06 per kWh. Older thermal assets such as NVVN (Bundled Power) and Hiranmaye Energy Limited report significantly high generation costs of INR 6.78 and INR 5.37 per kWh, respectively. Similarly, the hydropower plant of NHPC's Teesta Low Dam-III (TLDP-III) reports a high generation cost of INR 5.57 per kWh.

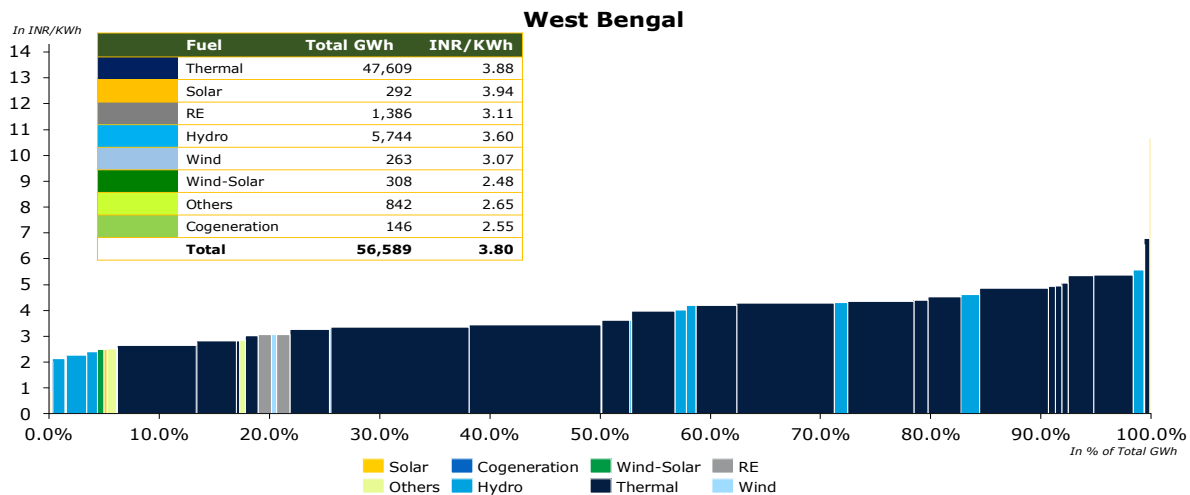


Figure 35: Merit Order Analysis of West Bengal

¹⁰⁰ Report on Resource Adequacy of West Bengal upto 2034-35, Central Electricity Authority, GoI

¹⁰¹ MYT 2023-24, West Bengal Electricity Regulatory Commission, 2023-24

2. Region: East

State: Jharkhand

Jharkhand's peak electricity demand for FY 2025 stood at 2.3 GW, which was fully met, resulting in minimal peak deficit of 0.1%³⁵. The state has an installed generation capacity of 2.37 GW¹⁰², with 87% sourced from fossil fuel and remaining 13% from renewable energy sources (RES), including hydro.

Jharkhand's electricity profile is heavily reliant on coal-based thermal power, which contributes over 85% of its utility scale generation.

Table 23: Energy Demand of Jharkhand

Particulars	Units	2021-22 (Actual) ¹⁰³	2024-25 (Actual) ¹⁰⁴	2026-27 (Projected) ¹⁰⁵	2031-32 (Projected) ¹⁰⁶
Energy Requirement	TWh	10.9	15.2	25.5	33.8
Peak Demand	GW	1.9	2.3	3.8	5.00

Fuel Mix of Jharkhand ¹⁰⁷

Jharkhand's contracted power capacity reached 3 GW by the end of March 2023, of this 63% is derived from fossil fuel-based sources, primarily coal while remaining 37% from non-fossil fuel-based energy sources, including hydro, solar, storage, wind and Distributed RE.

The projected contracted capacity for the year 2033-34 is 13 GW, with 57% from non-fossil fuel and remaining 43% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 10% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:

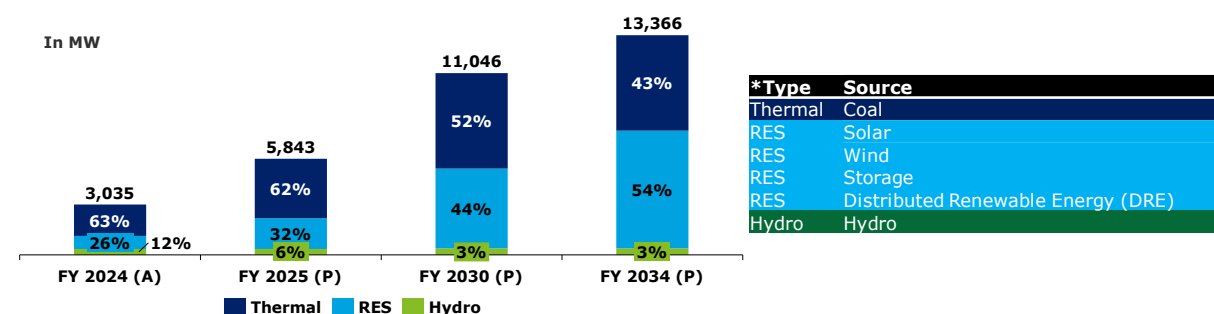


Figure 36: Fuel-wise capacity mix of Jharkhand¹⁰⁸

Financial Health of Jharkhand's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 3% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 97%, are those that are still within their respective due dates but have not yet been fully or partially paid:¹⁰⁹

¹⁰² All India Installed Capacity, Central Electricity Authority of India, 2025

¹⁰³ Load Generation Balance Report, 2022

¹⁰⁴ Load Generation Balance Report, 2025

¹⁰⁵ National Electricity Plan 2022-32

¹⁰⁶ National Electricity Plan 2022-32

¹⁰⁷ Resource Adequacy Report, Central Electricity Report, GOI, 2024

¹⁰⁸ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

¹⁰⁹ Dashboard for Jharkhand, PRAAPTI- as accessed on 27th May 2025

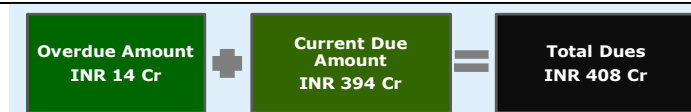


Figure 37: Snapshot of PRAAPTI - Jharkhand

Utility Name : Jharkhand Bijli Vitran Nigam Limited (JBVNL)

Jharkhand Bijli Vitran Nigam Limited (JBVNL) is the largest electricity distribution company in Jharkhand, formed after the unbundling of the Jharkhand State Electricity Board in 2013. JBVNL serves over 5.84 million consumers, serving majorly industrial consumers followed by domestic, commercial, agriculture and others.¹¹⁰

Performance History of Jharkhand:

Jharkhand's power sector exhibited a steady increase in both gross energy sold and input energy at CAGR of 11% and 6% respectively. The average cost of supply (ACS) remained consistently high, peaking at INR 11.27 per kWh in FY 2023-24, whereas average revenue realized (ARR) also improved during this period. The gap between ARR and ACS remained negative, while slightly narrowing in FY 2023-24.¹¹¹

Table 24: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ¹¹²	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	8,026	9,491	9,820	10,928
Gross Input Energy	GWh	13,290	14,369	14,468	15,785
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	10.09	9.06	10.93	11.27
Average Revenue Realized (ARR)	INR /kWh	7.35	6.91	7.14	8.90
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(2.74)	(2.15)	(3.79)	(2.37)
PAT (Profit/(Loss))	INR in Crore	(2,200)	(2,038)	(3,720)	(2,587)
Payable Days	Days	555	536	433	420

¹¹⁰ India Climate & Energy Dashboard¹¹¹ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24¹¹² Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Capacity Addition Plans of JBVNL:

Jharkhand's energy capacity is expected to grow significantly, with a notable shift toward more diversified energy mix. The share of coal expected to decline from 63% in FY 2023 to 43% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise, reflecting the state's commitment to cleaner and more sustainable energy solutions. JBVNL plans to procure additional capacity of 20 MW hydropower till 2034. Capacity Addition requirement has been illustrated below¹¹³:

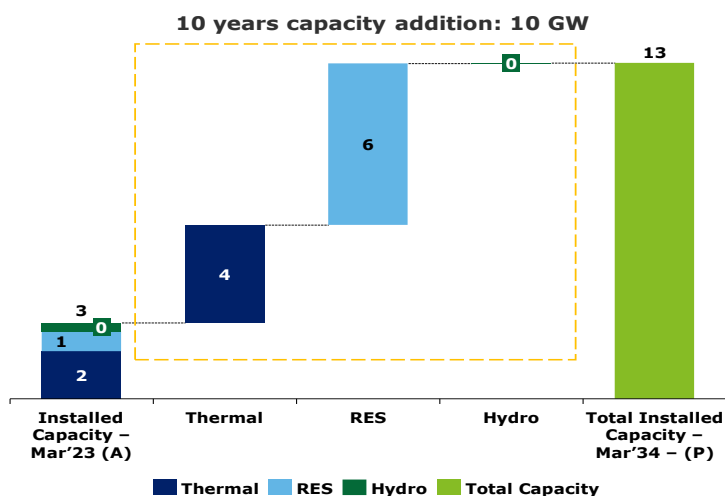


Figure 38: Capacity addition plan for next 10 years of Jharkhand

Merit Order Analysis¹¹⁴:

Jharkhand's energy generation landscape features a diverse mix of sources, with total generation reaching 14,838 GWh. Thermal power remains the dominant contributor, accounting for 11,049 GWh. As per the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal and solar sources, with an average cost of ₹7.28 per kWh. Notably, the state reports significantly high generation costs from State IPPs (MNRE-I) and SECI (MNRE-II), at ₹17.95 per kWh and ₹6.49 per kWh respectively. These elevated costs are largely attributed to high fixed charges and annual tariff escalations applicable in FY 2024–25.

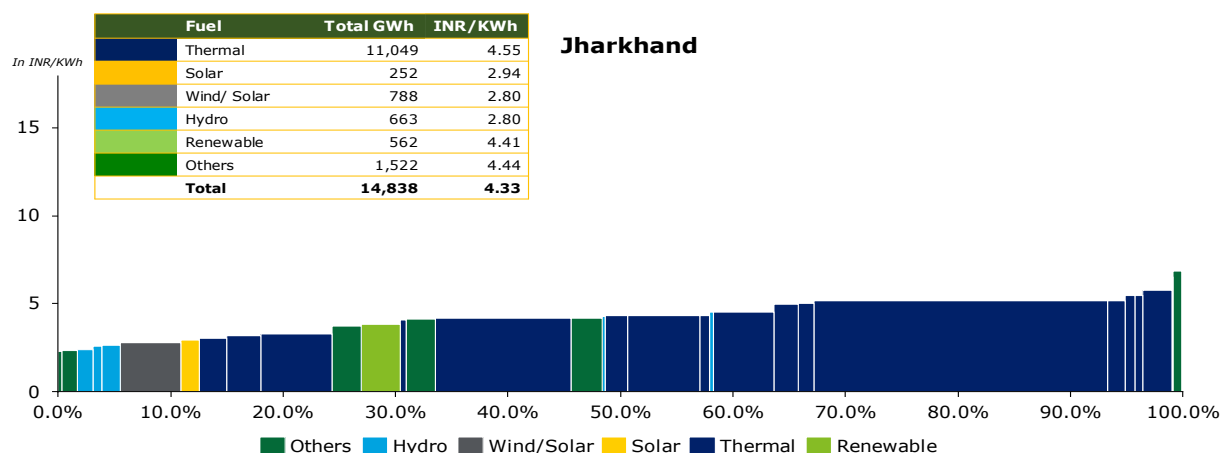


Figure 39: Merit Order Analysis of Jharkhand

¹¹³ Report on Resource Adequacy of Jharkhand upto 2033-34, Central Electricity Authority, GoI

¹¹⁴ Tariff 2024-25, Jharkhand Electricity Regulatory Commission, 2025

2. Region: East

State: Bihar

Bihar's peak electricity demand for FY 2025 was 8.08 GW of with 7.85 GW was met, resulting in a peak deficit of 2.8%¹¹⁵. The state has an installed generation capacity of 7.90 GW¹¹⁵, with 92% from fossil fuel-based sources and remaining from renewable energy sources (RES), including hydro. Despite historical challenges in power infrastructure, Bihar is steadily expanding its renewable energy footprint and currently contributes around 2.1% of India's total renewable energy capacity.

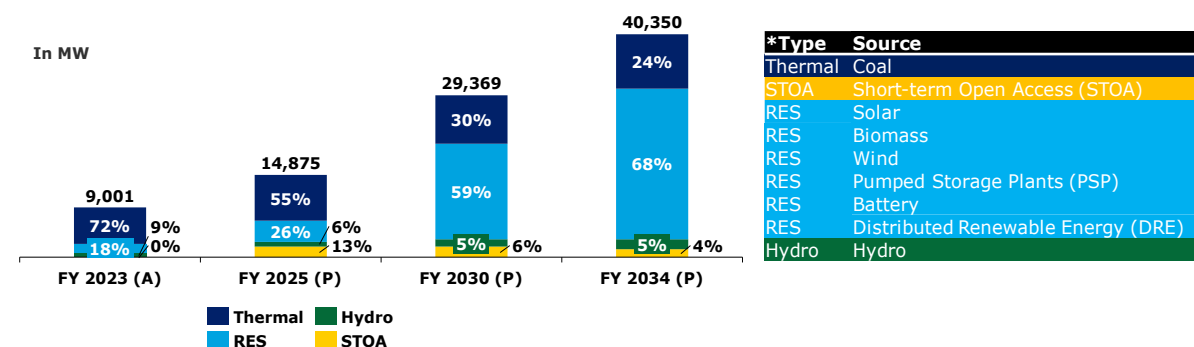
Table 25: Energy Demand of Bihar

Particulars	Units	2021-22 (Actual) ¹¹⁶	2024-25 (Actual) ¹¹⁷	2026-27 (Projected) ¹¹⁸	2031-32 (Projected)
Energy Requirement	TWh	36	44	58	83
Peak Demand	GW	7	8	11	15

Fuel Mix of Bihar ¹¹⁹

Bihar's contracted power capacity reached 9 GW by the end of March 2023. Of this, 72% is derived from fossil fuel-based sources, primarily coal and remaining 28% from non-fossil fuel-based energy, including hydro, solar, battery, wind, PSP and other RE.

The projected contracted capacity for the year 2033-34 is 40 GW, with 76% from non-fossil fuel and 24% from fossil fuel based. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:



Financial Health of Bihar's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 6% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 94%, are those that are still within their respective due dates but have not yet been fully or partially paid:¹²¹

¹¹⁵ All India Installed Capacity, Central Electricity Authority of India, 2025

¹¹⁶ Load Generation Balance Report, 2022

¹¹⁷ Load Generation Balance Report, 2025

¹¹⁸ National Electricity Plan 2022-32

¹¹⁹ Resource Adequacy Report, Central Electricity Report, GOI, 2024

¹²⁰ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

¹²¹ Dashboard for Bihar, PRAAPTI- as accessed on 27th May 2025

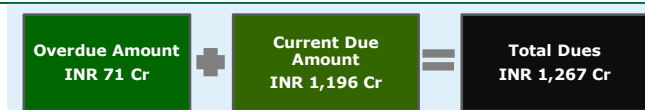


Figure 41: Snapshot of PRAAPTI -Bihar

Utility Name : Bihar State Power Holding Company Limited (BSPHCL)

Bihar's electricity distribution is managed by two State-owned DISCOMs under Bihar State Power Holding Company Limited (BSPHCL):

- North Bihar Power Distribution Company Ltd. (NBPDC)
- South Bihar Power Distribution Company Ltd. (SBPDC)

Together, they serve over 19.08 million¹²² consumers, with the domestic sector accounting for the largest share, followed by industrial, agricultural and commercial sectors.

Performance History of BSPHCL:

Bihar's power sector demonstrated steady increase in gross energy sold at 18% and gross input energy at 11%. The average cost of supply (ACS) showed a gradual decline from ₹9.06/kWh in FY 2020-21 to INR 8.42 per kWh in FY 2023-24 meanwhile, the average revenue realized (ARR) steadily increased, reaching INR 8.79 per kWh in FY 2023-24. This led to a positive gap between ARR and ACS in the latest year, marking a financial turnaround after years of deficits. PAT followed a similar trend, shifting from significant losses in FY 2020-22 to a modest profit in FY 2022-23 and a substantial profit of INR 1,411 crore in FY 2023-24. Payable days, though still relatively high, remained stable, ranging over the four-year period.¹²³

Table 26: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ¹²⁴	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	23,475	31,311	37,685	38,333
Gross Input Energy	GWh	34,205	40,799	47,731	47,296
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	9.06	8.31	8.52	8.42
Average Revenue Realized (ARR)	INR /kWh	7.99	7.63	8.52	8.79
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(1.07)	(0.68)	-	0.37
Profitability (Surplus/(Deficit))	INR in Crore	(2,523)	(2,128)	7	1,411
Payable Days	Days	104	148	104	109

Capacity Addition Plans of BSPHCL:

Bihar's energy capacity is projected to grow significantly, with a notable shift toward more diversified energy mix. The share of coal expected to decline from 72% in FY 2023 to 24% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. NBPDC plans to procure additional capacity of 1,000 MW through hydropower till 2034. Capacity Addition requirement has been illustrated below¹²⁵:

¹²² India Climate & Energy Dashboard

¹²³ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

¹²⁴ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

¹²⁵ Report on Resource Adequacy of Bihar, Central Electricity Authority, GoI

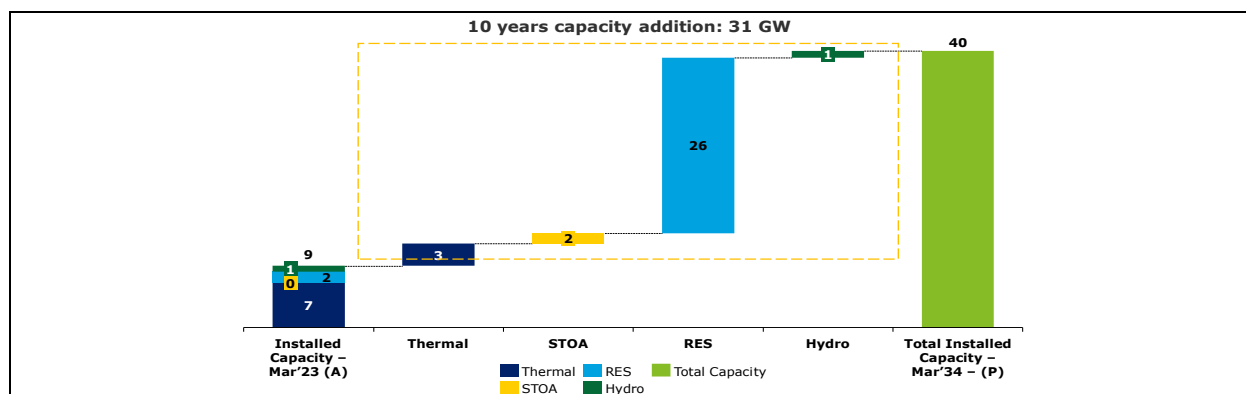


Figure 42: Capacity addition plan for next 10 years of Bihar

Merit Order Analysis¹²⁶:

Bihar's energy generation landscape features a diverse mix of sources, with a total generation of 46,422 GWh. Thermal power remains the dominant contributor, accounting for 33,404 GWh. According to the approved tariff order for FY 2024–25, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal power plants, along with some solar and biomass facilities, at an average cost of INR 8.14 per kWh. Certain thermal assets, such as Barh Stage I Unit III and Barh Stage II, report significantly high generation costs of INR 16.29 and INR 10.95 per kWh, respectively, largely due to elevated fixed costs. Similarly, solar-based projects like the Acme Cleantech and Welspun Renewables report high generation costs of INR 8.74 and INR 8.60 per kWh, respectively. Additionally, the Siddhashram Rice Mill Cluster Pvt. Ltd., a biomass-based plant, records generation cost among renewables at approximately INR 9.22 per kWh.

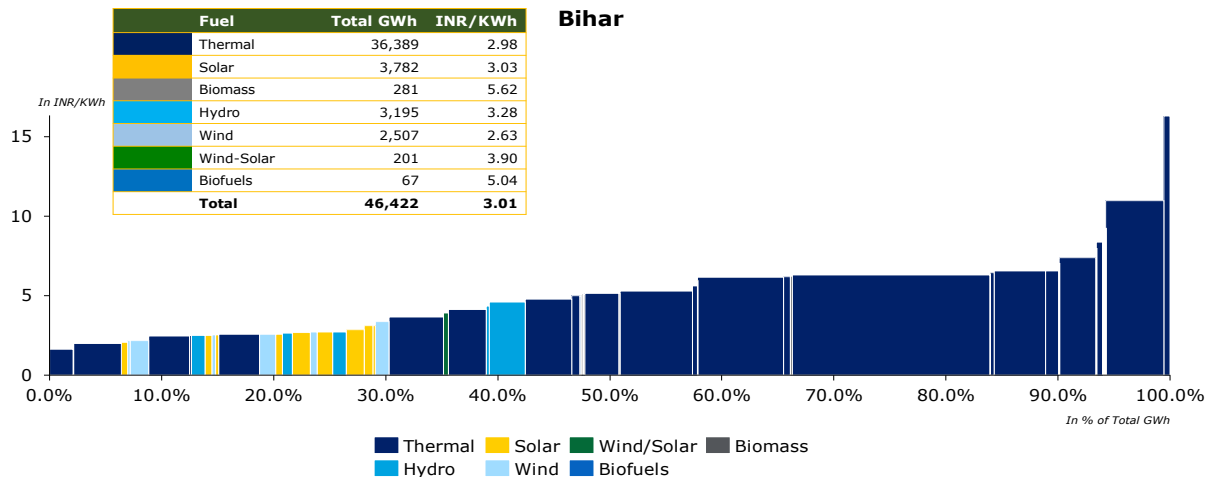


Figure 43: Merit Order Analysis of Bihar

¹²⁶ True-Up for FY 2022-23, Annual Performance Review for FY 2023-24, Annual Revenue Requirement (ARR) and Determination of Retail Tariff for FY 2024-25, Bihar Electricity Regulatory Commission

2. Region: East

State: Odisha

Odisha's peak electricity demand for FY 2025 was 6.9 GW³⁵. The state has an installed generation capacity of 8.33 GW¹²⁷, with 64% sourced from fossil fuel-based generation and remaining 36% from renewable energy sources (RES), including hydro. Despite its current dependence on coal, Odisha is actively working to diversify its energy mix by expanding its renewable energy portfolio. The state plans to shift its power mix from the current 45:55 ratio of of renewable to conventional energy to 55:45 over the next decade ¹²⁸.

Table 27: Energy Demand of Odisha

Particulars	Units	2021-22 (Actual) ¹²⁹	2024-25 (Actual) ¹³⁰	2026-27 (Projected) ¹³¹	2031-32 (Projected)
Energy Requirement	TWh	38	43	49	59
Peak Demand	GW	5.6	7	8	10

Fuel Mix of Odisha¹³²

Odisha's contracted power capacity reached 8 GW by the end of March 2023, with 53% sourced from fossil fuel-based generation, primarily coal and the remaining 47% from non-fossil fuel-based sources, including hydro, solar, battery, wind, PSP and other RE. The state's projected contracted capacity for FY 2033-34 is expected to rise significantly to 26 GW. Of this, 76% is anticipated from non-fossil fuel sources while 24% from fossil fuels. This shift highlights Odisha's strategic focus on sustainability and energy diversification. The state's contracted capacity is projected to grow at CAGR of 6% over the ten-year period. Capacity wise generation mix from FY 2023-34 has been illustrated below:

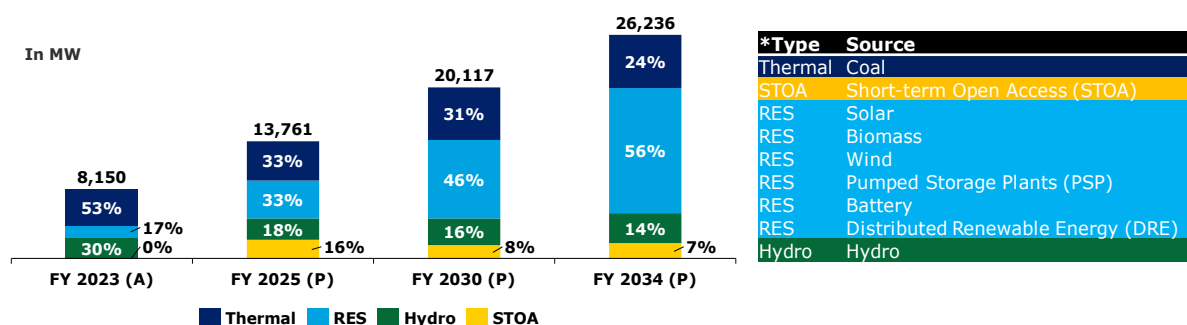


Figure 44: Fuel-wise capacity mix of Odisha¹³³

Financial Health of Odisha's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: Overdue and Current dues. Overdue dues, which constitute 16% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 84%, are those that are still within their respective due dates but have not yet been fully or partially paid :

¹²⁷ All India Installed Capacity, Central Electricity Authority of India, 2025

¹²⁸ Green Initiative of Government of Odisha

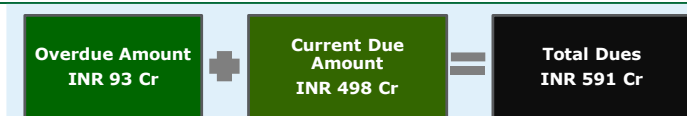
¹²⁹ Load Generation Balance Report, 2022

¹³⁰ Load Generation Balance Report, 2025

¹³¹ National Electricity Plan 2022-32

¹³² Resource Adequacy Report, Central Electricity Report, GOI, 2024

¹³³ As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

Figure 45: Snapshot of PRAAPTI - Odisha¹³⁴**Utility Name 1: Grid Corporation of Odisha Limited (GRIDCO)**

GRIDCO plays a central role in ensuring power availability and financial stability in Odisha's electricity sector. It supplies power to the four Tata Power-led DISCOMs (TPCODL, TPSODL, TPNODL, TPWODL) and manages power purchase agreements, renewable energy obligations, and grid coordination. Odisha's power distribution sector is managed by four regional Tata Power-led distribution companies, operating as joint ventures between Tata Power and the Government of Odisha:

- TP Central Odisha Distribution Ltd. (TPCODL)
- TP Southern Odisha Distribution Ltd. (TPSODL)
- TP Northern Odisha Distribution Ltd. (TPNODL)
- TP Western Odisha Distribution Ltd. (TPWODL)

These DISCOMs serve both urban and rural consumers across the State, covering around 9.89 million consumers, with the industrial sector forming the largest share, followed by domestic, commercial, agricultural and others.

Performance History of GRIDCO:

Odisha's power sector has demonstrated steady growth, with gross energy sold increasing at a CAGR of 15% and gross input energy at 13% over the past four years. The average cost of supply (ACS) rose consistently from INR 5.81 per kWh in FY 2020-21 to INR 6.47 per kWh in FY 2023-24, reflecting rising operational costs. Meanwhile, the average revenue realized (ARR) also improved, contributing to better financial performance. Profit after tax (PAT) remained positive for the last three consecutive years, indicating sustained profitability. Additionally, payable days declined significantly from 106 in FY 2020-21 to just 54 in FY 2023-24, highlighting improved financial discipline and timely settlement of dues.¹³⁵

Table 28: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ¹³⁶	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	18,717	21,500	26,851	28,710
Gross Input Energy	GWh	24,532	27,443	33,573	35,444
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	5.81	5.88	6.13	6.47
Average Revenue Realized (ARR)	INR /kWh	5.35	6.35	6.44	6.34
Gap between ARR vs ACS (Surplus/(Deficit))	INR /kWh	(0.46)	0.47	0.31	(0.12)
PAT (Profit/(Loss))	INR in Crore	(909)	236	253	307
Payable Days	Days	106	56	56	54

Capacity Addition Plans of Odisha:

¹³⁴ Dashboard for Odisha, PRAAPTI- as accessed on 27th May 2025

¹³⁵ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

¹³⁶ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Odisha's energy capacity is projected to grow significantly with diversified energy mix. The share of coal expected to decline from 54% in FY 2023 to 24% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. TPCODL plans to procure additional capacity of 1,101 MW through hydropower till 2034. Capacity Addition requirement has been illustrated below¹³⁷:

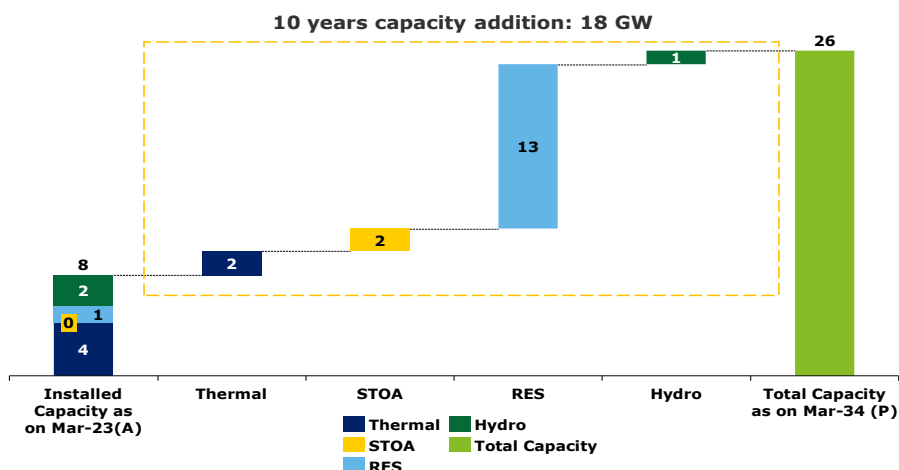


Figure 46: Capacity addition plan for next 10 years of Odisha

Merit Order Analysis¹³⁸:-

Odisha's energy generation landscape features a diverse mix of sources, with total generation reaching 39,652 GWh. Thermal power remains the dominant contributor, accounting for 22,081 GWh. As per the approved tariff order for FY 2023–24, the fourth quartile (75%–100%) of the state's power procurement is primarily composed of thermal and solar and hydro sources, with an average cost of ₹5.15 per kWh. Notably, the state reports significantly high generation costs from Barh I, at ₹9.17 per kWh. These elevated costs are largely attributed to high fixed charges.

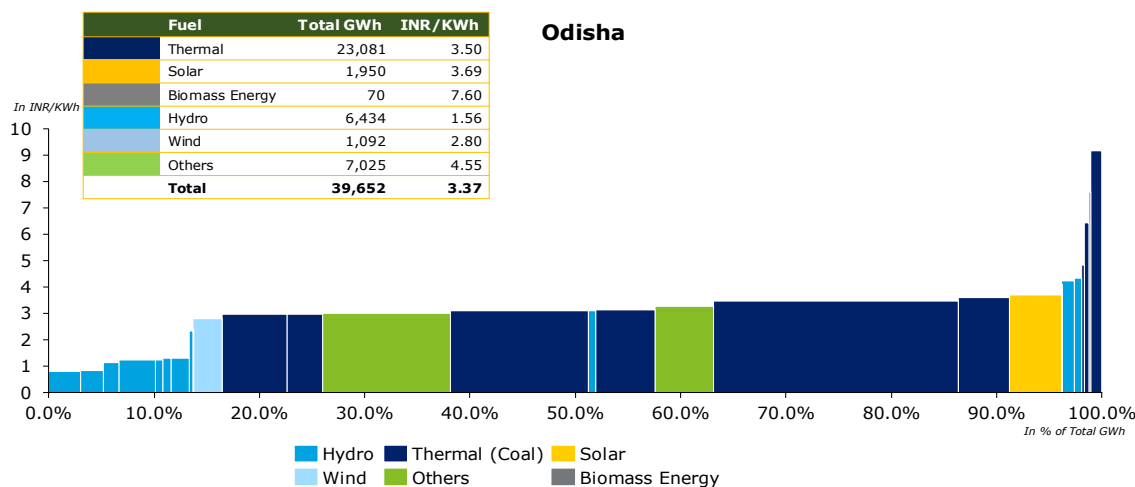


Figure 47: Merit Order Analysis of Odisha

¹³⁷ Report on Resource Adequacy of Odisha, Central Electricity Authority, GoI

¹³⁸ True-Up for FY 2023-24, Odisha Electricity Regulatory Commission, 2023-24

6.1.2. North-Eastern Region

The North-East region plays a strategic role in India's energy sector due to its rich hydropower potential and border proximity. Assam and Tripura have relatively better-developed infrastructure, while other states face challenges like limited grid access and high transmission losses. The region is gradually adopting solar and small hydro projects to enhance energy access. Its integration with the national grid and proximity to Bhutan and Bangladesh also position it well for cross-border electricity trade.

List of All States of Eastern & North-Eastern Region have been Stated below¹³⁹:

Table 29: State-wise Gross Input Energy¹⁴⁰

State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Assam	13,344	56%
Tripura	3,174	13%
Meghalaya	2,633	11%
Arunachal Pradesh	1,577	7%
Manipur	1,174	5%
Nagaland	1,036	4%
Mizoram	886	4%

Among the above-mentioned States, Assam has been selected from North-Eastern Region for further for detailed level Analysis. Refer below mentioned table:

3. Region: North-East

State: Assam

Assam's peak electricity demand for FY 2025 was 2.81 GW, with 2.68 GW met, resulting in a peak deficit of 4.4%.³⁵ Assam has an installed generation capacity of 2.37 GW¹⁴¹, with 68% from fossil fuel-based projects, 32% from renewable energy sources (RES), including hydro. Assam is committed to increasing its renewable energy adoption, aiming to achieve an additional 11,700 MW¹⁴² of renewable power projects by 2030.

Assam is the largest State of North-Eastern part of India, which has approximately 6.4 million¹⁴³ electricity consumers. As of March 2023, Assam's total contracted capacity stands at 2,430 MW¹⁴⁴, with 31% derived from hydro and 9% from solar Energy.

Table 30: Energy Demand of Assam

Particulars	Units	2021-22 (Actual) ¹⁴⁵	2024-25 (Actual) ¹⁴⁶	2026-27 (Projected) ¹⁴⁷	2031-32 (Projected)
Energy Requirement	TWh	11	13	15	20
Peak Demand	GW	2	3	3	4

¹³⁹ Central Electricity Authority

¹⁴⁰

Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

¹⁴¹ All India Installed Capacity, Central Electricity Authority of India, 2025

¹⁴² Assam: Policy targets 11,700-MW renewable energy in State, 2025

¹⁴³ APDCL

¹⁴⁴ Resource Adequacy Report of Assam, Central Electricity Authority, 2024

¹⁴⁵ Load Generation Balance Report, 2022

¹⁴⁶ Load Generation Balance Report, 2025

¹⁴⁷ National Electricity Plan 2022-32

Fuel Mix of Assam ¹⁴⁸

Assam’s contracted power capacity reached 25 GW by the end of March 2024. Of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2034-35 is 118 GW, out of which 72% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 17% over the period of next 10 years. Capacity wise generation mix from FY 2024-35 has been illustrated below:

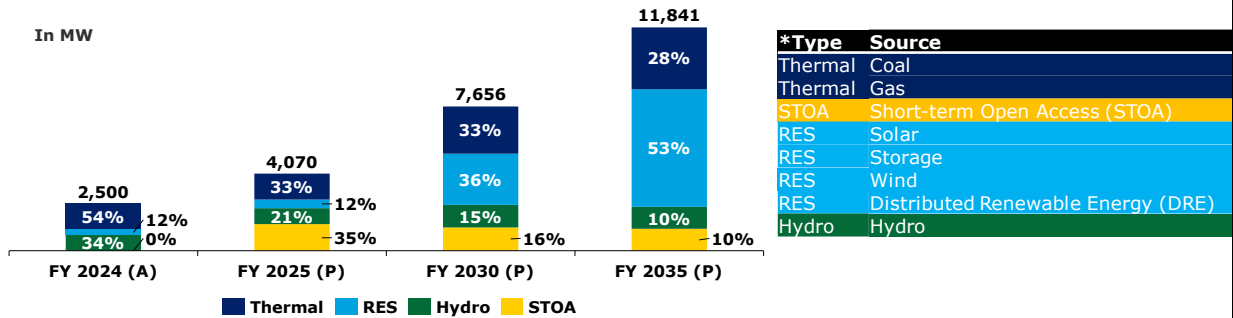


Figure 48: Fuel-wise capacity mix of Assam*

Financial Health of Assam’s Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 1% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 99%, are payments still within their respective due dates but not yet fully or partially paid:¹⁴⁹

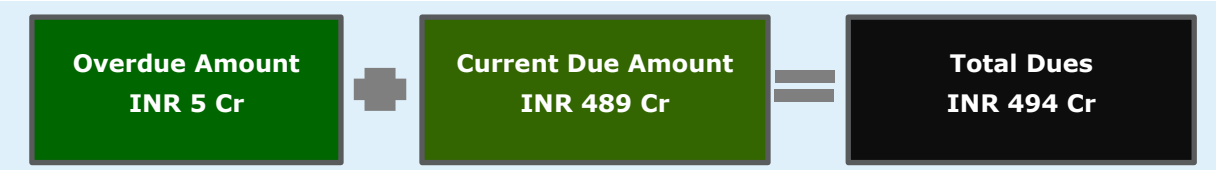


Figure 49: Snapshot of PRAAPTI - Assam

Utility Name : Assam Power Distribution Company Limited (APDCL)

Assam Power Distribution Company Limited (APDCL) is a pivotal entity in the state’s electricity sector, catering to approximately 6.7 million consumers. APDCL’s advancement of solar energy projects has enhanced energy security and socio-economic growth, positioning the company as a key player in Assam’s sustainable energy development. Electricity consumers are distributed unevenly across sectors, with domestic usage at 42.99%, followed by industrial (26.3%), commercial (13.62%), and the remaining in other categories.

¹⁴⁸ Resource Adequacy Report, Central Electricity Report, GOI, 2024

¹⁴⁹ Dashboard for Assam, PRAAPTI- as accessed on 27th May 2025

* As per the Government of India’s classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type

Performance History:

In FY 2023–24, APDCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold also demonstrated steady growth of 8%. The Average Cost of Supply (ACS) declined marginally to INR 9.51 per kWh, while the Average Revenue Realized (ARR) increased to INR 9.77 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite losses having quadrupled over the past few years, an increase was again observed in FY 2023–24. Payable days have steadily decreased over the past four years and remain favorable compared to the national average (excluding private DISCOMs) of 130 days in FY 2023–24.¹⁵⁰

Table 31: Performance Report Snapshot

Key Parameters	Units	FY 2020-21 ¹⁵¹	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	8,786	9,535s	10,290	10,955
Gross Input Energy	GWh	11,004	11,953	12,813	13,344
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.48	7.98	10.03	9.51
Average Revenue Realized (ARR)	INR/kWh	8.15	8.33	9.25	9.77
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.33)	0.35	(0.78)	0.26
Profitability (Surplus/(Deficit))	INR in Crore	(292)	336	(800)	384
Payable Days	Days	65	37	59	41

Capacity Addition Plans of APDCL

As stated above, Assam's energy capacity is projected to grow significantly and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 54% in FY 2024 to 28% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. APDCL plans to purchase additional capacity through short term open access (STOA) whereas no such plan for Hydropower has been noted till 2035. Capacity Addition requirement has been illustrated below¹⁵²:

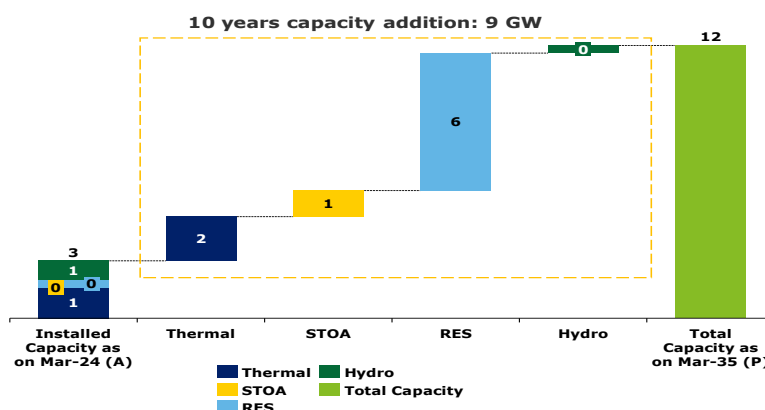


Figure 50: Capacity Addition Plan for next 10 years of Assam

¹⁵⁰ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

¹⁵¹ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

¹⁵² Report on Resource Adequacy of Assam upto 2034-35, Central Electricity Authority, GoI

Merit Order Analysis¹⁵³:-

Assam's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 16,300 GWh. Of this, thermal power accounts for approximately 48% of the total output. According to the provisional true-up order for FY 2024-25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 6.15 per kWh. Additionally, the hydropower plant at DHEP has a generation cost of INR 6.19 per kWh.

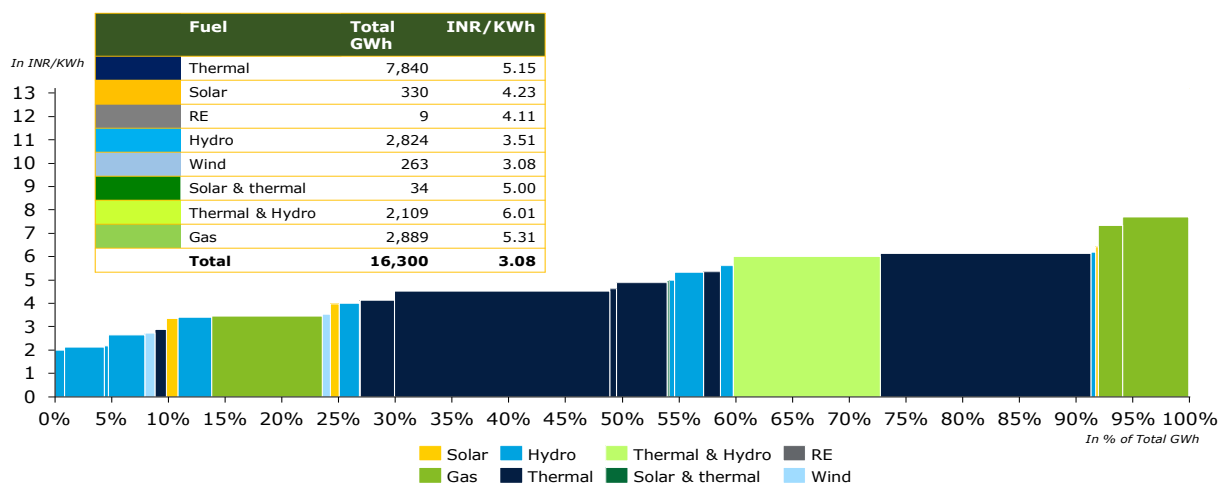


Figure 51: Merit Order Analysis of Assam

¹⁵³ True-Up for FY 2023-24, Assam Electricity Regulatory Commission, 2023-24

6.1.3. Northern Region

The Northern region of India presents strong potential for offtake from Bhutan's Dorjilung Hydropower Project. The region has high and seasonal electricity demand, particularly during summer peaks, making it well-suited for reliable peaking power. Additionally, the Hydro Purchase Obligation (HPO) mandates clean energy procurement, aligning with Dorjilung's profile. Financially stable DISCOMs in Delhi and Haryana further enhance the region's viability for long-term power offtake agreements. List of All States of Northern Region have been Stated below¹⁵⁴:

Table 32: State-wise Gross Input Energy¹⁵⁵

State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
Uttar Pradesh	1,34,269	29%
Rajasthan	1,12,663	25%
Punjab	71,876	16%
Haryana	66,375	15%
Delhi	37,767	8%
Uttarakhand	17,233	4%
Himachal Pradesh	15,551	3%
Ladakh	315	0%
Chandigarh	-	0%
Jammu & Kashmir	-	0%

Among the above-mentioned States, Uttar Pradesh, Rajasthan, Punjab, Haryana, and Delhi has been selected further for detailed level Analysis as it covers approximately 93% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 33: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Uttar Pradesh	Pashchimanchal Vidyut Vitran Nigam Ltd	39,108	28%
	Purvanchal Vidhyut Vitran Nigam Ltd	32,930	24%
	Dakshinanchal Vidyut Vitran Nigam Ltd	29,628	21%
	Madhyanchal Vidyut Vitran Nigam Ltd	28,314	21%
	Kanpur Electricity Supply Company	4,289	3%
	Noida Power Company Limited	3,615	3%
	Total of Uttar Pradesh	1,37,884	
Rajasthan	Jaipur Vidyut Vitran Nigam Limited	42,907	38%
	Jodhpur Vidyut Vitran Nigam Limited	38,762	34%
	Ajmer Vidyut Vitran Nigam Limited	30,994	28%
	Total of Rajasthan	1,12,663	

¹⁵⁴ Central Electricity Authority

¹⁵⁵

Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Punjab	Punjab State Power Corporation Limited	71,876	100%
	Total of Punjab	71,876	
Haryana	Dakshin Haryana Bijli Vitran Nigam	39,880	60%
	Uttar Haryana Bijli Vitran Nigam	26,495	40%
	Total of Haryana	66,375	
Delhi	BSES Rajdhani Power Limited	15,835	40%
	Tata Power Delhi Distribution Limited	12,686	32%
	BSES Yamuna Power Limited	9,246	24%
	New Delhi Municipal Council	1,568	4%
	Total of Delhi	39,335	

4. Region: North

State: Uttar Pradesh

Uttar Pradesh's peak electricity demand for FY 2025 was about 30.62 GW. The state has an installed generation capacity of 33.82 GW, with 71.8% from fossil fuel-based projects, 27.4% from renewable energy sources (RES), including hydro, and the remaining 0.8% from nuclear projects. The state is actively investing in renewable energy through various schemes like the Kusum Yojana and the Village Electrification Program, aiming to enhance its capacity and reduce reliance on fossil fuels.

Table 34: Energy Demand of Uttar Pradesh

Particulars	Units	2021-22 (Actual) ¹⁵⁶	2024-25 (Actual) ¹⁵⁷	2026-27 (Projected) ¹⁵⁸	2031-32 (Projected) ¹⁵⁹
Energy Requirement	TWh	129	165	191	254
Peak Demand	GW	25	31	33	44

Fuel Mix of Uttar Pradesh ¹⁶⁰

Uttar Pradesh's contracted power capacity reached 31 GW by the end of March 2023. Of this capacity, 68% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 147 GW, out of which 75% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 11% over the next 10 years. Capacity wise generation mix from FY 2023-34 has been illustrated below:

¹⁵⁶ [Load Generation Balance Report, 2022](#)

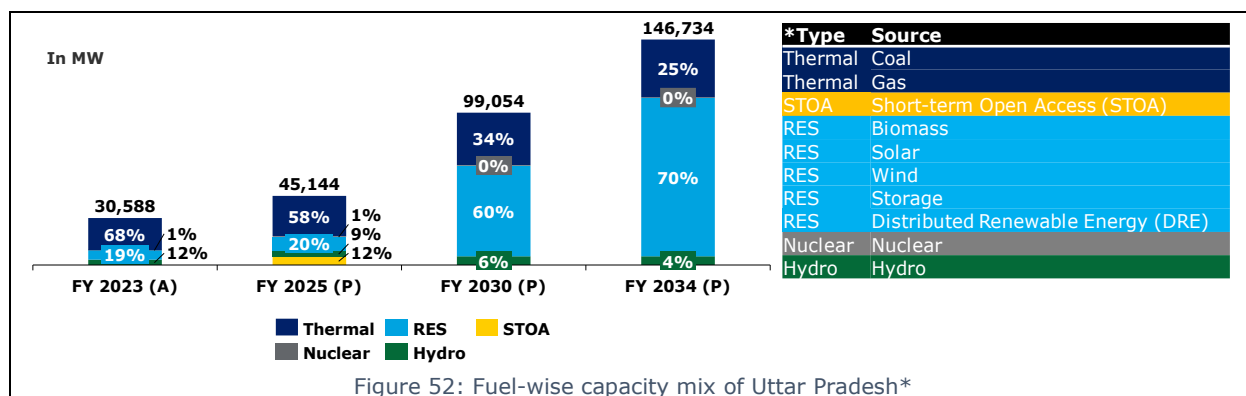
¹⁵⁷ Load Generation Balance Report, 2025

¹⁵⁸ [National Electricity Plan 2022-32](#)

¹⁵⁹ [National Electricity Plan 2022-32](#)

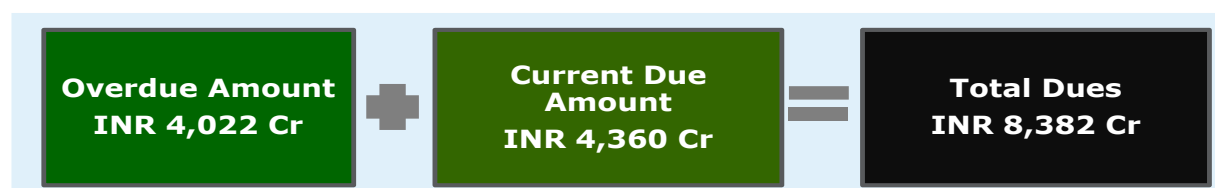
¹⁶⁰ [Resource Adequacy Report, Central Electricity Report, GOI, 2024](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.



Financial Health of Uttar Pradesh's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 48% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 52%, are payments still within their respective due dates but not yet fully or partially paid: ¹⁶¹



Utility Name: Uttar Pradesh Power Corporation Limited (UPPCL)

Uttar Pradesh Power Corporation Limited (UPPCL) is a state government-owned utility responsible for electricity transmission and distribution across Uttar Pradesh. Established in 2000, UPPCL ensures reliable power supply to millions of consumers through its extensive network and advanced technologies. For efficient operation & management, UPPCL is divided further divisions:

- Dakshinanchal Vidyut Vitaran Nigam Limited (DVVNL)
- Madhyanchal Vidyut Vitaran Nigam Limited (MVVNL)
- Pashchimanchal Vidyut Vitaran Nigam Limited (PVVNL)
- Purvanchal Vidyut Vitaran Nigam Limited (PUVVNL)
- Kanpur Electricity Supply Company (KESCO)

UPPCL serves 30.88 million consumers with a diverse consumer base, where the domestic sector accounts for approximately 41% of total electricity consumption. This is followed by the agricultural sector at 16%, industrial consumers at 13%, and the commercial segment at 7%, with the remaining share attributed to other sectors

Performance History:

In FY 2023–24, UPPCL reported consistent increase in gross input energy over the past four years at CAGR of 6% similarly, gross energy sold has also demonstrated steady growth of 8%. The Average Cost of Supply (ACS) declined marginally to INR 9.20 per kWh while Average Revenue realized (ARR) has increased to INR 8.57 per kWh in FY 2023-24, narrowing the gap between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past four years, a decline was

¹⁶¹ [Dashboard for Uttar Pradesh, PRAAPTI](#) - as accessed on 27th May 2025

observed over in FY 2023-24. Payable Days, while still high, they have steadily decreased over the past three years and unfavourable compared to the national average (excluding private DISCOMs) of 130 days in FY 2023-24.¹⁶²

Table 35: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ¹⁶³	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	90,372	93,745	108,410	113,477
Gross Input Energy	GWh	113,859	116,885	130,059	134,269
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	8.64	9.25	9.74	9.20
Average Revenue Realized (ARR)	Rs/kWh	7.46	8.55	8.14	8.57
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	(1.18)	(0.70)	(1.60)	(0.63)
PAT (Profit/(loss))	Rs in Crore	(10,660)	(6,498)	(15,512)	(7,058)
Payable Days	Days	208	203	172	157

Capacity Addition Plans of UPPCL:-

Uttar Pradesh's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 68% in FY 2023 to 25% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. UPPCL has planned to procure 1,808 MW of hydropower till 2034. The state's planned capacity addition for next 10 years has been stated below¹⁶⁴:

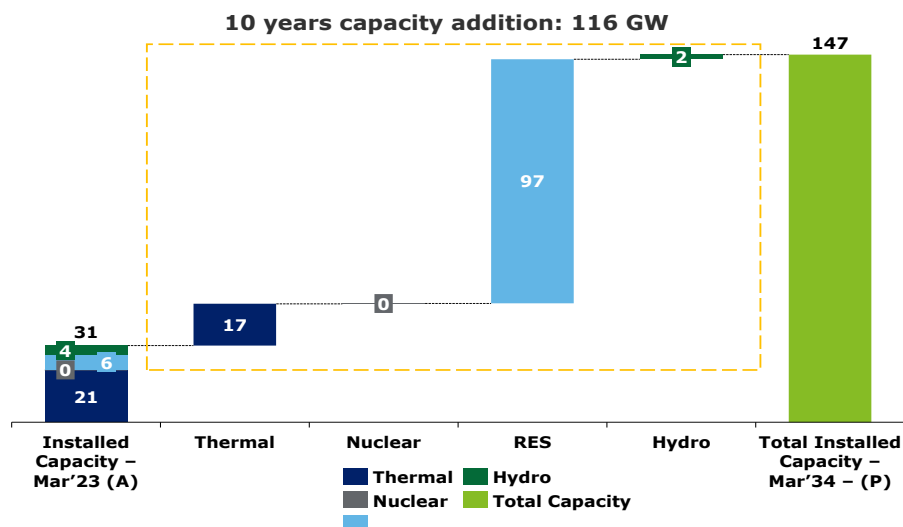


Figure 54: Capacity Addition Plan for next 10 years of Uttar Pradesh

Merit Order Analysis¹⁶⁵:-

Uttar Pradesh's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 147,902 GWh. Of this, thermal power accounts for approximately 74% of the total output. According to the provisional true-up order for FY 2024-25 as approved by the Commission,

¹⁶²

Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

¹⁶³ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

¹⁶⁴ Report on Resource Adequacy of Uttar Pradesh, Central Electricity Authority, GoI

¹⁶⁵ True-Up for FY 2022-23, Uttar Pradesh Electricity Regulatory Commission, 2024

the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 7 per kWh. However, some older thermal assets, such as KSK Mahanandi and Ghatampur, report significantly higher generation costs of INR 11.58 and INR 6.53 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 7.07 per kWh.

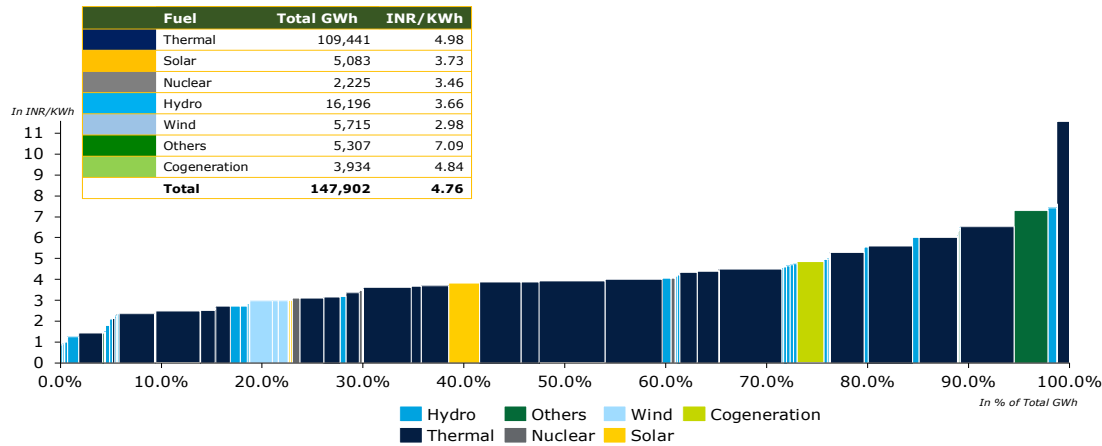


Figure 55: Merit Order Analysis of Uttar Pradesh

4. Region: North

State: Rajasthan

Rajasthan's peak electricity demand for FY 2025 was approximately 19.17 GW. The state has an installed generation capacity of approximately 51.29 GW, with about 29.2% from fossil fuel-based sources, 69.7% from renewable energy sources (RES), including hydro, and the remaining 1.1% from nuclear power. Rajasthan is a front-runner in renewable energy adoption, contributing over 11.5% to India's total renewable energy capacity, with a strong focus on solar and wind power.

Table 36: Energy Demand of Rajasthan

Particulars	Units	2021-22 (Actual) ¹⁶⁶	2024-25 (Actual) ¹⁶⁷	2026-27 (Projected) ¹⁶⁸	2031-32 (Projected) ¹⁶⁹
Energy Requirement	TWh	90	114	134	172
Peak Demand	GW	16	19	21	27

Fuel Mix of Rajasthan¹⁷⁰

Rajasthan's contracted power capacity reached 24 GW by the end of March 2023. Of this capacity, 60% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted Capacity for the year 2031-32 is 63 GW, out of which 68% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 8 years. Capacity wise generation mix from 2023 to 2032 has been illustrated below:

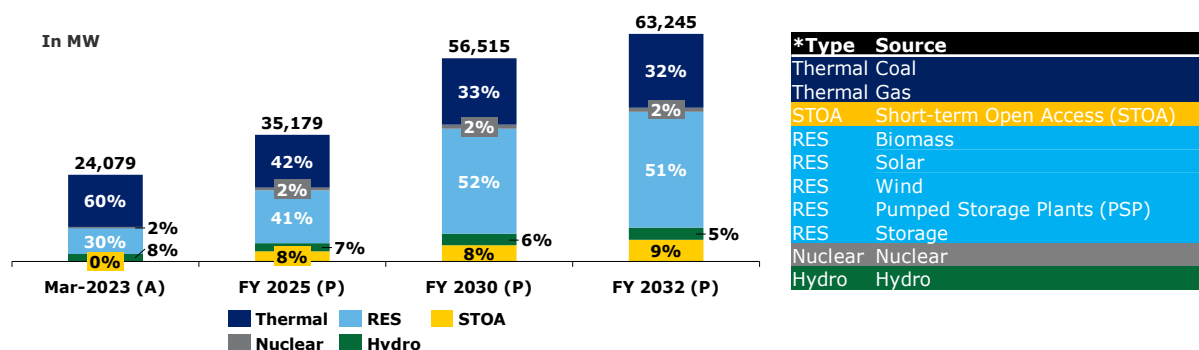


Figure 56: Capacity Mix of Rajasthan*

Financial Health of Rajasthan's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 33% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 67%, are payments still within their respective due dates but not yet fully or partially paid:¹⁷¹

¹⁶⁶ [Load Generation Balance Report, 2022](#)

¹⁶⁷ [Load Generation Balance Report, 2025](#)

¹⁶⁸ [National Electricity Plan 2022-32](#)

¹⁶⁹ [National Electricity Plan 2022-32](#)

¹⁷⁰ [Resource Adequacy Report of Rajasthan](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

¹⁷¹ [Dashboard for Rajasthan, PRAAPT1](#) - as accessed on 27th May 2025

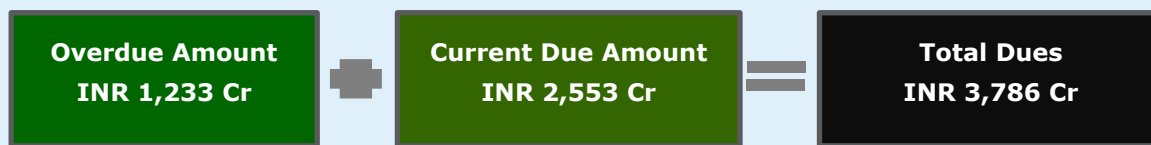


Figure 57: Snapshot of PRAAPTI - Rajasthan

Utility Name : Rajasthan State Power Corporation Limited (RSPCL)

Rajasthan State Power Corporation Limited (RSPCL) is a state government-owned utility responsible for electricity transmission and distribution across Rajasthan. Established in 1999, RSPCL ensures reliable power supply to millions of consumers through its extensive network and advanced technologies. For efficient operation & management, RSPCL is divided further divisions

Rajasthan's power distribution is managed primarily by three State-owned DISCOMs:

- Jaipur Vidyut Vitran Nigam Ltd (JVVNL)
- Ajmer Vidyut Vitran Nigam Ltd (AVVNL)
- Jodhpur Vidyut Vitran Nigam Ltd (JdVVNL)

These utilities collectively serve 16.32 million¹⁷² consumers in the state vast and diverse consumer base majorly to Agriculture followed by Industrial, Domestic, Commercial and Others.

Performance History:

In FY 2023–24, RSPCL reported consistent increase in gross input energy over the past four years at CAGR of 9% similarly, gross energy sold has also demonstrated steady growth of 9%. The Average Cost of Supply (ACS) has increased to INR 8.81 per kWh and Average Revenue realized (ARR) has increased to INR 8.86 per kWh in FY 2023-24, improving the margin between ACS and Average Revenue Realized (ARR). Despite losses having quadrupled over the past three years, a turnaround was observed in FY 2023–24, with the company returning to profitability. Payable Days, while still high, they have steadily decreased over the past four years and improve compared to the national average (excluding private DISCOMs) of 130 days in FY 2023-24¹⁷³.

Table 37: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ¹⁷⁴	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	66,553	73,189	80,951	87,023
Gross Input Energy	GWh	87,083	94,302	1,03,662	112,663
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.73	8.20	8.44	8.81
Average Revenue Realized (ARR)	INR/kWh	8.39	8.14	8.13	8.86
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.34)	(0.06)	(0.31)	0.05
Profitability (Surplus/(Deficit))	INR in Crore	(2,217)	(4720)	(2,514)	505
Payable Days	Days	256	203	88	63

Capacity Addition Plans of RSPCL:-

¹⁷² [India Climate & Energy Dashboard](#)

¹⁷³ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

¹⁷⁴ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Rajasthan's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 60% in FY 2023 to 32% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. RSPCL plans to procure 1,342 MW of hydropower till 2032. The state's planned capacity addition for next 10 years has been stated below.¹⁷⁵

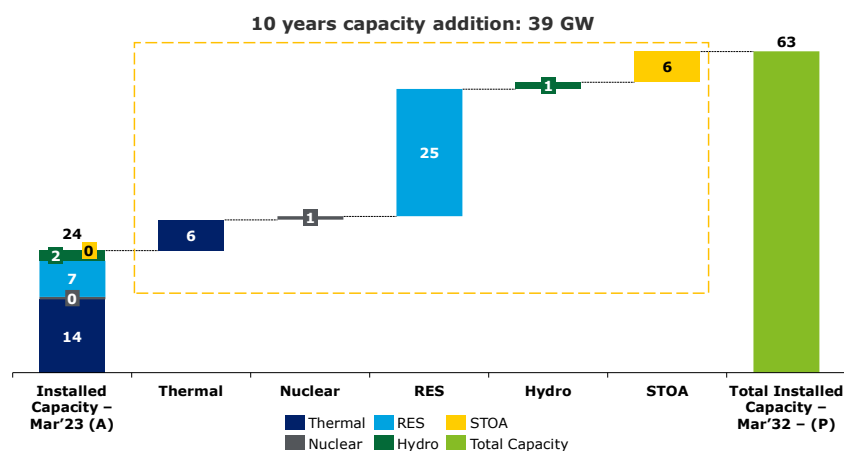


Figure 58: Capacity addition plan for next 10 years of Rajasthan

Merit Order Analysis ¹⁷⁶:-

Rajasthan's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 106,169 GWh. Of this, thermal power accounts for approximately 71% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 5.03 per kWh. However, Coastal Gujarat report significantly higher generation costs of INR 5.21 per kWh. Additionally, the Hydropower plant at PTC(Sikkim Urja Ltd) has a generation cost of INR 4.89 per kWh.

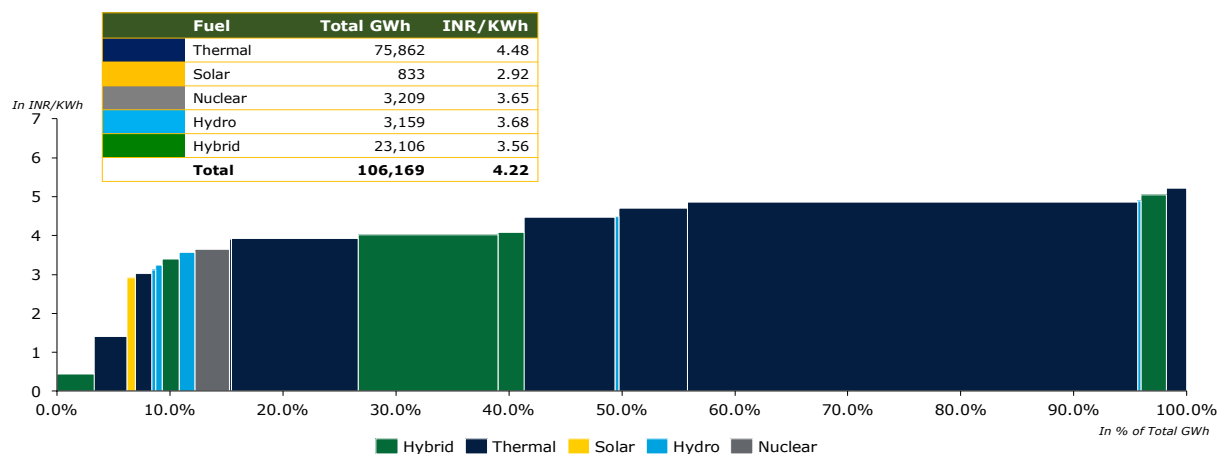


Figure 59: Merit Order Analysis of Rajasthan

¹⁷⁵ Resource Adequacy Report, Central Electricity Report, GOI, 2024

¹⁷⁶ MYT Order, Rajasthan Electricity Regulatory Commission, Jaipur

4. Region: North

State: Punjab

Punjab's peak electricity demand for FY 2025 was 16.06 GW. The state has an installed generation capacity of approximately 14.56 GW, with around 57.4% from fossil fuel-based sources, 41.2% from renewable energy sources (RES), including hydro, and the remaining 1.4% from nuclear power. Punjab is steadily increasing its renewable energy footprint and currently accounts for about 2.5% of India's total renewable energy capacity, with a strong focus on solar and wind power.

Table 38: Energy Demand of Punjab

Particulars	Units	2021-22 (Actual)	2024-25 (Actual)	2026-27 (Projected)	2031-32 (Projected)
Energy Requirement	TWh	63	77	82	105
Peak Demand	GW	14	16	17	21

Fuel Mix of Punjab ¹⁷⁷

Punjab's contracted power capacity reached 15 GW by the end of March 2023. Of this capacity, 57% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted Capacity for the year 2029-30 is 39 GW, out of which 65% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 17% over the next 6 years. Capacity wise generation mix from 2023 to 2030 has been illustrated below:

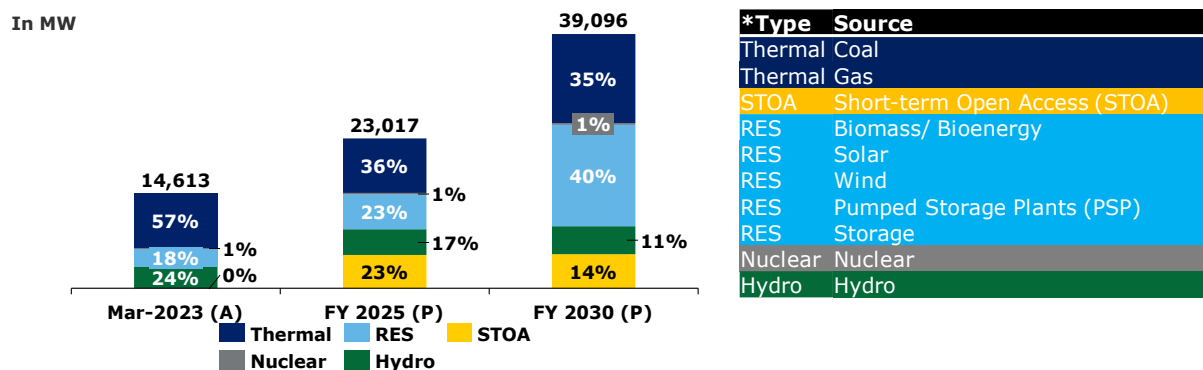


Figure 60: Fuel-wise capacity mix of Punjab*

Financial Health of Punjab's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 25% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 75%, are payments still within their respective due dates but not yet fully or partially paid:¹⁷⁸

¹⁷⁷ [Resource Adequacy Report of Punjab](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

¹⁷⁸ [Dashboard for Punjab, PRAAPT](#) - as accessed on 27th May 2025

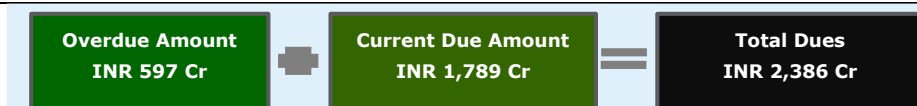


Figure 61: Snapshot of PRAAPTI - Punjab

Utility Name : Punjab State Power Corporation Limited (PSPCL)

Punjab State Power Corporation Limited (PSPCL) is a government-owned entity responsible for electricity generation and distribution in Punjab. Established in 2010, PSPCL operates several thermal and hydropower plants, ensuring a reliable energy supply. It plays a crucial role in maintaining a stable and efficient electricity network, supporting the state's economic growth. The major consumer base in Punjab comprises industrial consumers, followed by domestic, agricultural, commercial, and other categories.

Performance History:

In FY 2023–24, PSPCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 4%. Similarly, gross energy sold also demonstrated steady growth of 4%. The Average Cost of Supply (ACS) decreased to INR 7.06 per kWh, while the Average Revenue Realized (ARR) increased to INR 7.19 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite incurring losses in FY 2022–23, a turnaround was observed in FY 2023–24, with the company returning to profitability. Payable days increased in FY 2022–23 but decreased again in FY 2023–24, which is an improvement compared to the national average (excluding private DISCOMs) of 130 days in the same year.¹⁷⁹

Table 39: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ¹⁸⁰	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	49,729	53,166	59,089	61,136
Gross Input Energy	GWh	58,137	62,154	69,269	71,876
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	6.60	6.59	7.46	7.06
Average Revenue Realized (ARR)	Rs/kWh	6.89	6.79	6.65	7.19
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	0.29	0.20	(0.81)	0.13
Profitability (Surplus/(Deficit))	Rs in Crore	1,446	1,069	(4,776)	800
Payable Days	Days	65	45	55	40

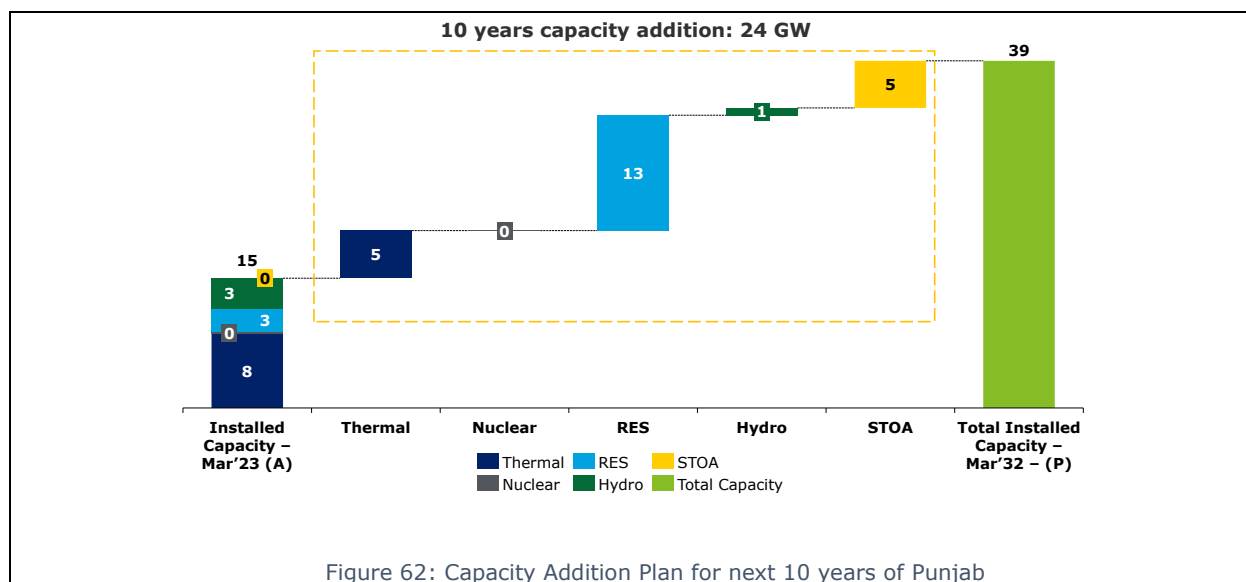
Capacity Addition Plans of PSPCL:-

Punjab's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 57% in FY 2023 to 35% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below¹⁸¹:

¹⁷⁹ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

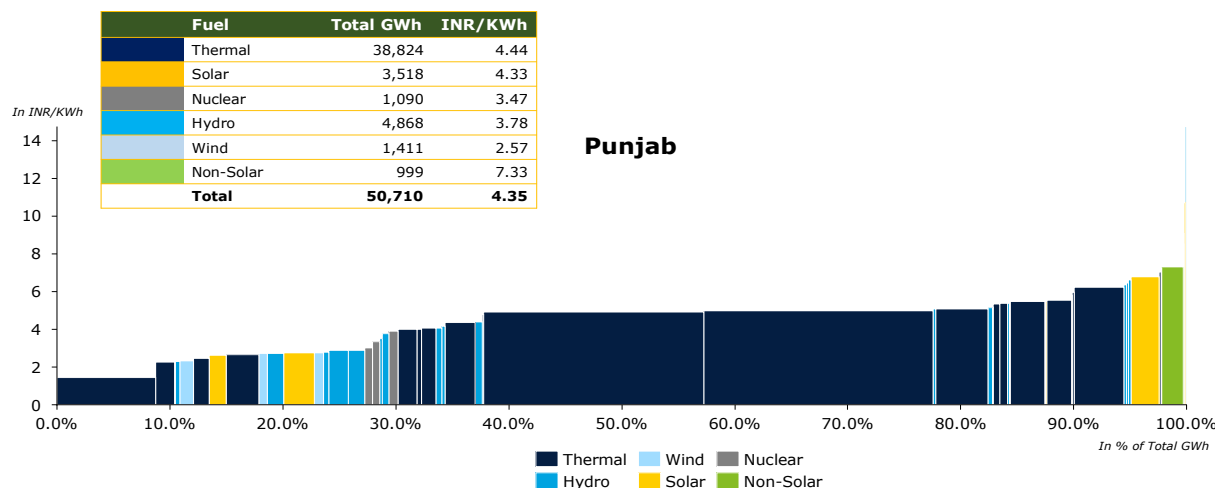
¹⁸⁰ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

¹⁸¹ Report on Resource Adequacy of Punjab, Central Electricity Authority, GoI



Merit Order Analysis¹⁸²:-

Punjab's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 50,710 GWh. Of this, thermal power accounts for approximately 77% of the total output. According to the provisional true-up order for FY 2024–25 as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an average generation cost of INR 5.96 per kWh. However, some older thermal assets, such as NPL UI and Unchahar-1, report significantly higher generation costs of INR 8.53 and INR 7.24 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 7.33 per kWh, reflecting a moderate cost profile within the state's energy mix.



¹⁸² True-Up for FY 2023-24 and ARR for 3rd MYT Control Period, Punjab Electricity Regulatory Commission, 2024

4. Region: North

State: Haryana

Haryana's peak electricity demand for FY 2025 was approximately 14.66 GW. The state has an installed generation capacity of 14.15 GW, with about 65.2% from fossil fuel-based sources, 34.2% from renewable energy sources (including hydro), and the remaining 0.7% from nuclear power. The state is steadily expanding its renewable energy footprint and currently hosts around 2.3% of India's total renewable energy capacity.

Table 40: Energy Demand of Haryana

Particulars	Units	2021-22 (Actual) ¹⁸³	2024-25 (Actual) ¹⁸⁴	2026-27 (Projected) ¹⁸⁵	2031-32 (Projected) ¹⁸⁶
Energy Requirement	TWh	55	70	83	115
Peak Demand	GW	12	15	16	22

Fuel Mix of Haryana ¹⁸⁷

Haryana's contracted power capacity reached 14 GW by the end of March 2024. Of this capacity, 62% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The total projected contracted Capacity for the year 2034-35 is 49 GW, out of which 71% through non-fossil fuel. The state is projected to grow substantially, with contracted capacity rising at a CAGR of 13% over the next 10 years. Capacity wise generation mix from 2024-35 has been illustrated below:

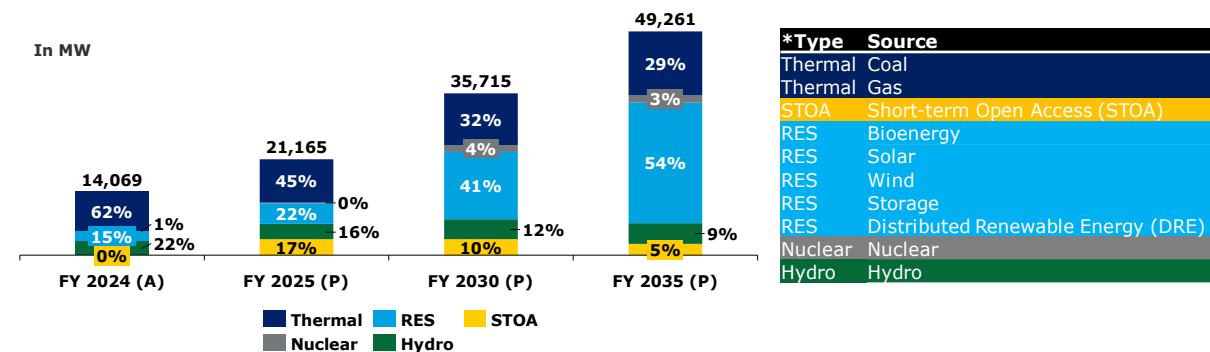


Figure 64: Fuel-wise capacity mix of Haryana*

Financial Health of Haryana's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 5% of the total, refer to payments that have surpassed their due dates and remain either partially or fully

¹⁸³ [Load Generation Balance Report, 2022](#)

¹⁸⁴ [Load Generation Balance Report, 2025](#)

¹⁸⁵ [National Electricity Plan 2022-32](#)

¹⁸⁶ [National Electricity Plan 2022-32](#)

¹⁸⁷ [Resource Adequacy Report of Haryana](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

unpaid. In contrast, current dues, accounting for the remaining 95%, are payments still within their respective due dates but not yet fully or partially paid:¹⁸⁸

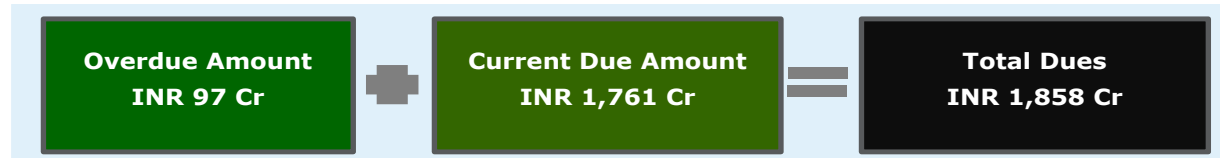


Figure 65: Snapshot of PRAAPTI - Haryana

Utility Name : Haryana Power Generation Corporation Limited (HPGCL)

Haryana Power Generation Corporation Limited (HPGCL) is a government-owned entity responsible for electricity generation in Haryana. Established in 1997, HPGCL operates several thermal and hydropower plants, ensuring a reliable energy supply. Haryana's electricity sector is managed by two main distribution companies: Uttar Haryana Bijli Vitran Nigam (UHBVN) and Dakshin Haryana Bijli Vitran Nigam (DHBVN). Both the Utilities serve 7.94 million consumers¹⁸⁹ majorly to Industrial consumers, followed by Domestic, Agriculture, Commercial and Others.

Performance History:

In FY 2023–24, HPGCL reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold demonstrated steady growth of 10%. The Average Cost of Supply (ACS) decreased to INR 7.16 per kWh, and the Average Revenue Realized (ARR) also decreased to INR 7.21 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite increasing profits in FY 2022–23, a turnaround was observed in FY 2023–24, with the company experiencing reduced profitability. Payable days increased in FY 2022–23 but decreased again in FY 2023–24, improving compared to the national average (excluding private DISCOMs) of 130 days in the same year.¹⁹⁰

Table 41: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ¹⁹¹	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	43,165	47,501	54,445	57,433
Gross Input Energy	GWh	53,762	56,856	63,286	66,375
Average Cost of Supply (ACS) on Energy Sold	Rs/kWh	6.50	6.72	7.88	7.16
Average Revenue Realized (ARR)	Rs/kWh	6.64	6.89	8.06	7.21
Gap between ARR vs ACS (Surplus/(Deficit))	Rs/kWh	0.15	0.18	0.18	0.05
Profitability (Surplus/(Deficit))	Rs in Crore	637	849	975	276
Payable Days	Days	55	45	62	38

Capacity Addition Plans of HPGCL:-

Haryana's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 62% in FY 2025 to

¹⁸⁸ [Dashboard for Haryana, PRAAPTI](#)- as accessed on 27th May 2025

¹⁸⁹ [India Climate & Energy Dashboard](#)

¹⁹⁰ [Report on Performance of Power Utilities, Power Finance Corporation, 2023-24](#)

¹⁹¹ [Report on Performance of Power Utilities, Power Finance Corporation, 2022-23](#)

29% by FY 2035. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below¹⁹²:

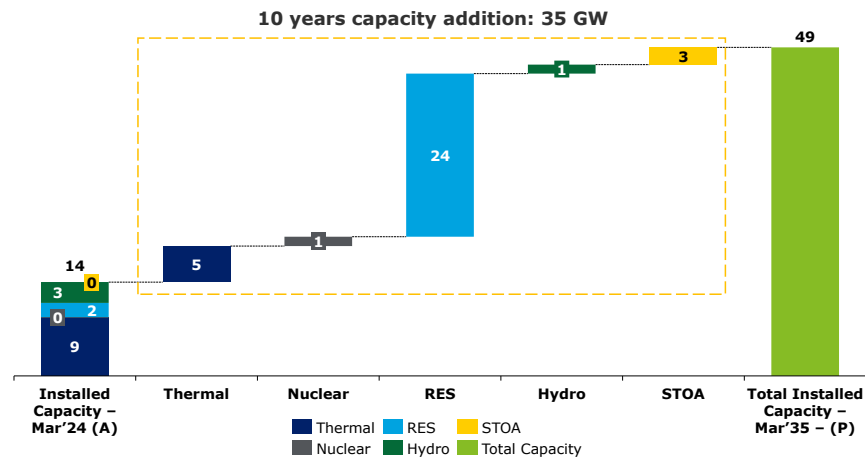


Figure 66: Capacity Addition Plan for next 10 years of Haryana

Merit Order Analysis¹⁹³:-

Haryana's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 73,109 GWh. Of this, thermal power accounts for approximately 69% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of thermal power plants, with an average generation cost of INR 8.05 per kWh. However, some older thermal assets, such as Feroz Gandhi Unchahar-3 TPS and Feroz Gandhi Unchahar-1 TPS, report significantly higher generation costs of INR 11.95 and INR 11.27 per kWh, respectively. Additionally, the nuclear plant has an average generation cost of INR 6.16 per kWh. Other non-solar renewable sources contribute at an average tariff of INR 6.84 per kWh, reflecting a moderate cost profile within the state's energy mix.

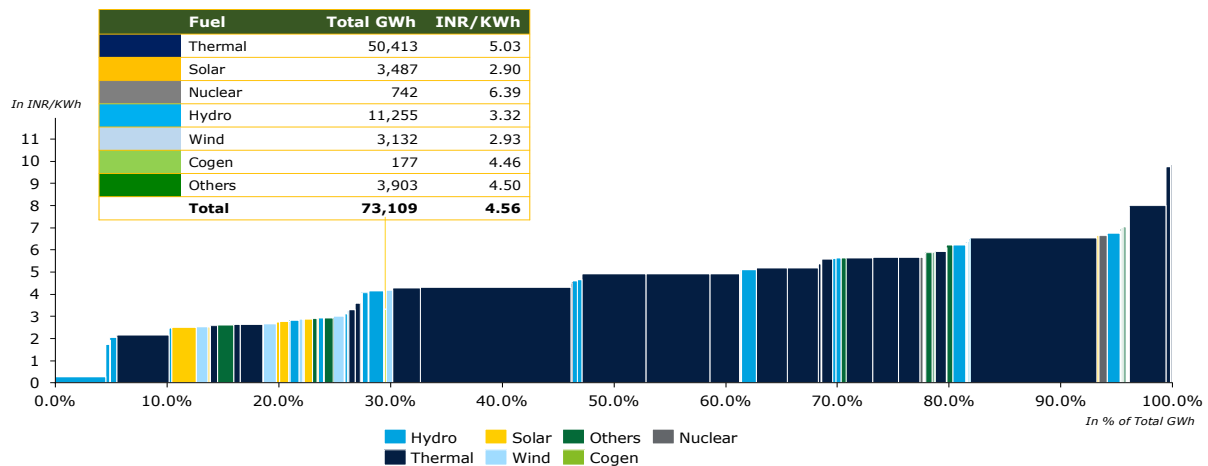


Figure 67: Merit Order Analysis of Haryana

¹⁹² Report on Resource Adequacy of Haryana, Central Electricity Authority, GoI

¹⁹³ True-Up for FY 2022-23, Haryana Electricity Regulatory Commission, 2024

4. Region: North

State: Delhi

Delhi's peak electricity demand for FY 2025 was 8.66 GW. The city has an installed generation capacity of about 7.1 GW, with approximately 82.2% sourced from fossil fuel-based projects, 16.4% from renewable energy sources (RES), including solar and waste-to-energy, and the remaining 1.5% from nuclear power. The city contributes around 2.5% of India's total renewable energy capacity, supported by a strong policy push toward clean energy integration, electric mobility, and demand-side management.

Table 42: Energy Demand of Delhi

Particulars	Units	2021-22 (Actual) ¹⁹⁴	2024-25 (Actual) ¹⁹⁵	2026-27 (Projected) ¹⁹⁶	2031-32 (Projected) ²¹
Energy Requirement	TWh	31	38	42	53
Peak Demand	GW	7	9	9	12

Fuel Mix of Delhi

BRPL's contracted power capacity reached 32 GW by the end of March 2024. Of this capacity, 57% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 106 GW, out of which 65% through non-fossil fuel. BRPL is projected to grow substantially, with contracted capacity rising at a CAGR of 14% over the next 9 years Capacity wise generation mix from 2024-34 has been illustrated below:

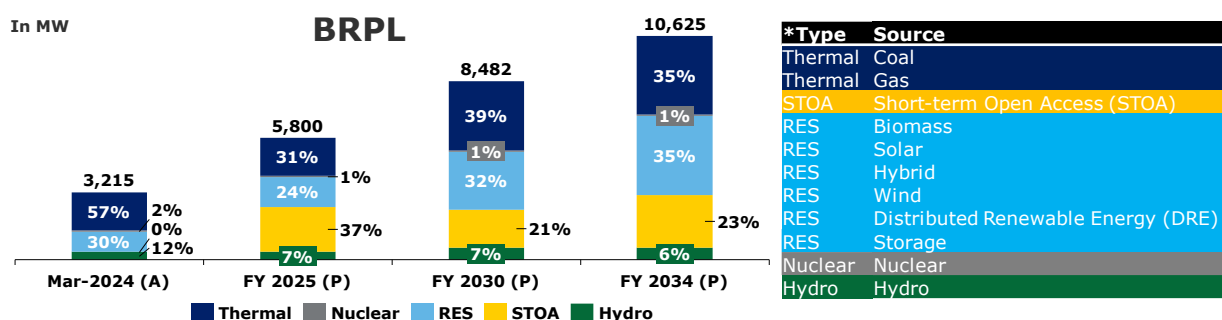


Figure 68: Capacity Mix of BRPL – Delhi^{197*}

BYPL's contracted power capacity reached 20 GW by the end of March 2023. Of this capacity, 70% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 41 GW, out of which 59% through non-fossil fuel. BYPL is projected to grow substantially, with contracted capacity rising at a CAGR of 8% over the next 9 years Capacity wise generation mix from 2024 to 2034 has been illustrated below:

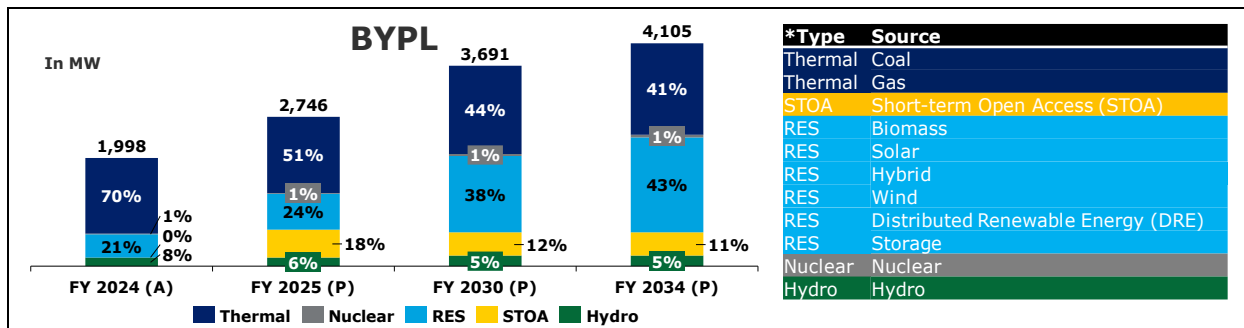
¹⁹⁴ Load Generation Balance Report, 2022

¹⁹⁵ Load Generation Balance Report, 2025

¹⁹⁶ National Electricity Plan 2022-32

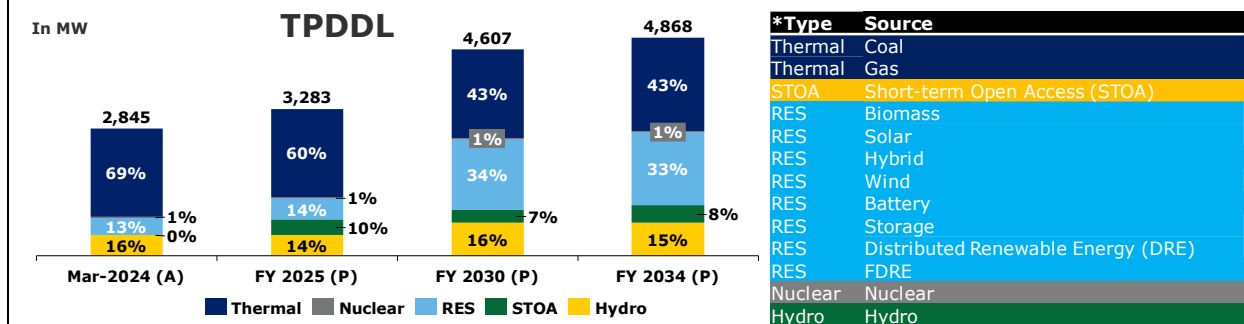
¹⁹⁷ Resource Adequacy Report of BRPL-Delhi, FY 2025-34

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

Figure 69: Capacity Mix of BYPL- Delhi^{198*}

TPDDL's contracted power capacity reached 28 GW by the end of March 2024. Of this capacity, 69% is derived from fossil fuel-based sources, primarily coal and natural gas, while the remaining share is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE).

The projected contracted capacity for the year 2033-34 is 49 GW, out of which 57% through non-fossil fuel. TPDDL is projected to grow substantially, with contracted capacity rising at a CAGR of 6% over the next 9 years. Capacity wise generation mix from 2024 to 2034 has been illustrated below:

Figure 70: Capacity Mix of TPDDL - Delhi^{199*}

Financial Health of Delhi's Distribution Companies:

The analysis of payable dues serves as a crucial tool for evaluating the financial position, payment behavior, and creditworthiness of the respective state DISCOMs. As of 27th May 2025, the dues have been classified into two categories: overdue and current dues. Overdue dues, constituting 60% of the total, refer to payments that have surpassed their due dates and remain either partially or fully unpaid. In contrast, current dues, accounting for the remaining 40%, are payments still within their respective due dates but not yet fully or partially paid.²⁰⁰

¹⁹⁸ [Resource Adequacy Report of BYPL- Delhi, FY 2025-34](#)

¹⁹⁹ [Resource Adequacy Report of TPDDL-Delhi, FY 2025-34](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

²⁰⁰ [Dashboard for Delhi, PRAAPTI](#)- as accessed on 27th May 2025

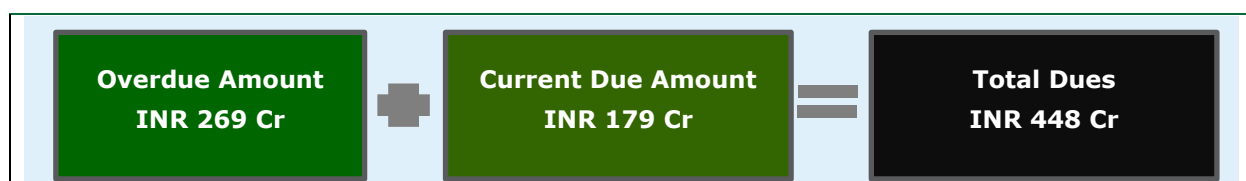


Figure 71: Snapshot of PRAAPTI - Delhi

Utility Name 1: BSES Rajdhani Power Limited (BRPL)
Utility Name 2: BSES Yamuna Power Limited (BYPL)
Utility Name 3: Tata Power Delhi Distribution Limited (TPDDL)

Delhi's electricity distribution is managed by three main companies: BSES Rajdhani Power Limited (BRPL), BSES Yamuna Power Limited (BYPL), and Tata Power Delhi Distribution Limited (TPDDL). BRPL and BYPL serve the southern and eastern parts of Delhi, respectively, while TPDDL caters to the northern and north-western regions. They are regulated by the Delhi Electricity Regulatory Commission (DERC), which ensures compliance with standards and addresses consumer grievances. These Utilities together serve approx. 7.16 million²⁰¹ consumers, majorly catering to Domestic category at 53%, followed by Commercial, Industrial, Agriculture and Others.

Performance History:

In FY 2023–24, Delhi reported a consistent increase in gross input energy over the past four years, with a CAGR of 7%. Similarly, gross energy sold demonstrated steady growth of 8%. The Average Cost of Supply (ACS) decreased to INR 8.33 per kWh, while the Average Revenue Realized (ARR) increased to INR 8.99 per kWh in FY 2023–24, improving the margin between ACS and ARR. Despite increasing profits in FY 2022–23, a turnaround was observed in FY 2023–24, with the company experiencing reduced profitability. Payable days, while still high, steadily decreased until FY 2022–23 but increased again in FY 2023–24, remaining higher than the national average (excluding private DISCOMs) of 130 days in FY 2023–24.²⁰²

Table 43: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ²⁰³	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	27,410	29,728	32,748	34,082
Gross Input Energy	GWh	30,512	33,259	36,602	37,767
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.07	7.70	8.50	8.33
Average Revenue Realized (ARR)	INR/kWh	8.86	7.96	8.49	8.99
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	0.79	0.26	(0.01)	0.58
Profitability (Surplus/(Deficit))	INR in Crore	4,858	1,047	2,096	1,681
Payable Days	Days	343	335	271	277

Capacity Addition Plans:-

BRPL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 57% in FY 2024 to 35% by FY

²⁰¹ [India Climate & Energy Dashboard](#)

²⁰²

[Report on Performance of Power Utilities, Power Finance Corporation, 2023-24](#)

²⁰³ [Report on Performance of Power Utilities, Power Finance Corporation, 2022-23](#)

2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.²⁰⁴

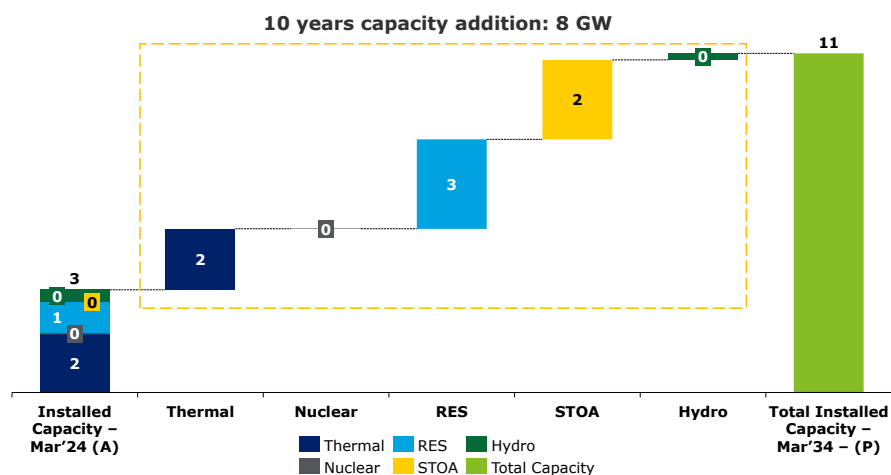


Figure 72: Capacity Addition Plan for next 10 years of BRPL

BYPL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 70% in FY 2024 to 41% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.²⁰⁵

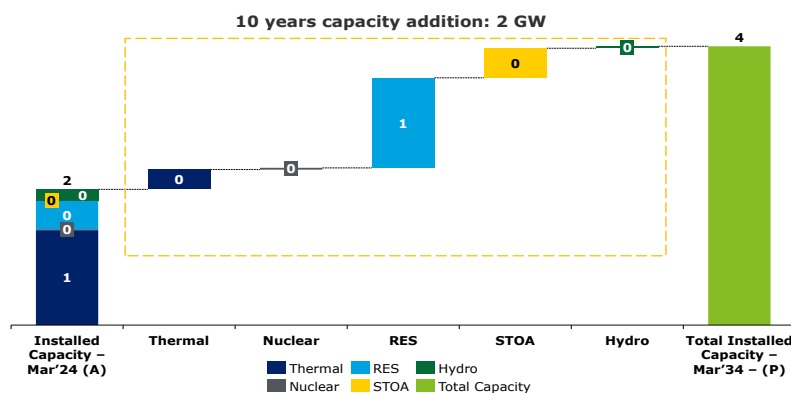


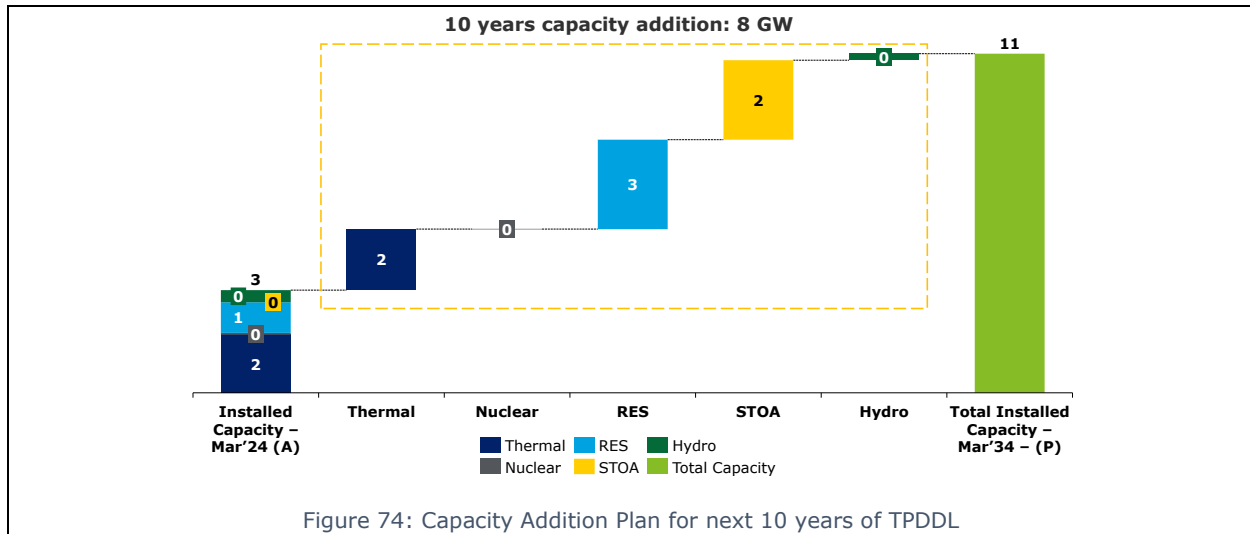
Figure 73: Capacity Addition Plan for next 10 years of BYPL

TPDDL's energy capacity is projected to grow significantly, and the energy mix is becoming increasingly diversified, with the share of coal and gas expected to decline from 69% in FY 2024 to 43% by FY 2034. In contrast, the contribution of non-fossil sources and RE sources is set to rise. The state's planned capacity addition for next 10 years has been stated below.²⁰⁶

²⁰⁴ Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI

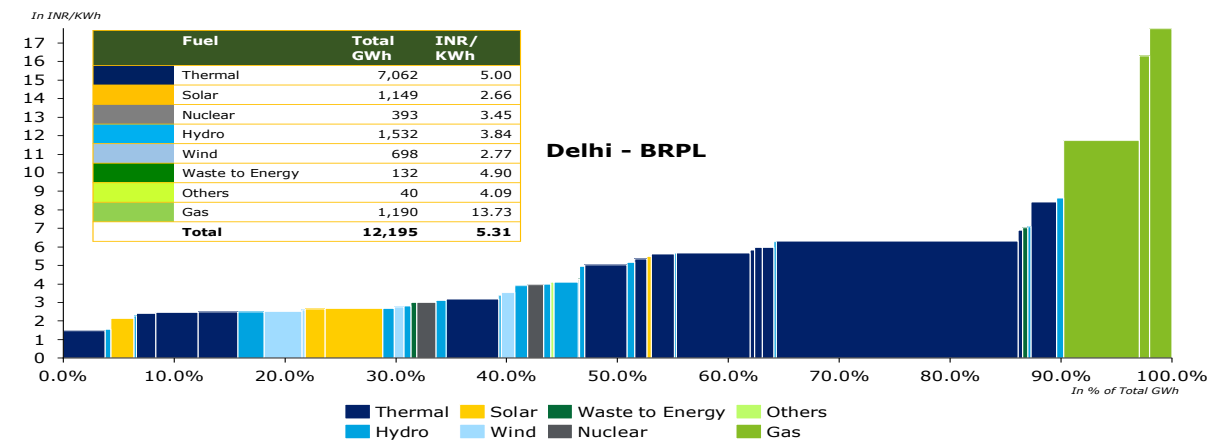
²⁰⁵ Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI

²⁰⁶ Report on Resource Adequacy of Delhi, Central Electricity Authority, GoI



Merit Order Analysis²⁰⁷:-

BRPL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 12,195 GWh. Of this, thermal power accounts for approximately 58% of the total output. According to the provisional true-up order for FY 2024-25, as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of Solar power plants, with an average generation cost of INR 3.31 per kWh. However, some solar assets, such as Acme and Thyagraj solar, report significantly higher generation costs of INR 5.50 and INR 3.58 per kWh, respectively. Other non-solar renewable sources contribute at an average tariff of INR 4.09 per kWh, reflecting a moderate cost profile within the state's energy mix.



BYPL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 8,733 GWh. Of this, thermal power accounts for approximately 74% of the total output. According to the provisional true-up order for FY 2024-25, as approved by the Commission, the state's fourth quartile (75%-100%) is predominantly composed of thermal power plants, with an

²⁰⁷ Aggregate Approval of True Up of Expenses upto FY 2022-23 & Aggregate Revenue Requirement (ARR) for FY 2024-25, Delhi Electricity Regulatory Commission, 2024

average generation cost of INR 1.47 per kWh. Other non-solar renewable sources contribute at an average tariff of INR 2.76 per kWh, reflecting a moderate cost profile within the state's energy mix.

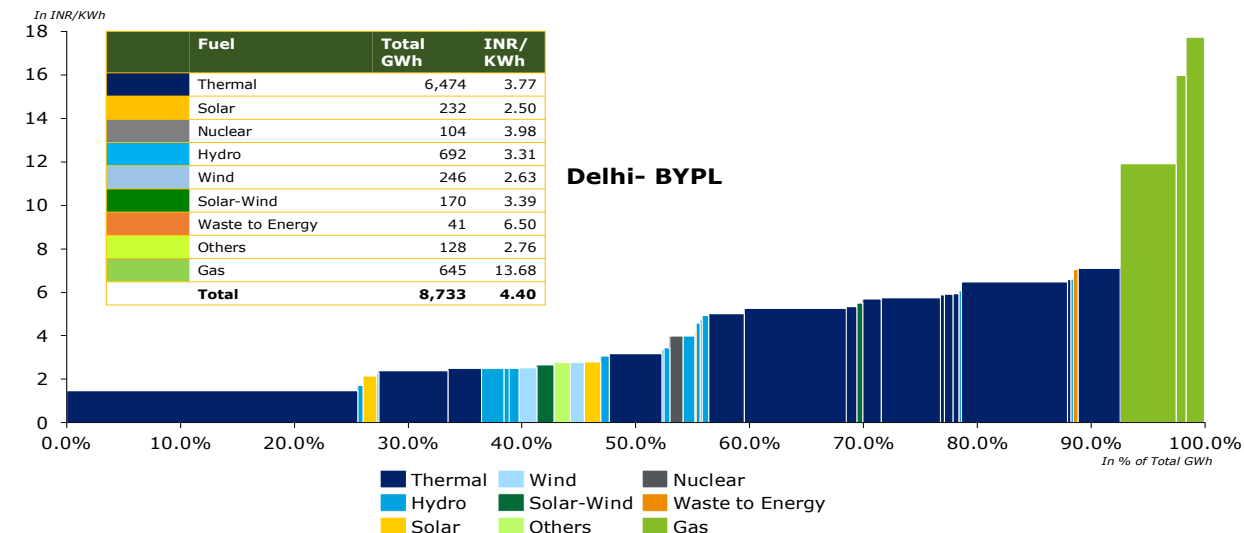


Figure 76: Merit Order Analysis – BYPL

TPDDL's electricity generation portfolio demonstrates a diverse mix of energy sources, with a total generation of 10,216 GWh. Of this, thermal power accounts for approximately 90% of the total output. According to the provisional true-up order for FY 2024–25, as approved by the Commission, the state's fourth quartile (75%–100%) is predominantly composed of thermal power plants, with an average generation cost of INR 6.09 per kWh. Hydropower contributes at an average tariff of INR 8.13 per kWh, reflecting a moderate cost profile within the state's energy mix.

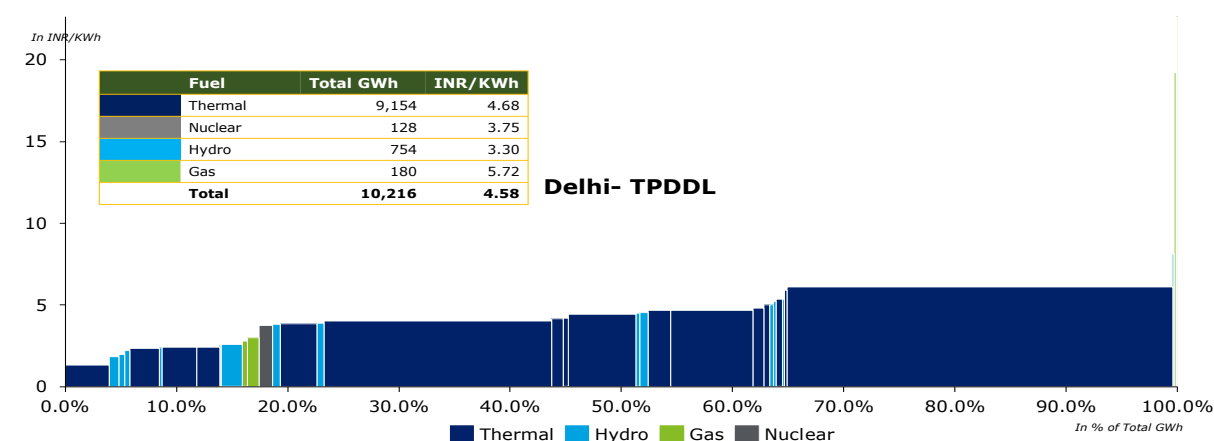


Figure 77: Merit Order Analysis - TPDDL

6.1.4. Southern Region

The Southern region of India showcases a well-diversified electricity profile, marked by a balanced mix of energy sources. It relies significantly on thermal power for consistent base load supply, while hydropower supports peak demand and enhances sustainability. The region has also made substantial progress in integrating renewable energy, particularly solar and wind, reflecting its commitment to clean energy. Nuclear power, though limited in scale, contributes to energy security and grid stability. Overall, the Southern region demonstrates a forward-looking approach to energy management, emphasizing efficiency, reliability, and environmental responsibility.

List of All States of Southern Region have been Stated below²⁰⁸:

Table 44: State-wise Gross Input Energy²⁰⁹

Region	State	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy of Region FY 2023-24 (%)
South	Tamil Nadu	1,10,695	27%
South	Andhra Pradesh	80,664	20%
South	Telangana	86,163	21%
South	Karnataka	91,324	22%
South	Kerala	32,868	8%
South	Puducherry	4,499	1%
South	Lakshadweep	-	0%

Among the above-mentioned States, Tamil Nadu, Karnataka, Andhra Pradesh and Telangana has been selected further for detailed level Analysis as it covers approximately 91% of total gross input energy of the region. DISCOM list of the selected states has been attached herewith:

Table 45: State-level DISCOM's Gross Input Energy

State	Distribution Company (DISCOMs)	Gross input Energy FY 2023-24 (GWh)	% Share of gross input energy FY 2023-24 (%)
Andhra Pradesh	Andhra Pradesh Eastern Power Distribution Company Limited	30,038	37%
	Andhra Pradesh Southern Power Distribution Company Limited	32,781	41%
	Andhra Pradesh Central Power Distribution Company Limited	17,845	22%
	Total of Andhra Pradesh	80,664	
Karnataka	Bangalore Electricity Supply Company Limited	42,587	47%
	Chamundeshwari Electricity Supply Corporation Limited	10,262	11%
	Gulbarga Electricity Supply Company Limited	11,273	12%
	Hubli Electricity Supply Company Limited	19,146	21%
	Mangalore Electricity Supply Company Limited	8,056	9%
	Total of Karnataka	91,324	

²⁰⁸ [Central Electricity Authority](#)

²⁰⁹ Report on Performance Utilities, Power Finance Corporation Limited, 2023-24

Telengana	Telangana State Northern Power Distribution Company Limited	24,908	29%
	Telangana State Southern Power Distribution Company Limited	61,255	71%
	Total of Telengana	86,163	
Tamil Nadu	Tamil Nadu Generation and Distribution Corporation Limited	110,695	100%
	Total of Tamil Nadu	110,695	

5. Region: South

State: Tamil Nadu

In FY 2025, Tamil Nadu's peak electricity demand reached 20.78 GW, with no reported surplus or deficit. The state boasts an installed generation capacity of approximately 42.77 GW²¹⁰, comprising around 37.5% from fossil fuel-based sources, 59.1% from renewable energy sources (RES) such as solar and waste-to-energy, and 3.4% from nuclear power. A national leader in renewable energy adoption—particularly in wind and solar—Tamil Nadu contributes over 14%²¹¹ of India's total renewable capacity. The state has also made significant strides in supporting its clean energy transition through investments in grid-scale energy storage, green energy corridors, and smart grid technologies.

Table 46: Energy Demand of Tamil Nadu

Particulars	Units	2021-22 (Actual) ²¹²	2024-25 (Actual) ²¹³	2026-27 (Projected) ²¹⁴	2031-32 (Projected) ²¹⁵
Energy Requirement	TWh	110	130	144	186
Peak Demand	GW	17	21	22	28

Fuel Mix of Tamil Nadu ²¹⁶

As of March 2024, Tamil Nadu's contracted power capacity reached approximately 37 GW, establishing the state as a key contributor to India's power infrastructure. Of this, 39% is sourced from fossil fuels—primarily coal and natural gas—while the remaining 61% comes from non-fossil fuel sources, including nuclear, biomass, hydro, solar, wind, and other renewable energy (RE) technologies. Looking ahead, the projected contracted capacity for FY 2034–35 is expected to reach around 114 GW, with 80% derived from non-fossil fuel sources and 20% from fossil fuels. The state's energy

²¹⁰ [All India Installed Capacity, Central Electricity Authority of India, 2025](#)

²¹¹ [Press Information Bureau, Government of India, 2025](#)

²¹² [Load Generation Balance Report, 2022](#)

²¹³ [Load Generation Balance Report, 2025](#)

²¹⁴ [National Electricity Plan 2022-32](#)

²¹⁵ [National Electricity Plan 2022-32](#)

²¹⁶ [Resource Adequacy Report, Central Electricity Report, GOI, 2024](#)

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

capacity is projected to grow at CAGR of approximately 9%, rising from 49 GW in FY 2025 to 115 GW by FY 2035. The evolution of the generation mix from FY 2024–35 is illustrated below:

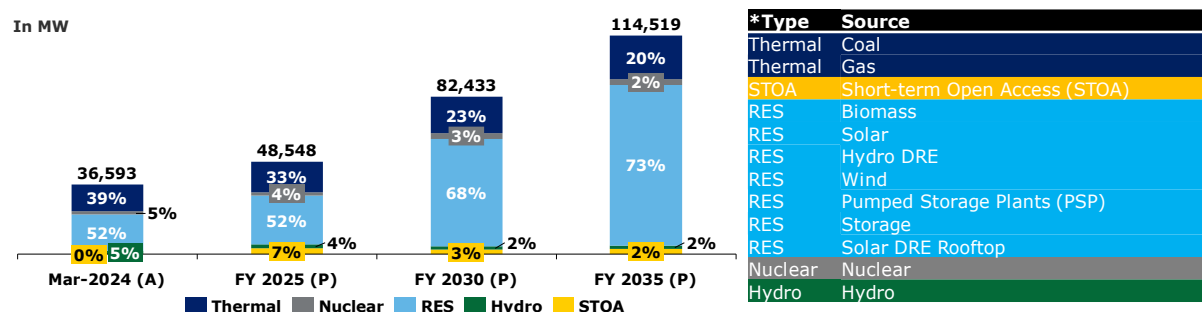


Figure 78: Fuel-wise capacity mix of Tamil Nadu*

Financial Health of Tamil Nadu's Distribution Companies:

The analysis of payable dues is a vital indicator of the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—comprising 41% of the total—represent payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, which account for the remaining 59%, are still within their respective due dates but have not yet been settled:²¹⁷

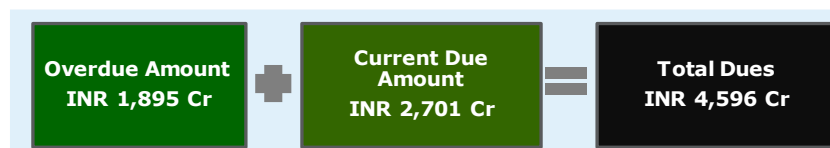


Figure 79: Snapshot of PRAAPTI Portal - Tamil Nadu

Utility Name 1: Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO)

Tamil Nadu's electricity distribution is managed by the Tamil Nadu Generation and Distribution Corporation Limited (TANGEDCO), a subsidiary of Tamil Nadu Electricity Board (TNEB). TANGEDCO oversee both power generation and distribution across the state. TANGEDCO serves over 3 crore consumers, including domestic, agricultural, commercial, and industrial categories.

Performance History of TANGEDCO:

From FY21 to FY24, energy input has increased at a compound annual growth rate (CAGR) of approximately 8%, reflecting the state's growing power demand. During the same period, the ACS-ARR gap has reduced by nearly 93%, indicating stronger revenue realization and improved billing and collection efficiencies. A similar positive trajectory is observed in Profit After Tax (PAT) due to decrease in ACS, increasing trend of ARR, high collection and billing efficiency, reduction in A&T losses among others. Although payable days remain relatively high, they have shown a consistent decline over the past three years. However, they still exceed the national average (excluding private DISCOMs) of 130 days recorded in FY 2023-24 ²¹⁸

Table 47: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ²¹⁹	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	73,622	82,076	87,917	92,735

²¹⁷ Dashboard for Tamil Nadu, PRAAPTI, as accessed on 27th May 2025

²¹⁸ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

²¹⁹ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Gross Input Energy	GWh	88,099	98,037	103,245	110,695	
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	9.14	8.76	10.42	10.79	
Average Revenue Realized (ARR)	INR/kWh	7.37	7.65	9.37	10.66	
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(1.77)	(1.11)	(1.05)	(0.13)	
PAT (Profit/(Loss))	INR in Crore	(13,066)	(9,130)	(9,192)	(1,196)	
Payable Days	Days	257	223	170	184	

Capacity Addition Plans of TANGEDCO:

TANGEDCO's installed capacity is projected to grow substantially in the coming years, with a progressively diversified energy mix. The share of renewable energy sources is expected to rise from 52% in FY 2024 to 73% by FY 2035, signaling a strategic shift toward cleaner and more sustainable power generation. According to the Resource Adequacy Report, hydropower capacity is set to increase by only 20 MW, indicating a relatively modest expansion in this segment compared to other renewables. The required capacity additions over this period are illustrated below ²²⁰

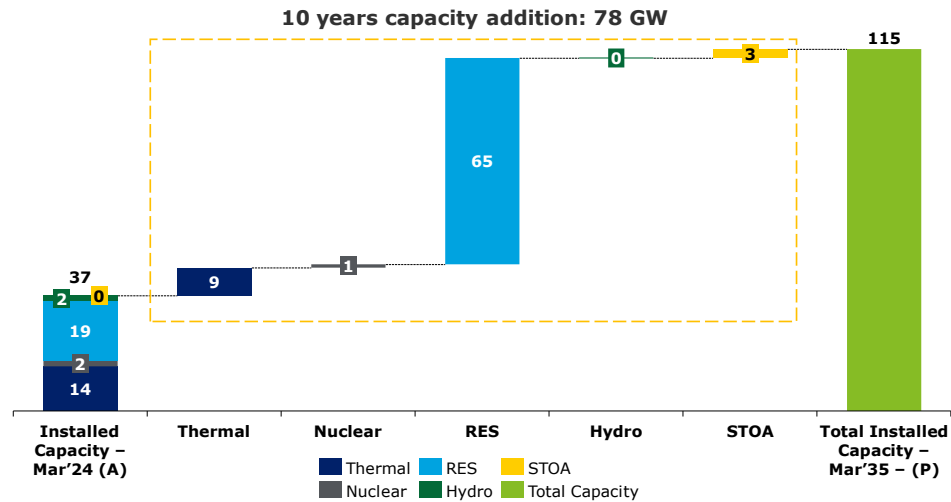


Figure 80: Capacity Addition for the next 10 years of Tamil Nadu

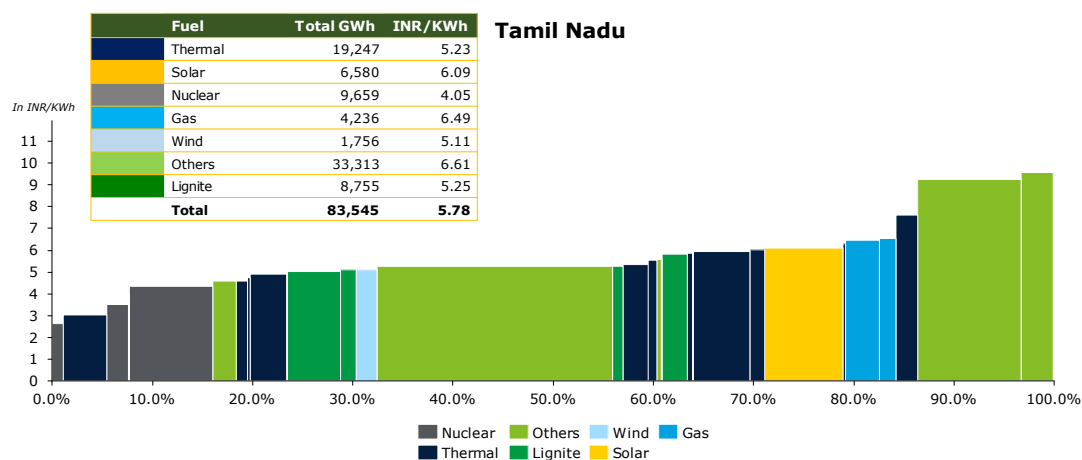
Merit Order Analysis : ²²¹

As per the true-up order for FY 2023-24 approved by the Commission, the state's fourth quartile (75%-100%) generation portfolio is primarily composed of solar, gas, and thermal power plants, with an average generation cost of ₹7.56 per kWh during FY23-24. Notably, cogeneration from sugar mills

²²⁰ Report on Resource Adequacy of Tamil Nadu, Central Electricity Authority, GoI

²²¹ Approval of True Up of TNPDC (formerly TANGEDCO) for FY 2023-24, Tamil Nadu Electricity Regulatory Commission, 2024

and short-term power transactions report significantly higher power purchase costs of ₹9.69/kWh and ₹9.56/kWh, respectively.



5. Region: South

State: Telangana

In FY 2025, Telangana's peak electricity demand was approximately 17 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 20 GW²²², with roughly 61% from fossil fuel-based sources, 38% from renewable energy sources such as solar and waste-to-energy, and around 1% from nuclear power.

Table 48 : Energy Demand of Telangana

Particulars	Units	2021-22 (Actual) ²²³	2024-25 (Actual) ²²⁴	2026-27 (Projected) ²²⁵	2031-32 (Projected) ²²⁶
Energy Requirement	TWh	71	88	93	121
Peak Demand	GW	15	17	20	27

Fuel Mix of Telangana ²²⁷

Telangana's contracted power capacity reached ~ 19 GW by the end of March 2024, positioning it as a key contributor to India's power infrastructure. Of this capacity, 54% is derived from fossil fuel-based sources, primarily coal and natural gas and 46% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2034-35 is ~ 73 GW, out of which 77% through non-fossil fuel and 23% from fossil fuel based. The state's energy capacity is projected to grow at CAGR of approximately 9%, rising from 19 GW in FY 2025 to 73 GW by FY 2035. Capacity wise generation mix from FY 2024-35 has been illustrated below:

²²² All India Installed Capacity, Central Electricity Authority of India, 2025

²²³ Load Generation Balance Report, 2022

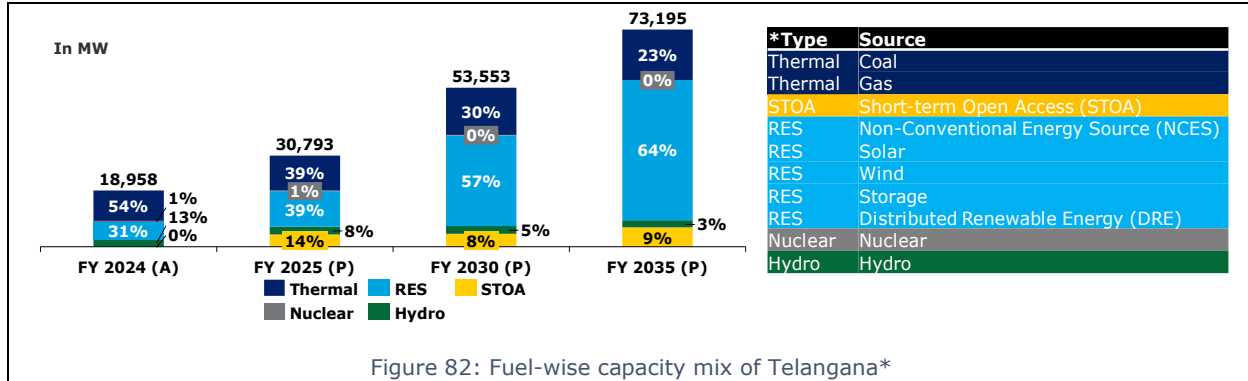
²²⁴ Load Generation Balance Report, 2025

²²⁵ National Electricity Plan 2022-32

²²⁶ National Electricity Plan 2022-32

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

²²⁷ Resource Adequacy Report, Central Electricity Report, GOI, 2024



Financial Health of Telangana's Distribution Companies:

The analysis of payable dues is a key metric for assessing the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—constituting 50% of the total—refer to payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, which make up the remaining 50%, are still within their respective due dates but have not yet been settled:²²⁸

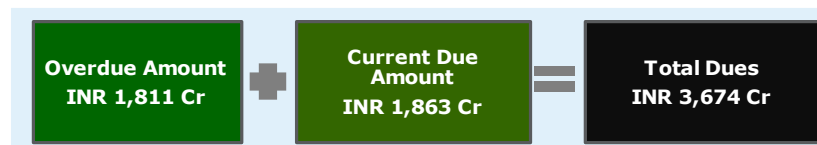


Figure 83: Snapshot of PRAAPTI Portal - Telangana

Utility Name 1: The Southern Power Distribution Company of Telangana Limited (TSSPDCL)

Utility Name 2: The Northern Power Distribution Company of Telangana Limited (TSNPDCL)

Telangana is managed by two State-owned DISCOMs. These utilities operate under the regulatory oversight of the Telangana State Electricity Regulatory Commission (TSERC) and serve both urban and rural regions across the State. Together, these utilities serve 17.20 million²²⁹ consumers serving majorly Agriculture sector, followed by Industrial, Domestic, Commercial and Others.

Performance History:

From FY21 to FY24, energy input has increased at a CAGR of ~ 9%, reflecting the state's growing power demand. During the same period, the ACS-ARR gap has reduced by nearly 20%, indicating enhanced revenue realization, improved billing and collection efficiencies. Although payable days remained relatively high in FY22, they have remained steady over the past two years. However, they still significantly exceed the national average (excluding private DISCOMs) of 130 days recorded in FY 2023-24²³⁰

Table 49: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ²³¹	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	59,451	65,900	69,136	75,179

²²⁸ [Dashboard for Telangana, PRAAPTI, as accessed on 27th May 2025](#)

²²⁹ [India Climate & Energy Dashboard](#)

²³⁰ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

²³¹ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Gross Input Energy	GWh	66,468	74,651	78,117	86,163
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.22	7.48	8.83	8.74
Average Revenue Realized (ARR)	INR/kWh	6.03	7.39	7.25	7.87
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(1.19)	(0.09)	(1.58)	(0.86)
PAT (Profit/(Loss))	INR in Crore	(6,686)	(831)	(11,103)	(6,351)
Payable Days	Days	297	380	310	310

Capacity Addition Plans of Telangana:

As outlined above, Telangana's energy capacity is set for substantial growth, accompanied by a major transformation in its energy mix. The share of coal and gas is projected to decline significantly—from 54% in FY 2023 to just 23% by FY 2035. In contrast, the contribution of non-fossil and renewable energy (RE) sources is expected to more than double, rising from 31% to 64%. This shift is driven by increasing energy demand and the accelerated adoption of renewable technologies across the state. The transition aligns closely with India's national clean energy targets for 2030. The required capacity additions to support this transformation are illustrated below ²³²:

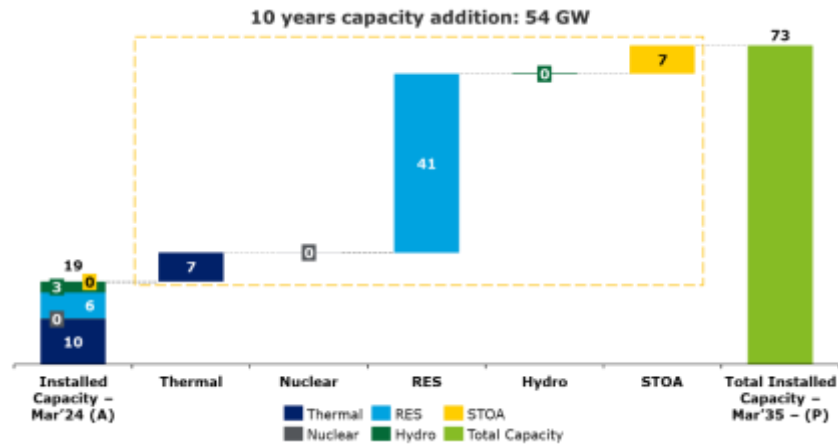


Figure 84: Capacity Addition for next 10 years of Telangana

Merit Order Analysis²³³:-

Telangana's power generation is predominantly reliant on thermal and solar energy, which remains one of the most expensive generation sources for the state. According to the revised Aggregate Revenue Requirement (ARR) order for FY 2025-26 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio primarily comprises thermal, solar, and biomass power plants, with an average generation cost of ₹7.63 per kWh during FY 2023-24. Notably, solar generation under JNNSM Phase I incurs significantly higher power purchase costs—₹10.4 per kWh—

²³² Report on Resource Adequacy of Telangana, Central Electricity Authority, GoI

²³³ Revised Aggregate Revenue Requirement, Telangana Electricity Regulatory Commission, 2024-25

due to legacy feed-in tariffs fixed by the Central Electricity Regulatory Commission (CERC) for a 25-year period.

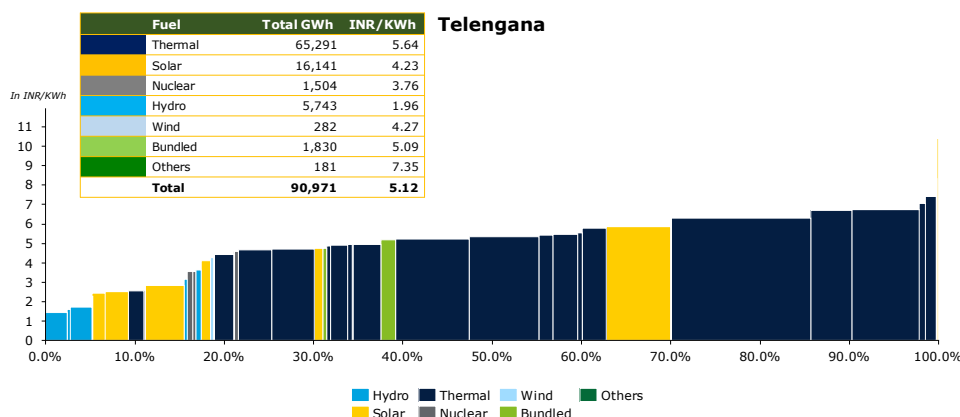


Figure 85: Merit Order Analysis of Telengana

5. Region: South

State: Andhra Pradesh

In FY 2025, Andhra Pradesh's peak electricity demand was approximately 14 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 28 GW²³⁴, with around 57% sourced from fossil fuel-based projects, 43% from renewable energy sources (RES) such as solar and waste-to-energy, and a small fraction—about 0.5%—from nuclear power. Andhra Pradesh has made notable progress in renewable energy adoption, contributing roughly 8.5% of India's total renewable capacity. The state is actively expanding its solar and wind infrastructure, with strategic plans to increase the share of renewables to 74% by 2035²²⁸. This positions Andhra Pradesh as a frontrunner in India's clean energy transition, aligning with national goals for sustainability and energy security.

Table 50: Energy Demand of Andhra Pradesh

Particulars	Units	2021-22 (Actual) ²³⁵	2024-25 (Actual) ²³⁶	2026-27 (Projected) ²³⁷	2031-32 (Projected)
Energy Requirement	TWh	69	79	98	137
Peak Demand	GW	12	14	18	24

Fuel Mix of Andhra Pradesh²³⁸

Andhra Pradesh's contracted power capacity reached ~ 21 GW by the end of March 2024, positioning it as a key contributor to India's power infrastructure. Of this capacity, 53% is derived from fossil fuel-based sources, primarily coal and natural gas and 47% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2034-35 is ~59 GW, out of which 74% through non-fossil fuel and 26% from fossil fuel based. The's energy capacity is projected to grow at CAGR of

²³⁴ All India Installed Capacity, Central Electricity Authority of India, 2025

²³⁵ Load Generation Balance Report, 2022

²³⁶ Load Generation Balance Report, 2025

²³⁷ National Electricity Plan 2022-32

²³⁸ Resource Adequacy Report, Central Electricity Report, GOI, 2024

* As per the Government of India's classification, hydroelectric power is included under Renewable Energy Sources (RES); however, for the purposes of this report, it has been considered as a separate fuel type.

approximately 9%, rising from 24 GW in FY 2025 to 59 GW by FY 2035. Capacity wise generation mix from FY 2024-35 has been illustrated below:

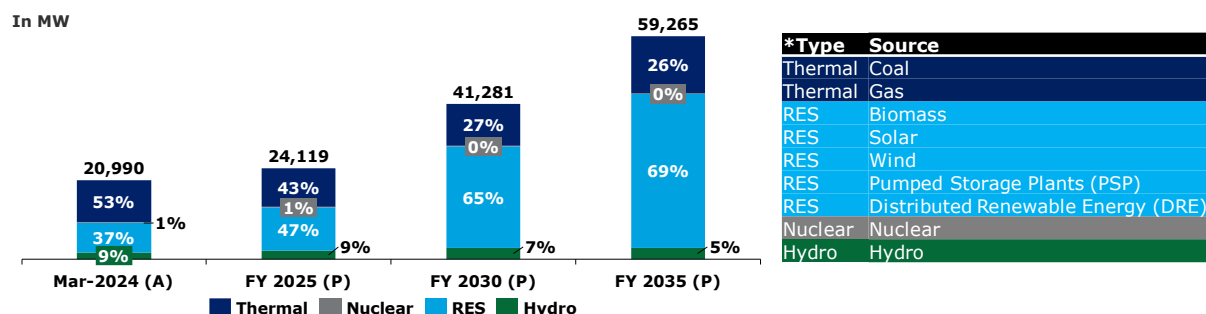


Figure 86: Fuel-wise capacity mix of Andhra Pradesh*

Financial Health of Andra Pradesh's Distribution Companies:

The analysis of payable dues is a critical tool for assessing the financial health, payment behavior, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments: Overdue and Current. Overdue dues—comprising 49% of the total—represent payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues, accounting for the remaining 51%, are still within their respective due dates but have not yet been settled.²³⁹

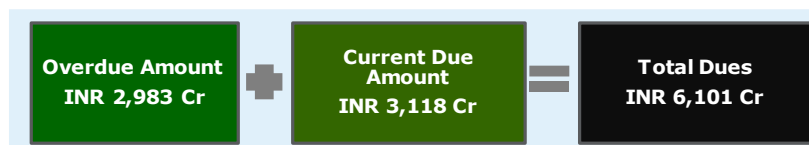


Figure 87: Snapshot of PRAAPTI portal – Andhra Pradesh

Utility Name1: Southern Power Distribution Company of Andhra Pradesh Limited (APSPDCL)

Utility Name2: Eastern Power Distribution Company of Andhra Pradesh Limited (APEPDCL)

Utility Name3: Central Power Distribution Company of Andhra Pradesh Limited (APCPDCL)

Electricity distribution in Andhra Pradesh is managed by three state-owned DISCOMs, each responsible for serving distinct geographical regions and consumer segments across the state. These DISCOMs play a vital role in ensuring reliable power supply, managing demand, and implementing both state and central energy policies. Collectively, they serve approximately 15.4 million consumers²⁴⁰, with the industrial sector accounting for the largest share at 43%, followed by domestic, agricultural, commercial, and other categories

Performance History:

Between FY21 and FY24, energy input grew at a compound annual growth rate (CAGR) of approximately 6%, reflecting the state's rising power demand. Over the same period, the ACS-ARR gap narrowed by only 8%, indicating modest improvements in revenue realization and billing and collection efficiency. However, the Average Cost of Supply (ACS) has continued to rise since FY21, pointing to increasing power purchase costs. On a positive note, payable days have steadily declined over the past three years and now remain below the national average (excluding private DISCOMs) of 130 days recorded in FY 2023–24²⁴¹

²³⁹ Dashboard for Andra Pradesh, PRAAPTI, as accessed on 27th May 2025

²⁴⁰ [India Climate & Energy Dashboard](#)

²⁴¹ Report on Performance of Power Utilities, Power Finance Corporation, 2023–24

Table 51: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ²⁴²	FY 2021-22	FY 2022-23	FY 2023-24
Gross Energy Sold	GWh	62,493	69,745	71,463	72,628
Gross Input Energy	GWh	68,564	76,122	78,424	80,664
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	7.14	6.89	7.87	9.24
Average Revenue Realized (ARR)	INR/kWh	7.14	6.54	7.17	8.86
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	-	(0.35)	(0.70)	(0.38)
PAT (Profit/(Loss))	INR in Crore	44	(2,458)	1,736	169
Payable Days	Days	217	166	91	105

Capacity Addition Plans of Andhra Pradesh:

Andhra Pradesh's energy capacity is projected to grow significantly, accompanied by a marked diversification of its energy mix. The share of coal and gas is expected to decline from 53% in FY 2024 to 26% by FY 2035. In contrast, the contribution of non-fossil and renewable energy sources is set to rise from 46% to 74% over the same period, driven by increasing demand and the rapid adoption of renewable technologies across the state. This transition aligns closely with India's national clean energy targets for 2030. The required capacity additions to support this transformation are illustrated below.²⁴³

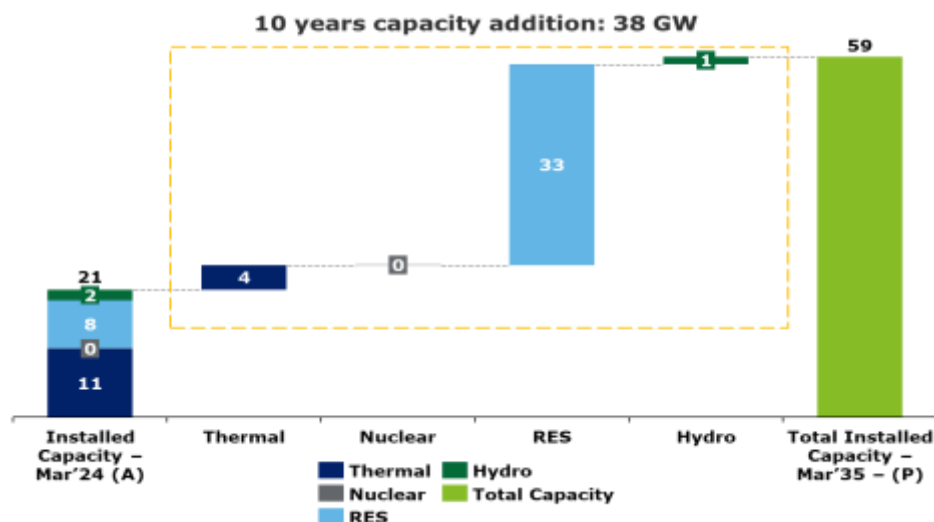


Figure 88: Capacity Addition Plan for next 10 years of Andhra Pradesh

Merit Order Analysis²⁴⁴:-

Andhra Pradesh's electricity generation is dominated by thermal power, followed by solar and wind. While thermal energy has the highest share, it is also among the costlier sources. As per the revised Retail Supply Tariffs order for FY 2025-26 approved by the Commission, the state's fourth quartile (75%-100%) generation portfolio is primarily composed of thermal and hydropower plants, with an average generation cost of ₹7.82 per kWh during FY25-26. Notably, generation from PABRHES report

²⁴² Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

²⁴³ Report on Resource Adequacy of Andhra Pradesh, Central Electricity Authority, GoI

²⁴⁴ RST Order, Andhra Pradesh Electricity Regulatory Commission, 2025-26

significantly higher power purchase costs of ₹20.5/kWh, due to the high fixed cost of the hydropower plant

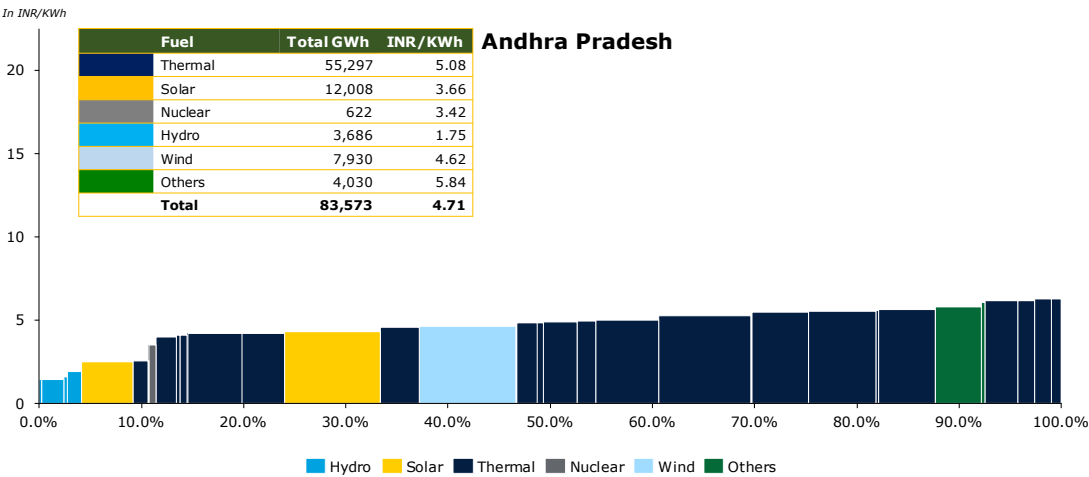


Figure 89: Merit Order Analysis of Andhra Pradesh

5. Region: South

State: Karnataka

In FY 2025, Karnataka's peak electricity demand was approximately 18 GW, with no reported surplus or deficit. The state has an installed generation capacity of about 35 GW²⁴⁵, with around 31% from fossil fuel-based sources, 67% from renewable energy sources such as solar and waste-to-energy, and about 2% from nuclear power.

Karnataka has been a frontrunner in solar energy, home to some of India's largest solar parks, including the Pavagada Solar Park. The state contributes over 12%²⁴⁶ of the country's total renewable energy capacity, ranking among the top three states in renewable energy deployment.

Table 52: Energy Demand of Karnataka

Particulars	Units	2021-22 (Actual) ²⁴⁷	2024-25 (Actual) ²⁴⁸	2026-27 (Projected) ²⁴⁹	2031-32 (Projected) ²⁵⁰
Energy Requirement	TWh	72	93	88	106
Peak Demand	GW	15	18	18	22

Fuel Mix of Karnataka²⁵¹

Karnataka's contracted power capacity reached ~ 30 GW by the end of March 2023, positioning it as a key contributor to India's power infrastructure. Of this capacity, 32% is derived from fossil fuel-based sources, primarily coal and 68% is sourced from non-fossil fuel-based energy, including nuclear, biomass, hydro, solar, wind and Other RE.

The projected contracted Capacity for the year 2031-32 is ~ 47 GW, out of which 68% through non-fossil fuel and 32% from fossil fuel based. The state's energy capacity is projected to grow at a CAGR of approximately 4%, increasing from 37 GW in FY 2025 to 47 GW by FY 2032. This growth is driven by rising energy demand and the accelerated adoption of clean technologies. Capacity wise generation mix from FY 2023-32 has been illustrated below:

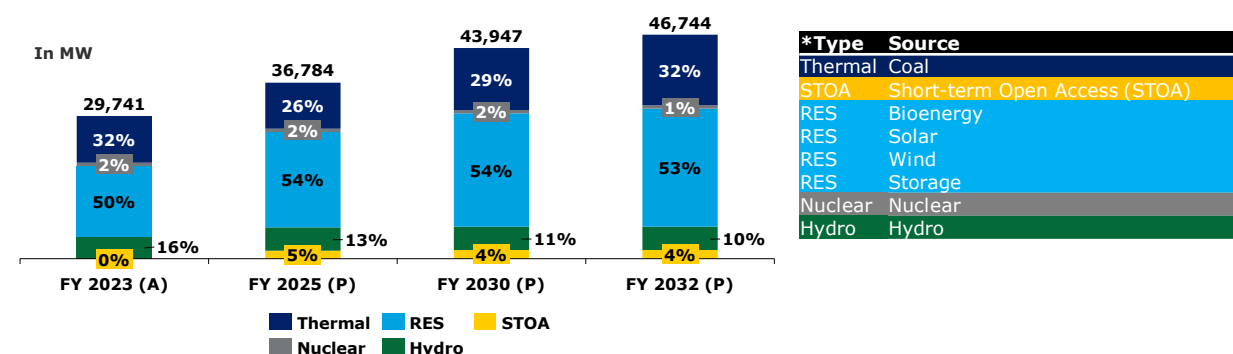


Figure 90: Fuel-wise capacity mix of Karnataka*

Financial Health of Karnataka's Distribution Companies:

The analysis of payable dues is a key indicator of the financial health, payment discipline, and creditworthiness of state DISCOMs. As of 27th May 2025, dues are categorized into two segments:

²⁴⁵ All India Installed Capacity, Central Electricity Authority of India, 2025

²⁴⁶ Resource Adequacy Report, Karnataka

²⁴⁷ Load Generation Balance Report, 2022

²⁴⁸ Load Generation Balance Report, 2025

²⁴⁹ National Electricity Plan 2022-32

²⁵⁰ National Electricity Plan 2022-32

²⁵¹

Resource Adequacy Report, Karnataka

Overdue and Current. Overdue dues, which make up 50% of the total, refer to payments that have exceeded their due dates and remain either partially or fully unpaid. In contrast, Current dues—accounting for the remaining 50%—are still within their respective due dates but have not yet been settled:²⁵²

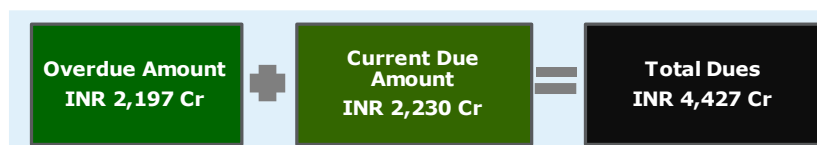


Figure 91: Snapshot of PRAPTI portal – Karnataka

Utility Name 1: Electricity Supply Companies (ESCOs)

Karnataka's electricity distribution is managed by five regional Electricity Supply Companies (ESCOs):

- Bangalore Electricity Supply Company Limited (BESCOM)
- Mangalore Electricity Supply Company (MESCOM)
- Hubli Electricity Supply Company (HESCOM)
- Gulbarga Electricity Supply Company (GESCOM)
- Chamundeshwari Electricity Supply Company (CHESCOM)

distinct geographic regions and consumer segments across the state. Together, these DISCOMs form the backbone of Karnataka's power distribution network, playing a critical role in ensuring reliable electricity supply, minimizing losses, and advancing the state's clean energy transition. Collectively, they serve approximately 30.9 million consumers²⁵³, with the majority being industrial, followed by agricultural, domestic, commercial, and other categories.

Performance History:

Between FY 2021 and FY 2024, energy input in the state grew at a compound annual growth rate (CAGR) of approximately 12%, reflecting a steady rise in power demand. During the same period, the ACS-ARR gap narrowed by only 13%, indicating modest improvements in revenue realization and billing and collection efficiency. However, the Average Cost of Supply (ACS) has continued to increase since FY 2021, largely due to rising power purchase costs. While payable days have been declining over the years, they still remain above the national average (excluding private DISCOMs) of 130 days recorded in FY 2023–24²⁵⁴

Table 53: Performance Snapshot Report

Key Parameters	Units	FY 2020-21 ²⁵⁵	FY 2021-22	FY 2022-23	FY 2022-23
Gross Energy Sold	GWh	54,784	58,462	62,270	76,911
Gross Input Energy	GWh	65,116	69,139	73,493	91,324
Average Cost of Supply (ACS) on Energy Sold	INR/kWh	8.49	8.58	9.55	9.97
Average Revenue Realized (ARR)	INR/kWh	7.67	8.19	9.18	9.24
Gap between ARR vs ACS (Surplus/(Deficit))	INR/kWh	(0.82)	(0.39)	(0.37)	(0.72)
PAT (Profit/(Loss))	INR in Crore	(4,175)	(4,592)	(3,146)	(8,550)

²⁵² Dashboard for Karnataka, PRAPTI, as assessed on 27th May 2025

²⁵³ India Climate & Energy Dashboard

²⁵⁴ Report on Performance of Power Utilities, Power Finance Corporation, 2023-24

²⁵⁵ Report on Performance of Power Utilities, Power Finance Corporation, 2022-23

Payable Days	Days	201	186	169	177
--------------	------	-----	-----	-----	-----

Capacity Addition Plans of Karnataka:

From FY 2023 to FY 2032, Karnataka's projected energy mix indicates that thermal power will consistently hold a dominant share above 50%, while Renewable Energy Sources (RES) are expected to contribute between 63% and 67% from FY 2024 to FY 2032. This trend reflects a measured energy transition, shaped by the need to balance energy security, cost-effectiveness, and infrastructure readiness. The sustained reliance on thermal power highlights its role in providing stable base-load capacity, even as the state gradually integrates cleaner and more flexible energy alternatives. The required capacity additions to support this transformation are illustrated below²⁵⁶:

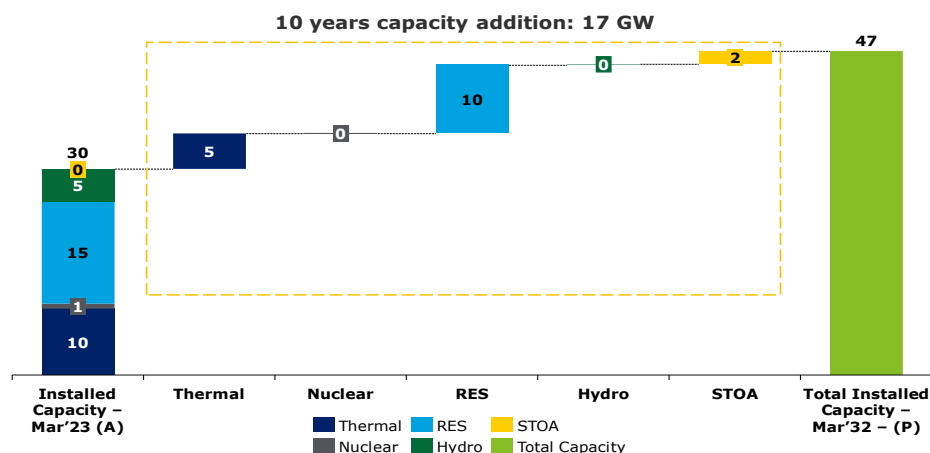


Figure 92: Capacity Addition Plan for the next 8 years of Karnataka

Merit Order Analysis²⁵⁷:-

Karnataka's power generation is mainly dependent on thermal energy, which is also among the costliest sources. As per the revised APR order for FY 2024–25 approved by the Commission, the state's fourth quartile (75%–100%) generation portfolio is primarily composed of thermal and bundled solar, with an average generation cost of ₹8.86 per kWh during FY23–24. Notably, generation from Bundled Power Solar report significantly higher power purchase costs of ₹11.9/kWh, due to the feed in tariffs for old solar plants locked by CERC for 25 years.

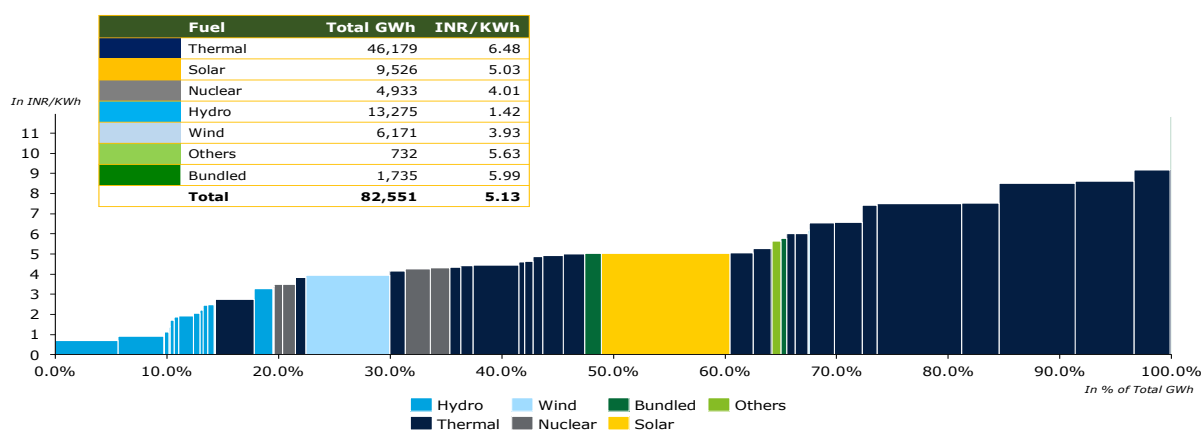


Figure 93: Merit Order Analysis of Karnataka

²⁵⁶ Report on Resource Adequacy of Karnataka, Central Electricity Authority, GoI

²⁵⁷ Annual Performance Review for FY 23 and Approved Revised ARR for FY 25, Karnataka Electricity Regulatory Commission, 2024

6.1.5. Key Takeaways from DISCOM analysis

The Offtake Analysis conducted above provides valuable insights on the potential for select State and Private Discoms to absorb electricity generated from DHPP. A snapshot of the analysis is provided below:

Table 54: Summary of Discom Offtake Analysis

State	Forward Year Projections for Capacity Available till	Total Capacity Addition Requirement (from current levels) (in GW)	Total Hydro Capacity Addition Required (in GW)	Average PP Cost (INR/kWh)	Average Cost of Supply - FY 2023-24 (INR/kWh)	Average Revenue Realized - FY 2023-24 (INR/kWh)
Western Region						
Maharashtra	2034	64.89	1.06	4.73	8.96	8.67
Gujarat	2035	86.07	1.10	4.01	7.12	7.78
Madhya Pradesh	2035	57.54	0.34	3.35	8.22	7.54
Chhattisgarh	2035	26.03	0.57	3.74	6.13	6.14
Eastern Region						
West Bengal	2035	44.52	0.09	3.8	7.86	7.65
Jharkhand	2034	10.33	0.02	4.33	11.27	8.90
Bihar	2034	31.35	1.00	3.01	8.42	8.79
Odisha	2034	18.09	1.10	3.37	6.47	6.34
North-Eastern Region						
Assam	2035	9.34	0.33	4.95	9.51	9.77
Northern Region						
Uttar Pradesh	2034	116.15	1.81	4.76	9.20	8.57
Rajasthan	2032	39.17	1.34	4.22	8.81	8.86

State	Forward Year Projections for Capacity Available till	Total Capacity Addition Requirement (from current levels) (in GW)	Total Hydro Capacity Addition Required (in GW)	Average PP Cost (INR/kWh)	Average Cost of Supply - FY 2023-24 (INR/kWh)	Average Revenue Realized - FY 2023-24 (INR/kWh)
Punjab	2030	24.48	0.83	4.36	7.06	7.19
Haryana	2035	35.19	1.35	4.56	7.16	7.21
Delhi - BRPL	2034	7.41	0.20	5.31	8.33	8.99
Delhi - BYPL	2034	2.11	0.03	4.40		
Delhi - TPDDL	2034	2.02	0.27	4.58		
Southern Region						
Tamil Nadu	2035	77.93	0.02	5.78	8.96	10.66
Telengana	2035	54.24	0.00	5.12	8.74	7.85
Andhra Pradesh	2035	38.28	1.19	4.71	9.24	8.86
Karnataka	2032	17.00	0.14	5.13	9.97	9.24

States such as Maharashtra, Gujarat, Bihar, Odisha, Uttar Pradesh, Rajasthan, Haryana and Andhra Pradesh are projected to need more than 1 GW of hydropower capacity to meet their Resource Adequacy needs over the next decade. Further States of Tamil Nadu, Telengana, Karnataka and BRPL in Delhi have an average power procurement cost of over Rs. 5 / kWh providing an opportunity to rationalize their power procurement costs through replacement of expensive fuel based power in their over power portfolio.

DHPP can be a viable alternative for the discoms subject to the tariff from the project assisting them to lower their power procurement costs as well as providing them with a base load in wet months to meet the variability in their renewable energy generation portfolio.

6.2. Offtake Analysis in Short/Medium Term Power Market through Indian Power Exchanges and Bilateral trade

The short and medium term power market in India has been evolving rapidly since the last few years due to changing customer preferences, favourable regulations and higher margin potential. Some of the key drivers are enumerated below:

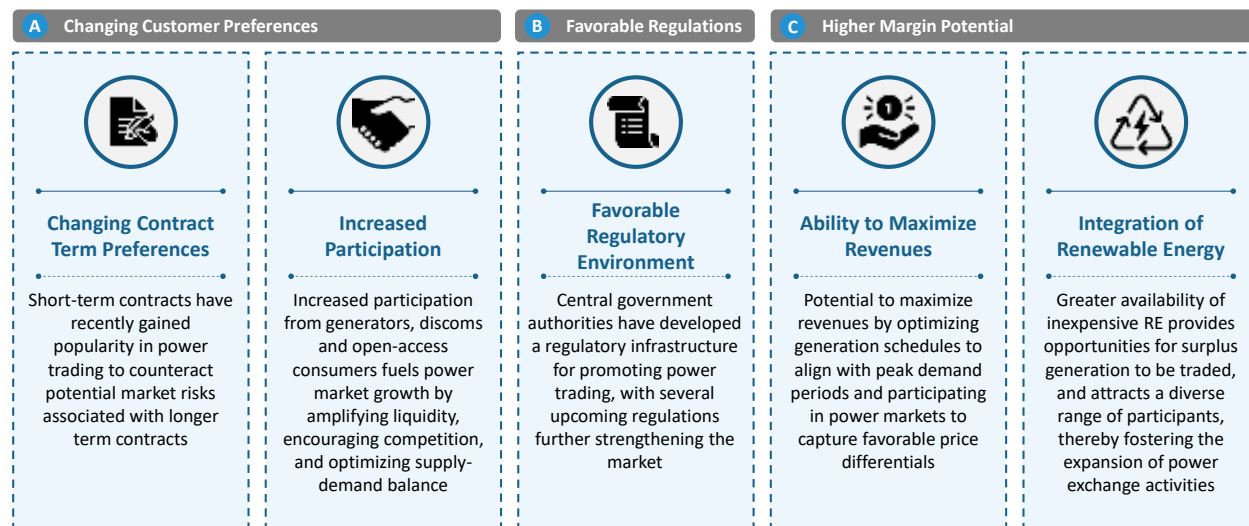


Figure 94: Key Drivers impacting Short/Medium Term Power Markets

A snapshot of market segments in the short/medium term market is shown:

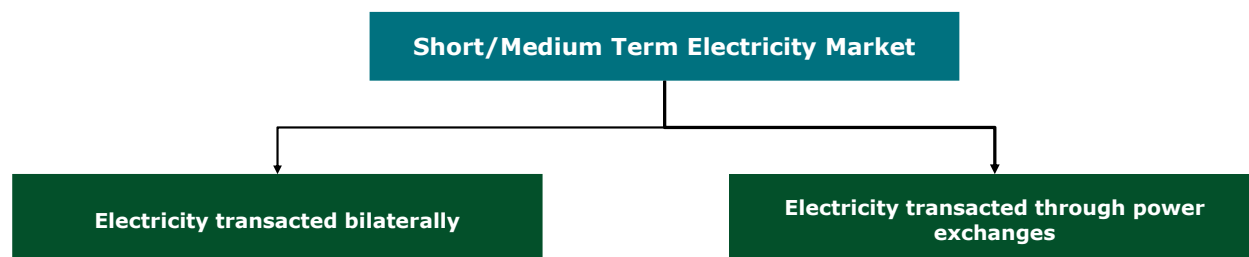


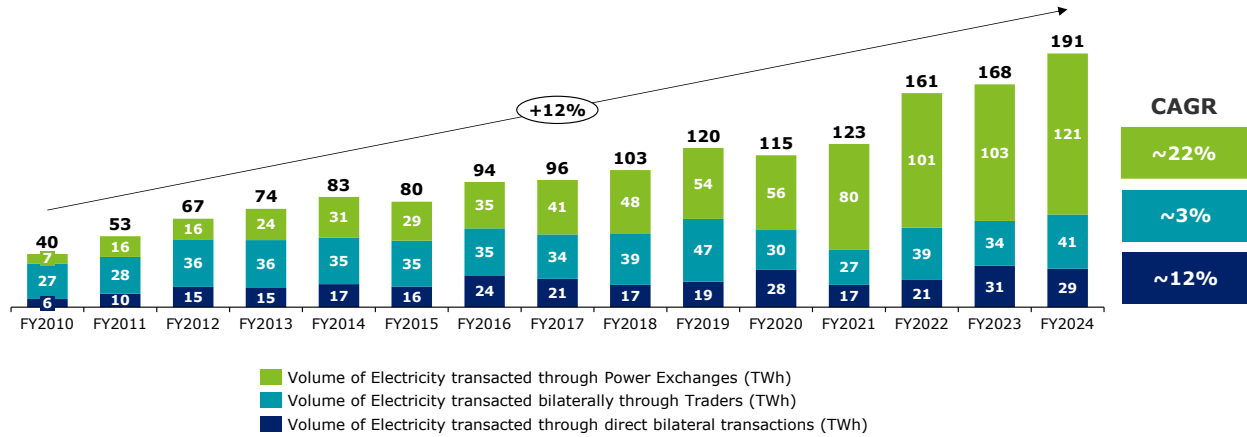
Figure 95: Market Segments in the Short/Medium -Term Power Market

In 2023-24, the short-term market constituted about 12% of the total electricity generation in India.²⁵⁸ Of the total short-term transactions in 2023-24, the volume of electricity transacted through power exchanges was maximum (63%), followed by bilateral transactions through traders (21%), and direct bilateral transactions between entities (15%).



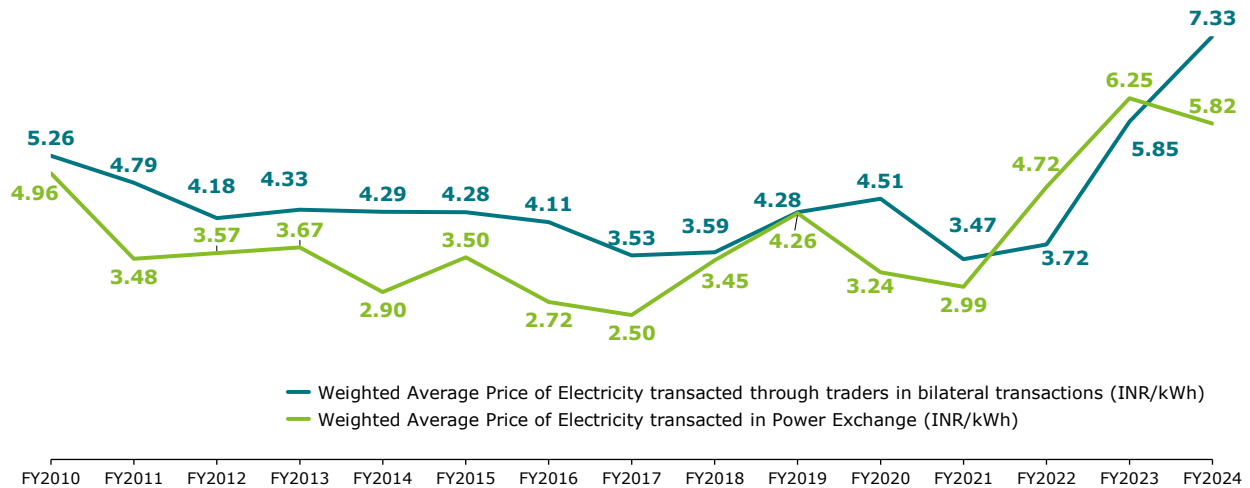
Figure 96: Share of different market segments in Short-Term Power Transactions in 2023-24

The volume of electricity transacted through power exchanges increased at a CAGR of 22.4%, and the volume of electricity transacted through traders bilaterally increased at a CAGR of 3.1% in the last 15 years.

Figure 97: Trend of volume of transaction through Traders and Power Exchanges²⁵⁸

Electricity trading from 2009 to 2019 was dominated by bilateral transactions. Since 2020, power exchanges have taken the lead, with transaction volumes exceeding those of the bilateral transactions.

However, the weighted average price of transaction through traders bilaterally was higher than the transactions recorded in the power exchange in 2023-2024 and this has broadly followed a similar trend in the last 15 years.

Figure 98: Trend of weighted average price of transaction through Traders and Power Exchanges^{258 258}

6.2.1. Offtake through Power Exchanges

Power exchanges in India are electronic platforms that facilitate the trading of electricity. They operate under the regulatory oversight of the Central Electricity Regulatory Commission (CERC) and play a crucial role in the short-term power market. These exchanges provide a transparent, efficient, and competitive environment for buyers and sellers to trade electricity through various market segments.

Power exchanges in India offer several market segments to cater to diverse trading needs.

²⁵⁸ [Report on Short Term Power Market in India 2023-2024, Central Electricity Regulatory Commission](#)

However, for Cross border Power Market Trading, only trading in Day-Ahead Market (DAM) and Real Time Market (RTM) is allowed by the regulations²⁵⁹. Any Indian trader, after obtaining approval from Designated Authority, can trade in Indian Power Exchanges on behalf of any entity of neighboring country complying with CERC regulations. There are three major exchanges in India for Power Trading – Indian Energy Exchange (IEX), Power Exchange of India Ltd. (PXIL) and Hindustan Power Exchange Limited (HPX).

The Cross-Border Trade of Electricity was commenced in the Day Ahead Market of IEX in 2021-22. The trade with Nepal commenced on 17.04.2021, whereas the trade with Bhutan commenced on 01.01.2022. Cross-Border Trade in Real Time Market commenced from October 2023 in Nepal and in December 2023 in Bhutan. As per CERC market monitoring reports, no cross-border transactions were recorded in PXIL and HPX.

The below table shows the transactions by Bhutan in the IEX in the DAM and RTM segments in the last 4 years :

Table 55: Power Exchange Transactions of Bhutan recorded in IEX (DAM and RTM)²⁶⁰

Financial Year	Sell		Buy	
	Volume Sold (GWh)	Weighted Average Price (INR/kWh)	Volume Purchased (GWh)	Weighted Average Price (INR/kWh)
2021-22	-	-	240	2.89
2022-23	-	-	319	4.39
2023-24	40	5.78	1299	3.74
2024-25 (till Feb 2025)	240	4.94	902	3.02
Total	280		2,760	

Bhutan started selling power in the Indian power exchange from 2023-24. Previously, Bhutan used to only purchase power from the exchange. The below illustration shows the power sold in IEX by Bhutan and Nepal in the last 3 years :

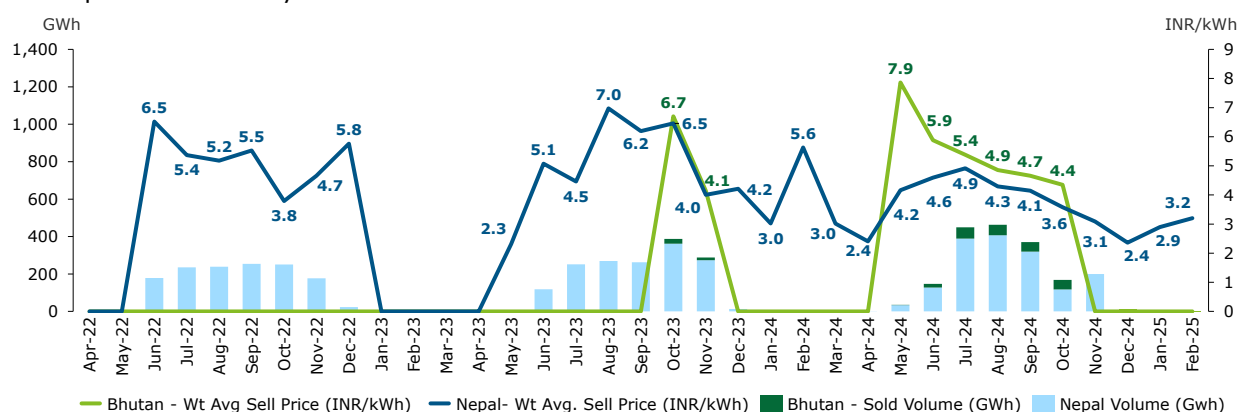


Figure 99: Power sold in IEX (DAM and RTM) by Nepal and Bhutan

²⁵⁹ [Amendment-1 to the Procedure for approval and facilitating Import/Export \(Cross Border\) of electricity by the Designated Authority](#)
²⁶⁰ [CERC Monthly and Annual Market Monitoring Reports](#)

Bhutan has been selling power in the exchange only during the wet season which is from May to November. Although, the volume sold by Bhutan has been lower in the exchanges but the price at which the electricity sold is higher when compared with Nepal.

6.2.1.1. Price and Volume Trends in Indian Energy Exchange (IEX)

The below illustrations showcase the Market Clearing Price (MCP) and the Final Scheduled volume (FSV) in the DAM and the RTM segments of IEX in the last 3 years to visualize the trends during the months allotted for export of power from DHPP which is from May to November.

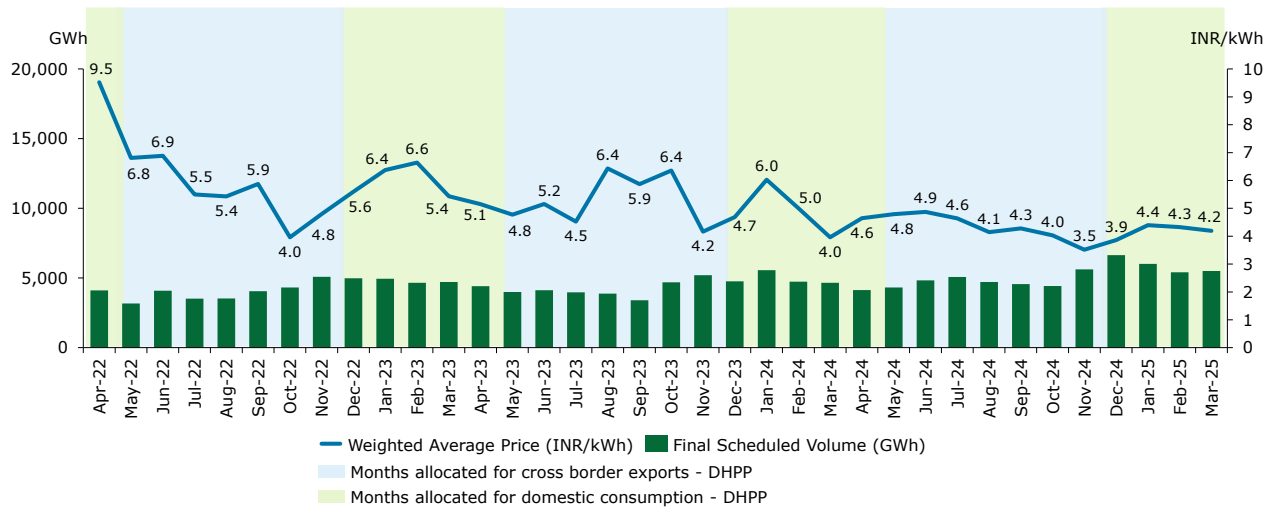


Figure 100: Monthly FSV and MCP in DAM of IEX during last 3 financial years

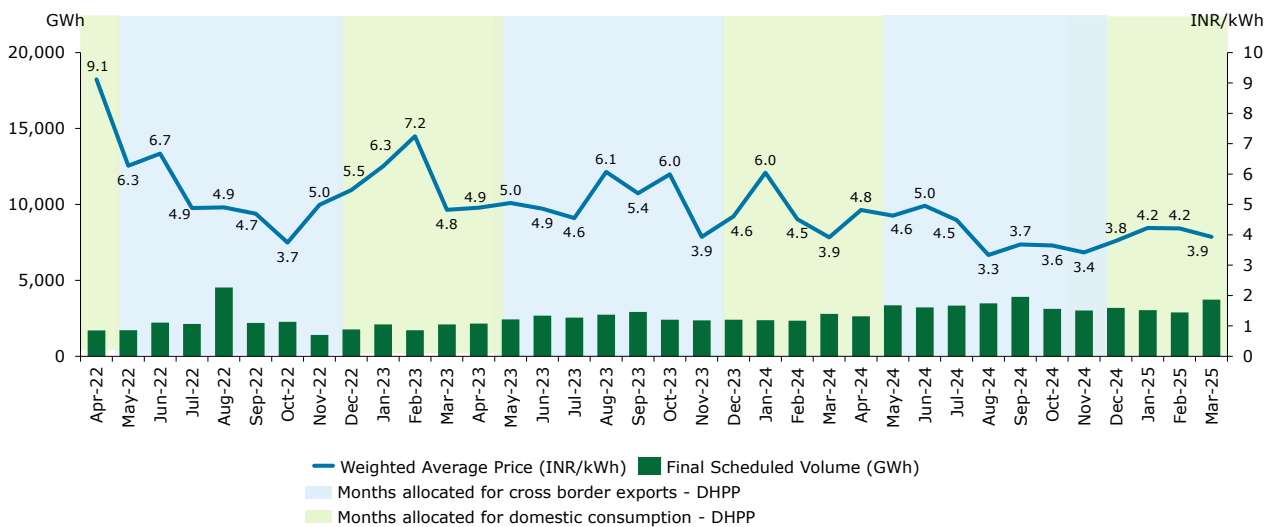


Figure 101: Monthly FSV and MCP in RTM of IEX during last 3 financial years

The MCP shows significant volatility throughout the period, indicating a dynamic interplay of supply and demand. However, there's a noticeable pattern in MCP during the DHPP export months (May to November) :

Pre-Monsoon/Summer Peaks : MCP rises during the pre-monsoon and summer months (May-June), likely due to the higher demand for cooling requirements during summer in India.

Monsoon/Post-Monsoon Dip: MCP is lower during the monsoon months (July-August) likely due to increased domestic hydropower generation.

The MCP in both DAM and RTM has decreased over the years. The transactions and price in the DAM has been higher when compared with RTM.

The hourly trend during the May-November period is shown below for the last 3 financial years for DAM and RTM :

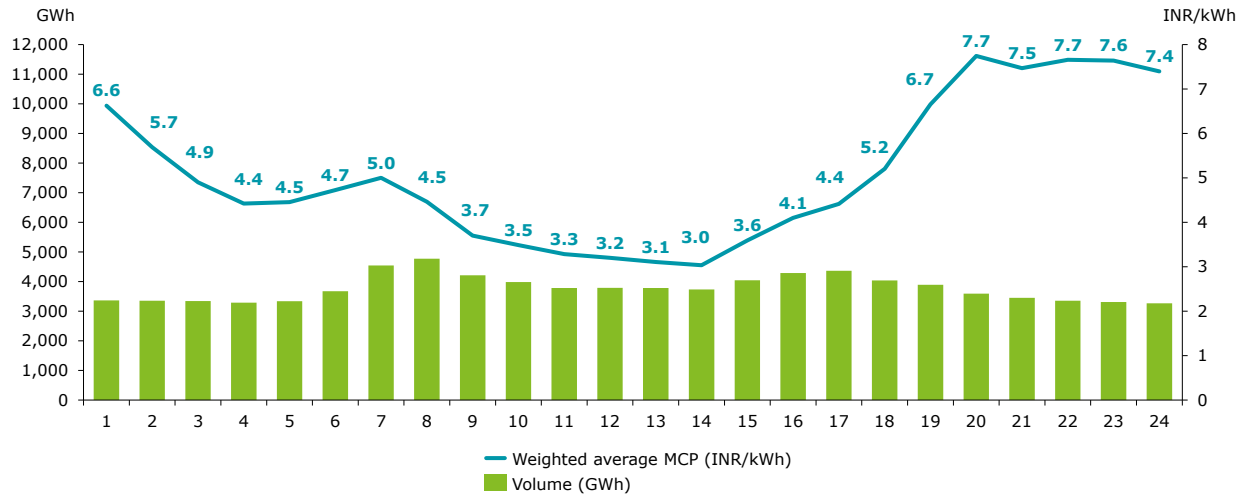


Figure 102: Hourly FSV and MCP in DAM of IEX during last 3 financial years from May to November

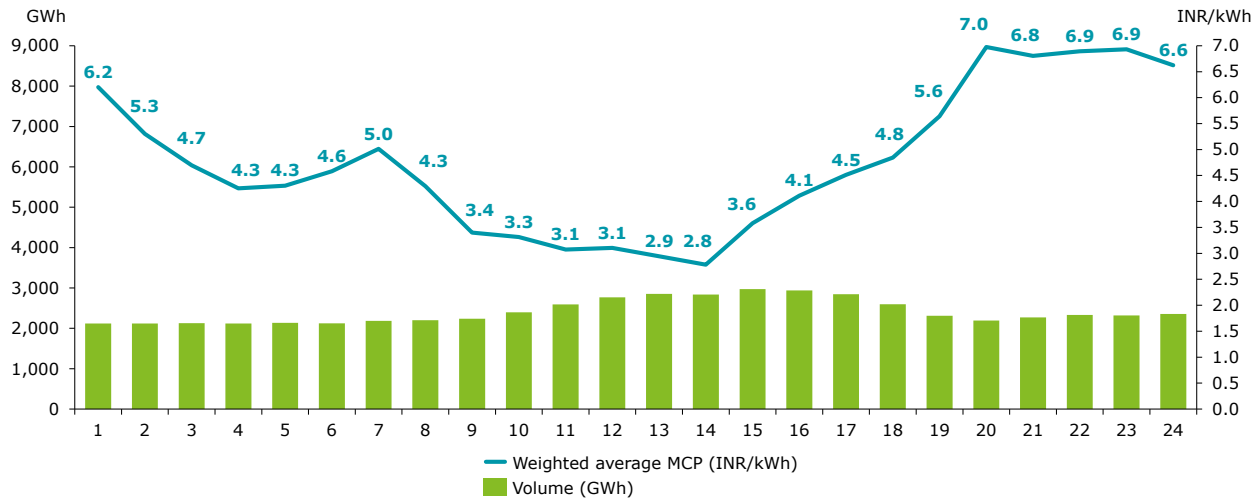


Figure 103: Hourly FSV and MCP in RTM of IEX during last 3 financial years from May to November

The hourly trend shows MCP peaking in the evening after 8 pm for both DAM and RTM segment and lower during the solar hours. This is primarily because during daylight, abundant solar power generation significantly increases electricity supply and due to its low cost, the overall MCP is decreased. As Solar energy decreases, post sunset, the grid relies more on expensive conventional power sources to meet the increased demand resulting from residential sector.

6.2.1.2. Offtake through Bilateral Transactions

The Bilateral transactions happen through the traders or directly between the utilities. To increase transparency, the Government of India has introduced DEEP (Discovery of Efficient Electricity Price) e-Bidding & e-Reverse Auction portal for procurement of short and medium-term power²⁶¹. The portal can be accessed through the MSTC website.

Majority of the utilities prefer the DEEP portal for procurement of short and medium-term power. The below table showcases the power procured for short and medium term from the DEEP portal :

Table 56: Bilateral transactions in short and medium term²⁶²

States	Total Volume Procured (MW)	Short Term*		Medium Term**											
		(0-1) years		(1-2) years		(2-3) years		(3-4) years		(4-5) years		(5-6) years		(6-7) years	
		Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)	Volume %	Price (INR/kWh)
Andhra Pradesh	29,103	100%	9.30	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Himachal Pradesh	27,661	100%	8.16	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
West Bengal	26,263	98%	7.64	1%	5.88	0%	0.00	0%	0.00	1%	3.88	0%	2.24	0%	0.00
Tamil Nadu	9,206	88%	9.02	0%	0.00	0%	0.00	0%	0.00	12%	4.95	0%	0.00	0%	0.00
Kerala	10,806	90%	8.00	0%	0.00	2%	7.48	0%	0.00	7%	6.47	0%	0.00	0%	0.00
Delhi	16,243	99%	7.18	0%	0.00	0%	0.00	0%	0.00	1%	5.99	0%	0.00	0%	0.00
Rajasthan	10,951	99%	7.16	0%	0.00	0%	0.00	0%	0.00	1%	5.58	0%	0.00	0%	0.00
Haryana	12,247	82%	6.90	0%	0.00	1%	5.61	0%	0.00	8%	5.65	8%	6.03	0%	0.00
Maharashtra	11,928	74%	6.00	0%	0.00	7%	5.71	0%	4.95	16%	4.83	0%	0.00	3%	6.95
Gujarat	9,900	90%	6.15	10%	3.68	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Uttar Pradesh	7,245	90%	4.99	1%	4.25	6%	5.13	0%	0.00	3%	5.40	0%	0.00	0%	0.00
Bihar	1,805	100%	9.99	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Chhattisgarh	1,493	100%	9.85	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Odisha	3,349	85%	9.01	6%	5.49	0%	0.00	9%	6.23	0%	0.00	0%	0.00	0%	0.00
Punjab	370	100%	8.41	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Madhya Pradesh	400	100%	6.69	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Assam	4,184	98%	6.63	2%	7.98	0%	0.00	0%	0.00	0%	0.00	0%	0.00	0%	0.00
Uttarakhand	3,682	91%	5.96	1%	6.51	0%	0.00	8%	7.77	0%	0.00	0%	0.00	0%	0.00

The volume and price of bilateral transactions in the short term (upto 1 year) is higher when compared with the medium term.

The Top 10 States with greater requirement of power and higher price in the short term are: Andhra Pradesh, Himachal Pradesh, West Bengal, Tamil Nadu, Kerala, Delhi, Rajasthan, Haryana, Maharashtra, and Gujarat.

Among these 10 States, 80% of them have price higher than INR 7/kWh.

Among the medium-term contracts, 4-5 years is the most preferred duration for procurement among the States. The Top 3 States with greater requirement of power and higher price in the 4-5 years duration are Maharashtra, Tamil Nadu and Kerala.

Since the volume of bilateral transactions in the short term market is higher, the power procurement pattern of the entities in the short term is analyzed. The following table captures the pattern of power procurement by the states in the short term:

²⁶¹ [Guidelines for short term procurement of power by Distribution Licensees through tariff based bidding process](#)

²⁶² The Price mentioned is in INR/kWh; The Short term Data is taken from bids floated in FY 2025 and the Medium term Data is taken from bids floated 2017-2025

Table 57: Power procurement pattern in the Short term bilateral transactions²⁶³

States	Short Term (0-1 years)														
	Short Term Volume (MW)	Price (INR/kWh)	Total No of Bids	Total No of Bids											
				Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Andhra Pradesh	29,103	9.30	44	5	4	3	3	3	3	1	4	4	5	4	5
Himachal Pradesh	27,661	8.16	187	36	36	34							13	33	35
West Bengal	25,823	7.64	85	2	1	1	12	11	11	9	8	8	20	1	1
Tamil Nadu	8,089	9.02	21		2	4	7	8							
Kerala	9,746	8.00	33	3	3	3	5	7	1	2	1	1	2	2	3
Delhi	16,146	7.18	71				5	14	10	13	14	15			
Rajasthan	10,791	7.16	14	1	1			1	2	1	1	2	2	2	1
Haryana	10,094	6.90	15					1	3	3	3	3	2		
Maharashtra	8,831	6.00	97	9	6	7	12	6	10	13	6	6	9	6	7
Gujarat	8,900	6.15	24	1	1	2	2	3	3	1	1	3	3	2	2
Uttar Pradesh	6,485	4.99	98	6	6	6	6	9	9	10	11	11	10	7	7
Bihar	1,805	9.99	18				2	2	2	4	4	4			
Chattisgarh	1,493	9.85	6			2	2	2							
Odisha	2,854	9.01	16				4	6	6						
Punjab	370	8.41	5						1	1	1	1	1		
Madhya Pradesh	400	6.69	1										1		
Assam	4,084	6.63	27				6	5	4	4	4	4			
Uttarakhand	3,350	5.96	21	2	2	2	1	1	1	1	1	1	3	3	3
Sum			783	65	62	64	67	79	66	63	59	63	71	60	64

The power procurement pattern clearly shows that highest number of bids is floated in May, followed by October. This is likely due to the increasing demand for cooling requirements in the Summer (April-June) and increasing demand during the festival season (September – November).

Further during the export months of DHPP (May – November), the top 3 states with maximum number of bids and price greater than INR 6/kWh are West Bengal, Delhi and Maharashtra.

²⁶³ The Price mentioned is in INR/kWh; The Short term Data is taken from bids floated in FY 2025

6.3. Offtake Analysis through C&I

The Commercial and Industrial (C&I) sector in India plays a vital role not only in driving the country's economic growth but also in supporting its broader energy transition. Over the past eight years, ie., from FY 2016 to FY 2024, the sector has maintained a steady growth rate of approximately 5%. It currently accounts for around 40% of the total electricity consumption in India and is also responsible for ~25%²⁶⁴ of the total CO₂ emissions. The details of the same is provided in the figure below:-

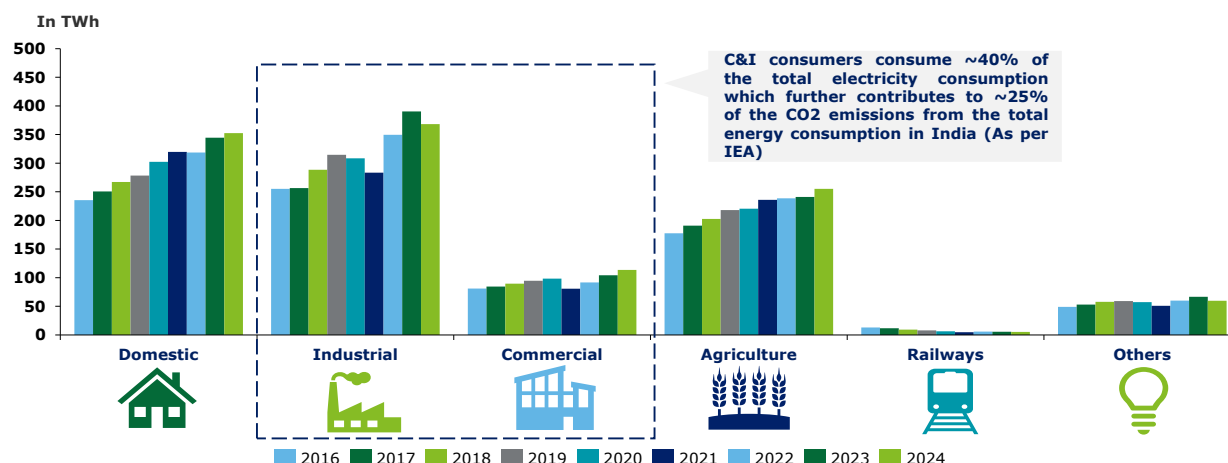


Figure 104: Category wise Electricity Consumption in TWh in India²⁶⁵

At COP26, India announced its ambitious target to achieve net-zero emissions by 2070—a critical milestone in the global fight against climate change, especially given India's position as one of the world's fastest-growing economies. The Commercial and Industrial (C&I) segment is expected to play a pivotal role in this decarbonization journey. Many major C&I players have already begun aligning with net-zero pathways, incorporating a defined share of renewable energy into their overall energy consumption mix. The table below highlights the current share of renewables in their energy portfolios, along with their respective net-zero target timelines.

Table 58: Share of Renewable Energy of major C&I consumers in India

S. N	Company Name	Share of Renewable Energy		Net-Zero Emission target	Source
		Non Renewable Sources	Renewable Sources		
1	Reliance Industries Ltd	99%	1%	2035	BRSR Report 2023-24
2	Vedanta	79%	21%	2050	BRSR Report 2023-24
3	Aditya Birla Group	12%	88%	2050	BRSR Report 2023-24

²⁶⁴ Source: [IEA, 2022](#)

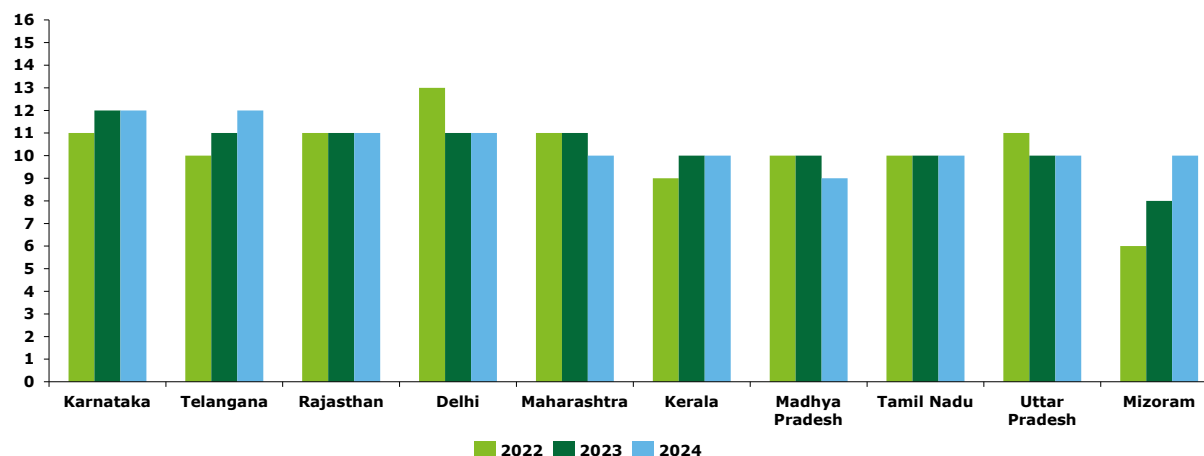
²⁶⁵ Source: [India's Climate and Energy Dashboard](#)

S. N	Company Name	Share of Renewable Energy		Net-Zero Emission target	Source
		Non Renewable Sources	Renewable Sources		
4	ITC limited (in TJ)	48%	52%	2040	BRSR Report 2023-24
5	Tata Motors	67%	33%	2030	BRSR Report 2023-24
6	Sunpharm a	72%	28%	2050	BRSR Report 2023-24
7	Ambuja Cement	94%	6%	2050	BRSR Report 2023-24
8	Adani Enterprise	72%	28%	2050	BRSR Report 2023-24
9	Infosys	67%	33%	2040	Annual Report 2023-24
10	Asian Paints	45%	55%	2070	BRSR Report 2023-24

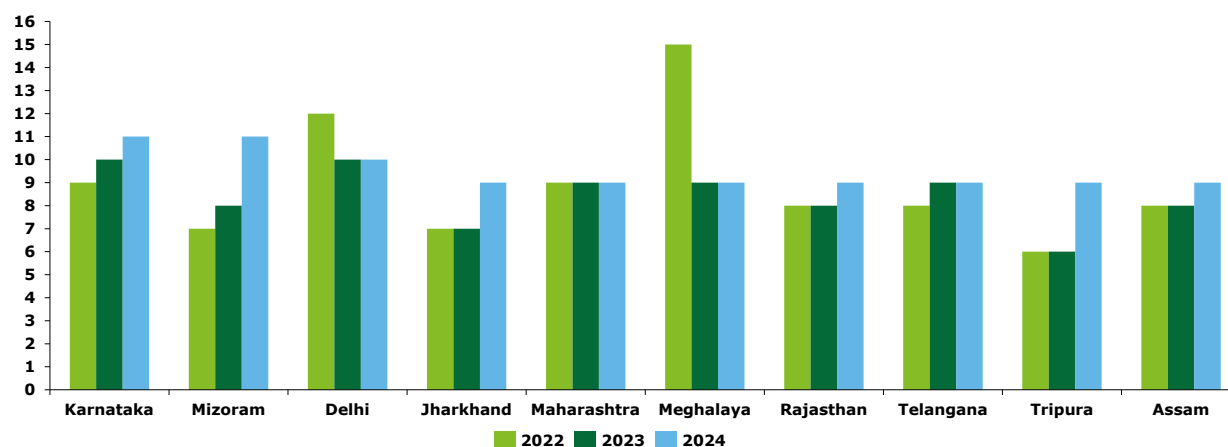
The Commercial and Industrial (C&I) segment is one of the largest open access consumer bases in India, allowing these consumers to procure power directly from generators, bypassing traditional distribution companies (DISCOMs). In many cases, open access consumers in India are subject to Renewable Purchase Obligation (RPO) targets, which require them to source a specified percentage of their electricity from renewable energy. These targets are established by the Central and State Electricity Regulatory Commissions (CERC and SERC).

With India's ambitious net-zero goals, RPO targets are expected to become more stringent, driving increased renewable energy procurement within the C&I segment. Procuring renewable energy is particularly advantageous for this segment, as DISCOM tariffs for C&I consumers are typically higher compared to other consumer categories.

We have identified the top ten states where average billing rates for C&I consumers are highest which provides opportunity for C&I consumers located in these states to look for alternative source of supply.

Figure 105: ABR (in INR/kWh) of top 10 States for Commercial Consumers²⁶⁶

As can be seen that for the top ten states based on average billing rate of FY 2023-24, the tariffs charged by the state DISCOMs is in the range of INR 10-12 in FY 2023-24.

Figure 106: ABR (in INR/kWh) of top 10 States for Industrial Consumers²⁶⁷

For Industrial consumers, the average billing rate (ABR) charged by the state DISCOMs is in the range of INR 9-11 per unit in FY 203-24.

It is therefore crucial for major corporate houses to procure cheaper sources of energy, such as renewables including hydropower, to reduce their electricity costs. This not only positively impacts profitability but also enhances process efficiency through the use of cleaner fuels. However, power procured via open access is subject to additional charges, including cross-subsidy surcharge (CSS), additional surcharge, SLDC charges, distribution and wheeling charges, reactive energy charges, and Green Cess.

In the context of DHPP, it is important to engage with major corporate houses to better understand their specific requirements for hydropower integration.

²⁶⁶ [India Climate and Energy Dashboard](#)

²⁶⁷ [India Climate and Energy Dashboard](#)

6.4. Benchmarking of Offtake Strategy for Hydropower Projects

The offtake arrangements of major hydro projects across India have been analyzed to evaluate how offtake structuring varies based on project ownership—ranging from fully government-owned to fully private entities. Based on this structuring, the associated risk of participating in the short-term power market has been assessed. This analysis can help inform DHPP's approach to balancing long-term PPAs with short-term market participation.

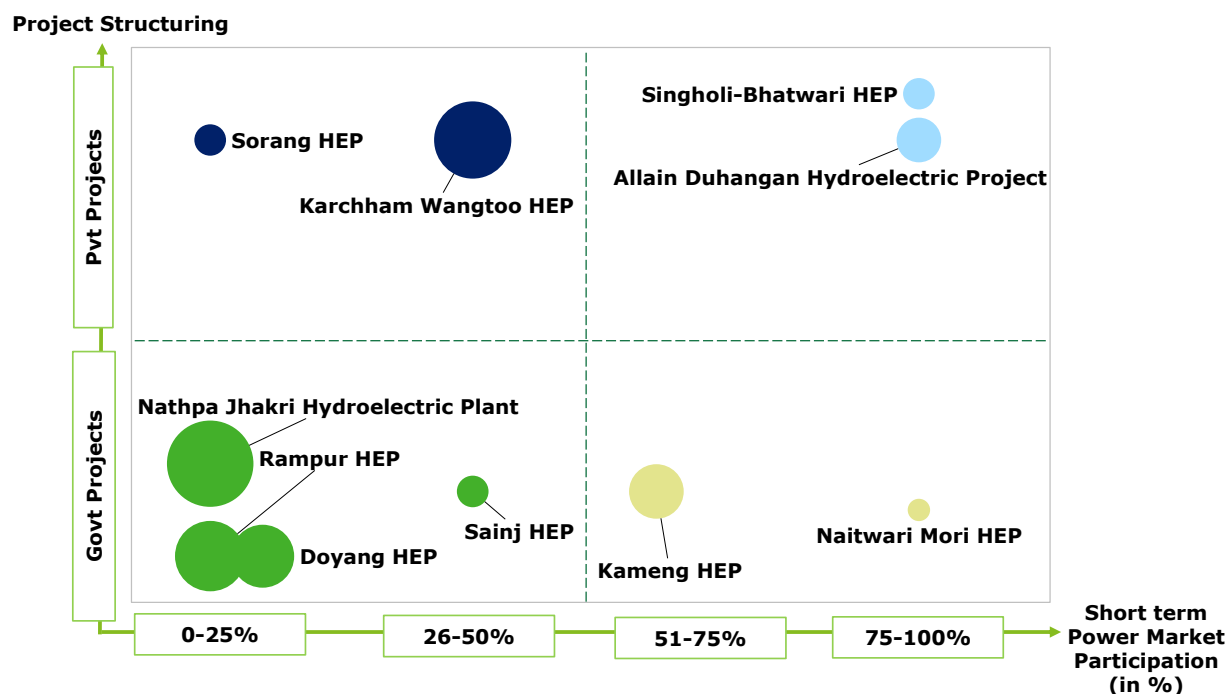


Figure 107: List of Offtake Arrangement of Hydro Projects

The Study of various offtake models across different project structures reveals that most hydro projects are fully committed (100%) under long-term Power Purchase Agreements (PPAs) with State DISCOMs or through central government allocations. Among the remaining government-owned projects, short-term market participation is generally limited to up to 30%. In contrast, private hydro projects which is driven by a higher risk appetite exhibits significantly greater exposure, with some participating up to 100% in the merchant power market. These trends are illustrated in the graph below

Table 59: The detailed list of key Hydro project is provided in the table below

S.N	Project	Capacity (MW)	Location	Ownership	Mode of Sale	Offtake
1	Naitwar Mori HEP	60	Uttarakhand	Government – SJVN Ltd.	Merchant (short-term sale)	Sold via exchanges (no long PPA)
2	Singholi-Bhatwari HEP	99	Uttarakhand	Private – ReNew Power Ltd.	Merchant (short-term sale)	No long PPA; entire generation sold on exchange.
3	Sainj HEP ²⁶⁸	100	Himachal Pradesh	Government- HPPCL	Long term and Merchant	50% Sold via exchanges (Remaining Long term PPA)

²⁶⁸ As per petition for determination of tariff from COD to FY 2023-24

S.N	Project	Capacity (MW)	Location	Ownership	Mode of Sale	Offtake
					(short-term sale)	
4	Kameng HEP	600	Arunachal Pradesh	Government – NEEPCO Ltd.	Merchant (short-term sale)	345 MW untied (out of 600 MW) sold on exchange
5	Doyang HEP	75	Nagaland	Government – NTPC NEEPCO Ltd.	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
6	Nathpa Jhakri Hydroelectric Plant	1500	Himachal Pradesh	Government-SJVNL	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
7	Karchham Wangtoo HEP ²⁶⁹	1000	Himachal Pradesh	Private-JSW Energy	Long term and Merchant (short-term sale)	70.4% with PTC under Long term PPA, remaining-Short term
8	SORANG HEP	100	Himachal Pradesh	Private-HSPL (subsidiary of Greenko East Coast Power Projects Pvt Ltd.)	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
9	RAMPUR HEP	412	Himachal Pradesh	Government-SJVNL	Long term and Merchant (short-term sale)	Tied to Long term PPAs, Surplus power sold via Short-term markets.
10	Allain Duhangan Hydroelectric Project ²⁷⁰	192	Himachal Pradesh	Private-Aditya Birla Group	Merchant (short-term sale)	Short term PPAs (1-3 years) and Power Exchange

Based on the analysis of offtake structures for hydro projects in India, it is observed that most of these projects, whether privately or government-owned, are tied up in long-term PPAs or allocated through central government mechanisms. However, participation in the short-term power market varies depending on the risk appetite. Government-owned projects typically participate in the short-term market up to approximately 30% in addition to their long-term commitments, whereas private projects, driven by higher risk tolerance, can have up to 100% exposure in the merchant power market.

²⁶⁹ As per True Up Order 2014-19 of KW plant

²⁷⁰ As per IFC disclosure

7. Bangladesh Power Sector Landscape

Bangladesh is one of Asia's fastest growing economies and envisages to be in the 'high-income' country by 2041²⁷¹. The electricity sector in Bangladesh has made significant progress over the last decade in terms of generation capacity addition, transmission and distribution network strengthening and enhancing access to electricity as outcomes of various reform initiatives since the 1990s.

A snapshot of Bangladesh's power sector as of June 2024 is set out as follows:

Figure 108 : Power Sector Snapshot of Bangladesh²⁷²

Generation	• 28,098 MW installed capacity • Gas: 44%, Furnace Oil: 23%, Coal: 18%, Import: 9%, Solar: 2%, Diesel: 2%, Hydro: 0.8% • Public Share : 49% • 95,996 GWh Energy Generation
Transmission	• 15,624 Ckt. Km transmission lines • 66,869 MVA Grid capacity • ~3% Transmission Loss
Distribution	• Key Distribution Entities: DPDC, DESCO, WZPDCL, REB, NESCO, BPDB • 56% Domestic Consumption, 27% Industrial consumption • ~8% Distribution Loss
Consumers	• 640 kWh per capita generation (including captive and off-grid renewable energy) • ~4 million consumers • 100% access to electricity

Bangladesh's electricity sector is navigating a period of rapid transformation, characterized by increasing demand. The electricity peak demand has grown at a CAGR of ~7% between FY15 and FY24, backed by high economic development and the Government of Bangladesh's ambitious Power for All program. Although, the demand shortfall has significantly declined over the years since FY15 but from FY20 onwards, the shortfall has been following an increasing trend.

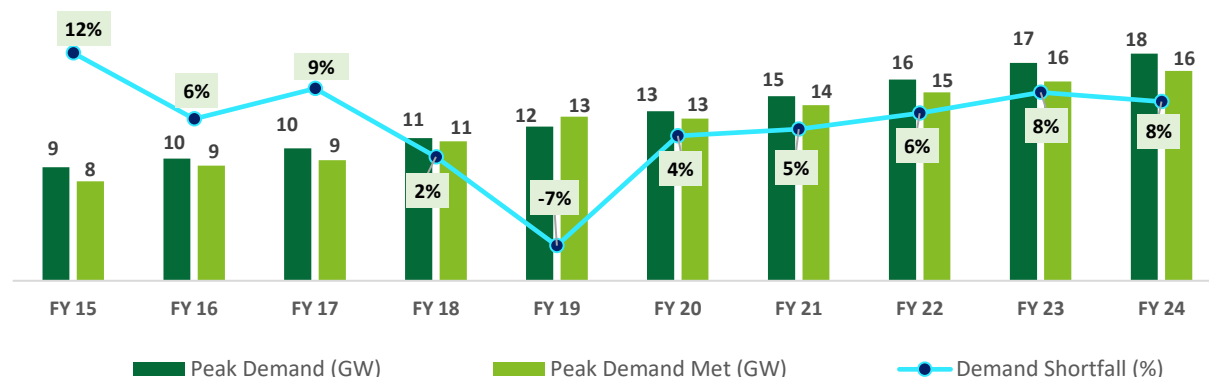


Figure 109 : Electricity Peak Demand and Peak Supply (GW)²⁷²

²⁷¹ [Making Vision 2041 a Reality PERSPECTIVE PLAN OF BANGLADESH 2021-2041](#)

²⁷² [Annual Report 2023-2024, Bangladesh Power Development Board \(BPDB\)](#)

To meet the burgeoning electricity demand, Bangladesh has also been increasing its power supply. The country has historically been reliant on gas which contributes over ~50% of its total generation capacity. Considering the depleting natural gas reserves and to ensure energy security for the country the Government of Bangladesh has also been increasing power imports from neighbouring countries over the years. Bangladesh has also made a commitment to decrease GHG emissions by almost 22% below the “business as usual” level by 2030 (89.47 MT CO_{2e}). Therefore, over the years, the country has also scaled up the domestic renewable energy capacity.

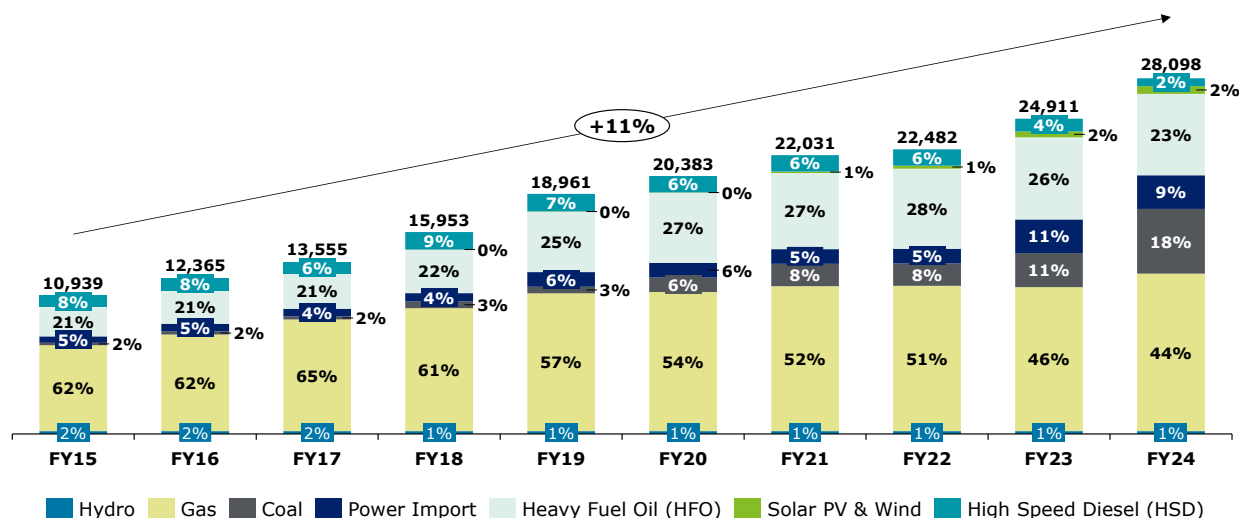


Figure 110 : Installed Capacity trend of Bangladesh by fuel type (MW)

In FY24, Bangladesh imported 2,656 MW (~9% of total installed capacity) of power. All the imported power was from India. The below table shows further details on the imported power :

Table 60: Import of power by Bangladesh in FY2024²⁷³

Particulars	Capacity (MW)	Units (GWh)	Price (INR/kWh)
NVVN Ltd. - India	250	1880	2.93
NVVN Ltd. - India (Tripura)	160	972	6.42
NVVN Bharamara Ph. II - India	300	2278	5.66
PTC India Ltd.	200	1672	6.50
Sembcrop Energy India Ltd.	250	1845	7.30
Adani Power (Jharkhand) Limited, India	1,496	8167	10.41
Total	2,656	16,814	

Further, a landmark tripartite agreement was signed in October 2024 between Nepal, India, and Bangladesh, enabling Nepal to export 40 MW of electricity to Bangladesh via India’s transmission grid. The agreement is set for a five-year term, with the power to be traded at approximately 8.17 Bangladeshi Taka (about INR 9.30 or \$0.11) per unit²⁷⁴. The first transaction occurred on November 15, 2024, with Nepal exporting 40 MW to Bangladesh via India’s grid²⁷⁵. This successful transaction, which took place in November 2024, sets a clear precedent for similar trilateral transactions involving other countries in the region.

²⁷³ Annual Report 2023-2024, Bangladesh Power Development Board (BPDB)

²⁷⁴ South Asia Subregional Economic Cooperation (SASEC)

²⁷⁵ Ministry of External Affairs, Government of India

The Bangladesh Power Development Board (BPDB) serves as the single buyer in the power market of Bangladesh. Besides its own generation, BPDB also procures electricity from private companies, commonly referred to as Independent Power Producers (IPPs), rental power plants and public power plants to meet the rising demand and sells bulk electricity to all the distribution utilities. Most of the IPPs (>70%) are Gas and HFO based with just ~5% of renewable energy capacity. All the Rental plants are HFO (65%) and gas based (35%). Majority (~70%) of the public sector plants (BPDB and others) are gas based with just ~1% of renewable energy capacity.

The per unit generation cost of BPDB's own plants and electricity purchases from external sources for the last 3 financial years are shown below²⁷⁶:

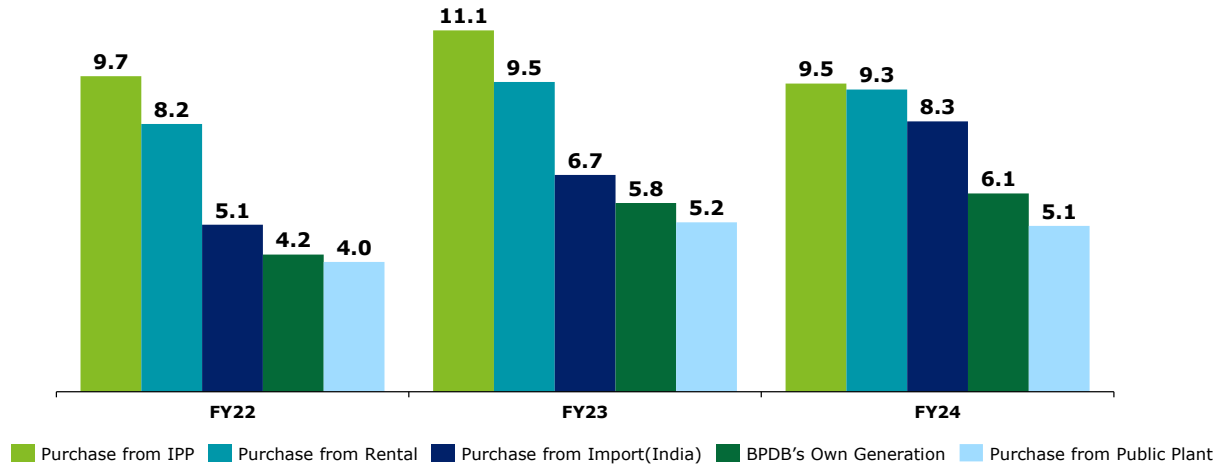


Figure 111: Per Unit cost of electricity generation and purchase by plant type in last 3 years (INR/kWh)

The fuel wise per unit generation cost for BPDB's own generation is given below by comparing with the import prices:

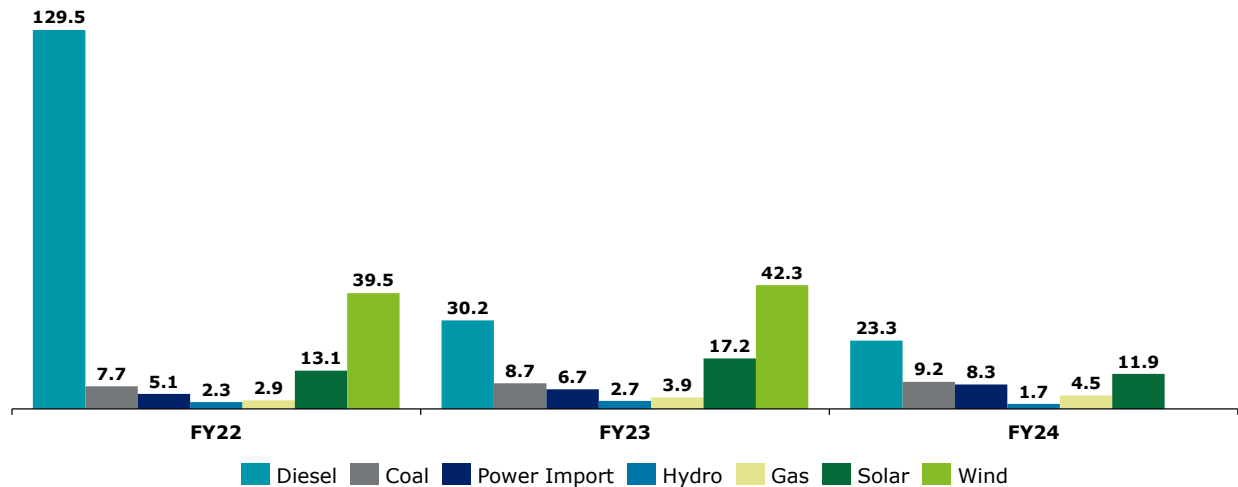


Figure 112 : Per Unit cost of fuel for BPDB's electricity generation in last 3 years (INR/kWh)

The cost of renewable energy generation (solar and wind) by BPDB is very high. Moreover, there is almost negligible hydro capacity in Bangladesh (1% of total installed capacity). Therefore, to increase

²⁷⁶ BPDB Annual reports. Average Exchange rate from BDT to INR for each year is taken from <https://www.exchangerates.org.uk/BDT-INR-spot-exchange-rates-history-2024.html>

renewable capacity, in addition to scaling up domestic renewables, Bangladesh needs to leverage regional cooperation to import clean electricity.

Further, as per Integrated Energy and Power Master Plan (IEPMP) 2023, overall electricity demand and supply projections under In-Between scenario (between BAU and Optimistic scenario) from 2030 to 2050 have been mapped (considering 1 Mtoe = 11.63 TWh for demand).

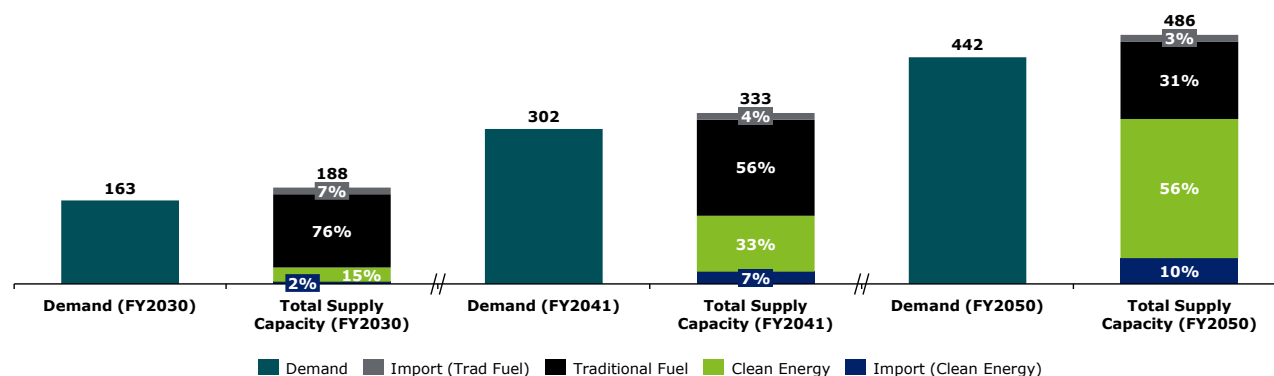


Figure 113 : Electricity Demand and Supply forecast (in TWh)

Bangladesh is significantly expanding its clean energy capacity, targeting 40% of total supply by 2041 and 66% by 2050. With imported sources expected to contribute roughly 20% of this clean energy, Bhutanese hydropower can be a key partner in achieving these ambitious governmental goals.

8. Regulatory Landscape in India and Bangladesh

8.1. Cross Border Electricity Trade Regulations

8.1.1. Background

Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. In 2021, the Central Electricity Authority (CEA) issued the "Procedure for Approval and Facilitating Import/Export (Cross Border) of Electricity by the Designated Authority," enabling power trading through the Indian Energy Exchange (IEX). By October 2021, Nepal began participating in the day-ahead market (DAM) on the IEX, initially trading around 39 MW (24 MW from the Trishuli Hydropower Plant and 15 MW from the Devighat Powerhouse). Subsequently Bhutan also began trading electricity in the Indian Energy Exchange (IEX) on January 1, 2022 onwards.

Over the past three years, cross border trade volume in the DAM and RTM segments has increased. The Nepal Electricity Authority (NEA) has since secured medium-term power sales agreements with distribution companies in Haryana and Bihar. Additionally, India now recognizes hydropower imports from other countries as part of the Hydropower Purchase Obligation (HPO), encouraging Indian buyers to source power from Bhutan and Nepal.

In a landmark development, a trilateral agreement was also signed between Nepal, India, and Bangladesh for export of 40 MW of electricity from Nepal to Bangladesh via India's grid. The Cabinet Committee on Public Purchase (CCPP) in Bangladesh has also approved importing approximately 500 MW from Nepal's proposed 900 MW Upper Karnali Hydroelectric Project, developed by GMR, which will flow to Bangladesh via NTPC Vidyut Vyapar Nigam Limited through Indian transmission corridors. These developments further strengthen the creation of an integrated South Asian power market

Some of the key framework agreement for Cross Border Electricity Trade in the region have been enumerated below:

BIMSTEC MoU for Establishment of BIMSTEC Grid Interconnection

The Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC) came into being on 6 June 1997 through the Bangkok Declaration.²⁷⁷ The BIMSTEC Energy Centre (BEC) was established on 22 January 2011.

A Memorandum of Understanding (MoU) for Establishment of the BIMSTEC Grid Interconnection was signed on 31 August 2018 at the Fourth BIMSTEC Summit held in Kathmandu, Nepal. The MoU came in to force on 07 April 2019. The MoU broadly provides, the framework for parties to cooperate towards the implementation of the grid interconnections for the trade in electricity with a view to promoting rational and optimal power transmission in the BIMSTEC Region.

Table 61: BIMSTEC MoU Agreements

Article	Details
Article3 (Institutional Arrangements)	For successful implementation of grid interconnections and trade in electricity, the BGICC may engage BIMSTEC Energy Sector Committee of Experts/Officials, Task Force for BIMSTEC Trans-Power Exchange and Development Projects, BIMSTEC Energy Center and other institutions to provide technical support. Furthermore, the BIMSTEC Grid Interconnection Coordination Committee (BGICC) shall determine modalities to implement BIMSTEC Grid Interconnections Master Plan and regional trade arrangements.

²⁷⁷ BIMSTEC member states are Bangladesh, Bhutan, India, Myanmar, Nepal, Sri Lanka, and Thailand.

Article	Details
Article 4 (Settlement of Disputes)	Member parties implementing the MoU will seek amicable resolution through negotiations/consultations, which would be further referred to BGICC. If BGICC is unable to resolve the differences, that issue shall be referred to BIMSTEC Senior Officials' Meeting on Energy Resolutions.
Article 8 (Relation to Other Agreements)	Member States are not restricted to enter into agreements with a non-BIMSTEC countries on bilateral/multilateral and would not impair a BIMSTEC Member State who is Party to such agreements from fulfilling its obligations under the MoU.

8.1.2. Salient Features of Indian Cross Border Electricity Trade Regulations

Cross-border electricity trade (CBET) in India is primarily managed by the Ministry of Power in coordination with key power sector entities, including the Central Electricity Regulatory Commission (CERC) and the Central Electricity Authority (CEA). The Ministry of Power oversees policy-level interventions, while CERC and CEA handle regulatory and procedural aspects as the central regulating and planning bodies, respectively. The major policies and guidelines for CBET in India include the following:

- Cross Border Trade of Electricity Regulations, 2019
- Guidelines for Import/Export (Cross Border) of Electricity – 2018 and its amendment in 2023
- Procedure for Approval and Facilitating Import/Export (Cross Border) of Electricity by the Designated Authority, 2021 and 2023

Additionally, power exchange operations and transactions are regulated by CERC through specific regulations, with the National Load Dispatch Centre (NLDC) ensuring grid stability during power flows on the exchange.

Under the "Procedure for approval and facilitating Import/Export (Cross Border) of Electricity" by the Designated Authority, 2021", Indian entities are not able to trade with any generating company that is, "owned, directly or indirectly by any natural/ legal personality(ies) whose effective control or source of funds or residence of beneficial owner, is situated in/ citizen of a third country with whom India shares land border and that third country does not have a bilateral agreement on power sector cooperation with India." For any relaxation in this provision, the Designated Authority must consult with the Indian Ministry of Power and Ministry of External Affairs.

The Government of India has introduced facilitating measures as well, including an amendment by the Central Electricity Authority (CEA) on 31st July 2023 to the procedure for cross-border electricity import/export approval, wherein Indian power traders can trade on Indian power exchanges (Day Ahead Market, Real-Time Market, or both) on behalf of entities from neighbouring countries, after obtaining approval from the designated authority. This is permitted up to a specified quantum and duration, provided the neighbouring country has a power sector cooperation agreement with India, and the generating asset involved is owned or controlled by that country. Both Bhutan and Nepal are trading in Indian Power Exchanges through this route.

However, on 12th August 2024, the Ministry of Power announced revisions to the "Guidelines for Import/Export (Cross Border) of Electricity - 2018" to regulate and streamline electricity exports from coal and gas-based plants in India. Under the updated guidelines, coal-based plants can only export electricity if the coal is sourced through imports, commercial mining, or spot e-auctions. Similarly, gas-based plants must use gas that is imported or sourced from approved suppliers. The guidelines also include provisions for Indian generating stations that supply electricity exclusively to neighbouring countries. These stations may build dedicated transmission lines to connect with the neighbouring country's transmission system, considering technical and strategic factors, with approval from the designated authority as per the Electricity Act, 2003. The costs for such construction will be outlined in the contract agreement between the Indian and neighbouring entities. Additionally, the guidelines allow these stations to connect to the Indian grid (interstate or intra-state) to sell power domestically if there

is a sustained non-scheduling of full or partial capacity, or if a default notice is issued by the generator for reasons such as delayed payments under the power purchase agreement (PPA).

Settlement Nodal Agency (SNA)

NTPC Vidyut Vyapar Nigam Ltd (NVVN) has been designated as the SNA²⁷⁸ for Bangladesh, Bhutan, Nepal and Myanmar, under the Guidelines for Import /Export (Cross Border) of Electricity 2018, from India's side. It is therefore responsible for settlement of grid operation related charges like operating charges, charges for deviation. NVVN's key functions include coordination with Load Dispatch Centers on behalf of Cross Border Entity, provide information on Open Access, undertake weekly meter reading among others.

Other Cost Components for Cross Border Power Trading

The other billing and payment related activities include raising bill to Cross Border Entity for deviation in scheduling, regional transmission deviation charges, reactive energy charge & congestion charge, system operation fees & charges, Settlement Nodal Agency charges / fees, other statutory and necessary charges and late payment surcharge (if applicable).

8.1.3. Bangladesh Rules, Regulation and Guidelines

Bangladesh's policies and regulatory framework to promote CBET are distributed across several instruments including:

- Bangladesh Electricity Act 2018
- Bangladesh Electricity Regulatory Commission Electricity Grid Code Regulations, 2018
- Power Purchase Agreements
- Joint Steering Committee / Working Groups

Some of the salient coverage are summarized below:

- 1 Bangladesh Electricity Act, 2018**
 - Provides for licensing regulations for electricity generation, transmission, distribution and supply
- 2 Bangladesh Electricity Regulatory Commission Electricity Grid Code Regulations 2018**
 - Provides for grid operation and connectivity standards, scheduling and dispatch procedures, standards for protection, metering and performance, and reporting requirements
- 3 Power Purchase Agreement**
 - Bangladesh currently only imports power from India. For Power Import, an electricity generator / trader needs to only sign a Power Purchase Agreement with BPDB for facilitating power trade between the countries
- 4 Joint Steering Committee / Working Groups**
 - Both India and Bangladesh have set JSC and JWG for cooperation in power sector which assists in development of transmission lines, plan for power trade among others

Figure 114: Key Instruments guiding Bangladesh Cross Border Power Trade

Some of the other regulations which may impact cross border trade are:

²⁷⁸ Source: <http://nvvn.co.in/wp-content/uploads/2020/11/Nomination-of-NVVN-as-SNA.pdf>

- Provisions related to CBET in the Quick Enhancement of Electricity and Energy Supply (Special Provisions) Act, 2010;
- Setting of fair rules and procedures for non-discriminatory open access in: Policy Guidelines for Enhancement of Private Participation, Electricity Act 1910, and BERC Power Transmission Tariff Regulations, 2016;
- Transmission pricing mechanisms in the Policy Guidelines for enhancement of Private Participation in the Power Sector; and
- Dispute resolution processes in BERC dispute settlement regulation 2014.

However, there is a lack of detailed and explicit regulations and institutions dedicated towards governing bilateral and multilateral CBET, arrangements for transmission planning and pricing, clarity of roles and responsibilities of institutions for CBET.

8.2. Tariff Regulations - India

The determination of tariff for hydropower generation plants in India is governed by a well-established regulatory framework administered by the Central Electricity Regulatory Commission (CERC) and the respective State Electricity Regulatory Commissions (SERCs). The objective of this framework is to ensure a fair, transparent, and balanced pricing mechanism that protects consumer interests while ensuring financial viability and reasonable returns for generating companies.

The jurisdiction for tariff determination is clearly defined. The CERC is responsible for determining tariffs for inter-state generating stations (ISGS), which supply electricity to more than one state. On the other hand, SERCs are responsible for determining tariffs for intra-state generating stations that operate within a single state, as well as for state-owned generating companies and distribution licensees within their respective states. While SERCs have the autonomy to frame their own regulations, they typically align their methodologies and principles with those established by the CERC, ensuring consistency and uniformity across the national and state regulatory frameworks.

As per the CERC (Terms and Conditions of Tariff) Regulations, 2024, the approach adopted by both CERC and generally followed by SERCs for tariff determination is based on a cost-plus methodology, regulated under a Multi-Year Tariff (MYT) framework. This ensures recovery of prudent costs and allows a fair return on investment for generating companies, including hydropower plants. As per CERC Tariff Regulations (2024-29)²⁷⁹, following are the Annual Revenue Requirement (ARR) components as determined for Hydro projects across India:

Table 62: Normative Parameters as per CERC Tariff Regulation, 2024-29

Component	Description
Return on Equity (RoE)	15.50% for run-of-river hydro generating station and 17% for storage type hydro generating stations (pre-tax) allowed return, subject to normative parameters (30% normative equity)
Interest on Loan Capital	On normative or actual debt as approved by the regulator (70% normative debt). If actual loan details are unavailable, a weighted average interest rate (capped at 14%) is used; if no loans exist, State Bank of India's 1-year MCLR as of April 1 of the relevant year is applied
Depreciation	Based on straight-line method at rates in Appendix-II; After 15 years, the remaining value is spread over the balance useful life; for new hydro stations, a lower rate may be used with beneficiary consent.
O&M Expenses	For hydro stations commissioned on or after 1.4.2024, O&M expenses for the first year are set at 3.5% (capacity >200 MW) or 5.0% ($\leq$ 200 MW) of project cost (excluding R&R, IDC, IEDC), with 5.47% annual escalation thereafter
Interest on Working Capital	Rate of interest on working capital shall be on a normative basis, based on the Reference Rate as of April 1 of each year during 2024–29. working capital includes 45 days of fixed cost, 15% of O&M (including security) as spares, and one month of O&M expenses
Additional Capitalization	Post-COD capex allowed for justified reasons
Debt-Equity Ratio	The normative debt-equity ratio is 70:30.

8.2.1. Details of Key Hydro projects in India-

The following table represents some hydropower plants for which tariff was determined by CERC for the control period 2014-19 and 2019-24. Although the CERC has issued the latest regulation, Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2024, the prevailing control period is 2024–29; however, only a limited number of large-scale hydro projects (with capacities

²⁷⁹ [Central Electricity Regulatory Commission \(Terms and Conditions of Tariff\) Regulations, 2024](#), issued by Central Electricity regulatory Commission, Dated 15th March, 2024

greater than 400 MW) have been commissioned over the past decade. Several of these projects were commissioned under the 2014 and 2019 tariff regulations, resulting in tariff determination across multiple control periods—namely, 2014–19, 2019–24, and continuing into 2024–29. The table below presents the tariffs as determined by the Central Electricity Regulatory Commission (CERC) under the respective control periods.-

Table 63: Key Parameters of similar Hydro Projects in India

Particulars	Kameng	Teesta III	Koldam	Parbati III	Rampur
Project Description					
Location	Arunachal Pradesh	Sikkim	Himachal Pradesh	Himachal Pradesh	Himachal Pradesh
Owner	NEEPCO	Sikkim Urja Limited	NTPC	NHPC	SJVNL
Type	ROR with Pondage	ROR with Pondage	Storage	ROR with Pondage	ROR with Pondage
COD	2021	2017	2015	2014	2014
Project Capacity (in MW)	600	1200	800	520	412
Tariff rate as per Tariff order 2014-19					
Claimed by Petitioner Tariff for FY 2018-19 (in INR/kWh)	-	5.09 ²⁸⁰	4.68 ²⁸¹	2.71 ²⁸²	4.15 ²⁸³
Approved by CERC (2018-19), INR/kWh	-	5.06	4.29	2.65	3.71
Tariff rate as per True-up order 2014-19					
Claimed (2018-19), INR/ kWh	-	5.11 ²⁸⁴	4.47 ²⁸⁵	2.60 ²⁸⁶	4.00 ²⁸⁷
Approved (2018-19), INR/ kWh	-	4.85	4.41	2.57	3.79
Tariff rate as per Tariff order 2019-24					
Claimed (2023-24), INR/ kWh	4 ²⁸⁸	4.75 ²⁸⁹	3.88 ²⁹⁰	2.35 ²⁹¹	4.02 ²⁹²
Approved (2023-24), INR/ kWh	4	4.36	3.83	2.34	3.59
Core Financial Parameters as per Tariff order 2019-24					
Date of CERC Tariff Order	8 th Aug 2023	15 th July 2024	14 th Jan 2024	31 st March 2024	24 th Jan 2022
Capital Cost Disallowed, in Crore (2019-24)		2,324	-	79	492
Return on Equity(Base Rate),%	-	16.5	16.5	16.5	16.5

²⁸⁰ Tariff Order 2014-19, CERC, 9th Jan, 2020

²⁸¹ Tariff Order 2014-19, CERC, 5th April, 2018

²⁸² Tariff Order 2014-19, CERC, 5th April, 2019

²⁸³ Tariff Order 2014-19, CERC, 15th Feb, 2017

²⁸⁴ True-up Order 2014-19, CERC, 15th July, 2024

²⁸⁵ True-up Order 2014-19, CERC, 5th Oct, 2023

²⁸⁶ True-up Order 2014-19, CERC, 23rd Apr, 2019

²⁸⁷ True-up Order 2014-19, CERC, 4th June, 2021

²⁸⁸ Tariff Order 2019-24, CERC, 8th Aug, 2023

²⁸⁹ Tariff Order 2019-24, CERC, 15th July, 2024

²⁹⁰ Tariff Order 2019-24, CERC, 14th Jan, 2024

²⁹¹ Tariff Order 2019-24, CERC, 31st Mar, 2024

²⁹² Tariff Order 2019-24, CERC, 24th Jan, 2022

Particulars	Kameng	Teesta III	Koldam	Parbati III	Rampur
Debt- Equity Ratio	-	80:20	70:30	70:30	70:30

*The tariff rate for the Kameng project has been fixed at a provisional rate of Rs. 4 per kWh as per Tariff Order No. 51/GT/2021 dated 8th August 2023.

The insights reflected in the table indicates a consistent trend of tariff rationalization across various control periods. In most instances, the approved tariff rates were lower than the corresponding claimed rates, both at the initial tariff determination stage and during the subsequent true-up process. The assessment indicates significant changes in the approved tariff rates between the tariff orders and subsequent true-up orders, highlighting the regulator's intent to align tariffs with verified costs and audited financial statements while ensuring controllable costs are kept within normative parameters. These consistent reductions in approved tariffs during true-up stages underline the role of regulatory oversight in ensuring that only justified and prudent expenditures are allowed to be recovered through tariffs, ultimately benefiting consumers through lower energy costs.

An in-depth analysis of the regulatory and financial frameworks governing tariff determinations for the FY 2014–19 and FY 2019–24 control periods is undertaken to underscore the challenges encountered and the key observations arising from tariff petitions and the regulatory approval process. The following case studies provide a detailed examination of these elements and their impact:

Case Study 1: Teesta Stage III Hydroelectric Project (1200 MW)

Project COD	28/02/2017
Design Energy (GWh)	Approved by Commission in Tariff Order for FY 2019-24: 5,214 GWh Claimed by Petitioner in Tariff Order for FY 2019-24: 5,214 GWh
Average Tariff (INR/kWh) (2023-24)	Approved by Commission: INR 4.36/kWh Claimed by Petitioner: INR 4.75/kWh
Major Disallowances	- Capital Cost Disallowance: A capital cost of Rs. INR 12,967 Crore was claimed in the tariff petition. However, the Central Electricity Regulatory Commission (CERC), in its final true-up order, approved INR 12,749 Crore, leading to a disallowance of INR 218 Crore. - Additional Capitalization: An additional capital expenditure of INR 159 Crore was proposed for approval. However, the Commission approved only INR 94 Crore, resulting in a disallowance of approximately 41% of the proposed amount.
Reason for Disallowance	- Procurement of spares beyond the cut-off date, capital spares for hydro generating stations shall be allowed separately as additional O&M expenses. - It has been observed that the submitted documentation does not include specific details regarding the plant's availability or an assessment of the improvement in Plant Availability Factor (PAF) following the capitalization of the claimed asset/work. - It has been observed that no information has been provided regarding claim.
Impact of Disallowances	A significant portion of the Annual Fixed Charges (AFC) initially claimed was not approved by the Commission. While a total AFC of INR 12,623 Crore was proposed, the Commission approved only INR 11,826 Crore, resulting in a disallowance of INR. 796 Crore, or approximately 6% of the total claimed amount. This disallowance affected key financial components including Interest on Loan, Depreciation, Return on Equity, Interest on Working Capital, and Operation & Maintenance (O&M) expenses—all of which were

proportionately reduced. As a result, the recoverable amount through tariffs was also correspondingly lowered.

Case Study 2: Rampur Hydropower Station (412 MW)

Project COD	13/05/2014
Design Energy (GWh)	Approved by Commission in Tariff Order for FY 2019-24: 1878 GWh Claimed by Petitioner in True Up Order for FY 2019-24: 1878 GWh
Average Tariff (INR/kWh) (2023-24)	Approved by Commission: INR 3.59/kWh Claimed by Petitioner: INR 4.02/kWh
Major Disallowances	- Capital Cost Disallowance: A capital cost of INR 4,149 Crore was claimed in the tariff petition. However, the Central Electricity Regulatory Commission (CERC), in its final true-up order, approved only Rs. INR 4,083 Crore, resulting in a disallowance of INR 65 Crore. - Additional Capitalization: An additional capital expenditure of INR 52 Crore was proposed for approval. However, the Commission approved only INR 15 Crore, resulting in a disallowance of approximately 70% of the proposed amount.
Reason for Disallowance	- Procurement of spares beyond the cut-off date, capital spares for hydro generating stations shall be allowed separately as additional O&M expenses. - No justification has been provided to demonstrate that the claim was made in response to any advice, direction, or mandate issued by the Appropriate Government Instrumentality or a statutory authority. - No justification has been submitted in support of the claim. - It has been observed that no information has been provided regarding claim.
Impact of Disallowances	A significant portion of the Annual Fixed Charges (AFC) initially claimed was not approved by the Commission. While a total AFC of INR 3,672 Crore were proposed, the Commission approved only INR 3,366 Crore, resulting in a disallowance of INR. 305 Crore, or approximately 8% of the claimed amount. This disallowance affected key financial components such as Interest on Loan, Depreciation, Return on Equity, Interest on Working Capital, and Operation & Maintenance (O&M) expenses—which were all proportionately reduced. As a result, the overall recoverable amount through tariffs was also correspondingly lowered.

Case Study 3: Tuirial Hydroelectric Power Plant (60 MW)

Project COD	30/10/2017
Design Energy (GWh)	Approved by Commission in True-up Order for FY 2014-19: 251 GWh Claimed by Petitioner in True-up Order for FY 2014-19: 251 GWh
Average Tariff (INR/kWh) (2018-19)	Approved by Commission: INR 5.21/kWh Claimed by Petitioner: 4.26/kWh
Major Disallowances	Time and cost overrun: A total project delay of 1,380 days was claimed. The Commission condoned 490 days of delay, attributing them to factors beyond the control of the project developer, such as poor working conditions, local disruptions, and logistical challenges. However, the remaining 890 days linked to issues like slope failure, road restrictions, and increased scope of work were not accepted due to the absence of adequate supporting evidence.

	Capital Cost ceiling: An additional capital expenditure of INR 18 Crore was claimed for the period up to March 31, 2019. However, the Commission did not approve this expenditure, as the ceiling capital cost of INR 1,117 Crore agreed upon earlier—was considered the maximum permissible cost for the original scope of the project. Since the proposed expenditure fell within this original scope and exceeded the approved ceiling, no further capital expenditure was allowed under this head.
Reason for Disallowance	- The disallowance of the delay is attributable to the lack of sufficient supporting evidence to substantiate the claims of delay. - The proposed expenditure remained within the original scope but exceeded the approved ceiling, and therefore, no additional capital expenditure was permitted under this category
Impact of Disallowances	These disallowances may affect the project's financial planning, timelines, and overall execution strategy

8.2.2. Key Challenges observed in determining Tariff for above Hydro projects:-

- 1. Project Delays and Cost Overruns-** The regulator assesses each project delay and resultant cost overrun on the basis of their nature controllable and non-controllable. Controllable costs such as construction delays, availability of manpower, material management related costs are allowed as per norms and prudence check by regulator. Any deviation in such costs are allowed in case-to-case basis subject to the satisfaction of the regulator on the reasonings provided by developers. Natural disasters, slope failures, and geological risks are some of the non-controllable costs which may be allowed by the regulator subject to appropriate technical studies and planned mitigation activities undertaken by the developers.
- 2. Capital Cost -** The disallowance majorly occurs due to reduction in the approved equity base on which regulated returns are calculated, inadequate justification for the additional capital cost provided by the petitioner and procurement of capital spares beyond the cut-off date. The disallowance has a direct financial impact as it reduces the approved capital cost used to calculate tariff components such as depreciation, interest on loan, and—most importantly—Return on Equity (RoE), lowering actual returns.
- 3. Return on Equity (RoE) -** Regulators allow RoE only on the approved equity portion of the capital cost (typically 30% of approved capital), since RoE is allowed only on the equity portion of the approved capital cost, any disallowed amount leads to a lower return, even though the actual equity investment remains unchanged. This results in reduced earnings from the project, lowers overall profitability, and increases the opportunity cost of capital for investors.
- 4. Fixed Return on Equity (RoE) Rates Without Risk Adjustment-** As per Tariff Regulation, RoE rates are fixed at 15.5% in India and do not always reflect market risks, project-specific risks, or inflation. Thus, it does not adequately compensate investors for the higher risk profile and long development horizon of hydro projects.
- 5. Under-approval of Annual Fixed Charges-** The petitioner's total Annual Fixed Charges (AFC) were not fully accepted by the Commission. The reduction in approved AFC significantly lowered the recoverable tariff and impacts the overall Annual Fixed Charges (AFC), reducing net project cash flows. This further compresses the margin available to meet equity return expectations.

These factors often lead to significant disallowances, which must be carefully reviewed, addressed, and incorporated both by the regulatory body and the project developers. It is crucial that the project team proactively engages with these regulatory expectations to ensure that all relevant aspects are transparently accounted for in the tariff determination process.

8.3. Tariff Regulations - Bangladesh

In recent years, Bangladesh has made significant progress in expanding its power generation capacity and enhancing electricity access across the country. However, when it comes to cross-border electricity trade, the country is still in the early stages of development. Unlike many other nations with established power markets, power trade in Bangladesh is not yet recognized as a distinct or standalone commercial activity. Instead, electricity import and export activities are primarily managed and executed through the Bangladesh Power Development Board (BPDB), which remains the central agency for power procurement and distribution. While Bangladesh has a regulatory framework in place, including institutions such as the Bangladesh Energy Regulatory Commission (BERC), the regulatory maturity and market mechanisms are still evolving and are not yet comparable to the advanced regulatory regimes found in more developed power markets. For example, tariff determination methodologies and market-based trading mechanisms are still limited in scope and application.

As a result, electricity tariffs for cross-border trade are often determined through negotiated agreements, either via direct bilateral (B2B) discussions or through limited competitive bidding processes.

The Bangladesh Energy Regulatory Commission (BERC) introduced the Generation Tariff Regulations in 2008, adopting a “cost-plus” approach for determining tariffs for electricity generated from conventional power projects. The tariff structure is primarily designed to ensure recovery of the plant’s fixed (capacity) costs, with BERC prescribing specific norms and parameters for these components. These guidelines are particularly relevant for hydroelectric projects. A detailed breakdown of the allowable fixed cost elements is provided below:

1. **Return on Rate Base:** It is computed based on the following formula:

$$\text{Tariff Rate of Return} = \frac{[(\text{Equity Capital} \times \text{Cost of Equity}) + (\text{Debt Capital} \times \text{Cost of Debt})]}{[\text{Equity Capital} + \text{Debt Capital}]}$$

The debt component of the capital structure is evaluated using the effective interest rates on the licensee’s current borrowings. In contrast, the cost of equity does not follow a standard formula and is determined on a case-by-case basis, subject to regulatory discretion. For state-owned entities, the return on equity is equated to the prevailing government cost of capital, which is typically derived from the latest auction rate of Treasury Bills issued by the Central Bank.

2. **O&M Costs:** The Bangladesh Energy Regulatory Commission (BERC) does not set fixed norms for Operation and Maintenance (O&M) expenses. Instead, these costs are approved based on actual expenditures submitted by the licensee across various categories. Generation expenses include operational costs like supervision, engineering, electricity usage, rents, and maintenance. Customer account expenses cover billing, record-keeping, and bad debt provisions. Sales expenses involve marketing and promotional activities, while Administrative and General (A&G) expenses include salaries, office supplies, rents, insurance, pensions, and regulatory fees. This flexible approach ensures that actual costs are accounted for while maintaining regulatory oversight.
3. **Depreciation Cost:** Straight Line Method is used for computing depreciation costs based on the useful life of the asset.
4. **Income Tax:** When determining the tariff rates for the financial year, the amount of income tax included is the actual income tax paid to the government. Other types of taxes, such as VAT and customs duty, are considered part of the cost of the asset. The proposed increase in tariff, based on the revenue gap approved by the Commission, is grossed up using the prevailing income tax rate to compensate the licensee for the tax paid.

The policy of Government of Bangladesh for setting the electricity tariff stipulates the achievement of the cost reflective tariff and generates a surplus to expand coverage and supply but maintain least cost to the consumers. The tariff setting policy addresses the social commitments of the Government as well.

The tariff structure for the end customers is based on principles of cross-subsidies from commercial consumers to residential consumers. With the uniform tariffs across the country, some distribution companies achieve higher profits while others unable to make profit and struggle to achieve their commercial targets. The distribution costs and the average revenues differ significantly between rural and urban areas due to the different load density and different customer mix.

9. Electricity Export Risk Analysis

Cross Border Electricity Trade is subject to a host of export related risks. Some of the key risks and mitigation measures which may impact DHPP have been enumerated below:

Table 64: Electricity Export Risk Analysis

Risk Category	Types of risk	Risk Description	Risk Mitigation
Curtailment Risk	Grid Infrastructure Limitations	India's transmission network may sometimes lack the capacity to evacuate the full power generated, especially during peak generation periods or in regions with underdeveloped transmission. This leads to forced curtailment of power.	- Collaborate with central and state transmission utilities to ensure timely development and strengthening of evacuation infrastructure. - Secure grid connectivity approvals early and factor transmission upgrades into project timelines. - Explore storage or hybridization (e.g., solar + hydro) to shift generation to off-peak hours when grid availability is higher.
	Load Demand Variability	Hydropower output may not always align with regional or national demand cycles. During low demand or oversupply situations, grid operators may curtail power to maintain grid stability.	- Integrate advanced forecasting tools for demand and generation alignment. - Negotiate flexible PPAs that allow for scheduling adjustments. - Explore participation in ancillary services markets or demand response programs to utilize surplus generation.
	Prioritization of Domestic Generation	In some cases, domestic power plants or renewable sources like solar and wind might be given priority dispatch over imported or foreign-backed hydropower, increasing curtailment risk for foreign projects.	- Engage with regulators to ensure equitable dispatch rules that include foreign projects under "must-run" status where applicable. - Structure long-term PPAs with firm offtake commitments to reduce reliance on merit-order dispatch.
	Regulatory and Contractual Constraints	Power purchase agreements (PPAs) and grid codes may include clauses permitting curtailment under certain conditions, such as grid emergencies, transmission congestion, or force majeure events.	- Carefully review and negotiate PPA clauses related to curtailment and compensation. - Ensure inclusion of deemed generation provisions to recover revenue in curtailment scenarios. - Maintain continuous dialogue with regulators to stay informed of policy updates and influence equitable regulations.
	Cross-Border Transmission Issues	For projects where power is transmitted across borders or through multiple states, delays or disruptions in transmission infrastructure upgrades can cause curtailment.	- Coordinate with central and regional transmission planning agencies to prioritize cross-border infrastructure. - Diversify offtake routes (e.g., multiple grid interconnection points) to reduce dependence on a single corridor.

Risk Category	Types of risk	Risk Description	Risk Mitigation
			Establish bilateral coordination frameworks for smoother cross-border electricity trade.
	Market and Policy Changes	Policy shifts or changes in renewable energy incentives might alter dispatch priorities, affecting foreign hydro projects' ability to supply power continuously.	- Monitor regulatory developments closely and participate in public consultations to shape policy outcomes. - Build flexibility into project financing models to adapt to changing market conditions.
Currency Risk	Revenue and Payment Risk	Cross-border infrastructure projects with a mismatch between the currency of project revenues and that of liabilities or costs can face asset liability mismatch if foreign exchange rates move unfavourably.	- Hedging instruments may be a solution to currency risk in these circumstances, but they rarely offer a cost-effective solution in many markets due to the costs involved and the lack of long-term hedging options for many local currencies. - Another alternate would be to create reserve to cater to adverse foreign exchange fluctuation thereby smoothening the impact over time.
	Price Peg Risk	Since Bhutan's currency Ngultrum is pegged with Indian currency INR, Bhutan is exposed to currency impact of Indian currency which will directly impact the local currency of Bhutan.	- Use financial derivatives (e.g., futures, options) to hedge against price fluctuations when possible. - Though limited in some markets, exploring available instruments can offset exposure.
PPAs with DISCOM	Credit Risk	Many Indian DISCOMs face financial stress and liquidity issues, which can lead to delays in payments or defaults, affecting the cash flow and financial viability of the project.	- Include payment security mechanisms in Power Purchase Agreements (PPAs) such as Letters of Credit (LC), escrow accounts, or bank guarantees to ensure timely payments. - Incorporate penalty clauses for delayed payments to incentivize prompt settlement.
	Regulatory and Policy Risk	Changes in government policies, tariffs, or regulations can alter the terms of PPAs or introduce additional compliance burdens, impacting project returns.	- Diversify project portfolio across different states or segments to reduce dependency on a single regulatory regime. - Balance long-term fixed contracts with shorter-term agreements or merchant market exposure to maintain flexibility.
	Scheduling Risk	DISCOMs may curtail power procurement due to grid constraints or demand fluctuations, limiting the ability to sell the contracted power fully and reducing revenue	- Include deemed generation clauses in PPAs that compensate the generator for curtailed power. - Negotiate minimum off-take guarantees or take-or-pay provisions to secure revenue even during curtailment. - Collaborate closely with transmission utilities and DISCOMs to prioritize grid upgrades and reduce congestion.

Risk Category	Types of risk	Risk Description	Risk Mitigation
	Force Majeure and Political Risk	Events such as natural disasters, political instability, or changes in cross-border relations (especially for projects sourcing water or power from neighboring countries) can affect project operations and contractual obligations.	- Negotiate bilateral or multilateral treaties and agreements ensuring water flow, power transmission, and operational cooperation with neighboring countries. - Include dispute resolution mechanisms (such as international arbitration) specifically designed for cross-border conflicts.
Merchant market risk	Price Volatility Risk	As observed in last three years, the MCP in exchange during peak hours is very low due to excess supply of solar power. Thus, it is important to align the generation profile of DHPP with non-peak hours for better returns.	- Optimize the hydro project's generation schedule to maximize output during non-peak hours when market prices are relatively higher. - Use flexible operation strategies (e.g., pumped storage or reservoir management) to shift generation away from low-price peak solar periods. - Consider hybrid models combining hydro with solar or wind to balance supply and improve revenue stability.
	Energy storage solutions	As India is expecting a huge quantum of energy storage solutions to come online by 2030, it is expected that present attractive MCPs during non-peak hours to come down leading to lesser profitability.	- Capping participation in merchant power market. - Explore aligning of DHPP's generation with non-peak hours for maximum returns.
	Market Liquidity Risk	At times, liquidity in the power exchange may be limited, affecting the ability to sell the full volume of generated power at competitive prices.	Capping participation in merchant power market.
C&I segment Risk	Average PPC of DISCOMs	With the increasing integration of renewable energy (RE) into the portfolios of State DISCOMs, the average power purchase cost is expected to decline, which should, in turn, reduce the average billing rates for the Commercial and Industrial (C&I) segment in India. Given that this segment is highly price-sensitive, corporate houses are likely to prefer procuring power directly from DISCOMs as their tariffs become more competitive.	- Offer C&I consumers additional benefits such as guaranteed supply reliability, green energy certification, and customized energy solutions to differentiate from DISCOM tariffs. - Provide flexible contract terms and superior customer service to retain open access consumers.

10. Transmission Infrastructure between India, Bhutan and Bangladesh

The South Asian countries like Bhutan, India, Bangladesh and Nepal have growing energy demands and an increasing emphasis on sustainable energy. Enhanced regional cooperation is critical to boosting energy security and overall climate resilience in the region. Moreover, the seasonal demand peaks of these nations are complementary. The peak demand seasons of India and Bangladesh correspond to generation surpluses in run-of-river hydropower systems in Bhutan and Nepal. Similarly, when Bhutan and Nepal have supply shortages (during their dry season) there is excess capacity in India and Bangladesh.

Currently, the regional power trade of around 7 GW is bilateral in nature. Until 2023, trade was facilitated through bespoke long-term power purchase agreements. The table below sets out details of the existing cross-border electricity transactions between Bhutan, India and Bangladesh.

Table 65: Power Trade Agreements between South Asia

Particulars	Source	Agreement Type
Bhutan → India (2380 MW)	1020 MW Tala	Government-to-Government
	336 MW Chukha	Government-to-Government
	60 MW Kurichhu	Government-to-Government
	126 MW Dagachhu	Commercial
	720 MW Mangdechhu	Government-to-Government
	118 MW Nikachhu	Government-to-Government
India → Bangladesh (2856 MW²⁹³)	250 MW NVVN Ph-I	Government-to-Government
	200 MW PTC Ph-II	Commercial
	160 MW Tripura	Government-to-Government
	300 MW NVVN Bharamara Ph-II	Government-to-Government
	250 MW Sembcorp	Commercial
	1496 MW Adani Godda	Commercial
	200 MW PTC India	Commercial
India → Nepal (600-800 MW)	237 MW India	Government-to-Government
	400-500 MW Market	Commercial
Nepal → India (800-1000 MW)	941 MW from 28 hydro projects	Commercial
Nepal → Bangladesh (40 MW)	40 MW (through Indian grid)	Government-to-Government (Tripartite Agreement)

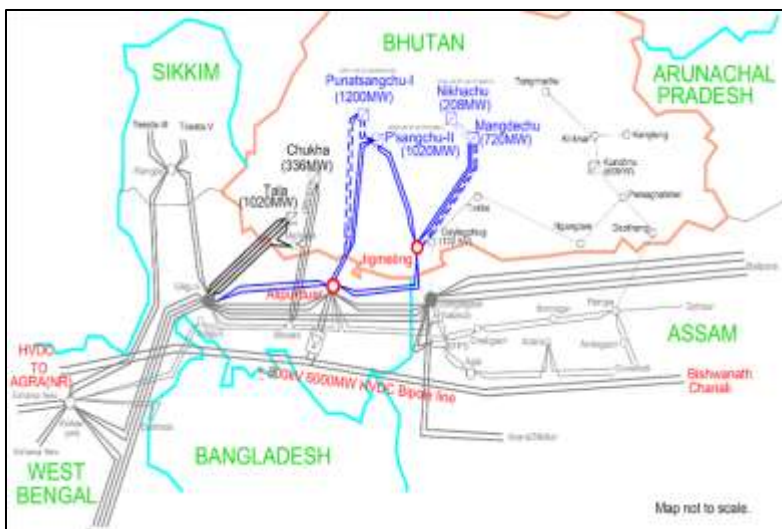
To facilitate the trade, at present the interconnection lines amongst South Asian countries is provided in the table below:

Bhutan-India

Currently, the power system of Bhutan is connected to the Indian grid, through 13 interconnecting lines with eight feeders at 400kV and three feeders at 220kV voltage levels from western Bhutan, and two S/C 132kV lines from eastern Bhutan as below:

²⁹³ Source: BPDB Annual report 2023-24

- 2xD/C, 400 kV Twin Moose from Tala to Silliguri with one circuit LILOed at Malbase Substation;
- 1xD/C, 400 kV Quad Moose from Jigmeling to Alipurduar;
- 1xD/C, 400kV Quad Moose from Lhamozingkha to Alipurduar;
- 1xD/C, 220 kV from Chukha to Birpara;
- 1xS/C, 220kV from Malbase to Birpara;
- 1xS/C, 132 kV from Gelephu to Salakati; and
- 1xS/C, 132 kV from Motanga to Rangia



With additional planned hydropower and solar projects, following new cross-border transmission lines are required to be built for export of surplus power:

- 2xD/C, 400kV Quad Moose Durungri - Rangia/Rowta/Bornagar
- 1xD/C, 400 kV Twin Moose Phuntshothang - Rangia/Rowta/Bornagar

As per the National Transmission Grid Master Plan (NTGMP-2012), the power from Bhutan is to be pooled at Alipurduar and Rangia/Rowta/Bornagar pooling stations which will further be evacuated to the deficit regions of India (NR/WR) via ± 800 kV, 3,000 MW HVDC sub-station at Alipurduar, and establishment of a ± 800 kV, HVDC bipole line with converter capacity of 6,000 MW at Rangia/Rowta/Bornagar.

India-Bangladesh

Currently, Bangladesh's power system is connected to the Indian grid via two interconnecting lines operating at 400 kV, utilizing both HVDC and HVAC technologies. The proposed Katihar–Porbotipur–Bornagar transmission line is expected to have an evacuation capacity of 1,000 MW. Therefore, DHPP can evacuate power to Bangladesh through this line, as it has sufficient capacity to accommodate the transfer. The details of present status and upcoming line is provided in the table below:-

Table 66: Details of Transmission Infrastructure between India and Bangladesh

Sno.	Interconnection Project	Voltage and Configuration	Intercn. Type	Evac. Capacity (MW)	Intercn. Point in India	Intercn. Point in Bangladesh
Status – Existing Projects						
1.	Baharampur – Bheramara Line	400 kV D/C	HVDC	1000	West Bengal (Baharampur)	Bheramara
2.	Surjyamaninagar – North Comila – South Comila	400 kV (operated at 132 kV)	HVAC	160	Tripura (Suraj Maninagar)	North Commila
Status – Proposed Projects						
1.	Katihar – Parbotipur - Bornagar	765 kV D/C	HVDC	1,000	- Bihar (Katihar) - Assam (Bornagar)	Parbotipur

11. Evolving Tariff Trends in the Indian Electricity Market

In this section we have assessed the historical tariff trends in India as well as the major developments in the sector governing future pricing outlooks. Historically from FY 2016-24, the tariffs from thermal energy have grown at a steady growth rate of 5% for both coal and oil & gas this is majorly due to rising coal prices and domestic coal shortages. In the same period due to focus on Renewable energy, the tariffs of solar based power plants have decreased by ~7% as the equipment cost has fallen significantly. The average tariff for hydro-based power plants is approximately INR 2.5 per unit; however, this rate primarily reflects costs associated with older and depreciated assets. The details of the growth trend of source wise electricity supply are provided in the figure below:-

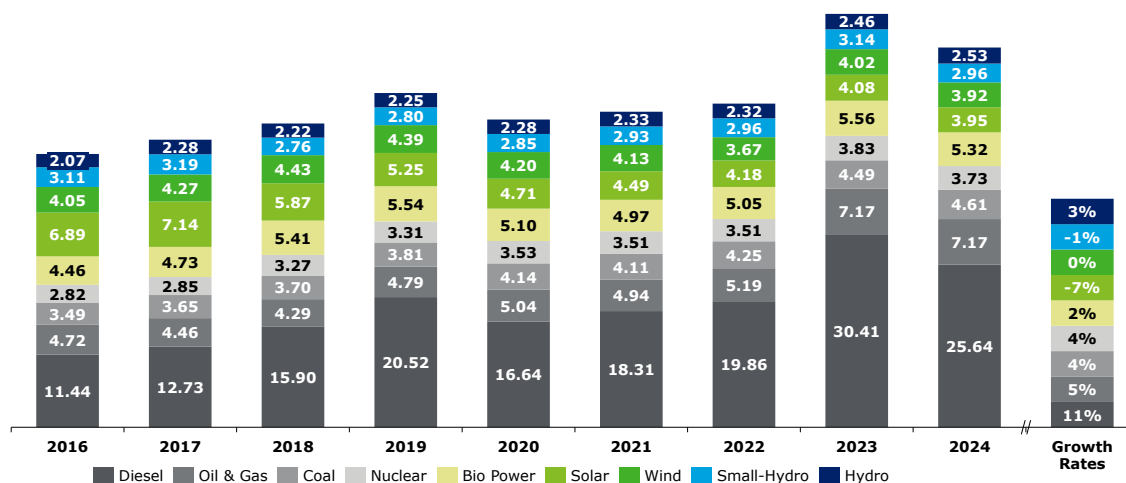


Figure 115: Source wise Electricity tariff (INR/kWh) in last 8 years²⁹⁴

The electricity market is evolving rapidly and dynamically, driven by technological advancements, changing regulatory frameworks, and shifting consumer demands. Several key trends are emerging that are expected to shape the future landscape of the market and resultant tariffs. These include increasing integration of renewable energy sources, greater adoption of energy storage solutions, the rise of decentralized generation, and the implementation of more sophisticated pricing and trading mechanisms. Understanding these trends is crucial for stakeholders to navigate and capitalize on the opportunities in the evolving electricity sector.

Table 67: Evolving Trends affecting Pricing In India

Addition of Thermal Power Plants	To meet the estimated electricity demand by the year 2031-32, generation planning studies have been carried out by Central Electricity Authority (CEA). As per the study results, it is envisaged that to meet the base load requirement of the country in 2032, the required coal & lignite based installed capacity would be 283 GW against the present installed capacity of 240 GW as on May 2025 ²⁹⁵ . Based on the recent bids, the discovered tariff of upcoming thermal plant is expected to be ~INR 5.4 per unit ²⁹⁶ . These are for the coal plants coming online in 2030-31.
Delayed Hydro Project Installation	At present, ~9.9 GW are under construction in the country as on April 2025 ²⁹⁷ . The hydro capacity is projected to increase from 47 GW in FY 2024-25 to ~54-62 GW by 2029-30 (under different scenarios), marking possibility of substantial increase in capacity. Currently, PSPs with aggregate capacity of ~4 GW are under

²⁹⁴ [India's Climate and Energy Dashboard](#)

²⁹⁵ [As per MoP, GoI](#)

²⁹⁶ Source: [Reuters News Report](#)

²⁹⁷ Source: [Hydroelectric Projects under Construction, CEA, April 2025](#)

	construction in the county while CEA in its revised report on optimal generation mix in 2030, has projected installed capacity of around ~19-25 GW of PSP by 2029-30 under different scenarios. Historically, the average tariff for hydro-based power plants is approximately INR 2.5 per unit; however, this rate primarily reflects costs associated with older and depreciated assets. However, the tariffs of projects commissioned in the last decade ranges between INR 4-5.5 per unit.
Declining Solar Tariffs	The Government of India (GoI) has set an ambitious target of achieving 500 GW of renewable energy capacity by 2030, with approximately 220 GW already installed—of which nearly 50% is contributed by solar energy. The project cost of solar plant has also decreased from INR 4.5 crore to INR 4.1 crore which has resulted in reduction of tariff by 10-12%. Additionally, as discussed in Chapter 5, an analysis of the Indian Energy Exchange (IEX) over the past three years reveals that Market Clearing Prices (MCP) consistently peak and reaches even up to ~INR 8 per unit during the evening hours after 8 p.m. and early morning hours between 1 a.m. and 3 a.m. in both the Day-Ahead Market (DAM) and Real-Time Market (RTM) segments. However, in solar hours, the MCP is in the range of INR 2.5 to 3.5 per unit. Recently, MCPs have fallen to near zero during afternoon hours, primarily due to a surplus of solar generation coinciding with low electricity demand during the pre-monsoon season.
Energy Storage Solutions	In response to the falling tariffs during solar hours, the market is beginning to reward flexibility and time-shifting capabilities. Energy storage systems, particularly battery storage and PSPs, are emerging as critical assets. Projects integrated with storage can store excess solar energy generated during the day and dispatch it during peak periods, allowing developers to capture higher tariffs during peak hours. This arbitrage potential is altering the revenue model of solar projects—pushing the sector toward hybrid and round-the-clock (RTC) configurations. In energy storage systems, 13,050 MW/78,300 MWh Pumped Storage Projects are under construction/concurred and 14,970 MW/54,803 MWh Battery Energy Storage System are currently under various stages of construction/bidding²⁹⁸. Out of this, 6,853 MW/ 36,592 MWh of energy storage system (3,180 MW/19,080 MWh Pumped Storage Projects and 3,673 MW/ 17,512 MWh of Battery Energy Storage System) is likely to be added by 2025-26.²⁹⁹ As per the recent price bids, the discovered tariffs from The recent bids has fallen to INR 3.33 per unit³⁰⁰.

Based on tariff trends, solar energy continues to be the most cost-effective energy source. However, energy storage solutions are gaining momentum due to their role in enhancing grid stability and integrating variable renewable energy (VRE). Storage projects (BESS, PSP) are expected to be a key focus area for investments to leverage their peak shift capabilities. Additionally, recently commissioned hydro tariffs remain competitive compared to thermal power plants. With the proposed Hydropower Purchase Obligations and inclusion of cross border hydropower projects in meeting HPO obligations, the segment has an important role in meeting regional decarbonation and grid stability requirements.

²⁹⁸ [PIB, March 2025](#)

²⁹⁹ [PIB, March 2025](#)

³⁰⁰ [As per Tariff Discovery](#)

12. Identifying the trends of hydropower financing and investment in Bhutan

12.1. Considerations for Indian/International Lenders:

a. Key appraisal requirements

Banks / Financiers typically require a thorough and comprehensive appraisal of under-construction hydropower projects to assess their financial viability and risk profile before extending loans or investments.

Multiple teams of lending institutions are generally involved in appraising various categories of risks / project aspects. This appraisal involves evaluating various aspects including but not limited to technical feasibility of the project, environmental impact, hydrological and seismological risks, socio-economic considerations and commercial outcomes (profitability and sustainability dependent on market dynamics).

i. Technical Appraisal of the Project

Lenders' consideration:

- **Hydrological studies:** Assessing water availability, flow rates, and potential for storage or run-of-river projects.
- **Geotechnical investigations:** Evaluating soil and rock conditions at the dam site and surrounding areas to ensure stability and prevent landslides.
- **Design and Engineering:** Reviewing the project design, including dam type, turbine selection, and power plant layout, to ensure it is optimized for power generation and cost-effective.
- **Power generation capacity:** Evaluating the potential power output and assessing the 90% dependable output to ensure reliability and meet demand.

DHPP's preparedness:

- A robust Detailed project report (DPR) has been prepared for DHPP, encompassing a comprehensive study of all critical elements required for technical evaluation of the project. The report includes thorough hydrological studies to validate water availability and dependable power generation estimates. It incorporates extensive geotechnical and geological investigations to confirm site stability and inform sound engineering decisions. The design and layout of project components—such as the dam, turbines, and powerhouse—are optimized for performance and cost-efficiency.

ii. Financial Appraisal

Lenders' consideration:

- **Project cost estimation:** Accurately estimating project costs including construction costs, land acquisition, plant and machinery components, and environmental mitigation, soft costs (viz. preliminary and pre-operative expenses, interest during construction, margin money for working capital requirement etc.) to ensure they are realistic and achievable.
- **Revenue projections:** Forecasting power generation revenue through proposed mix of domestic and export of energy along-with assessment of respective tariff rates, ongoing regulatory scenario (for tariff and/or free power), and delivery point along with appropriate

sensitivity analysis to fairly assess project's ability to generate sufficient profit and cash for servicing of debt availed by the project.

- **Operating cost projections:** This analysis includes elements viz. assessment of operational and maintenance cost, periodic repair and maintenance costs, employee costs, insurance cost, administrative overhead allocation and other operating costs along with consideration for suitable escalation rates dependent on contractual and/or inflation related estimations.
- **Financial Modelling:** Based on above analysis, a detailed financial model is prepared to analyze the project's cash flows for the period of asset life thereby assessing its debt servicing capabilities, and financial sustainability.
- **Sensitivity analysis:** Assessing the project's vulnerability to changes in key parameters, such as tariff rates, operating costs, and power demand, to understand the potential range of outcomes.

DHPP'S preparedness:

- Comprehensive financial model shall be prepared to provide a clear picture of the project's financial viability and resilience. Model will include detailed projections of project cost also incorporating sensitivity analysis to assess the potential impact of key risks, including variations in hydrology, construction cost overruns, and project delays. These stress tests will help identify financial vulnerabilities and guide risk mitigation strategies, ensuring that the project remains robust under different scenarios and aligns with lender expectations for financial soundness.

iii. Environmental and social impact:

Lenders' consideration:

- **Environmental impact assessment:** Evaluating the environmental impacts of the project, including those on water resources, ecosystems, and biodiversity, and assessing mitigation measures.
- **Social impact assessment:** Evaluating the social impacts of the project, including resettlement, displacement, and livelihood impacts on local communities, and developing mitigation measures.
- **Resettlement and rehabilitation:** Assessing the need for resettlement and rehabilitation of project-affected populations and ensuring fair and equitable compensation and support.
- **Stakeholder engagement:** Conducting consultations with local communities, affected populations, and other stakeholders to ensure their concerns are addressed.

DHPP's preparedness:

- Key Environmental and social risks have already been identified keeping in view similar projects afflicted with these issues. Accordingly, a comprehensive and detailed risk mitigation plan has been developed by the company and it shall be implemented accordingly. Statutory approvals for such Risk mitigation plans have also been obtained.

iv. Legal and regulatory appraisal

Lenders' consideration:

- **Permits and licenses:** Ensuring the project obtains all necessary permits and licenses from relevant government agencies including but not limited to environment and forest clearance, transmission related approvals, land acquisition and project implementation related approvals.

- **Compliance with regulations:** Ensuring the project complies with all applicable environmental regulations, social impact regulations and other legal requirements.
- **Risk management:** Identifying and managing potential legal and regulatory risks, such as changes in environmental regulations or land acquisition disputes.

DHPP's preparedness:

- Forestry Clearance from Department of Forest and Park Services obtained on 11th and 22nd April 2025.
- No Objection Clearances (NOCs) duly obtained from all the PAHs on 9th August 2024, 18th August 2024, 14th August 2024, 12th August 1st August 2024 from Jarray, Saling, Tsakaling, Tsamang, Tsenkhar gewogs respectively.
- Administrative Clearance from Mongar and Lhuentse Dzongkhags have been obtained on February 25, 2025, and March 27, 2025, respectively.
- Statutory departments like DoST, MoIT, DoE, MoENR have granted the necessary technical and administrative approvals.

v. Commercial Considerations – Off-take related market dynamics

Lenders' consideration:

- **Demand assessment:** Assessing the demand for electricity and the potential market for power generated by the project.
- **Tariff rates:** Evaluating the tariff rates and power purchase agreements (PPAs) to ensure they are favorable and sustainable.
- **Power market analysis:** Analyzing the competitive landscape of the power market and assessing the project's ability to compete effectively.
- **Transmission and distribution:** Assessing the feasibility of connecting the project to the grid and ensuring reliable transmission and distribution of power.

DHPP's preparedness:

- We understand that:
 - DHPP is in advanced stages of discussion for tie-up of long-term power off-take agreement.
 - DHPP will be following international competitive bidding process for selecting vendors on key contracts for civil works and hydro-mechanical components
- Market reports are being prepared with the help of advisors which are aimed at providing suitable analysis for commercial consideration of DHPP.

b. Security enforcement by Indian / International Lenders

We understand that DGPC has obtained a legal opinion dated 14th Oct 2014 for Tangsibji Hydro Energy Limited (THyE) Project from Bhutan Law Services regarding enforceability and validity of the proposed security package in Bhutan.

We have perused this legal opinion to summarize the key observations on enforceability of security by foreign lenders to DHPP. These are detailed out hereunder:

- **Borrowing from Foreign Lenders** - DHPP is within its right to borrow from foreign lenders, provided that DHPP obtains prior written permission of the Royal Government of Bhutan as required

under section 13(1) of Moveable and Immovable Property Act (MIPA) of Bhutan for the proposed borrowing before executing any financing agreement with the lenders.

- **Creating Security** - It is permissible under MIPA to create security interest in movable and immovable property. However, for the foreign lenders, the rights and remedies relating the securities are only good when the Borrowing entity has the pre-requisite written permission of the Bhutanese government permitting it to borrow as required under Section 13(1) of MIPA.
 - Land Act of Bhutan doesn't permit lessee of leasehold land in encumbering any charges hereon the leased land other than the use of the leasehold land as per lease terms. This difficulty spills over the charges that the lenders may have on the structures built on the same leasehold land. Thus, it will be critical to obtain written consent from Ministry of Economic Affairs (MoEA) and National Land Commission (NLC) for DHPP to have a right to transfer its remaining lease tenure to the lenders, in the event of its default.
- **Step-in-Right and Security Enforcement** - The Bhutanese counsel has advised, inter alia, that once the right to use the leasehold rights as security for the borrowings is perfected by making a suitable provision in the lease agreement, the step-in-rights of the lenders, in the event of default, are also concurrently perfected. He has also advised that as long as the rights and remedies are expressly provided by the security agreement, the same read with the applicable provisions of MIPA are sufficient to secure the step-in-rights of the lenders.
 - It is possible to create charge/ assignment of the right, title and interest of the borrower company in project contracts, insurance contracts, approvals, etc.
 - A non-Bhutanese person cannot buy immovable property in Bhutan under any circumstances except by being a proprietor of a FDI incorporated company under the Companies Act of Bhutan. Thus the enforcement of securities and disposal of assets by a foreign lender can take place only through recovery of debt from the sale of proceeds of the defaulting company's immovable and movable assets. Hence, although there is no restriction in selling movable assets of a company to any nationals, the immovable properties can only be sold to Bhutanese nationals.
 - In terms of enforcement procedure, the secured party shall have the right to take peaceful possession of the collateral without first obtaining the court order provided the taking of the possession does not involve confrontation or breach of the peace.

c. Financing Timeline

For commercial lenders, **the financing process for DHPP is expected to take approximately 5 to 7 months following the finalization of the Power Purchase Agreement (PPA).**

This timeline accounts for lender due diligence, credit assessments, internal approvals, and the negotiation of loan terms and security arrangements.

Given the complexity and scale of the project, this duration is considered standard for securing debt financing from commercial institutions.

13. Project Risks - Benchmarking and Mitigations

Hydropower projects typically face a range of material risks, including Land acquisition issues, Geological surprises, construction delays leading to time and cost overruns, and regulatory challenges.

For the Dorjilung project, key risks and their mitigation measures have been carefully identified and addressed by DGPC. The following section outlines these risks along with the strategies implemented to manage them effectively.

Land Acquisition and Right of way:

Key factor / risk element considered by Lenders:

- Availability of land for construction of plant.
- In case, private land is to be acquired, whether compensation has been paid and No-Objection-Certificate has been obtained from private stakeholders.
- In case forest land is to be acquired, whether regulatory process and compensatory afforestation measures have been planned for implementation.

Sample project (outside Bhutan) impacted by above mentioned risk:

- **1,500 MW Tipaimukh Hydroelectric project in Manipur, India, initially entrusted to NEEPCO and later switched to NHPC** has been stalled for multiple years, due to issues with land acquisition due to concerns over displacement of indigenous people.
- **World Bank funded project example – 2,160 MW Dasu Hydropower Stage 1 project in Pakistan** has been having land acquisition related delays.

Mitigation in place for DHPP:

The DHPP project requires 900.08 acres of State Reserved Forest and 18.89 acres of private and institutional land, impacting 58 plots owned by 50 households, one business lessee, and two institutions—totaling 53 affected parties. Only one household will be physically displaced, and the overall resettlement impact is minimal compared to similar projects.

To mitigate these impacts, a comprehensive Land Acquisition and Livelihood Restoration Plan (LALRP) has been finalized. It includes detailed assessments, compensation frameworks, and livelihood support measures developed through stakeholder consultations. **Necessary NOCs and clearances have been obtained, and implementation costs are fully budgeted, with a 10% contingency for unforeseen needs.**

Statutory Approvals:

Key factor / risk element considered by Lenders:

- Whether Cabinet / Government approval, Forest and National Park clearances, Wildlife clearance, Culture clearance, District administrative approval, Environment Clearance, Department of Roads approval, NOC from the affected households losing land to the Project etc. and any other relevant approvals are in place.

Sample project (outside Bhutan) impacted by above mentioned risk:

- **Delay of c. 21 months in construction start of 1,091 MW Karcham Wangtoo Hydroelectric Project** of Jaiprakash Power Ventures Group on account of delay in grant of environment and forest clearance.
- **World Bank funded 444 MW Vishnugad Pipalkoti Hydroelectric Project of THDC India Limited-** Cabinet approval was obtained in 2008, works could not be awarded due to Forest clearance / diversion of forest land. Forest land was acquired in January-14 and subsequently works awarded in January-14. Further, there was disruption of work by local people.

Mitigation in place for DHPP:

DHPP has either already obtained or initiated the process for getting material requisite approvals from the concerned ministries and departments. A list of these approvals has been presented below:

Statutory approvals obtained:

Particulars	Concerned Authority	Approval Date
Approval for Implementation	Department of Energy, Bhutan Ministry of Energy and Natural Resources, Bhutan	August 9, 2024
Approval for Implementation	Ministry of Finance	September 16, 2024
Approval for Implementation	Druk holding and Investments	October 14, 2024
Technical Clearance	Department of Energy, Bhutan Ministry of Energy and Natural Resources, Bhutan	February 12, 2025
Access Road Construction	DoST, MoIT	February 19, 2025
District Administrative Clearance	Mongar and Lhuentse Dzongkhags	February 25, 2025 and March 27, 2025
No Objection Clearances from all Project affected Households	Project affected Households	August 2024
Forestry Clearance	Department of Forest and Park Services	11 th and 22 nd April 2025

Clearances under process:

Application for Environment Clearance (EC) applied to Department of Environment and Climate Change (DECC) on 5th March 2025 and is at the advanced stage of review process, considering 100 days of turnaround time (TAT). The EC is expected by June 2025 before the start of access road construction works.

Geological uncertainties

Key factor / risk element considered by Lenders:

- Geological uncertainties in hydropower projects such as unexpected rock types, faults, groundwater inflows, and unstable slopes, which can impact design, safety, and cost. These conditions are difficult to predict and can lead to construction delays or failures to implement.

Sample

Projects impacted by above mentioned risk:

- 2,000 MW under-construction Lower Subansiri Hydroelectric project of NHPC India** has encountered serious geological challenges including weak and ruptured rock strata on dam's right bank. This raised significant safety concerns and led to multiple re-design, resulting in significant time and cost over-run.
- Work on the **1200 MW Punatsangchhu-I hydroelectric project** commenced in 2008 and was initially expected to commission by 2016 but the project **is running behind schedule**. In July 2013, the project witnessed a geological surprise – the right bank of the project site had sunk, with the loose rock-face gradually moving down to the base of the dam site being excavated. The sinking riverbank necessitated stabilizing the riverbank and design changes in the project. The strengthening work has compelled the project to make technological changes in constructing the dam.

Mitigation in place for DHPP:

Extensive geological and geotechnical investigations were conducted during the 2015 DPR and 2024 DPR Update, including boreholes, seismic and resistivity profiles, exploratory drifts, and expert site inspections. The geology has been thoroughly vetted by consultants, the Panel of Experts, and World Bank specialists.

Given the depth of prior studies, adverse geological scenarios are not anticipated. The headrace tunnel design accounts for worst-case geological conditions, and a 20% cost contingency has been included. Any unforeseen geological issues during construction will be addressed with site-specific mitigation measures.

Further, sensitivity analysis for cost and time overrun would be carried out to ascertain impact on further fund requirement, debt serviceability and financial stability of the project.

Hydrology shocks

Key factor / risk element considered by Lenders:

- Hydrological shocks from seasonal variations or glacial outbursts etc. can cause water flows to fluctuate, potentially resulting in reduced power generation or flooding. These effects can undermine the project's financial viability and increase the risk for lenders.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- Hydropower plants are prone to hydrology shocks triggered by earthquakes, floods, siltation, droughts etc. Historically, numerous hydropower plants have faced damages or shutdowns due to these hydrology shocks including but not limited to:

- **6,400 MW Sayano-Shushenskaya Hydroelectric Power Station, Russia** which faced shutdown due to floods in 2009.
- **2013 floods in Uttarakhand region of India** lead to shutdown/damages in multiple Indian plants including 400 MW Vishnuprayag Hydroelectric projects of Jaiprakash Power Ventures Limited, India and 280 MW Dhauliganga Hydro project of NHPC, India.

Mitigation in place for DHPP:

To address flood risks, future Probable Maximum Flood (PMF) values have been assessed under different climate scenarios through the DPR's Climate Resilience Study. **The project design accounts for projected PMF increases, with significant safety margins for the far future (2060–2079).**

Energy production impacts due to climate variability (2020–2059) have also been analysed, with expected variations ranging from -7% to +18%, and **appropriate design measures have been incorporated to mitigate these risks.**

Environmental and ecological concerns

Key factor / risk element considered by Lenders:

- Impacts on aquatic ecosystems, biodiversity loss, and greenhouse gas emissions from reservoirs.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- **1,070 MW Nam Theun 2 Hydropower Project (Laos)** - Despite being considered a success, has faced criticism for methane emissions and long-term ecological impacts on the Xe Bang Fai River.

Mitigation in place for DHPP:

The DPR identifies potential impacts on aquatic ecosystems from construction activities such as land clearing, earthworks, and river diversion. These are addressed through dedicated plans like the Biodiversity Management Plan (BMP) and Environmental Flow Management Plan (EFMP). These plans outline strategies to protect aquatic life and water quality, supported by regular performance monitoring.

Impacts on plants, birds, and mammals are assessed as **low risk based on ecological data** and are addressed in the BMP. **The DHPP Biodiversity Unit is also working with government agencies to develop Operational Conservation Plans for each identified Critical Habitat.**

Social resettlement and livelihood disruptions

Key factor / risk element considered by Lenders:

- Large hydropower projects often require relocating entire communities, impacting their housing, agriculture, culture, and access to resources. Poorly managed resettlement plan can lead to long-term social unrest, reduced quality of life, and impoverishment of affected people.
- Failure to involve local communities, NGOs, and civil society groups from the early planning stage can lead to misinformation, protests, and even legal challenges.
- Such instances may lead to litigations thereby resulting time and cost overrun.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- 44 Hydropower projects on Siang belt of Sikkim, India.
- 444 MW Vishnugad Pipalkoti Hydroelectric project in Uttarakhand of THDC India Limited.

Mitigation in place for DHPP:

Based on feedback from DHPP, we understand that all affected stakeholders have already been identified through a detailed mapping exercise during the ESIA process. **Consultations were held at local to national levels, and concerns were integrated into mitigation instruments such as the Cultural Heritage Management Plan (CHMP) and Biodiversity Management Plan (BMP).** Ongoing engagement is ensured through a dedicated Stakeholder Engagement Plan (SEP).

Cost overruns and delays

Key factor / risk element considered by Lenders:

- Cost overruns and delays are the culmination/results of all the risks explained above. Many hydropower projects exceed initial budget and timeline estimates due to unforeseen geological conditions, technical complexities, Environment and social objections and procurement or contractor inefficiencies. This escalates financial risk and affects project returns.

Mitigation in place for DHPP:

In earlier DGPC projects, such as the Dagachhu Hydropower Project, cost overruns were shared between the equity partners—DGPC and TATA. A similar structure is proposed for the DHPP, with any cost overruns to be jointly financed by the equity partners. To further mitigate financial risks, the cost estimates include contingency allowances: 20% for Civil and Hydromechanical Works, and 5% for Electromechanical Work

Further, sensitivity analysis for cost and time overrun would be carried out to ascertain impact on further fund requirement, debt serviceability and financial stability of the project.

Vendor related risk

Key factor / risk element considered by Lenders:

- Financial stability and experience of vendors (Civil contractors, Electromechanical component suppliers and Hydro-mechanical component suppliers) is essential for ensuring quality and timely construction of project.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- Delay in completion of **444 MW Vishnugad Pipalkoti HEP** and **1,000 MW Tehri Pumped Storage Project** of THDC India is also attributable to financial health of main civil contractor. THDC had to approve special arrangement of gap funding to contractor for expediting project progress.

Mitigation in place for DHPP (based on feedback from DGPC):

The implementation of two hydropower projects with **ADB as lead financier**, alongside commercial banks, provided DGPC with critical exposure to **external commercial lending requirements**, compliance standards, and rigorous monitoring/reporting obligations. These experiences enhanced DGPC's capacity to manage financing risks and strengthened internal human resources. **DGPC/DHPP will apply these insights to anticipate and mitigate challenges in future project execution.**

Key Lessons from Dagachhu and Nikachhu Hydropower Projects to be applied on DHPP:

1. **EPC Oversight:** Effective implementation of EPC contracts requires active client oversight; quality control cannot be left solely to the contractor.
2. **Capacity Building:** Deep involvement of DGPC staff during execution led to valuable hands-on experience, strengthening institutional capacity for future projects.
3. **Technical Evaluation:** For complex components (e.g., tunneling, diversions), construction methodology should be a key evaluation criterion during procurement.
4. **Financial Controls:** Use of escrow accounts ensures contractor payments are directly used to pay material and service vendors, improving financial discipline.
5. **Working Capital Management:** Facilitation of working capital advances helps address short-term cash flow mismatches between receipts and payments.
6. **Variation Management:** A structured mechanism is needed for the **early identification and settlement** of variations and claims to prevent delays and cost escalation.

Foreign exchange related risk

Key factor / risk element considered by Lenders:

- Cross-border infrastructure projects with a mismatch between the currency of project revenues and that of liabilities or costs can face asset liability mismatch if foreign exchange rates move unfavorably. Hedging instruments may be a solution to currency risk in these circumstances, but they may not offer a cost-effective solution in certain markets due to the costs involved and the lack of long-term hedging options for local currencies. Another alternate would be to create balance sheet reserve to cater to adverse foreign exchange fluctuation thereby smoothening the impact over time.

Sample projects (outside Bhutan) impacted by above mentioned risk:

- World bank funded **1,070 MW Nam Theun 2 hydropower project in Laos**, faced a similar issue as there was a mismatch between the currency denomination for revenue stream, project cost and debt financing. Currency risk was mitigated in this project by structuring the currency profile of the financing to match that of the project costs (pre-completion of the project) and the revenues (post-completion of the project). This also provided a natural hedge against the tariff structure, which required half of the underlying long-term debt structure to be denominated in Thai baht, and the other half in US dollars.

Mitigation in place for DHPP

In the absence of an approved hedging policy in Bhutan, **the financing agreement may include a provision for establishing a reserve to manage foreign exchange fluctuations**, as has been implemented in previous completed projects.

Same would need to be factored in the financial projections for DHPP.

14. Evaluation of Financing Structure models for DHPP

14.1.1. Evolution of Hydropower financing in Bhutan

Bhutan's hydropower development journey has undergone significant transformations over the decades, reflecting changes in its funding models and strategic partnerships. This evolution can be divided into two distinct phases:

Phase 1: Bilateral Funding Model with India as Primary Partner (1988-2010s)

Indo-Bhutan hydropower cooperation began in 1961 with the signing of the Jaldhaka agreement. The Jaldhaka project is situated on the Indian side of Indo-Bhutan border in West Bengal. The major part of power produced at Jaldhaka hydropower plant was exported to southern Bhutan.

Bhutan's first mega power project, **Chukha Hydropower Project (CHP) with a capacity of 336 MW, was commissioned in 1988**. This was the first project to be developed on a bilateral funding model with India as the partner. Project was funded by India through a mix of Grant (60%) and Concessional Debt (40%), serving as the model for subsequent collaboration with India in Hydropower sector.

Successful implementation of CHP paved way for other bilaterally funded projects including **Kurichhu Hydropower plant (KHP) which became operational in 2002**. KHP has an installed capacity of 60 MW and was funded by grant and concessional debt from India.

Basochhu Hydropower plant (BHP) with a capacity of 64 MW, was the first notable hydropower plant in Bhutan to be developed without India's assistance. Plant became **operational in 2004** and was developed with equity from **Royal Government of Bhutan and support from Austria in form of Grants and Concessional Debts**.

The **Tala Hydropower Plant** with a capacity of 1,020 MW became another milestone in Bhutanese Hydropower landscape. **Commissioned in 2006**, this project followed the successful implementation of the Chukha project and reinforced the Bhutan-India hydropower cooperation model. Tala was funded by a combination of 60% grant and 40% concessional loans from India.

A significant restructuring of the hydropower sector took place in 2007, when the Chhukha, Kurichhu, and Basochhu project companies were consolidated into the newly formed Druk Green Power Corporation (DGPC). This was followed by the integration of the Tala Hydropower Project into DGPC in 2009.

Later in this phase, **commissioned in 2019**, the Mangdechhu Hydroelectric Project (720 MW) was also developed with Indian financial assistance, though with a modified structure of **70% loan and 30% grant**. This shift reflected evolving dynamics in bilateral funding arrangements, as Bhutan took on more financial responsibility while still leveraging Indian backing.

Though there were a few non-Indian-funded exceptions—such as the Basochhu Project (64 MW), supported by Austria—the overwhelming majority of hydropower investment during this phase was India-led, with Bhutan heavily reliant on this funding model for infrastructure expansion.

Phase 2: Diversification (2010s-Now)

By the early 2010s, DGPC actively started diversifying its financing structures to reduce reliance on bilateral sovereign funding. While India remained a vital partner, Bhutan began exploring multilateral institutions, private investors, and export credit agencies to broaden its funding base and reduce dependency.

The most notable milestone in this period was the **Dagachhu Hydroelectric Project (126 MW), commissioned in 2015**. It became Bhutan's first public-private partnership (PPP) in the power sector, co-developed by Druk Green Power Corporation (DGPC), Tata Power (India), and the Asian Development

Bank (ADB). Project was developed at a total project cost of \$237.6 million funded at a debt-equity ratio of 65:35 involving financing from the Multilateral Organisations, Commercial bank, local public stakeholders and Strategic Private Investors.

Commissioned in 2019, Nikachhu Hydropower Project (NHP) marked another advancement in the evolution of the hydropower development models employed by Bhutan. NHP is Bhutan's **first hydropower project to be funded on a fully public model** with 100% of the equity being held by DGPC. Project was developed at a cost of Nu. 16,007 million with major investments of \$120.5 Million from Asian Development Bank comprising of USD 50.5 million loan and grant from the ADB's Asian Development Fund (ADF), and a USD 70 million loan from ADB's ordinary capital resources (OCR) which was extended directly to the project company. Additionally, part debt was funded by a syndicate of Indian Commercial Banks, contributing Nu. 3,274 million to the project. Remaining funds were brought in by DGPC as its equity contribution to the project.

Bhutan has continued diversifying its funding sources for its upcoming projects as well. Similar to Dagachhu Hydropower Plant, **the Khorluchhu Power Plant with envisaged capacity of 600 MW, is being developed under a Public-Private partnership model** with Tata Power Company Limited (TPCL) as the off-taker and strategic investor. Project is envisaged to be developed at a project cost of Nu. 68,992.38 million with Debt-Equity Ratio of 70:30. Equity shareholding will be divided between DGPC and TPCL in the ratio of 60% and 40% respectively. The required debt funding is expected to be sourced from Indian Commercial Banks.

14.1.2. Financing Structure options for development of Dorjilung Hydropower Plant

This section outlines the various financing structure options available to DGPC for the development of the Dorjilung Hydropower Plant (DHPP). Drawing reference from existing DGPC projects and examples from international market, DGPC has three main available structures for development of DHPP (defined below). Each of these models offers distinct strategic advantages while also presenting certain limitations and risks.

The following section presents a comparative analysis of these financing structures, highlighting their key advantages and disadvantages while considering their suitability with respect to development of Dorjilung Hydropower Plant.

▪ Bilateral Sovereign Support

Financing Structure: Financing solely through mix of government loans and grants from bilateral sovereign Partners countries

Precedent Project: Initial hydro projects of DGPC funded by mix of grant and concessionary debt from India and Austria. Notable Projects includes: Chhukha Hydropower Plant, Basochhu Hydropower Plant, Kurichhu Hydropower Plant, Tala Hydropower Plant, Mangdechhu Hydropower Plant.

Key Benefits:

1. **No Drainage of financial resources:** Projects funded primarily through grants or highly concessional loans reduce the fiscal burden on the national budget, enabling Bhutan to pursue capital-intensive projects without upfront financial outflows.
2. **Gaining experience of operating plants without investing:** Early exposure to large-scale hydropower development allowed Bhutan to build technical, managerial, and institutional capabilities without bearing full investment risk.
3. **Access to concessional financial terms:** Bilateral arrangements often include below-market interest rates, longer repayment periods, and grace periods, which make the financing terms more favorable compared to commercial borrowing.

4. **Accelerated Infrastructure Development:** Reliable funding from sovereign partners enabled rapid construction of key hydropower infrastructure in Bhutan which was critical for national revenue generation.

Drawbacks:

1. **Limiting Bhutan's Autonomy in Project development:** Heavy reliance on external sovereign partners can result in reduced control over project design, timelines, and implementation priorities.
2. **Affecting Fiscal Space and Economic Dependence:** Even concessional debt adds to Bhutan's external debt profile, and recurring dependence on a single country may constrain fiscal flexibility and increase economic vulnerability.
3. **Less Advantageous Terms in Power Purchase Agreements (PPAs):** In case of bilateral funding, power is typically exported to the country supporting the project. Such funding linked PPAs can limit Bhutan's negotiation power, often leading to long-term contracts with fixed rates or limited pricing flexibility, especially when the financier is also the buyer.
4. **Restricted Market and Offtake Options:** Tying projects to specific bilateral partners may confine power exports to a single market, limiting Bhutan's ability to diversify energy buyers and optimize revenues.
5. **Limited Private Sector Participation and Innovation:** Sovereign-funded models often exclude private or multilateral stakeholders, reducing opportunities for innovative financing, risk-sharing mechanisms, and technology upgrades.

▪ Public ownership model (Nikachhu Model)

Funding Structure: 100% equity with DGPC with funding support from International Financial Institutions/Multilateral Organizations and Commercial bank.

Precedent Project: Nikachhu Hydropower Plant

Key Benefits:

1. **Ability to control the design and construction of the project:** As the sole owner, DGPC has complete autonomy over design, construction, and operational decisions-enabling optimization of technical solutions and maximization of long-term economic benefits.
2. **Retained Revenues and Strategic Asset Ownership:** 100% public ownership ensures that all project returns attributable to equity stakeholders remain with DGPC while strategically holding the asset within the country, enhancing long-term fiscal returns.
3. **Enhanced Institutional Capacity:** Managing projects in-house builds the project development and financial management capabilities of DGPC, reducing future dependence on external developers.
4. **Potential for Greater Export Value:** With control over power sales, Bhutan can potentially secure more favorable Power Purchase Agreements (PPAs) and explore diversified regional export markets.

Key Drawbacks:

1. **Limited Availability of Public Funds:** Fully publicly owned models require significant upfront capital. This poses challenge for Bhutan as it will have to deplete its own financial resources or depend on multilaterals or commercial bank for debt, both of which can strain fiscal capacity and potentially erode Bhutan's credit profile over the long term. Further, with DGPC or the government bearing the full financial risk, cost overruns, construction delays, or underperformance directly impact national finances.

2. **Reduces the public sector's ability to pursue other opportunities:** Allocating large sums to a single infrastructure project can restrict the government's ability to fund future projects or to support other critical sectors such as health and education.
 3. **Lack of Private Sector Participation:** Excluding the private sector limits innovation, risk-sharing, and operational efficiency. It also means Bhutan misses opportunities to attract private capital and expertise.
 4. **Requirement for Government Guarantees:** In absence of a reputed strategic investor, commercial financiers may require sovereign guarantee from the Royal Government of Bhutan to back the debt. While this may secure better financing terms, it exposes the government to contingent liabilities, potentially affecting Bhutan's credit profile and limiting its ability to raise debt for its future projects.
- **Public-Private Partnership (PPP) Model (Dagachhu Model)**

Funding Structure: Investment from Strategic partners with funding support from International financial Institutions/Multilateral Organizations and Commercial banks.

Precedent Project: Dagachhu Hydropower Plant

Key Benefits:

1. **Eases Fiscal Burden on the Royal Government of Bhutan (RGoB):** By leveraging private or strategic partner equity, the government can undertake large-scale hydropower development without committing significant public funds or sovereign borrowing.
2. **Access to International Expertise and Best Practices:** Strategic partners often bring global experience in engineering, project management, and sustainability standards, improving the overall efficiency and quality of project execution.
3. **Risk Sharing Between Public and Private Stakeholders:** Project risks—financial, construction, operational—are distributed between DGPC, the strategic partner, and lenders, reducing the government's exposure to potential losses.
4. **Improved Access to Commercial Financing:** The presence of a reputable private or international partner improves project bankability, making it easier to attract commercial loans without sovereign guarantee from RGOB.

Key Drawbacks:

1. **Reduced Control for DGPC and RGoB:** Sharing ownership and decision-making authority with a strategic partner means DGPC and RGoB may have less autonomy in key areas such as project design, procurement, tariff setting, and power sales negotiations.
2. **Potential Strategic Dependency:** Overreliance on external partners for capital and technical leadership could constrain Bhutan's strategic maneuverability in the hydropower sector, especially if partnerships influence project priorities or restrict market access.
3. **Profit-Sharing Dilutes Long-Term Returns:** Since profits are shared among stakeholders, Bhutan may forgo a portion of long-term revenue in exchange for upfront capital and risk-sharing, reducing the national benefit from its hydropower resources create a dependency.

For selection of an appropriate financing structure for Dorjilung Hydropower plant, DGPC must weigh multiple factors - ranging from fiscal impact and risk exposure to technical capability and access to commercial financing. The structure should take into consideration both Bhutan's evolution in Hydropower development and the complexity and scale associated with development of Dorjilung Hydropower plant.

14.1.3. Financing Structures for other International Hydropower Projects

Ruzizi III Hydroelectric PowerStation - Africa

Ruzizi III Hydroelectric power station is a 147 MW hydropower plant being developed via a public private partnership (PPP) model in Africa. Plant will be one of the largest infrastructure projects in the region comprising Burundi, the Democratic Republic of Congo, and Rwanda hereafter referred to as "Contracting States".

Ruzizi III Energy Limited (REL), a special purpose vehicle, has been established for design, construction, funding, operation and maintenance of the project over a 25-year concession period post operationalization of the project, commencing in either 2026 or 2027.

Industrial Promotion Services (IPS), the infrastructure and industrial development arm of Aga Khan Fund for Economic Development (AKFED) and SN Power AS (SNP) wholly owned by Scatec ASA shall hold 70% equity in the project company while remaining 30% is being held by the contracting States.

Stakeholders involved:

1. Public Stakeholders: Government of Burundi, the Democratic Republic of Congo, and Rwanda
2. Multilateral Organization: African Development Bank Group (ADFB), World Bank, European Investment Bank (EIB), KfW Development Bank, European Union
3. Strategic Partners: Industrial Promotion Services and SN Power AS
4. Commercial Banks

Funding Structure:

- Project is envisaged to be funded at gearing ratio of 70% debt and 30% equity.
- Equity for the project shall be contributed by the shareholders in their ownership ratio. Strategic Partners i.e. IPS and SN Power will bring in equity from its own sources while the contracting States may use grants and concessionary debt received from development agencies/multilateral organizations to fulfill their equity commitment.
- Debt component will be in the form of blended finance consisting of grants, commercial debt and concessionary debt on lent by contracting States to the project company.

Envisaged funding structure of the project has been enumerated below:

Table 68: Envisaged Funding Structure of the Ruzizi III Hydroelectric PowerStation Project

S.No	Description	% Capital Class	Amount (Million USD)	% Capital Outlay
1	Equity	100%	143	24%
1.1	SN Power/IPS	70%	100	17%
1.2	Three Sovereigns	30%	43	7%
2	Debt	100%	461	76%
2.1	Concessional	80%	369	61%
2.2	Commercial	20%	92	15%
	Total	100%	604	100%

Funding commitment by Development Agencies/Multilateral Organizations

About 80% of Ruzizi III's debt requirements (i.e. USD 369 million) as well as the three Countries' equity contributions (i.e. USD 43 million) will be co-financed by six development partners who have committed

concessional loans and grants to the Project. Their commitments, which will be subject to the completion of due diligence process, are summarised in the table below:

Table 69: Funding Commitments by Development Agencies/ Multilateral Organizations

Organisation	Committed Funds (Million USD)	Funding Type
African Development Bank	138	Concessional Loans and Grants
World bank	150	Concessional Loans
European Investment Bank	108	Concessional Loans and Grants
KfW Development Bank	31	Grants
European Union	35	Grants
African Development Bank	15	Concessional Loans
Total	477	

The envisaged financing structure of the project is outlined in the chart below:

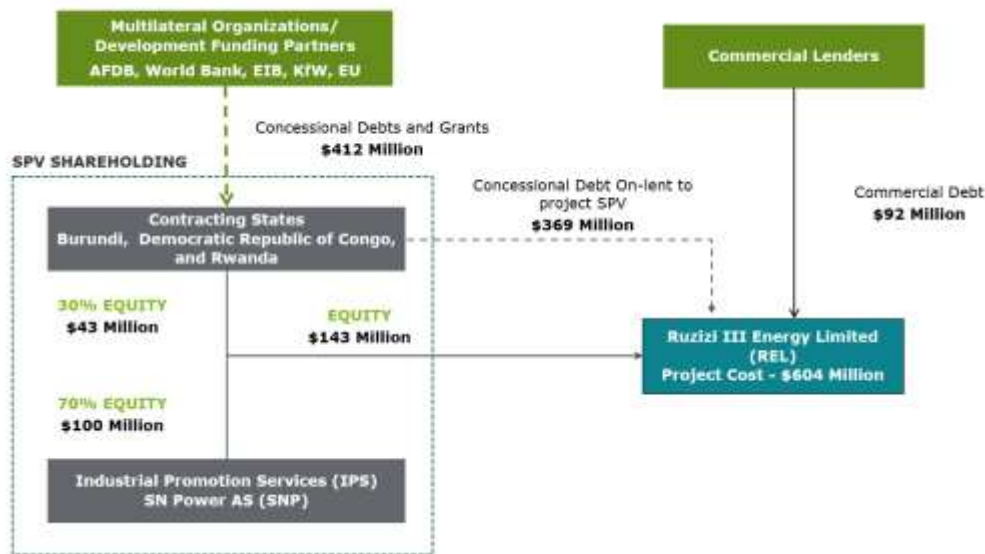


Figure 116: Funding structure of Ruzizi III Hydroelectric PowerStation - Africa

Nam Ngiep 1 Hydropower Project – Laos

Nam Ngiep 1 is a 290-megawatt hydropower project under operation in Bolikhamxay and Xaysomboun provinces of the Lao PDR. Power station generates around 272 MW of electricity for export to Thailand and releases water to a regulating pond where a second dam and power station generates around 18 MW of electric

ity for local use.

The project was implemented at a total cost of \$980 million, with a gearing ratio of 66:34. It was executed under a Public-Private Partnership (PPP) model, supported by strategic equity investments from corporations in Japan, Thailand, and Laos. The debt component was financed through a combination of funding from International Financial Institutions (IFIs) and commercial banks based in Japan and Thailand.

Stakeholders involved:

- **Public Stakeholders:** Lao Holding State Enterprise (a State corporation of Laos)

- **Strategic Investors:** EGAT International (Thailand) and KPIC Netherlands (Japan)
- **Multilateral Organization / IFIs:** Asian Development bank (ADB), Japan Bank for International Cooperation (JBIC)
- Commercial Banks from Thailand and Japan

Table 70: Funding structure of Nam Ngiep 1 Hydropower Project- Laos

S.No	Description	% Capital Class	Amount (Mn USD)	% Capital Outlay
1	Equity	100%	336	34%
1.1	Lao Holding State Enterprise	25%	84	9%
1.2	KPIC Netherlands (Japan)	45%	151.2	15%
1.3	EGAT International (Thailand)	30%	100.8	10%
2	Debt	100%	644	66%
2.1	ADB	23%	145	15%
2.2	JBIC	31%	200	20%
2.3	Thai Commercial Banks	35%	227	23%
2.4	Japanese Commercial Bank	11%	72	7%
	Total Capital Outlay	100%	980	100%

Further, financing structure of the project is outlined in the chart below:

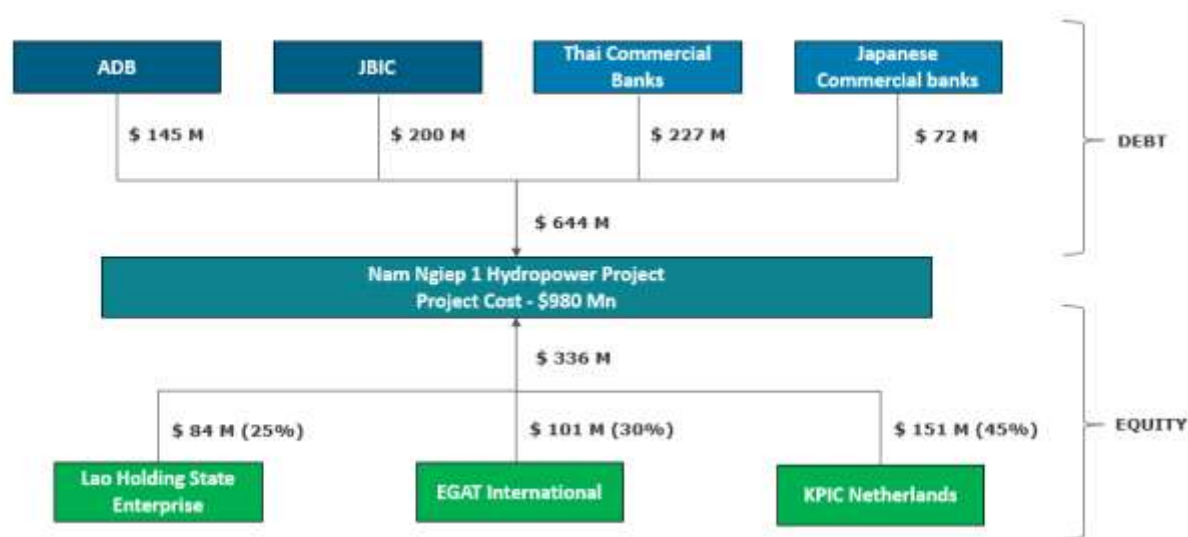


Figure 117: Funding structure of Nam Ngiep 1 Hydropower Project – Laos

Xayaburi Dam Project - Laos

The **Xayaburi Dam** is a run-of-river hydroelectric dam located on the Lower Mekong River in northern Laos. The power plant houses seven 175 MW Kaplan turbine-generators and one 60 MW Kaplan turbine generator. The total installed capacity is 1,285 megawatts commissioned in Oct 2019, with a total annual energy production of 7,406 GWh. Around 95% of the power is supplied to Electricity Generating

Authority of Thailand ("EGAT") pursuant to the Power Purchase Agreement for a period of 25 years from the Commercial Operation Date.

Project was implemented on a PPP basis by Xayaburi Power Company Limited (XPCL) a JV between CK Power Public Company Limited (a Thai Construction Company), Natee Synergy Company Limited, Electricity Generating Public Company Limited (EGCO), PT Sole Company Limited and EDL Generation Public Company (Lao Government Entity). The debt component was financed through a combination of funding from Thai Commercial Banks.

Plant was developed at a total project cost of \$3.8 billion at a gearing ratio of 84:16 with equity infusion from the JV partners of XPCL and debt participation from a consortium of Thai Commercial banks including Bangkok Bank, Kasikorn Bank, Krung Thai Bank, Siam Commercial Bank and other small Thai banks.

Stakeholders involved at time of project implementation:

- **Public Stakeholders:** EDL - Generation Public Company (a State corporation of Laos)
- **Strategic Investors:** CK Power Public Company Limited, Electricity Generating Public Company Limited, PT Sole Company Limited and Natee Synergy Company Limited
- **Commercial Banks** from Thailand

Table 71: Funding structure of Xayaburi Dam Project – Laos

S.No	Description	% Capital Class	Amount (Mn USD)	% Capital Outlay
1	Equity	100%	600	16%
1.1	CK Power Public Company Limited	37.5%	225	6%
1.2	Natee Synergy Company Limited	25%	150	4%
1.3	EDL - Generation Public Company	20%	120	3%
1.4	Electricity Generating Public Company Limited	12.5%	75	2%
1.5	PT Sole Company Limited	5%	30	1%
2	Debt	100%	3,200	84%
2.1	Syndicate of Thai Commercial banks	100%	3,200	84%
	Total Capital Outlay	100%	3,800	100%

Further, financing structure of the project is outlined in the chart below:

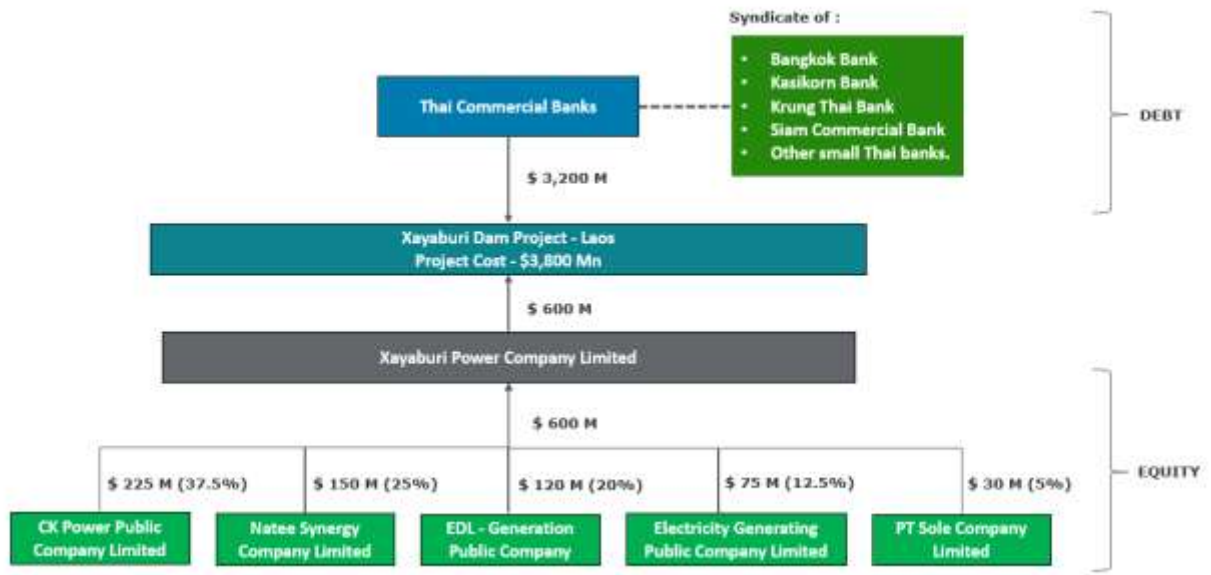


Figure 118: Funding structure of Xayaburi Dam Project - Laos

14.1.1. Key factors that should be considered while selecting the optimal structure:

1. **Fiscal Impact and Financing Flexibility:** Selected structure should minimize strain on Bhutan's public finance and reduce the need for large sovereign borrowing or guarantees.
2. **Risk Distribution:** Given the scale DHPP, financing structure shall allow construction, financing and operational risks to be distributed among multiple stakeholders which can help safeguard DGPC and the government from concentrated exposure. Further, hydropower projects are prone to cost and time overruns and it would be essential to have a reputed strategic partner / stakeholder that has capacity to bring in additional proportionate capital as the need arises in future.
3. **Access to Technical Expertise:** Bringing in experienced partners can enhance project design, implementation, and management. This becomes increasingly important for large scale projects like DHPP.
4. **Autonomy and Long-term value:** Funding models such as PPP may involve revenue-sharing or shared governance. In such cases, it is important to maintain strategic oversight to safeguard and enhance the national interest. Bhutan's Sustainable Hydropower Development Policy 2020 requires any RGOB entity i.e. DGPC in this case, to hold minimum of 51% stake in a PPP structure. **This stipulation makes sure that national interests remain central in decision-making,** even when external partners are involved.
5. **Bankability of the project:** Selected funding structure should be likely to improve investor confidence which would help in accelerating the financial closure without need of RGOB guarantee.

Suitability of different financing structures for DHPP's development considering these:

Table 72: Suitability of Different Financing Structures for DHPP's Development

Financial Objectives	Suitability for DHPP (High/Low/Neutral)		
	Bilateral Sovereign Support	Public Ownership	PPP
Fiscal Impact	High	Low	High
Risk Distribution	High	Low	High
Access to Technical Expertise	Low	Low	High
Autonomy and Long-term value	Low	High	Neutral
Bankability of the project	Not applicable	Neutral	High
Recommendations	Not Suitable	Not Suitable	Suitable

Conclusion:

After evaluating the different funding structures, each model has its strengths and limitations. **Bilateral Sovereign Support**, while offering lower immediate fiscal impact due to grants and concessional loans, can lead to long-term dependency on a single partner and limits strategic flexibility. **Public Ownership** provides greater autonomy and control over the project, but it requires substantial government resources and may lack access to the technical expertise needed for optimal project execution. In contrast, a **Public-Private Partnership (PPP)** offers a balanced approach, facilitating risk-sharing, improving project bankability, and providing access to technical expertise and private sector efficiency. Therefore, a PPP model may be considered as a suitable option for the Dorjilung Hydropower Plant, as it enables Bhutan to leverage external investment while retaining oversight and aligning with long-term development goals.

15. Next Steps

The team has initiated the development of the next set of deliverables after conducting stakeholder consultation and incorporating necessary feedback from them. These deliverables have been clearly defined to ensure alignment with project goals and stakeholder expectations, with a continued emphasis on quality, precision, and adherence to established timelines. The deliverables have been enlisted below:

Financial Analysis Report

- Constructing a financial model for the project based on proposed financing plan
- Identifying suitable financing plan
- Prepare Scenario analysis for different offtake arrangement
- Assessing level of royalties and tax that may accrue to Bhutan, as well as overall financing return to project participant.
- Assessing the availability of public, donor financing commercial debt financing
- Examining risk mitigation tools for obtaining commercial funding
- Proposing an implementation structure for the project
- Analyzing the interplay between financing arrangement, offtake arrangement and tariffs
- Assessing the viability and preparing holistic roadmap for DGPC financing
- Propose a work program to conduct a preliminary market soundness assessment

Information memorandum & RFP

- Determining indicative terms to ensure optimal financing.
- Preparation of Information Memorandum document



Deloitte refers to one or more of Deloitte Touche Tohmatsu Limited, a UK private company limited by guarantee ("DTTL"), its network of member firms, and their related entities. DTTL and each of its member firms are legally separate and independent entities. DTTL (also referred to as "Deloitte Global") does not provide services to clients. Please see www.deloitte.com/about for a more detailed description of DTTL and its member firms.

This material has been prepared by Deloitte Touche Tohmatsu India LLP ("DTTILLP"), a member of Deloitte Touche Tohmatsu Limited, on a specific request from you and contains proprietary and confidential information. This material may contain information sourced from publicly available information or other third-party sources. DTTILLP does not independently verify any such sources and is not responsible for any loss whatsoever caused due to reliance placed on information sourced from such sources. The information contained in this material is intended solely for you. Any disclosure, copying or further distribution of this material or its contents is strictly prohibited.

Nothing in this material creates any contractual relationship between DTTILLP and you. Any mutually binding legal obligations or rights may only be created between you and DTTILLP upon execution of a legally binding contract. By using this material and any information contained in it, the user accepts this entire notice and terms of use.

©2025 Deloitte Touche Tohmatsu India LLP. Member of Deloitte Touche Tohmatsu Limited